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PROSPECTUS

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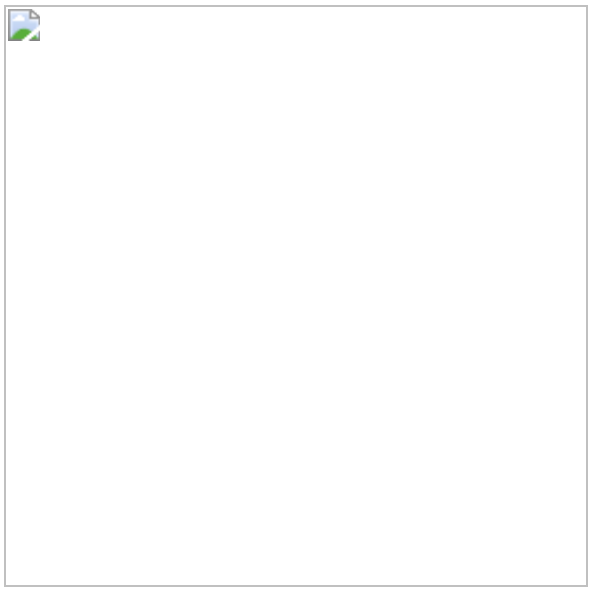

Cadrenal Therapeutics, Inc.

1,400,000 Shares

Common Stock

This is the initial public offering of our common stock and no public market currently exists for our common stock. We are offering 1,400,000 shares of our common stock, par value $0.001 per share ("common stock"), at the initial public offering price of $5.00 per share. Our common stock has been approved for listing on The Nasdaq Capital Market ("Nasdaq") under the symbol "CVKD" and will begin trading on January 20, 2023.

In addition, the Selling Stockholders (as defined herein) are offering 1,704,783 shares of common stock to be sold pursuant to a separate resale prospectus. We will not receive any proceeds from the sale of the common stock to be sold by the Selling Stockholders.

We are an "emerging growth company" as defined under the federal securities laws and, as such, will be subject to reduced public company reporting requirements. See "Prospectus Summary — Implications of Being an Emerging Growth Company" for additional information.

After the completion of this offering, our Chief Executive Officer, Quang Pham, and entities controlled by him, will own approximately 53.93% (or 52.98% if the underwriters exercise their over-allotment option in full) of the outstanding voting power of our common stock. As a result, we will be a "controlled company" within the meaning of the corporate governance standards of Nasdaq. We currently do not intend to avail ourselves of the controlled company exemption under the Nasdaq corporate governance standards. See "Management-Controlled Company" and "Principal Stockholders."

Investing in our common stock involves a high degree of risk. Please read "Risk Factors" beginning on page 14 of this prospectus.

Neither the Securities and Exchange Commission nor any state securities commission has approved or disapproved of these securities or determined if this prospectus is truthful or complete. Any representation to the contrary is a criminal offense.

	Per Share	Total

Public offering price	$	5.00	$	7,000,000
Underwriting discounts and commissions[1]	$	0.35	$	490,000
Proceeds, before expenses, to Cadrenal	$	4.65	$	6,510,000

(1) We have agreed to pay the underwriters a cash fee equal to seven percent (7.0%) of the aggregate gross proceeds from the sale of the common stock. The underwriters will also be entitled to a one percent (1%) non-accountable expense allowance, which amount is not included above. We have also agreed to reimburse the underwriters for certain expenses incurred by them and the underwriters will receive compensation in addition to underwriting discounts and commissions. See "Underwriting" beginning on page 129 of this prospectus for more information about the compensation payable to the underwriters, including a warrant and reimbursable expenses.

Delivery of the shares is expected to be made on or about January 24, 2023. We have granted the underwriters an option for a period of 45 days to purchase up to an additional 210,000 shares of common stock. If the underwriters exercise the option in full, the total underwriting discounts and commissions payable by us will be $563,500, and the total proceeds to us, before expenses, will be $7,468,500.

Boustead Securities, LLC **Sutter Securities, Inc.**

Revere Securities LLC

Prospectus dated January 19, 2023

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We have not, and the underwriters have not, authorized anyone to provide any information to you or to make any representations other than those contained in this prospectus, any amendment or supplement to this prospectus, or in any free writing prospectuses prepared by or on behalf of us or to which we have referred you. We take no responsibility for, and can provide no assurance as to the reliability of, any other information that others may give you. This prospectus is an offer to sell only the shares offered hereby, and only under circumstances and in jurisdictions where offers and sales are permitted. The information contained in this prospectus is accurate only as of the date on the front cover of this prospectus regardless of the time of delivery of this prospectus or of any sale of common stock. Our business, financial condition, results of operations and prospects may have changed since the date on the front cover of this prospectus.

Neither we nor the underwriters have taken any action to permit this offering or possession or distribution of this prospectus in any jurisdiction where action for that purpose is required, other than in the United States. Persons in jurisdictions outside of the United States who come into possession of this prospectus must inform themselves about and observe any restrictions relating to this offering and the distribution of this prospectus.

For investors outside the United States: We have not, and the underwriters have not, done anything that would permit this offering or possession or distribution of this prospectus or any applicable free writing prospectus in any jurisdiction where action for that purpose is required, other than in the United States.

Persons outside the United States who come into possession of this prospectus and any applicable free writing prospectus must inform themselves, and observe any restrictions relating to, the offering of the common stock and the distribution of this prospectus outside the United States.

PUBLIC OFFERING PROSPECTUS SUMMARY

This summary highlights selected information contained elsewhere in this prospectus and does not contain all of the information that you should consider in making your investment decision. Before deciding to invest in our common stock, you should read this entire prospectus carefully, including the sections of this prospectus entitled "Risk Factors" and "Management's Discussion and Analysis of Financial Condition and Results of Operations" and our financial statements and the related notes included elsewhere in this prospectus.

Unless the context otherwise requires, references in this prospectus to the "Company," "Cadrenal," "Cadrenal Therapeutics," "we," "us" and "our" refer to Cadrenal Therapeutics, Inc.

This prospectus contains trade names, trademarks and service marks of other companies that are the property of their respective owners. Solely for convenience, trademarks and tradenames referred to in this prospectus appear without the ® and ™ symbols, but those references are not intended to indicate that we will not assert, to the fullest extent under applicable law, our rights, or that the applicable owners will not assert their rights, to these trademarks and trade names. We do not intend our use or display of other companies' trade names, trademarks or service marks to imply a relationship with, or endorsement or sponsorship by us of, these other companies.

Our website address is *www.cadrenal.com*. The information contained on, or that can be accessed through, our website is not incorporated by reference into this prospectus, and you should not consider any information contained on, or that can be accessed through, our website as part of this prospectus or in deciding whether to purchase our common stock.

Our Business

We are focused on developing tecarfarin, a novel therapy with orphan drug indication, designed for the prevention of systemic thromboembolism (blood clots) of cardiac origin in patients with end-stage renal disease, or ESRD, and atrial fibrillation (irregular heartbeat), or AFib. We secured the rights to tecarfarin on April 1, 2022 via an asset purchase agreement from HESP LLC, a wholly owned subsidiary of Horizon Technology Finance Corporation. HESP LLC acquired the assets of Espero BioPharma, Inc., or Espero, including tecarfarin, in an assignment for the benefit of creditors in which the creditor, Horizon Technology Finance Corporation and Horizon Credit II LLC (collectively, Horizon), a secured lender of Espero, designated HESP LLC as the assignee of Espero's assets.

Tecarfarin is an anticoagulant that uses a drug design process which targets a different pathway than the most commonly prescribed drugs used in the treatment of thrombosis and AFib. Tecarfarin has been evaluated in eleven (11) human clinical trials conducted by its previous owners and other third parties in over 1,003 individuals (269 patients were treated for at least six months and 129 patients were treated for one year or more). In Phase 1, Phase 2 and Phase 2/3 clinical trials, tecarfarin has generally been well-tolerated in both healthy adult subjects and patients with chronic kidney disease, or CKD. In the Phase 2/3 trial, EMBRACE-AC, the largest tecarfarin trial with 607 patients having completed it, only 1.6% of the blinded tecarfarin subjects suffered from major bleeding and there were no thrombotic events. Five patients died during the trial, but only one death due to intracerebral hemorrhage was considered to be possibly related to the tecarfarin.

In 2019, the United States Food and Drug Administration, or FDA, provided input on the Phase 3 trial design for tecarfarin, which was submitted by Espero, the previous owner of tecarfarin. We intend to submit our Phase 3 trial design to the FDA using the same protocol that was submitted by Espero. Assuming the FDA accepts our Phase 3 trial design, we intend to commence the Phase 3 pivotal trial in the second half of 2023. However, there can be no assurance that the trial design will be accepted by the FDA. We are pursuing regulatory approval of tecarfarin as an individual treatment, although we might evaluate, in consultation with the FDA, other potential uses in the future.

In March 2019, the FDA granted orphan drug designation, or ODD, for tecarfarin for the prevention of systemic thromboembolism of cardiac origin in patients with ESRD and AFib. The FDA grants ODD status to drugs that are intended for the treatment, diagnosis, or prevention of rare diseases or conditions, which are defined as a disease or condition that affects fewer than 200,000 people in the U.S. The ODD program

provides a drug developer with certain benefits and incentives, including a seven-year period of U.S. marketing exclusivity from the date of marketing authorization, waiver of FDA user fees, and tax credits for clinical research. The granting of an orphan drug designation

does not alter the FDA's regulatory requirements to establish safety and effectiveness of a drug through adequate and well-controlled studies to support approval and commercialization. Furthermore, orphan drug designation does not indicate or guarantee FDA approval of the New Drug Application, or NDA, and we might not receive exclusivity.

Tecarfarin was developed by researchers using a retrometabolic drug design process which targets a different metabolic pathway than the most commonly prescribed drugs for the treatment of thrombosis and AFib. "Drug metabolism" refers to the process by which a drug is inactivated by the body and rendered easier to eliminate or to be cleared by the body. Most approved drugs, including warfarin, the only FDA-approved Vitamin K antagonists, or VKAs, which is a prescribed drug for the treatment of thrombosis, are metabolized in the liver through a pathway known as the Cytochrome CYP450 system, or CYP450, by the enzymes known as CYP2C9 and CYP3A4. By using a different metabolic pathway, tecarfarin eliminates or minimizes the CYP450 metabolism in the liver. Patients taking multiple medications that interact with CYP2C9, or CYP3A4 or those with impaired kidney function, can experience an overload in the pathway, creating a bottleneck that often leads to insufficient clearance, which results in a toxic build-up of one or more drugs. In some instances, patients taking multiple medications metabolized by the same CYP450 pathway may experience decreased efficacy of one or more of the medications due to rapid metabolism or increased drug effect and/or toxicity due to enzyme induction. Patient-specific genetic differences can also hinder drug clearance in the CYP450 pathway. Our product candidate tecarfarin was designed to follow a metabolic pathway distinct from the CYP450 pathway and is metabolized by both CYP450 and non-CYP450 pathways. We believe this may allow elimination by large capacity and non-saturable tissue esterase pathways that exist throughout the body rather than just in the liver.

Tecarfarin is an orphan designated, vitamin K antagonist, oral, once-daily anticoagulant in the same drug class as warfarin designed for use in patients requiring chronic VKA anticoagulation, to prevent systemic thromboembolism of cardiac origin in patients with ESRD and AFib. The prevailing treatment for thrombosis is with an oral anticoagulant, either a VKA, like warfarin, or non-vitamin K oral anticoagulant ("NOAC"). VKAs block the production of vitamin K-dependent blood clotting factors, such that the blood is "thinned," preventing clots, while NOACs directly block the activity of certain of these clotting factors. Tecarfarin, like warfarin, is a VKA.

Vitamin K epoxide Reductase Complex subunit 1 (VKORC1) is a significant enzyme for effective clotting. VKORC1 reduces vitamin K epoxide to its active form (Vitamin K), which is the rate-limiting step in the physiological process of vitamin K recycling. Vitamin K serves as a cofactor for normal function of several clotting/anticoagulation factors including Factors II, VII, IX and X and Proteins C, S, and Z. VKORC1 genetic deficiencies result in increased sensitivity to VKAs, which results in an increase in the risk of significant hemorrhaging. We believe tecarfarin has similar potency for VKORC1 inhibition as warfarin, but it is an investigational new drug, and we must demonstrate it is safe and effective for its proposed indication.

AFib is the most common arrhythmia, with its incidence and prevalence increasing over the last 20 years. AFib is associated with an approximate five-fold increased risk of stroke. The risk of developing AFib increases in patients with CKD. According to 2021 estimates by the Centers for Disease Control and Prevention, or CDC, approximately 15% of the U.S. adult population, or 37 million people, have CKD. An estimated 0.4% of people in the U.S. suffer from Stage 4 CKD and 0.1% of people in the U.S. have ESRD.

Patients with ESRD and AFib represent a spectrum of disorders involving both the heart and kidneys (known as cardiorenal syndrome or CRS) in which acute or chronic dysfunction in one organ may induce acute or chronic dysfunction in the other organ. These patients have typically been excluded from randomized clinical trials because the approved therapies for AFib have metabolic profiles that may increase drug exposures thereby increasing the known risks and challenges in managing these patients. The presence of either CKD or AFib, increases the risk of serious thromboembolic adverse clinical outcomes, such as stroke and death. Antithrombotic therapy is typically recommended to decrease this risk in AFib patients, but there are no approved treatment options for patients with ESRD and AFib. Warfarin may cause substantial harm in these patients. Low-dose apixaban (Eliquis) was approved by the FDA for use in ESRD patients on hemodialysis based upon limited pharmacokinetic data by 8 subjects, despite that randomized trials to date of apixaban

versus warfarin for AFib excluded patients with severe and end-stage kidney disease. The RENAL-AF (Trial to Evaluate Anticoagulation Therapy in Hemodialysis Patients With Atrial Fibrillation) was terminated early in 2019 by its sponsor.

There are more than 809,000 Americans with ESRD, with approximately 70% on dialysis, according to the United States Renal Data System. Approximately 150,000 ESRD patients also have AFib. AFib nearly doubles the anticipated mortality and increases the stroke risk by approximately five-fold in these patients. There is evidence that AFib is an independent risk factor for developing ESRD in CKD patients. Both diseases share common risk factors including hypertension, diabetes, vascular disease, and advancing age. Cardiovascular disease contributes to more than half of all deaths among patients with ESRD. According to the Annual Data Report published by the United States Renal Data System, total Medicare spending for patients with ESRD reached $51 billion in 2019, accounting for approximately 7% of the Medicare paid claims costs.

We have licensed out the rights to tecarfarin for several Asian markets including China, to Lee's Pharmaceutical Holdings Limited, an integrated research-driven and market-oriented biopharmaceutical publicly listed company based in Hong Kong with over 25 years' experience in the pharmaceutical industry in China. Lee's Pharmaceutical Holdings Limited is developing tecarfarin as an anti-thrombotic for patients with mechanical heart valves. In 2020 and 2021, Lee's Pharmaceutical Holdings Limited completed two Phase 1 studies in China and Hong Kong and is currently preparing for its Phase 2 trial.

As more fully set forth in our risk factors, we are a clinical development biopharmaceutical company with a limited operating history. We have a history of operating losses and expect to continue to incur substantial losses for the foreseeable future. Our independent registered public accounting firm has expressed substantial doubt about our ability to continue as a going concern. Our cash and the proceeds of this offering will only fund our operations for a limited time. The proceeds from this offering will be insufficient to allow us to fully fund our planned pivotal Phase 3 clinical trial. We will need to raise additional capital for the initiation of enrollment of patients and completion of the planned pivotal Phase 3 trial.

With respect to tecarfarin, we have two issued U.S. patents directed to tecarfarin. While the patents currently expire in 2024, we expect to seek extensions of patent terms. In the United States, the Drug Price Competition and Patent Term Restoration Act of 1984 permits a patent term extension of up to five years beyond the normal expiration of the patent, which is limited to the approved indication (or any additional indications approved during the period of extension). We also intend to seek exclusivity for our proprietary product candidates through market and data exclusivity granted by regulatory agencies in the United States and other countries. Further, as discussed above, the ODD program provides a drug developer with certain benefits and incentives, including a seven-year period of U.S. marketing exclusivity from the date of marketing authorization.

While we have engaged intellectual property counsel to assist in protecting our patent ownership rights, to date, we have not had intellectual property counsel conduct a freedom to operate analysis regarding our tecarfarin product. As a result, we cannot be certain that we will not be exposed to third party legal claims, liabilities and/or litigation actions when we seek to develop, make and market products using our tecarfarin technology.

Clinical Trials

Tecarfarin has been evaluated in 11 human clinical trials in over 1,003 individuals which includes eight Phase 1 trials, two Phase 2 trials and one Phase 2/3 trial evaluating the efficacy and safety of tecarfarin.

In a Phase 2/3 randomized and blinded trial sponsored by ARYx Therapeutics, Inc. in 2008, 607 patients with indications for chronic anticoagulation were treated with either tecarfarin or warfarin. The Time in Therapeutic Range, or TTR, with tecarfarin was similar to that with well-managed warfarin and tecarfarin appeared to have a favorable safety profile and be well tolerated with only 1.6% of the blinded tecarfarin subjects suffering from major bleeding and no thrombotic events. When thrombotic and major bleeding events during the blinded period were combined, a numerical imbalance favoring tecarfarin over warfarin was seen (warfarin 11 subjects, 3.6%; tecarfarin 5 subjects, 1.6%). The trial however did not meet its primary endpoint as superiority of tecarfarin over warfarin as measured by TTR was not demonstrated.

In a subsequent Phase 1 study with 23 patients with CKD sponsored by Armetheon, Inc. in 2016, the metabolism of warfarin was inhibited, but not tecarfarin. The safety of repeated dosing of tecarfarin in CKD

patients remained unknown. However, if the pharmacokinetic findings of this single-dose study are present with repeated dosing, tecarfarin may lead to dosing that is more predictable than warfarin in CKD patients who require anticoagulation therapy.

Recent Events

Asset Purchase Agreement

On April 1, 2022, we entered into an asset purchase agreement with HESP LLC, the assignee of tecarfarin and related assets (the "Asset Purchase Agreement"). Pursuant to the terms of the Asset Purchase Agreement, on April 1, 2022 we acquired all of the assets of HESP LLC, including all intellectual property and other right related to tecarfarin, the tecarfarin IND 77041, all rights under the license, development and commercialization agreement dated as of September 16, 2015 by and between Armetheon, Inc. ("Armetheon") (which was later assigned to Espero BioPharma, Inc. ("Espero")) and China Cardiovascular Focus Ltd, an affiliate of Lee's Pharmaceutical Holdings Limited ("Lee's Pharmaceutical"), relating to tecarfarin and related trademarks. In consideration of the purchase of the assets we paid HESP LLC $100,000 on the closing date and agreed to pay an additional $100,000 60 days after the closing date. On June 1, 2022, we made the second $100,000 payment. In addition, we agreed to pay HESP LLC the following development payments.

Development Milestones		Milestone Payments
Completion of enrollment of Lee's Pharmaceutical Phase 3 clinical trial	$	250,000
First MAA submitted in the People's Republic of China	$	350,000
First Commercial Sale to a Third Party	$	1,200,000

As additional consideration, the Company agreed to pay the following amounts, up to $2,000,000, upon each financing milestone as follows: (i) 35% of any proceeds received from any licensing or partnering revenue; and (ii) out of the net proceeds of our initial public offering. The aggregate payments under the development milestone payments and financing milestone payments shall not exceed $2,000,000.

On August 18, 2022, we entered into an amendment (the "Amendment") to the Asset Purchase Agreement, whereby in lieu of the $1,800,000 cash payment that would have been due to HESP LLC pursuant to the Asset Purchase Agreement as a result of this offering, HESP LLC agreed to accept shares of our common stock, such number of shares to be calculated based upon a 40% discount to the price of our common stock sold in this offering.

Private Placements

On March 1, 2022, we issued a convertible promissory note (the "March 2022 Note") in the amount of $500,000 to John Murphy, a director nominee, which bears interest at 5% and matures on March 1, 2025. The note, as amended in December 2022, is convertible into shares of our common stock upon the consummation of this offering at a conversion price equal to $1.00 per share.

On June 13, 2022, we issued a convertible promissory note in the amount of $50,000 (the "June 2022 Note"). The June 2022 Note bears interest at 6% and matures on June 13, 2025 ("June Note Maturity Date"). The principal amount due under the June 2022 Note (and, at our option, any accrued but unpaid interest under the June 2022 Note) will be automatically converted upon the consummation of this offering into shares of common stock at a conversion price, as amended in December 2022, equal to $1.00 per share.

On July 16, 2022, we closed the first tranche of a private placement (the "Private Placement") pursuant to which we issued convertible promissory notes in the principal amount of $450,000 (the "First Tranche Private Placement Notes") and received gross proceeds of $450,000. The private placement offering was for up to $2,000,000 of convertible promissory notes and was to terminate upon the sale of all $2,000,000 of convertible promissory notes or July 31, 2022 subject to extension until August 31, 2022, in the sole discretion of the placement agent and us. We, together with the placement agent, determined to extend the Private Placement to August 31, 2022. On August 22, 2022, we issued a convertible promissory note pursuant to the Private Placement in the principal amount of $50,000 (the "Second Tranche Private Placement Note") and received gross proceeds of $50,000. The Private Placement was further extended to September 30, 2022, after obtaining the consent of the investors in the Private Placement to date. On September 13, 2022, we issued convertible promissory notes in the aggregate principal amount of $75,000 (together with the First

Tranche Private Placement Notes and the Second Tranche Private Placement Note the "Private Placement Notes") and received gross proceeds of $75,000. The Private Placement Notes bear interest at 6% and mature on the third anniversary of the date of the last closing of the Private Placement Notes, i.e. September 13, 2025. The principal amount due under the Private Placement Notes (and, at our option, any accrued but unpaid interest under the Private Placement Notes) will be automatically converted into shares of our common stock upon the consummation of this offering. The Private Placement Notes are convertible into shares of common stock at a conversion price, as adjusted,

equal to $1.00 per share. The holders of the Private Placement Notes entered into an Investor Rights Agreement that provides for "drag along" and "tag along" rights and permits the holders to participate in subsequent securities offerings, including this offering, in a percentage amount of such securities offering equal to the percentage of the amount invested by such holder in the Private Placement Notes as compared to the amount invested by all holders in the Private Placement Notes; provided, however, that in no event shall the dollar amount of the participation right of any holder exceed 100% of original principal amount of the Private Placement Notes purchased by such holder.

Boustead Securities, LLC acted as the placement agent for the Private Placement and received a commission equal to 7% of the gross proceeds received by us from the sale of the Private Placement Notes, a non-accountable expense allowance equal to 1% of the gross proceeds received by us from the sale of the Private Placement Notes and five-year warrants, as amended, to purchase an aggregate of 11,500 shares of our common stock at an exercise price equal to 60% of the initial public offering price per share.

On November 30, 2022, we closed a $250,000 note offering (the "November Private Placement") pursuant to which we sold to two accredited investors, both of whom had a pre-existing relationship with us and Boustead Securities, LLC, units consisting of (i) notes in the aggregate principal amount of $250,000, which bear interest at the rate of 10%, repayable at the time of the completion of this offering (the "November Notes") and (ii) warrants to purchase up to an aggregate of 250,000 shares of common stock exercisable at $1.00 per share which are automatically exercisable into shares of our common stock upon the consummation of this offering (the "November Warrants"). The principal amount of the November Notes will be repaid upon the consummation of this offering out of the proceeds of the automatic exercise of the November Warrants, with any interest owed on the November Notes to be payable in cash.

Boustead Securities, LLC acted as the placement agent for the November Private Placement and received a commission equal to 7% of the gross proceeds received by us from the sale of the November Notes, a non-accountable expense allowance equal to 1% of the gross proceeds received by us from the sale of the November Notes and five-year warrants to purchase shares of our common stock in an amount equal to 6% of the shares of common stock underlying the securities issued in the November Private Placement at an exercise price of $1.00 per share.

Amended and Restated Certificate of Incorporation and Amended and Restated Bylaws

On December 5, 2022, our Board of Directors approved an amendment to our certificate of incorporation to increase our authorized capital stock to 82,500,000 shares, consisting of 75,000,000 shares of common stock, par value $0.001 per share, and 7,500,000 shares of preferred stock, par value $0.001 per share, which was also approved by a majority of our shareholders on the same day. On December 5, 2022, we filed an Amended and Restated Certificate of Incorporation with the Secretary of State of the State of Delaware, which became effective on December 5, 2022, at which time our authorized capital stock consisted of 75,000,000 shares of common stock, par value $0.001 per share, and 7,500,000 shares of preferred stock, par value $0.001 per share.

On December 5, 2022, our Board of Directors approved the Amended and Restated Bylaws, which became effective as of December 5, 2022 upon the filing and effectiveness of our Amended and Restated Certificate of Incorporation.

Fast Track Designation

On January 13, 2023, the FDA designated as a Fast Track development program the investigation of tecarfarin for the prevention of systemic thromboembolism of cardiac origin in patients with ESRD and AFib. Fast Track is a process designed to facilitate the development and expedite the review of drugs to treat serious conditions and fill an unmet medical need.

Our Strategy

Our goal is to build a biopharmaceutical company with a foundation of product candidates that significantly advance patient care in cardiorenal diseases. Key elements of our strategy are as follows:

- Complete the clinical development of and seek FDA approval for tecarfarin. We intend to initiate our pivotal Phase 3 clinical trial in the fourth quarter of 2023, subject to completion of this offering and funding from additional financings, which we believe will be our remaining pivotal trial based upon the latest feedback that the prior owner of tecarfarin had with the FDA in 2019. ACTOR AF: Anti-Coagulation with Tecarfarin on Outcomes in Renal disease and Atrial Fibrillation is designed as a Phase 3, 492-patient, Randomized, Double-Blind, Placebo-Controlled Outcomes Study of Tecarfarin vs. Placebo in Subjects with End-Stage

Renal Disease and Atrial Fibrillation not Currently Treated with Chronic Oral Anticoagulation. If we are able to complete the Phase 3 clinical trial and we are able to obtain FDA approval of our NDA, we believe tecarfarin can be an alternative treatment for patients who are currently without an approved treatment.

- If we obtain FDA approval of our NDA for our first indication, we intend to seek to expand the label for tecarfarin through a supplemental NDA. We intend to explore the full potential of tecarfarin in additional indications, including the treatment of patients with Left Ventricular Assist Devices (LVADs) and Mechanical Heart Valves (MHVs) who require chronic anticoagulation. A LVAD is an implantable pump attached to the heart, connecting the apex of the left ventricle to the ascending aorta. LVADs are intended to treat patients suffering from advanced heart failure. MHVs are a type of prosthetic heart valve constructed of durable materials such as titanium or carbon that can potentially last a patient's lifetime. Despite their high durability, MHVs are associated with an increased risk of developing blood clots that may interrupt blood flow and travel to various organs (thromboembolism). Oral anticoagulation therapy with a vitamin K antagonist (warfarin) continues to be the recommended treatment option for patients implanted with an MHV based on the 2014 American Heart Association/American College of Cardiology Guidelines.
- We intend to partner and/or in-license and/or acquire clinical and pre-clinical stage cardiorenal products to augment our current pipeline which consists of one investigational product.
- Create a commercial infrastructure for our product candidates. If tecarfarin is approved by the FDA, we intend to expand our commercial infrastructure and hire and train a focused and dedicated specialty cardiorenal salesforce which we believe can efficiently cover the top prescribing physicians and approximately 3,000 anticoagulation clinics in the U.S., which presently monitor patients on warfarin. If approved, we intend to initially target our tecarfarin commercialization efforts at the two largest dialysis providers in the U.S., DaVita Kidney Care and Fresenius Medical Care. DaVita maintains more than 2,800 brick-and-mortar dialysis centers in the U.S., while Fresenius' North America subsidiary owns more than 2,500 on the continent — together accounting for more than 70% of the entire U.S. market.

Our Resale Offering

Certain of our shareholders will be selling through a separate prospectus (the "Resale Prospectus") 1,704,783 shares of common stock consisting of: (i) 600,000 shares of common stock to be issued on the effective date of this offering pursuant to the terms of the Amendment to Asset Purchase Agreement, as described in more detail herein; (ii) 250,000 shares of common stock issuable upon the exercise of the November Warrants, which November Warrants will be exercised upon consummation of the initial public offering; (iii) 575,908 shares of common stock to be issued upon consummation of the initial public offering pursuant to the conversion of the June 2022 Note and the Private Placement Notes held by non-affiliates; and (iv) 278,875 shares of common stock, as set forth in the Resale Prospectus. We will not receive any proceeds from the sales by the Selling Stockholders of the securities set forth in the Resale Prospectus.

The Resale Prospectus is substantively identical to the Public Offering Prospectus, except for the following principal points:

- they contain different outside and inside front covers and back covers;
- they contain different Offering sections in the Prospectus Summary section beginning on page 1;
- they contain different Use of Proceeds sections on page 54;
- a "Selling Stockholder" section is included in the Resale Prospectus;
- a "Selling Stockholder Plan of Distribution" is included in the Resale Prospectus; and
- the "Legal Matters" section in the Resale Prospectus on page Alt-18 deletes the reference to counsel for the underwriter.

Summary of Risks Associated with Our Business

Our business is subject to numerous risks and uncertainties, including those highlighted in the section entitled "Risk Factors" immediately following this prospectus summary. These risks include, among others, the following:

Risks Related to Our Financial Position and Need for Capital

- We have a limited operating history;
- We have a history of operating losses and expect to continue to incur substantial losses;
- Our independent registered public accounting firm has expressed substantial doubt about our ability to continue as a going concern;
- Our cash and the proceeds of this offering will only fund our operations for a limited time;
- We will need to raise additional capital.

Risks Related to Product Development, Regulatory Approval, Manufacturing and Commercialization

- Our business is dependent upon the success of tecarfarin, which requires additional clinical testing before we can seek regulatory approval;
- All of our current data for tecarfarin are the results of clinical trials conducted by third parties;
- We cannot be assured that clinical trials conducted by third parties were in compliance with applicable laws, rules and regulations;
- Our development efforts may not generate data sufficient to support regulatory approval;
- Even if we complete our clinical trials, we may not receive regulatory approval for tecarfarin;
- Fast Track designation by the FDA does not assure FDA approval of our product candidate;
- Even if we obtain regulatory approval, we may face future development and regulatory difficulties;
- Clinical trials are very expensive, time-consuming and difficult to design and implement.
- We may experience delays in the enrollment of patients in any or all of our clinical trials;
- Even if approved, tecarfarin may not have labeling that allows us to successfully commercialize it;
- If our product candidate is approved, our success depends on our commercialization efforts;
- Tecarfarin may fail to achieve the degree of market acceptance necessary for commercial success;
- We have never submitted an NDA to the FDA or comparable applications to other regulatory authorities, and we may not be successful in achieving approval of our product candidates;
- Orphan Drug Designation does not translate to approval and, even if we obtain FDA approval, we may not enjoy marketing exclusivity or other expected benefits;
- After approval of tecarfarin, it will remain subject to regulatory obligations and review;
- After approval, our products could be subject to labeling and other restrictions;
- We are subject to federal and state obligations and regulations applicable to our marketing practices;
- We currently do not have an agreement with a third-party manufacturer to produce tecarfarin;

- If the manufacturer fails to comply with stringent regulations, we may face delays in development or commercialization, or be unable to meet demand;
- We face substantial competition;
- If serious adverse effects are identified with respect to our product candidates or our approved products, we may need to modify or abandon our development of that product candidate, discontinue sale of an approved product, or change our labeling;
- Recently enacted and future legislation may increase the difficulty and cost for us to obtain marketing approval of and commercialize tecarfarin and affect the prices we may obtain;

- If we market any of our products in a manner that violates healthcare fraud and abuse laws, or, if we violate government price reporting laws, we may be subject to civil or criminal penalties;
- Our ability to generate product revenues will be diminished if our products sell for inadequate prices;
- We will rely on third parties to conduct all of our clinical trials and, if they do not successfully carry out their contractual duties or meet expected deadlines, we may be unable to obtain regulatory approval;
- We currently have limited distribution, marketing, support and sales capabilities;
- Our employees, contractors, consultants, commercial partners and vendors may engage in misconduct;
- If we are not successful establishing a sales force, our ability to generate sales and profits will be limited;
- We plan to rely on collaborations and license arrangements with third parties to commercialize, market and promote our marketed products which may limit our ability to generate revenue;
- Our future growth depends, in part, on our ability to penetrate foreign markets;
- The ongoing COVID-19 global health crisis may impact our planned operations;
- Compliance with regulations regarding the treatment of animals could increase our costs.

General Company-Related Risks

- Our business depends upon our ability to attract and keep senior management and key scientific personnel;
- We will need to increase the size of our organization, and we may experience difficulties managing this;
- If product liability lawsuits are brought against us, we may incur substantial liabilities;
- Computer system failures could be costly and expose us to litigation and government enforcement actions;
- Acquisitions of other businesses could harm our operating results, dilute our stockholders' ownership, increase our debt or cause us to incur significant expense;
- Declining general economic or business conditions may have a negative impact on our business.

Risks Related to Our Intellectual Property

- We may be unable to obtain and maintain market exclusivity or patent protection for tecarfarin;
- To date, we have not engaged intellectual property counsel to conduct a freedom to operate analysis;
- We may become involved in lawsuits to protect or enforce our patents or other intellectual property;
- Litigation for intellectual property infringement could be costly and time consuming;
- Patent law changes in the United States and other jurisdictions could diminish the value of patents;
- Obtaining and maintaining our patent protection depends on compliance with various procedural, document submissions, fee payment and other requirements imposed by governmental patent agencies;

- We may not be able to enforce our intellectual property rights throughout the world;
- Patent terms may be inadequate to protect our competitive position on our products;
- We may be subject to third party intellectual property claims;

Risks Related to Ownership of Our Common Stock

- An active public trading market for our common stock may not develop or be sustained;
- We cannot be assured that we will be able to maintain our listing on the Nasdaq Capital Market;
- Our stock price may be extremely volatile;
- Stock prices in recent initial public offerings have been extremely volatile;

- If financial or industry analysts do not publish favorable research or reports about us our stock price and trading volume could decline;
- After this offering, our officers, directors, and principal stockholders will continue to exercise significant control over our Company;
- As a "controlled company" we may choose to be exempt from certain corporate governance requirements;
- Future sales of common stock by our officers, directors and principal stockholders or others of our common stock, or the perception that such sales may occur, could depress the market price of our common stock;
- The offering price of the shares and the other terms of the initial public offering have been determined through negotiations between the Company and the underwriter;
- The offering price of the primary offering and resale offering could differ;
- The resale by the selling stockholders in our resale offering may cause the market price of our common stock to decline;
- Our initial public offering price is substantially higher than the pro forma as adjusted net tangible book value per share of our outstanding common stock;
- Our charter documents will have anti-takeover provisions and provide that Delaware will be the exclusive forum for certain disputes between us and our stockholders;
- Our management has broad discretion in the use of the net proceeds from this offering;
- Claims for indemnification by our directors and officers may reduce our available funds;
- We do not intend to pay dividends in the foreseeable future;
- Certain members of our management team have limited experience managing a public company;
- We will incur significant increased costs as a result of operating as a public company; and
- We are an emerging growth company and may avail ourselves of reduced disclosure requirements or extended transition periods for complying with new or revised accounting standards.

Corporate History and Information

We were incorporated as a Delaware corporation on January 25, 2022 and secured the rights to tecarfarin on April 1, 2022 via an asset purchase agreement (the "Asset Purchase Agreement") from HESP LLC. HESP LLC acquired the assets of Espero BioPharma, Inc., or Espero, from Horizon Technology Finance Corporation and Horizon Credit II LLC (collectively, Horizon), a secured lender of Espero, including tecarfarin and the other assets of Espero in an assignment for the benefit of creditors. Quang Pham, our Chief Executive Officer, was the Chief Executive Officer of Espero and consultant to HESP LLC from July 2020 to December 2021.

Our principal executive offices are located at 822 A1A North, Suite 306, Ponte Vedra, Florida 32082, and our telephone number is (904) 300-0701. Our website address is *www.cadrenal.com*. The information contained on, or that can be accessed through, our website is not incorporated by reference into this prospectus, and you should not consider any information contained on, or that can be accessed through, our website as part of this prospectus or in deciding whether to purchase our common stock.

Implications of Being an Emerging Growth Company and a Smaller Reporting Company

We qualify as an "emerging growth company" as defined in the Jumpstart Our Business Startups Act of 2012, or the JOBS Act. For as long as we remain an emerging growth company, we may take advantage of specified reduced reporting requirements and other burdens that are otherwise applicable generally to other public companies. These provisions include, but are not limited to:

- Reduced obligations with respect to financial data, including presenting only two years of audited financial statements and selected financial data, and only two years of related Management's Discussion and Analysis of Financial Condition and Results of Operations disclosure in our initial registration statement;

- an exemption from the auditor attestation requirement in the assessment of our internal control over financial reporting pursuant to the Sarbanes-Oxley Act of 2002, as amended;
- reduced disclosure about executive compensation arrangements in our periodic reports, registration statements and proxy statements; and
- exemptions from the requirements to seek non-binding advisory votes on executive compensation or stockholder approval of any golden parachute arrangements.

We may take advantage of some or all of these provisions until we are no longer an emerging growth company. We will remain an emerging growth company until the earliest of (i) the last day the fiscal year following the fifth anniversary of the completion of this offering, (ii) the last day of the first fiscal year in which our annual gross revenues exceed $1.235 billion, (iii) the date on which we have, during the immediately preceding three-year period, issued more than $1.0 billion in non-convertible debt securities and (iv) the date on which we are deemed to be a large accelerated filer under the rules of the Securities and Exchange Commission, or the SEC. We may choose to take advantage of some but not all of these reduced burdens. For example, we have taken advantage of the reduced reporting requirements with respect to disclosure regarding our executive compensation arrangements, have presented only two years of audited financial statements and only two years of related "Management's Discussion and Analysis of Financial Condition and Results of Operations" disclosure in this prospectus, and have taken advantage of the exemption from auditor attestation on the effectiveness of our internal control over financial reporting. To the extent that we take advantage of these reduced burdens, the information that we provide stockholders may be different than you might obtain from other public companies in which you hold equity interests.

In addition, the JOBS Act permits emerging growth companies to take advantage of an extended transition period to comply with new or revised accounting standards applicable to public companies. We have elected to use this extended transition period. As a result of this election, our timeline to comply with new or revised accounting standards will in many cases be delayed as compared to other public companies that are not eligible to take advantage of this election or have not made this election. Therefore, our financial statements may not be comparable to those of companies that comply with the public company effective dates for these accounting standards.

We are also a "smaller reporting company" as defined in the Securities Exchange Act of 1934, as amended, or the Exchange Act, and have elected to take advantage of certain of the scaled disclosures available to smaller reporting companies. To the extent that we continue to qualify as a "smaller reporting company" as such term is defined in Rule 12b-2 under the Exchange Act, after we cease to qualify as an emerging growth company, certain of the exemptions available to us as an "emerging growth company" may continue to be available to us as a "smaller reporting company," including exemption from compliance with the auditor attestation requirements pursuant to SOX and reduced disclosure about our executive compensation arrangements. We will continue to be a "smaller reporting company" until we have $250 million or more in public float (based on our Common Stock) measured as of the last business day of our most recently completed second fiscal quarter or, in the event we have no public float (based on our Common Stock) or a public float (based on our Common Stock) that is less than $700 million, annual revenues of $100 million or more during the most recently completed fiscal year.

THE OFFERING

Common stock offered by us	1,400,000 shares
Common stock outstanding immediately before this offering	8,193,875 shares
Common stock to be outstanding immediately after this offering	11,634,575 shares which shares include: (i) 1,140,700 shares of common stock to be issued upon consummation of this offering upon the automatic conversion of outstanding promissory notes in the aggregate principal amount of $1,125,000; (ii) 600,000 shares of common stock to be issued upon the effective date of this offering pursuant to the terms of the Amendment to Asset Purchase Agreement; (iii) 50,000 shares of common stock to be issued to an executive officer upon consummation of this offering and (iv) 250,000 shares of common stock issuable upon the exercise of outstanding warrants at an exercise price of $1.00 per shares upon consummation of this offering (or 11,844,575 shares if the underwriters exercise their option to purchase additional shares in full).
Option to purchase additional shares	We have granted the underwriters an option for a period of 45 days from the date of this prospectus to purchase up to an additional 210,000 shares of common stock from us.
Use of proceeds	We estimate that the net proceeds to us from this offering will be approximately $5.7 million, or approximately $6.7 million if the underwriters exercise in full their option to purchase additional shares of common stock, after deducting underwriting discounts and commissions and estimated offering expenses payable by us. We intend to use the net proceeds from this offering as follows: approximately $3,300,000 for chemistry, manufacturing and controls ("CMC"), research and development and other trial preparation expenses and the balance of approximately $2,400,000 for working capital and general corporate purposes, which may include the acquisition or licensing of other products, businesses or technologies. See "Use of Proceeds" for additional information.
Representative's warrants	The registration statement of which this prospectus is a part also registers for sale warrants to purchase shares of our common stock that we will issue to Boustead Securities, LLC, as the representative of the underwriters (the "Representative") as a portion of the underwriting compensation payable to the underwriters in connection with this offering. The warrants will be exercisable, at an exercise price equal to 120% of the initial public offering price, for five years; however such warrants may not be sold, transferred, assigned, pledged or hypothecated for a period commencing 180 days following the commencement of sales in this offering. Please see "Description of Securities" and "Underwriting" for a description of these warrants.
Risk factors	See "Risk Factors" beginning on page 14 and the other information included in this prospectus for a discussion of

	factors you should consider carefully before deciding to invest in our common stock.
Nasdaq symbol	Our common stock has been approved for listing on the Nasdaq under the symbol "CVKD" and will begin trading on January 20, 2023.

Unless otherwise stated in this prospectus, the number of shares of common stock to be outstanding after this offering is based on 8,193,875 shares of common stock outstanding as of January 17, 2023, and excludes the following:

- 1,100,000 shares of common stock issuable upon the exercise of outstanding stock options at a weighted-average exercise price of $0.84 per share;
- 50,000 shares of common stock to be issued to an executive officer upon the consummation of this offering;
- 760,000 shares of common stock reserved for future issuance under our 2022 Successor Equity Incentive Plan, or the 2022 Plan, which 2022 Plan will become effective upon the effective date of the registration statement of which this prospectus forms a part; and
- 26,500 shares of common stock issuable upon exercise of outstanding warrants issued to Boustead Securities, LLC, as placement agent for the Private Placement and the November Private Placement.

Unless otherwise indicated, all information in this prospectus assumes and reflects the following:

- The automatic conversion upon consummation of this offering of all of the Private Placement Notes in the principal amount of $575,000 into an aggregate of 575,000 shares of our common stock, based on an as adjusted conversion price of $1.00 per share;
- The automatic conversion upon consummation of this offering of the June 2022 Note in the principal amount of $50,000, plus accrued interest through September 30, 2022 of $908, into 50,908 shares of our common stock, based on an as adjusted conversion price of $1.00 per share;
- The issuance subsequent to September 30, 2022 of 67,000 shares of common stock to consultants for the provision of services and the issuance of 50,000 shares of common stock to an executive officer upon consummation of this offering;
- The automatic conversion upon consummation of this offering of the March 2022 Note in the principal amount of $500,000, plus accrued interest through September 30, 2022 of $14,792, into 514,792 shares of our common stock, based on an as adjusted conversion price of $1.00 per share;
- The issuance of 600,000 shares of common stock, upon the effective date of this offering, pursuant to the terms of the Amendment to Asset Purchase Agreement, based on a $3.00 price per share which is sixty percent (60%) of the initial public offering price of $5.00 per share;
- The automatic exercise upon consummation of this offering of the November Warrants and the issuance of 250,000 shares of common stock upon such exercise at an exercise price of $1.00 per share upon consummation of this offering.
- No exercise of the underwriter's option to purchase up to an additional 210,000 shares of common stock; and
- No exercise of the warrants granted to the Representative of the underwriters upon completion of this offering.

SUMMARY FINANCIAL DATA

The following table summarizes the relevant financial data for our business and should be read with our financial statements, which are included in this prospectus.

The following tables present our statement of operations data and summary balance sheet data and should be read together with our financial statements and accompanying notes and "Management's Discussion and Analysis of Financial Condition and Results of Operations" appearing elsewhere in this prospectus. The statement of operations for the period ended September 30, 2022 and summary balance sheet data as of September 30, 2022 is derived from our unaudited financial statements, which are included elsewhere in this prospectus. The historical results are not necessarily indicative of the results to be expected in any future period.

	January 25, 2022 (Inception) to September 30, 2022
Operating expenses:	
General and administrative expenses	1,348,670
Research and development expenses	276,482
Depreciation and amortization expense	506
Total operating expenses	1,625,658
Loss from operations	(1,625,658)
Other expense:	
Interest expense	22,068
Interest expense, amortization of debt discount	29,302
Change in fair value of derivative liability	27,874
Total other expenses	79,244
Net loss and comprehensive loss	(1,704,902)
Net loss per common share, basic and diluted	(0.22)
Weighted average number of common shares used in computing net loss per common share, basic and diluted	7,778,778

Balance Sheet Data:	September 30, 2022	Pro Forma[1]	Pro Forma As Adjusted[2]
Working capital	$ (131,007)	$ 118,993	5,858,993
Total assets	$ 545,322	$ 795,322	6,535,322
Total liabilities	$ 1,725,145	$ 672,362	672,362
Stockholder's equity (deficit)	$ (1,179,823)	$ 210,877	5,950,877

(1) The pro forma gives effect to: (i) conversion of the Private Placement Notes in the aggregate principal amount of $575,000 into an aggregate of 575,000 shares of our common stock, upon consummation of the offering, based on a $1.00 as adjusted conversion price per share; (ii) conversion of the June 2022 Note in the principal amount of $50,000, plus accrued interest through September 30, 2022 of $908, into 50,908 shares of our common stock, upon consummation of the offering, based on a $1.00 as adjusted conversion price per share; (iii) conversion of the March 2022 Note in the principal amount of $500,000, plus accrued interest through September 30, 2022 of $14,792, into 514,792 shares of our common stock, upon consummation of the offering, based on a $1.00 as adjusted conversion price per share; (iv) elimination of the derivative liability related to the notes; (v) the issuance of 600,000 shares of common stock, based on a $3.00 price per share, which is sixty percent (60%) of the initial

public offering price of $5.00 per share, on the effective date of this offering pursuant to the terms of the Amendment to Asset Purchase Agreement; (vi) the issuance subsequent to September 30, 2022 of 67,000 shares of common stock to consultants for the provision of services; (vii) the issuance of 50,000 shares of common stock to an executive officer upon consummation of this offering; and (viii) the receipt of $250,000 in proceeds from the issuance of the November Notes and the November Warrants and the automatic exercise of the November Warrants upon consummation of this offering, which will result in the issuance of 250,000 shares of common stock and the repayment of the principal amount of the November Notes.

(2) The pro forma as adjusted balance sheet data in the table above reflects the items described in footnote (1) above and gives effect to the sale and issuance by us of 1,400,000 shares of our common stock in this offering, based upon the initial public offering price of $5.00, after deducting the underwriting discounts and commissions and estimated offering expenses payable by us.

RISK FACTORS

Investing in our common stock involves a high degree of risk. You should carefully consider the following risks and uncertainties, together with all the other information in this prospectus, including our financial statements and the related notes included elsewhere in this prospectus, before purchasing our common stock. If any of the following risks actually occurs, our business, operating results, financial condition, liquidity and prospects could be materially adversely affected. As a result, the trading price of our common stock could decline and you could lose part or all of your investment. Additional risks and uncertainties not currently known to us or that we currently deem to be immaterial may also materially adversely affect our business, operating results, financial condition, liquidity and prospects.

Risks Related to Our Financial Position and Need for Capital

We are a clinical development biopharmaceutical company with a limited operating history.

We are a newly formed company and have had limited operations to date. We have to complete clinical trials and receive regulatory approval of new drug applications, or NDAs, before commercial sales of our product candidates can commence. The likelihood of success of our business plan must be considered in light of the problems, substantial expenses, difficulties, complications and delays frequently encountered in connection with building and expanding clinical development pharmaceutical businesses and the regulatory and competitive environment in which we operate. Pharmaceutical product development is a highly speculative undertaking, involves a substantial degree of risk and is a capital-intensive business.

Accordingly, you should consider our prospects in light of the costs, uncertainties, delays and difficulties frequently encountered by companies in the later stage of development, especially clinical pharmaceutical companies such as ours. Potential investors should carefully consider the risks and uncertainties that a company with a limited operating history will face. In particular, potential investors should consider that we cannot assure you that we will be able to:

- successfully complete the clinical trials necessary to obtain regulatory approval for the marketing of our product candidate, tecarfarin;
- secure acceptance of our product candidate in the medical community and with third-party payors and consumers;
- if approved for commercial sale, launch commercial sales of our product candidate, whether alone or in collaboration with others;
- successfully build an internal sales force meeting our requirements for the marketing and sale of our product candidate, tecarfarin;
- successfully manufacture our clinical product and establish commercial drug supply;
- secure market exclusivity and/or adequate intellectual property protection for our product candidate;
- attract and retain an experienced management, board and scientific advisory team;
- successfully implement or execute our current business plan, and we cannot assure you that our business plan is sound; and
- raise sufficient funds in the capital markets to effectuate our business plan.

If we cannot successfully execute any one of the foregoing, our business may not succeed and your investment will be adversely affected.

We have a limited operating history upon which to evaluate our ability to commercialize our product candidate.

We are a development-stage company and our success is dependent upon our ability to obtain regulatory approval for and commercialize our product candidate, tecarfarin, and we have not demonstrated an ability to

perform the functions necessary for the approval or successful commercialization of any product candidate. We have yet to demonstrate our ability to overcome the risks frequently encountered in our industry and are still subject to many of the risks common

to such enterprises, including our ability to implement our business plan, market acceptance of our proposed business and lead product, under-capitalization, cash shortages, limitations with respect to personnel, financing and other resources, competition from better funded and experienced companies, and uncertainty of our ability to generate revenues. In fact, though individual team members have experience running clinical trials and our Chief Executive Officer has been involved with the development of tecarfarin for five years, as a company we have yet to prove that we can successfully run a clinical trial. There is no assurance that our activities will be successful or will result in any revenues or profit, and the likelihood of our success must be considered in light of the stage of our development. In addition, no assurance can be given that we will be able to consummate our business strategy and plans, or that financial, technological, market, or other limitations may force us to modify, alter, significantly delay, or significantly impede the implementation of such plans. We have insufficient results for investors to use to identify historical trends. Investors should consider our prospects in light of the risk, expenses and difficulties we will encounter as an early stage company. Our revenue and income potential is unproven and our business model is continually evolving. We are subject to the risks inherent to the operation of a new business enterprise, and cannot assure you that we will be able to successfully address these risks.

We have a history of operating losses and expect to continue to incur substantial losses for the foreseeable future. We may never become profitable or, if achieved, be able to sustain profitability.

To date, we have not generated any revenue from operations and we expect to continue to incur significant operating losses in connection with the development and sale of tecarfarin. We may continue to incur operating losses until such time, if ever, as we are able to achieve sufficient levels of revenue from operations. Our ability to achieve profitability will depend on regulatory approval of our product candidate and if approved, the market acceptance of our product offering and our capacity to develop, introduce and sell our product to our targeted markets. There can be no assurance that we will ever generate significant sales or achieve profitability. Accordingly, the extent of future losses and the time required to achieve profitability, if ever, cannot be predicted at this point.

Even if we succeed in developing and commercializing one or more product candidates, we expect to incur substantial losses for the foreseeable future and may never become profitable. We also expect to continue to incur significant operating and capital expenditures and anticipate that our expenses will increase substantially in the foreseeable future as we:

- continue to undertake the pivotal clinical trial for our product candidate;
- seek regulatory approvals for our product candidate;
- implement additional internal systems and infrastructure; and
- hire additional personnel.

We may not be able to generate revenue or achieve profitability in the future. Our failure to achieve or maintain profitability would likely negatively impact the value of our securities and could prevent us from continuing as a going concern.

Even if we can secure such arrangements, we may continue to have obligations and expenses that exceed the revenue generated by these marketed products. In addition, we could incur significant development and other expenses if we were to make alterations to the manufacturing process for tecarfarin, for preparation and submission of a supplemental NDA for such alterations, if required by the FDA, and in connection with the launch of tecarfarin, if approved. Further, as we pursue FDA approval for tecarfarin, we expect that our research and development expenses will continue to increase significantly as we advance our pivotal Phase 3 clinical trial.

Our independent registered public accounting firm has expressed substantial doubt about our ability to continue as a going concern.

As described in Note 1 of our accompanying audited financial statements, our independent registered public accounting firm has issued a going concern opinion on our financial statements, expressing substantial doubt that we can continue as an ongoing business for the next twelve months after issuance of their report based on our current development plans and our operating requirements and us having suffered recurring losses from operations and having a net capital deficiency. Our financial statements do not include any adjustments that may

result from the outcome of this uncertainty. If we cannot raise the necessary capital to continue as a viable entity, we could experience a material adverse effect on our business and our stockholders may lose some or all of their investment in us.

Our cash and the proceeds of this offering will only fund our operations for a limited time, and we will need to raise additional capital to fund our planned pivotal Phase 3 clinical trial and to support our development and commercialization efforts for our product candidate, tecarfarin.

If we do not succeed in raising additional funds on acceptable terms, we will be unable to commence our planned Phase 3 pivotal clinical trial or obtain approval of our product candidate from the FDA and other regulatory authorities. In addition, we could be forced to delay, discontinue or curtail product development, forego sales and marketing efforts, and forego licensing in attractive business opportunities. We estimate that we will require a total of $45 million for the completion of our planned pivotal Phase 3 clinical trial and other expenditures that we will need to incur in order to file our NDA. We intend to use approximately $3.3 million of the proceeds from this offering to fund CMC preparation, research and development and other trial preparation expenses required for the initiation of our planned pivotal Phase 3 clinical trial, therefore, we will require at least $42 million of additional funding to enroll patients and complete our first Phase 3 clinical trial. Additionally, we estimate that we will require $13 million for general and administrative expenses anticipated to be incurred over the next three years.

We will also need to raise additional capital to expand our business to meet our long-term business objectives.

Our current cash at September 30, 2022 will not be sufficient to fund operations for at least the next twelve months. However, we believe that our existing cash and the anticipated net proceeds from this offering will be sufficient in the aggregate to meet our anticipated cash requirements for at least the next twelve months. We will, however, require additional financing as we continue to execute our business strategy, including that we will require additional funds for the initiation of enrollment of patients and completion of the planned pivotal Phase 3 trial, which is the primary use of funds in this financing. Our liquidity may be negatively impacted as a result of a research and development cost increases in addition to general economic and industry factors. We anticipate that, to the extent that we require additional liquidity, it will be funded through the incurrence of other indebtedness, additional equity financings or a combination of these potential sources of liquidity. In addition, we may raise additional funds to finance future cash needs through grant funding and/or corporate collaboration and licensing arrangements. If we raise additional funds by issuing equity securities or convertible debt, our stockholders will experience dilution. Debt financing, if available, would result in increased fixed payment obligations and may involve agreements that include covenants limiting or restricting our ability to take specific actions, such as incurring additional debt, making capital expenditures or declaring dividends. If we raise additional funds through collaboration and licensing arrangements with third parties, it may be necessary to relinquish valuable rights to our products, future revenue streams or product candidates or to grant licenses on terms that may not be favorable to us. The covenants under future credit facilities may limit our ability to obtain additional debt financing. We cannot be certain that additional funding will be available on acceptable terms, or at all. Any failure to raise capital in the future could have a negative impact on our financial condition and our ability to pursue our business strategies.

Our present and future capital requirements will depend on many factors, including:

- the outcome, timing and cost of our Phase 3 clinical trial to obtain regulatory approval for tecarfarin in the United States;
- the degree and rate of market adoption of our products, if approved;
- the emergence of new, competing technologies and products;
- the costs of R&D activities we undertake to develop new products and indications;
- the costs of commercialization activities, including sales, marketing and manufacturing;
- the costs of building an internal sales force meeting our requirements for the marketing and sale of our product candidates, if approved;
- our ability to collaborate with third parties on the development and commercialization of our product candidates and products;
- the level of working capital required to support our growth; and

- our need for additional personnel, information technology or other operating infrastructure to support our growth and operations as a public company.

We do not currently have any arrangements or credit facilities in place as a source of funds, and there can be no assurance that we will be able to raise sufficient additional capital on acceptable terms, or at all. We anticipate that the additional funding we require will be funded through the incurrence of other indebtedness, additional equity financings or a combination of these potential sources of liquidity We may seek additional capital through a combination of private and public equity offerings, debt financings and strategic collaborations. Debt financing, if obtained, may involve agreements that include covenants limiting or restricting our ability to take specific actions, including issuing shares of our common stock or other securities and incurring additional debt, and could increase our expenses and require that our assets secure such debt. Equity financing, if obtained, could result in dilution to our then existing stockholders and/or require such stockholders to waive certain rights and preferences. If such financing is not available on satisfactory terms, or is not available at all, we may be required to delay, scale back or eliminate the development of business opportunities and our operations and financial condition may be materially adversely affected. We can provide no assurances that any additional sources of financing will be available to us on favorable terms, if at all. In addition, if we are unable to secure sufficient capital to fund our operations, we might have to enter into strategic collaborations that could require us to share commercial rights to our products or product candidates with third parties in ways that we currently do not intend or on terms that may not be favorable to us. If we choose to pursue additional indications and/or geographies for any of our products or product candidates or otherwise expand more rapidly than we presently anticipate, we may also need to raise additional capital sooner than expected.

Risks Related to Product Development, Regulatory Approval, Manufacturing and Commercialization

Our business is dependent upon the success of our investigational product candidate, tecarfarin, which requires additional clinical testing before we can seek regulatory approval and potentially launch commercial sales. We do not own any other product candidates or have any other products in clinical development.

Our business and future success depends upon our ability to obtain regulatory approval of and then successfully commercialize our product candidate, tecarfarin. Tecarfarin is in late clinical stage development. Our main focus and the investment of a significant portion of our efforts and financial resources is expected to be in the development of our only product candidate, tecarfarin, for which we are currently planning a Phase 3 clinical trial with approximately 492 patients in the United States. We believe that the proceeds from this offering will provide us with the necessary funding to complete the CMC preparation required for our planned pivotal Phase 3 clinical trial but will not provide us with sufficient funds to initiate or complete this pivotal Phase 3 clinical trial. Even though we are pursuing a registration pathway based on specific FDA input and guidance, there are many uncertainties known and unknown that may affect the outcome of the trial. These include adequate patient enrollment, adequate supply of our product candidate, potential changes in the regulatory landscape, the results of the trial being successful, and FDA acceptance of the data to support approval. We also rely on third parties to conduct the appropriate clinical trials, and their failure to perform in accordance with applicable law would have a negative effect on our regulatory submission.

Our future success depends heavily on our ability to successfully manufacture, develop, obtain regulatory approval, and commercialize tecarfarin, which may never occur. We currently generate no revenues from our product candidate, and we may never be able to develop or commercialize a marketable drug.

All of our current data for our product candidate are the results of clinical trials conducted by third parties and do not necessarily provide sufficient evidence that our products are viable as potential pharmaceutical products.

Through our acquisition of the asset of Espero, we possess toxicology, pharmacokinetic, and other preclinical data and clinical data on tecarfarin from studies and trials conducted several years ago. As of now, tecarfarin has completed eleven clinical trials and is now in preparations to enter a pivotal Phase 3 trial. There is no guarantee that Phase 1 or Phase 2 results can or will be replicated by the pivotal Phase 3 study. Further, as the clinical trials were conducted by third parties, and were completed prior to our ownership of the technology, we cannot be assured that such trials were conducted in compliance with applicable statutes, rules, regulations and guidelines applicable to such trials. Although the FDA concurred with our recommended endpoints in correspondence that it had with Espero in 2019, and Espero submitted the protocol we plan to use to FDA in 2019, we have not received FDA input on our Phase 3 protocol. The FDA agreed in principle that “a single registration trial that combines these populations may be acceptable.” However, there can be no assurance that the planned protocol will be accepted by FDA.

Previous clinical trials using tecarfarin have had different trial designs, doses, parameters and endpoints than the planned Phase 3 clinical trial that is expected to serve as a basis for approval of tecarfarin. We plan to use a fixed dose in future clinical trials that we believe provides good coverage given the dose ranges tested clinically; however, it is possible that the dose selected will not be the optimal dose and so drug effects may be limited or not be demonstrated sufficiently in clinical testing.

As all of our clinical trials to date were conducted by third parties, we cannot be assured that such clinical trials were in compliance with applicable laws, rules and regulations.

We did not acquire tecarfarin until April of 2022, and do not have first-hand knowledge of how the Phase 1 and Phase 2 clinical trials were completed. As such, we cannot be assured that such clinical trials were conducted in full compliance with applicable laws, rules and regulations. While we are not aware of any issues in relation to such trials and the performance thereof, we cannot be assured that we may learn in the future that there was a failure to abide by such laws, rules and regulations, which could potentially expose us to issues with regards to our Phase 3 clinical trials or otherwise create risks unknown to us with regards to our technology.

Our efforts to develop our product candidate may not generate data sufficient to support an application for regulatory approval.

Despite the global burden of cardiovascular disease, investment in cardiovascular drug development has stagnated over the past two decades, with relative underinvestment compared with other therapeutic areas. The reasons for this trend are multifactorial, but of primary concern is the high cost of conducting cardiovascular outcome trials in the current regulatory environment that demands a direct assessment of risks and benefits, using clinically meaningful cardiovascular endpoints. In addition, clinical trials are difficult to design and implement, can take many years to complete and are uncertain as to outcome. Success in early phases of pre-clinical and clinical trials does not ensure that later clinical trials will be successful, and interim results of a clinical trial do not necessarily predict final results. For example, in the Phase 2/3 EMBRACE-AC trial of tecarfarin conducted by ARYx Therapeutics, Inc., a predecessor company to Espero, tecarfarin did not reach statistical significance on its primary endpoint, superiority to warfarin as measured by time in TTR. There is no guarantee that later trials, including ACTOR AF, will reach statistical significance on their endpoints, or demonstrate superiority to warfarin or any other therapy. A failure of one or more of clinical trials can occur at any stage of testing. Our product candidate may prove to have undesirable or unintended side effects, toxicities or other characteristics that may preclude our obtaining regulatory approval or prevent or limit commercial use with respect to one or all intended indications. In addition, we may experience other numerous unforeseen events during, or as a result of, the clinical trial process that could delay or prevent our ability to continue development. Development stage risks include the following:

- although we have FDA minutes documenting Espero's correspondence with the FDA regarding tecarfarin, the FDA minutes are from 2019, are not binding on FDA, and our expectations regarding such plans may be out of date and not be in line with current market dynamics and the FDA or comparable foreign regulatory authorities or institutional review boards, or IRBs, may disagree with the design or implementation of our clinical trial and refuse to let them proceed;
- we may not be able to provide acceptable evidence of the safety and efficacy of our product candidates or an acceptable benefit/risk profile for our product candidate;
- we may not be able to successfully manufacture drug supplies for our clinical trial;
- the results of our clinical trial may not be satisfactory or may not meet the level of statistical or clinical significance required by the FDA, European Medicines Agency, or EMA, or other comparable foreign regulatory authorities to demonstrate effectiveness;
- we may not be able to determine the optimal dosing of our product candidates; and
- patients in our clinical trial may suffer adverse effects that are deemed related to our product candidates, leading us or regulatory authorities to stop clinical trial temporarily or permanently.

If unacceptable safety concerns or other adverse events arise in the development of a product candidate, our clinical trials could be suspended or terminated or the FDA or comparable foreign regulatory authorities could order us to cease clinical trials or deny approval of such product candidate for any or all targeted indications. Treatment-related side effects could also affect patient recruitment or the ability of enrolled subjects to complete the trial or result in potential product liability claims. Inadequate training in recognizing or managing the potential side effects of a product candidate could result in patient deaths. Any of these occurrences may harm our business, financial condition and prospects significantly.

Even if we successfully complete our clinical trials, we may not receive regulatory approval for tecarfarin, and we may not be able to commercialize our product candidate and our ability to generate revenue will be limited.

The research, testing, manufacturing, labeling, packaging, storage, approval, sale, marketing, advertising and promotion, pricing, export, import and distribution of drug products are subject to extensive regulation by the FDA and other regulatory authorities in the United States and other countries, which regulations differ from country to country. We are not permitted to market our product candidate in the United States until we receive approval of an NDA from the FDA and in non-U.S. markets until we receive the requisite approval from comparable regulatory agencies in such countries. Of the large number of drugs in development, only a small number are submitted for approval to the FDA through an NDA and even fewer are eventually approved for commercialization. In 2020 and 2021, the FDA approved only three new molecular entities to treat cardiovascular/vascular diseases. We may not succeed at gaining regulatory approval, which would materially harm our business.

Receipt of necessary regulatory approval is subject to a number of risks, including the following:

- the data collected from pre-clinical and clinical trials may not be sufficient to support the submission of an NDA or other submission or to obtain regulatory approval in the United States or elsewhere;
- the FDA or comparable foreign regulatory authorities may fail to approve the manufacturing processes or facilities of third-party manufacturers with which we contract for clinical and commercial supplies; and
- the relevant laws, approval policies or regulations of the FDA or comparable foreign regulatory authorities may significantly change in a manner rendering our clinical data insufficient for approval.

We cannot guarantee that regulators will agree with our assessment of the results of our clinical trials or that such trials will be considered by regulators to have shown safety or efficacy of our product candidates. For example, while we currently intend to conduct future clinical trials of tecarfarin with a primary endpoint of reduction of major cardiac adverse events (MACE), we cannot assure you that the FDA will consider it to be a clinically meaningful endpoint. The FDA, EMA and other regulators have substantial discretion in the approval process and may refuse to accept any application or may decide that our data are insufficient for approval and require additional clinical trials, or pre-clinical or other trials. In addition, varying interpretations of the data obtained from pre-clinical and clinical testing could delay, limit or prevent regulatory approval of a product candidate. Failure to obtain regulatory marketing approval for our product candidates in any indication will prevent us from commercializing the product candidate, and our ability to generate revenue will be materially impaired.

The process of obtaining regulatory approvals is expensive, often takes many years, if approval is obtained at all, and can vary substantially based upon, among other things, the type, complexity and novelty of the product candidates involved, the jurisdiction in which regulatory approval is sought and the substantial discretion of the regulatory authorities. Changes in regulatory review for a submitted product application may cause delays in approval or rejection of an application. Regulatory approval obtained in one jurisdiction does not necessarily mean that a product candidate will receive regulatory approval in all jurisdictions in which we may seek approval, but the failure to obtain approval in one jurisdiction may negatively impact our ability to seek or gain approval in a different jurisdiction. Obtaining foreign regulatory approvals and compliance with foreign regulatory requirements could result in significant delays, difficulties and costs for us and could delay or prevent the introduction of our products in certain countries. If we fail to comply with the regulatory requirements in international markets and/or fail to receive applicable marketing approvals, our target market will be reduced and our ability to realize the full market potential of tecarfarin or any future product candidates will be harmed.

Fast Track designation by the FDA may not actually lead to a faster development or regulatory review or approval process, and does not assure FDA approval of our product candidate.

If a product candidate is intended for the treatment of a serious or life-threatening condition and the product candidate demonstrates the potential to address unmet medical need for this condition, the sponsor may apply

for FDA Fast Track designation. However, a Fast Track designation does not ensure that the product candidate will receive marketing approval or that approval will be granted within any particular timeframe. As a result, while we have received Fast Track designation for tecarfarin for the prevention of systemic thromboembolism of cardiac origin in patients with ESRD and AFib, we may not experience a faster development process, review or approval compared to conventional FDA procedures. In addition, the FDA may withdraw Fast Track designation if it believes that the designation is no longer supported by data from our clinical development program. Fast Track designation alone does not guarantee qualification for the FDA's priority review procedures.

Even if we obtain regulatory approval, we will still face ongoing regulatory requirements and tecarfarin may face future development and regulatory difficulties.

Even if we receive regulatory approval of tecarfarin or any future product candidates, we will be subject to ongoing regulatory obligations, such as post market surveillance and current good manufacturing practice ("GMP") requirements, and continued regulatory review, which may result in significant additional expense. We may also be subject to penalties if we fail to comply with regulatory requirements or experience unanticipated problems with product candidates. In addition, third parties on whom we rely must comply with regulatory requirements, and any non-compliance on their part may negatively impact our business, assuming we obtain regulatory authorization at all.

Any regulatory approvals that we receive for product candidates will require surveillance to monitor the safety and efficacy of the product candidate. The FDA may also require a Risk Evaluation and Mitigation Strategy ("REMS") program in order to approve product candidates, which could entail requirements for a medication guide, physician communication plans or additional elements to ensure safe use, such as restricted distribution methods, patient registries and other risk minimization tools. The FDA could also require a boxed warning, sometimes referred to as a Black Box Warning on the product label to identify a particular safety risk, which could affect commercial efforts to promote and sell the product. In addition, if the FDA or a comparable foreign regulatory authority approves product candidates, the manufacturing processes, labeling, packaging, distribution, adverse event reporting, storage, advertising, promotion, import, export and recordkeeping for product candidates will be subject to extensive and ongoing regulatory requirements. These requirements include submissions of safety and other post-marketing information and reports, registration, as well as continued compliance with current GMPs and current good clinical practices ("GCPs") for any clinical trials that we conduct post-approval. We are also subject to certain user fees imposed by the regulatory agencies. Later discovery of previously unknown problems with product candidates, including adverse events of unanticipated severity or frequency, or with our third-party manufacturers or manufacturing processes, or failure to comply with regulatory requirements, may result in, among other things:

- import alerts or automatic detentions;
- restrictions on the marketing or manufacturing of product candidates, withdrawal of the product from the market, or product recalls;
- fines, warning letters or holds on clinical trials;
- refusal by the FDA to approve pending applications or supplements to approved applications filed by us or suspension or revocation of approvals;
- labeling changes;
- product seizure or detention, or refusal to permit the import or export of product candidates;
- injunctions or the imposition of civil or criminal penalties; and
- inability to obtain government contracts.

The FDA's and other regulatory authorities' policies may change, such as those required by the 21st Century Cures Act, and additional government regulations may be enacted that could prevent, limit or delay regulatory approval of tecarfarin or any future product candidates. In addition, it is unclear what changes, if any, the new presidential administration may bring. We cannot predict the likelihood, nature or extent of government regulation that may arise from future legislation or administrative action, either in the United States or abroad. If we are slow or unable to adapt to changes in existing requirements or the adoption of new requirements or policies, or if we are not able to maintain regulatory compliance, we may lose any marketing approval that we may have obtained and we may not achieve or sustain profitability.

Clinical trials are very expensive, time-consuming and difficult to design and implement.

As part of the regulatory process, we must conduct clinical trials for each product candidate to demonstrate safety and efficacy to the satisfaction of the FDA and other regulatory authorities. As we advance tecarfarin or any future product candidates we expect that our expenses will increase. The number and design of the clinical

trials that will be required varies depending upon product candidate, the condition being evaluated, current medical strategies and the trial results themselves. Therefore, it is difficult to accurately estimate the cost of the clinical trials. Clinical trials

are very expensive and difficult to design and implement, in part because they are subject to rigorous regulatory requirements. The clinical trial process is also time consuming. We estimate that clinical trials of product candidates including tecarfarin, will take at least several years to complete. Furthermore, failure can occur at any stage of the trials, and we could encounter problems that cause us to abandon or repeat clinical trials. The commencement and completion of clinical trials may be delayed or prevented by several factors, including:

- unforeseen safety issues;
- failure to determine appropriate dosing;
- greater than anticipated cost of our clinical trials;
- failure to demonstrate effectiveness during clinical trials;
- slower than expected rates of subject recruitment or difficulty obtaining investigators, particularly during COVID-19;
- subject drop-out or discontinuation;
- import delays of clinical trial materials;
- inability to monitor subjects adequately during or after treatment;
- third party contractors, including, without limitation, CROs and manufacturers, failing to comply with regulatory requirements or meet their contractual obligations to us in a timely manner
- reaching agreements with prospective CROs, and trial sites, both of which can be subject to extensive negotiation and may vary significantly among different CROs and trial sites;
- insufficient or inadequate supply or quality of product candidates or other necessary materials to conduct our trials;
- potential additional safety monitoring, or other conditions required by FDA or comparable foreign regulatory authorities regarding the scope or design of our clinical trials, or other studies requested by regulatory agencies;
- problems engaging Institutional Review Boards ("IRBs"), to oversee trials or in obtaining and maintaining IRB approval of studies;
- imposition of clinical hold or suspension of our clinical trials by regulatory authorities; and
- inability or unwillingness of medical investigators to follow our clinical protocols.

In addition, we or the FDA may suspend or terminate our clinical trials at any time if it appears that we are exposing participants to unacceptable health risks or if the FDA finds deficiencies in our Investigational New Drug, or IND, submissions or the conduct of these trials. Therefore, we cannot predict with any certainty when, if ever, future clinical trials will commence or be completed.

Delays in the enrollment of patients in any or all of our clinical trials could increase our development costs and delay completion of our clinical trials and associated regulatory submissions.

We may not be able to initiate or continue clinical trials for our product candidates if we are unable to locate and enroll a sufficient number of eligible patients to participate in these trials as required by the FDA or other regulatory authorities. COVID-19 will likely make this even more challenging. Even if we are able to enroll a sufficient number of patients in our clinical trials, if the pace of enrollment is slower than we expect, the development costs for our product candidates may increase, and the completion of our trials may be delayed or our trials could become too expensive to complete.

Even if approved, tecarfarin may not have labeling that allows us to successfully commercialize it.

The commercial success of tecarfarin and any of our future product candidates will depend in significant measure upon our ability to obtain approval from the FDA and other regulatory authorities of labeling describing a product candidate's expected features or benefits. Regulatory authorities may approve tecarfarin for fewer or more limited indications than we request or may approve tecarfarin with labeling that does not include the labeling claims necessary or desirable for the successful commercialization of that indication. Failure to achieve approval from the FDA or other regulatory authorities of product labeling containing certain types of information on features or benefits of our products will prevent or substantially limit our advertising and promotion of such features in order to differentiate our product candidates or any future product candidates from those products already existing in the market. This may make it difficult or impossible to achieve commercial success.

If our product candidate is approved, our success depends on our commercialization efforts, which may not be achieved. If we are unable to commercialize our product candidate, or experience significant delays in doing so, our business could be materially harmed.

We will invest a significant portion of our efforts and financial resources into the development and commercialization of tecarfarin. Product revenues from our product candidate, tecarfarin, which will not be realized until after regulatory approval, if ever, will depend on the successful development, regulatory approval and eventual commercialization of these product candidates. The success of our product candidate will depend on several factors, including the following:

- receipt of marketing approvals for our product candidate from the FDA and similar regulatory authorities outside the United States;
- obtaining product indications, other labeling information and product attributes that are acceptable and attractive to the medical community, third-party payors and patients;
- our ability to manufacture product commercially at acceptable costs;
- establishing and maintaining commercial manufacturing arrangements with third parties;
- successfully commercializing our product candidate, if approved, whether alone or in collaboration with others;
- a continued acceptable safety profile of the product candidate following approval; and
- obtaining, maintaining, enforcing and defending intellectual property rights and claims and available product exclusivities.

If we do not achieve one or more of these factors in a timely manner or at all, we could experience significant delays or an inability to successfully commercialize our product candidate, which would materially harm our business. In addition, even if we obtain regulatory approvals for tecarfarin, the timing or scope of any approval may prohibit or reduce our ability to commercialize tecarfarin successfully. For example, if the approval process takes too long, we may miss market opportunities and give other companies the ability to develop competing products or establish market dominance. Also, any regulatory approval we ultimately obtain may be limited or subject to restrictions or post-approval commitments that render tecarfarin not commercially viable. For example, regulatory authorities may grant approval contingent on the performance of costly post-marketing clinical trials or, outside the U.S., they may not accept or approve the price we intend to charge for tecarfarin. Further, the FDA or comparable foreign regulatory authorities may place conditions on approvals, such as risk management plans and Risk Evaluation and Mitigation Strategies, or REMS, to assure the safe use of the drug. If the FDA concludes a REMS is needed, the sponsor of the NDA must submit a proposed REMS; the FDA will not approve the NDA without an approved REMS, if required. A REMS could include medication guides, physician communication plans, and/or elements to assure safe use, such as restricted distribution methods, patient registries and other risk minimization tools. The FDA may also require a REMS for an approved product when new safety information emerges. Any of these limitations on approval or marketing could restrict the commercial promotion, distribution, prescription or dispensing of tecarfarin. Moreover, product approvals may be withdrawn for non-compliance with regulatory standards or if problems occur following the initial marketing of the product. Any of the foregoing scenarios could materially harm the commercial success of tecarfarin.

Our potential future product candidate, tecarfarin, may fail to achieve the degree of market acceptance by physicians, patients, healthcare payors and others in the medical community necessary for commercial success.

The commercial success of any potential future product candidates, including tecarfarin, for which we may obtain marketing approval from the FDA or other regulatory authorities will depend upon their acceptance by the medical community and third-party payors as clinically useful, cost-effective and safe. The degree of market acceptance of any drug depends on a number of factors, such as:

- effectively competing with other therapies;
- the prevalence and severity of any side effects;
- success of patients in well-controlled clinical trials compared to real-world success of patients post FDA approval;
- our ability to educate and increase physician awareness of the benefits of our products relative to competing drugs;
- the willingness of physicians and healthcare organizations to change their current treatment practices, especially with respect to warfarin, a drug that is dominant in the market and with which physicians and healthcare organizations have 60 years of familiarity;
- the willingness of hospitals and hospital systems to include our product candidates as treatment options;
- efficacy and potential advantages compared to alternative treatments;
- the price we charge for our product candidates;
- interpretations of the results of our clinical trials;
- the status of our products on the formularies of third-party payers;
- convenience and ease of administration compared to alternative treatments;
- the willingness of the target patient population to try new therapies and of physicians to prescribe these therapies;
- the willingness of the target patient population to pay for our products, including co-pays under their health coverage plans;
- the accuracy of the international normalized ratio, or INR, testing and whether such testing can be conducted at home or in a medical facility such as a doctor's office. A prothrombin time, or PT, is a test used to help detect and diagnose a bleeding disorder or excessive clotting disorder; the INR is calculated from a PT result and is used to monitor how well an anticoagulant medication such as tecarfarin is working to prevent thrombosis;
- the strength of marketing and distribution support; and
- the availability of third-party coverage and adequate reimbursement.

The failure to attain market acceptance among the medical community, patients and third-party payors may have an adverse impact on our operations and profitability.

We have never submitted an NDA to the FDA or comparable applications to other regulatory authorities and we may not be successful in achieving approval of our product candidates.

We have never submitted an NDA to the FDA or comparable applications to other regulatory authorities and expect to rely on consultants and third-party contract research organizations, or CROs, with expertise in this area to assist us in this process. Securing FDA approval requires the submission of pre-clinical, clinical and/or pharmacokinetic data, information about product manufacturing processes and inspection of facilities and supporting information to the FDA

for each therapeutic indication to establish a product candidate's safety and efficacy for each indication. Regulatory authorities in other jurisdictions impose similar requirements. If we are unable to successfully complete the approval process with the FDA or comparable applications of other regulatory authorities, our business will not be successful.

Orphan Drug Designation does not translate to approval and, even if we obtain FDA approval, we may not enjoy marketing exclusivity or other expected benefits.

Although we have been granted orphan drug designation for tecarfarin, this does not mean FDA will approve the NDA. Even if we obtain FDA approval, we may not be able to obtain or maintain orphan drug exclusivity for tecarfarin. We may not be the first to obtain marketing approval of tecarfarin designation for the orphan-designated indication due to the uncertainties associated with developing pharmaceutical products. In addition, exclusive marketing rights in the United States may be limited if we seek approval for an indication broader than the orphan-designated indication or may be lost if the FDA later determines that the request for designation was materially defective or if we are unable to assure sufficient quantities of the product to meet the needs of patients with the rare disease or condition. Further, even if we obtain orphan drug exclusivity for a product, that exclusivity may not effectively protect the product from competition because different drugs with different active moieties may be approved for the same condition, or the competitive product is otherwise outside the scope of exclusivity. Even after an orphan drug is approved, the FDA can subsequently approve the same drug with the same active moiety for the same condition if the FDA concludes that the later drug is clinically superior in that it is shown to be safer, more effective or makes a major contribution to patient care or the manufacturer of the product with orphan exclusivity is unable to maintain sufficient product quantity. Orphan drug designation neither shortens the development time or regulatory review time of a drug nor gives the drug any advantage in the regulatory review or approval process, nor does it prevent competitors from obtaining approval of the same product candidate for indications other than those in which orphan drug designation have been granted.

After approval of tecarfarin, tecarfarin will remain subject to ongoing regulatory obligations and continued regulatory review, which may result in significant additional risk and expense.

Drug products remain subject to the jurisdiction of the FDA and non-U.S. regulatory authorities after they have been approved. Even if we obtain regulatory approval of tecarfarin, the FDA and other regulatory authorities may impose significant restrictions on its indicated uses or marketing or the conditions of approval, or impose ongoing requirements for potentially costly and time-consuming post-approval trials, including Phase 4 clinical trials, and post-market surveillance to monitor safety and efficacy. Our product candidate, if approved, as well as our marketed products are subject to ongoing regulatory requirements governing the manufacturing, labeling, packaging, storage, distribution, safety surveillance, advertising, promotion, sampling, recordkeeping and reporting of adverse events and other post-market information. These requirements include registration with the FDA and continued compliance with current Good Manufacturing Practices requirements, or cGMPs, and current Good Clinical Practices requirements, or GCPs, for any clinical trials that we conduct post-approval.

After approval, our products could be subject to labeling and other restrictions and we may be required to withdraw from the market or be subject to penalties if we fail to comply with regulatory requirements.

The product labeling, advertising and promotion of our products and our product candidates, if approved, are subject to regulatory requirements and continuing regulatory review. Government authorities, including the FDA and the Office of the Inspector General of the Department of Health and Human Services, or OIG, strictly regulate the promotional claims and activities that may be made about prescription products. A drug product may not be promoted for uses that are inconsistent with the product's approved labeling. If we receive marketing approval for tecarfarin, physicians may nevertheless legally prescribe our products to their patients in a manner that is inconsistent with the approved labeling. However, if we are found to have promoted such off-label uses, we may become subject to significant liability and government fines. The federal government has extracted very large settlements and levied very large civil and criminal fines against companies for alleged improper promotion, has enjoined companies from engaging in off-label promotion, and made companies agree to onerous multi-year corporate integrity agreements. The FDA has also requested that companies enter into consent decrees of permanent injunctions under which specified promotional conduct is changed or curtailed.

The occurrence of any event or penalty described above may inhibit our ability to commercialize our product candidates and generate revenue. Adverse regulatory action, whether pre- or post-approval, can also potentially lead to product liability claims and increase our product liability exposure.

We are subject, directly or indirectly, to federal and state obligations and regulations applicable to our marketing practices. If we are unable to comply, or have not complied, with such laws, we could face substantial penalties.

Our marketing and sales operations are subject to various federal and state fraud and abuse laws, including, without limitation, the federal and state anti-kickback statutes and false claims laws. With respect to sales and marketing activities by us or any future partner, advertising and promotional materials must comply with FDA rules in addition to other applicable federal, state and local laws in the United States and similar legal requirements in other countries. We also are subject, directly or indirectly through our customers and partners, to various fraud and abuse laws, including, without limitation, the U.S. Federal Healthcare Program Anti-Kickback Statute, U.S. False Claims Act, and similar state laws, which impact, among other things, most of our interactions with customers, including our proposed sales, marketing, and scientific/educational grant programs. We are also subject to complex laws and regulation regarding reporting and payment obligations as a result of our participation in the U.S. Medicaid Drug Rebate Program, the Federal Supply Schedule of the U.S. Department of Veterans Affairs, and other government drug programs. All of these activities are also potentially subject to U.S. federal and state consumer protection and unfair competition laws. Similar requirements exist in many of these areas in other countries. If investigated, we could be forced to incur substantial expense responding to the investigation and defending our actions. If unsuccessful in our defense, we could be found to be in violation and subject to substantial fines and penalties,

We currently do not have an agreement with a third-party manufacturer for the production of tecarfarin and intend to rely upon third parties to produce our product candidate.

We currently do not have an agreement with any third-party manufacturers for the production of tecarfarin and there can be no assurance that we will be able to enter into an agreement on acceptable terms. If an agreement is not entered into, we may experience longer manufacturing lead times for any purchase orders we place with a manufacturer under purchase orders. We intend to rely on third-party manufacturers to produce tecarfarin for our clinical studies who are expected to purchase materials from third-party vendors and transport the materials necessary to produce tecarfarin, such as the required reagents and containers. If a third-party manufacturer was to experience any prolonged disruption for our manufacturing, or face enforcement scrutiny by regulatory authorities, we could be forced to seek additional third-party manufacturing contracts, thereby increasing our development costs and negatively impacting our timelines and any commercialization costs. If we change manufacturers at any point during the development process or after approval of a product candidate, we will be required to demonstrate comparability between the product manufactured by the old manufacturer and the product manufactured by the new manufacturer. If we are unable to do so, we may need to conduct additional clinical trials with product manufactured by the new manufacturer, thereby delaying our NDA submission or approval.

If the manufacturer upon which we rely fails to comply with stringent regulations, we may face delays in the development and commercialization of, or be unable to meet demand for, our product candidates and may lose potential revenues.

Any problems or delays our contract manufacturers experience in preparing for commercial-scale manufacturing of a product candidate or component may result in a delay in product development timelines and FDA or comparable foreign regulatory authority approval of the product candidate or may impair our ability to manufacture commercial quantities or such quantities at an acceptable cost and quality, which could result in the delay, prevention, or impairment of clinical development and commercialization of our product candidates and may materially harm our business, financial condition, results of operations, stock price and prospects.

In addition, manufacturers of drug products and their facilities are subject to continual review and periodic inspections by the FDA and other regulatory authorities for compliance with cGMP requirements relating to quality control, quality assurance and corresponding maintenance of records and documents. Although we do not have day-to-day control over our contract manufacturers' compliance with these requirements, we are responsible for ensuring compliance with such requirements. Our failure, or the failure of our contract manufacturers, to comply with applicable regulations could result in sanctions being imposed on us, including clinical holds, fines, injunctions, civil penalties, delays, suspension or withdrawal of approvals, revocation of licenses, seizures or recalls of product candidates, operating restrictions and criminal prosecutions, any of which

would significantly and adversely affect supplies of our product candidates and our business. If a contract manufacturer's facilities do not pass a pre-approval inspection or do not have a cGMP compliance status acceptable to the FDA or a comparable foreign regulatory authority, our product candidate will not be approved.

In addition, application holders must obtain FDA approval for product and manufacturing changes, depending on the nature of the change. Moreover, in the United States, the distribution of product samples to physicians must comply with the requirements of the U.S. Prescription Drug Marketing Act. In addition, our marketed products will have to comply with the Drug Supply Chain Security Act of 2013, which requires drug companies to enable electronic tracking of their products though the U.S. supply chain.

Any deviations from regulatory requirements may also require remedial measures that may be costly and/or time-consuming for us or a third-party to implement and that may include the temporary or permanent suspension of a clinical trial or the temporary or permanent closure of a facility. Any such remedial measures imposed upon us or third parties with whom we contract could materially harm our business. Any delays in obtaining products or product candidates that comply with the applicable regulatory requirements may result in delays to product approvals, and commercialization. It may also require that we conduct additional trials.

We face substantial competition, which may result in others discovering, developing or commercializing competing products more successfully than we do, or, perhaps obtaining approval before our product and, potentially delaying our approval.

The development and commercialization of new drugs is highly competitive. We face competition with respect to developing our current product candidates, and we will face competition with respect to any products that we may seek to develop or commercialize in the future, from major pharmaceutical companies, specialty pharmaceutical companies and biotechnology companies worldwide. We are seeking to develop tecarfarin as a marketable VKA, once-daily OAC designed to treat and prevent thrombosis for ESRD patients with AFib. If we succeed in developing tecarfarin, we shall face substantial competition. Traditionally, the drug of choice for CKD/ESRD patients has been warfarin. The reasoning for such therapy is that warfarin undergoes extensive hepatic metabolism into inactive compounds, theoretically posing less risk to the renally impaired patient. Randomized trials to date of apixaban versus warfarin for AFib excluded patients with severe and end-stage kidney disease, calculated creatinine clearance <25 mL per minute. The FDA cautiously extended apixaban use to patients with ESRD on hemodialysis based on limited pharmacokinetic data by 8 subjects. Additionally, the lower limit for creatinine clearance has been adjusted for rivaroxaban for stroke prevention in AFib, thereby permitting use in the ESRD population. The remaining two DOACs, dabigatran and edoxaban, are approved for use with dosage adjustment in moderate-severe CKD for stroke prevention. However, the net benefit of anticoagulation with apixaban or warfarin in the ESRD population is controversial.

In 2019, RENal Hemodialysis Patients Allocated Apixaban Versus Warfarin in Atrial Fibrillation (RENAL-AF) Randomized Clinical Trial was terminated early by the sponsor.

AXADIA is an investigator-driven, prospective, parallel-group, single country, multi-center phase IIIb trial to assess the safety of apixaban versus the VKA phenprocoumon in patients with non-valvular AFib and ESRD on hemodialysis treatment. The trial will be conducted in about 25-30 sites in Germany with an estimated completion date of July 2023.

Strategies for the Management of Atrial Fibrillation in patients Receiving Dialysis (SAFE-D) is a pilot trial in Toronto, Canada to evaluate the feasibility of conducting a randomized controlled trial comparing anticoagulation strategies in patients with AFib receiving dialysis (either hemodialysis or peritoneal dialysis). The trial's estimated completion date is December 31, 2022.

Many of these named products are marketed by some of the largest and most successful pharmaceutical companies. The companies that market these products have substantially more resources than we do and substantially more experience developing and marketing pharmaceuticals. We may not be able to successfully compete with these existing products. Potential competitors also include academic institutions, government agencies and other public and private research organizations that conduct research, seek patent protection and establish collaborative arrangements for research, development, manufacturing and commercialization of competing drugs and potentially competing drugs. Our competitors are or may be attempting to develop therapeutics for our target indications.

Factors affecting competition in these markets include the financial, research and development, testing, and marketing strengths of individual competitors, trends in industry consolidation, consumers' product options, product quality, price and technology, reputation, customer service capabilities and access to market partners and customers. Eliquis is manufactured and distributed by Bristol Myers Squibb, amiodarone is manufactured

and distributed by several companies including Sanofi, Baxter, and Pfizer, Pradaxa is manufactured and distributed by Boehringer Ingelheim, Xarelto is manufactured and distributed by Janssen Pharmaceuticals, and Savaysa is manufactured and distributed by Daiichi

Sankyo. Each of these organizations has a long operating history, extensive resources, strong brand recognition and large customer bases. As a result, we expect they will be able to devote greater resources than we can to the manufacture, promotion and sale of their products; receive greater resources and support than we will from market partners and independent distributors; initiate and withstand substantial price competition; and take advantage more readily than we could of acquisition and other strategic market opportunities. In addition, these or other organizations could succeed in developing new products that perform better or more cost-effectively than our products and product candidates in their respective markets. Moreover, changes in health trends, diet or other factors could substantially reduce the commercial attractiveness or viability of the markets for anti-anginal, anticoagulant, anti-arrhythmic and anti-platelet products.

The high level of competition in these markets could result in pricing pressure, reduced margins, the inability of our product candidates to achieve market acceptance and other impediments to commercial success. As a result, there can be no assurance that we will be able to complete the development of competitive products and commercialize them on a competitive basis.

Mergers and acquisitions in the pharmaceutical, biotechnology and diagnostic industries may result in even more resources being concentrated among a smaller number of our competitors. Smaller or early-stage companies may also prove to be significant competitors, particularly through collaborative arrangements with large and established companies. These companies compete with us in recruiting and retaining qualified scientific and management personnel, establishing clinical trial sites and patient registration for clinical trials, as well as in acquiring technologies complementary to, or necessary for, our programs.

If serious adverse effects are identified with respect to any of our product candidates or any of our approved products, we may need to modify or abandon our development of that product candidate, discontinue sale of an approved product, or change our labeling to reflect new safety risks.

It is impossible to guarantee when, or if, any of our product candidates will prove safe enough to receive regulatory approval. It is impossible to guarantee that safety issues that may arise during development will not significantly decrease the commercial potential of our product candidates. In addition, there can be no assurance that our clinical trials will identify all relevant safety issues. Known or previously unidentified adverse effects can adversely affect regulatory approvals or marketing of approved products. In such an event, we might need to abandon marketing efforts or development of that product or product candidate or enter into a partnership to continue development.

Serious adverse events have occurred in the clinical trials of our product candidate. For example, major hemorrhages occurred in 1.6% of the blinded tecarfarin patients randomized in the EMBRACE-AC trial. We expect that additional patients will experience serious adverse events in our future clinical trials and during marketing if our products are approved. Design features of ACTOR AF, such as using a smaller number of patients and using MACE clinical outcomes as an endpoint, may produce results that show an imbalance in adverse events between treatment groups, when no such imbalance truly exists, or may not permit an assessment of the risk of rare events (due to the overall reduced size of the safety database), either of which could lead the FDA to require additional studies to demonstrate the safety of tecarfarin.

If a regulatory agency discovers adverse events of unanticipated severity or frequency it may impose restrictions on that product or us, including requiring withdrawal of the product from the market. Among other legal and administrative actions, a regulatory agency may:

- mandate modifications to product labelling or promotional materials or require us to provide corrective information to healthcare practitioners;
- withdraw any regulatory approvals;
- place any ongoing clinical trials on clinical hold;
- refuse to approve pending applications or supplements to approved applications filed by us, our partners or our potential future partners;
- impose restrictions on operations, including costly new manufacturing, licensing or packaging requirements; or
- seize or detain products or require a product recall.

In addition, the occurrence of any of the foregoing, even if promptly remedied, could (1) negatively impact the perception of us or the relevant product among the medical community, patients or third-party payors and (2) result in product liability litigation that could result in the company paying substantial amounts of money in settlements or verdicts.

Recently enacted and future legislation may increase the difficulty and cost for us to obtain marketing approval of and commercialize tecarfarin and affect the prices we may obtain.

In the United States and some foreign jurisdictions, there have been a number of legislative and regulatory changes and proposed changes regarding the healthcare system that could prevent or delay marketing approval for tecarfarin, restrict or regulate post-approval activities and affect our ability to profitably sell tecarfarin. Legislative and regulatory proposals have been made to expand post-approval requirements and restrict sales and promotional activities for pharmaceutical products. We do not know whether additional legislative changes will be enacted, or whether the FDA regulations, guidance or interpretations will be changed, or what the impact of such changes on the marketing approvals of tecarfarin, if any, may be. In addition, increased scrutiny by the U.S. Congress of the FDA's approval process may significantly delay or prevent marketing approval, as well as subject us to more stringent product labeling and post-marketing testing and other requirements.

In the United States, under the Medicare Modernization Act, or MMA, Medicare Part D provides coverage to the elderly and disabled for outpatient prescription drugs by approving and subsidizing prescription drug plans offered by private insurers. The MMA also authorizes Medicare Part D prescription drug plans to use formularies where they can limit the number of drugs that will be covered in any therapeutic class. The Part D plans use their formulary leverage to negotiate rebates and other price concessions from drug manufacturers. Also under the MMA, Medicare Part B provides coverage to the elderly and disabled for physician-administered drugs on the basis of the drug's average sales price, a price that is calculated according to regulatory requirements and that the manufacturer reports to Medicare quarterly.

Both Congress and the Centers for Medicare & Medicaid Services, or CMS, the agency that administers the Medicare program, from time to time consider legislation, regulations, or other initiatives to reduce drug costs under Medicare Parts B and D. For example, under the 2010 Affordable Care Act, drug manufacturers are required to provide a 50% discount on prescriptions for branded drugs filled while the beneficiary is in the Medicare Part D coverage gap, also known as the "donut hole." The Bipartisan Budget Act of 2018 increased the manufacturer's subsidy under the program from 50% to 70% of the negotiated price, beginning in 2019. There have been legislative proposals to repeal the "non-interference" provision of the MMA to allow CMS to leverage the Medicare market share to negotiate larger Part D rebates. Further cost reduction efforts could decrease the coverage and price that we receive for tecarfarin and could seriously harm our business. Private payors often follow Medicare coverage policy and payment limitations in setting their own reimbursement rates, and any reduction in reimbursement under the Medicare program may result in a similar reduction in payments from private payors.

The 2010 Affordable Care Act is intended to broaden access to health Insurance and reduce or constrain the growth of healthcare spending. Further, the Affordable Care Act imposes a significant annual fee on companies that manufacture or import branded prescription drug products. It also increased the amount of the rebates drug manufacturers must pay to state Medicaid programs, required that Medicaid rebates be paid on managed Medicaid utilization, and increased the additional rebate on "line extensions" (such as extended-release formulations) of solid oral dosage forms of branded products. The law also contains substantial provisions affecting fraud and abuse compliance and transparency, which may require us to modify our business practices with healthcare practitioners, and incur substantial costs to ensure compliance.

Many members of the Republican Party have consistently opposed the Affordable Care Act since it was signed. Efforts to repeal the Act have been attempted numerous times and some portions of the Act have been amended in 2017 and 2018. It is unclear whether further amendments or repeal will be effectuated and what the effect on the healthcare sector will be. In addition to potential changes to the Affordable Care Act, there are indications that the Medicaid and Medicare programs may be restructured, which could lead to revisions in coverage and reimbursement of prescription drugs. While we are unable to predict what legislation, if any, may potentially be enacted, to the extent that future changes affect how our product candidates could be paid for and/or reimbursed by the government and private payers, our business could be adversely affected.

In addition, other legislative changes have been proposed and adopted in the United States since the Affordable Care Act was enacted. For example, the Budget Control Act of 2011 included, among other things, provisions that have led to 2% across-the-board reductions in Medicare payment amounts. Several states have adopted or are considering

adopting laws that require pharmaceutical companies to provide notice prior to raising prices and to justify price increases. We expect that additional healthcare reform measures will be adopted in the future, any of which could limit the amounts that federal and state governments will pay for healthcare products and services, and in turn could significantly reduce the projected value of certain development projects and reduce our profitability.

If we market any of our products in a manner that violates healthcare fraud and abuse laws, or if we violate government price reporting laws, we may be subject to civil or criminal penalties.

The FDA and other government authorities enforce laws and regulations that require that the promotion of pharmaceutical products be consistent with the approved prescribing information. While physicians may prescribe an approved product for a so-called "off-label" use under the practice of medicine, it is unlawful for a pharmaceutical company to promote its products in a manner that is inconsistent with its approved label and any company which engages in such conduct may be subject to significant liability. Similarly, industry codes in the European Union and other foreign jurisdictions prohibit companies from engaging in off-label promotion and regulatory agencies in various countries enforce violations of the code with civil penalties. While we intend to ensure that our promotional materials are consistent with our label, regulatory agencies may disagree with our assessment and may issue untitled letters, warning letters or may institute other civil or criminal enforcement proceedings. In addition to FDA restrictions on marketing of pharmaceutical products, several other types of state and federal healthcare fraud and abuse laws have been applied in recent years to restrict certain marketing practices in the pharmaceutical industry. These laws include the U.S. Federal Healthcare Program Anti-Kickback Statute, U.S. False Claims Act and similar state laws. Because of the breadth of these laws and the narrowness of the safe harbors, it is possible that some of our business activities could be subject to challenge under one or more of these laws.

The U.S. Federal Healthcare Program Anti-Kickback Statute prohibits, among other things, knowingly and willfully offering, paying, soliciting or receiving remuneration to induce, or in return for, purchasing, leasing, ordering or arranging for the purchase, lease or order of any healthcare item or service reimbursable under Medicare, Medicaid or other federally financed healthcare programs. This statute has been interpreted broadly to apply to arrangements between pharmaceutical manufacturers on the one hand and prescribers, purchasers and formulary managers on the other. Although there are several statutory exemptions and regulatory safe harbors protecting certain common activities from prosecution, the exemptions and safe harbors are drawn narrowly, and practices that involve remuneration intended to induce prescribing, purchasing or recommending may be subject to scrutiny if they do not qualify for an exemption or safe harbor. Our practices may not, in all cases, meet all of the criteria for safe harbor protection from anti-kickback liability. Moreover, recent health care reform legislation has strengthened these laws. For example, the Affordable Care Act, among other things, amends the intent requirement of the U.S. Federal Healthcare Program Anti-Kickback Statute and criminal health care fraud statutes; a person or entity no longer needs to have actual knowledge of this statute or specific intent to violate it. In addition, the Affordable Care Act provides that the government may assert that a claim including items or services resulting from a violation of the U.S. Federal Healthcare Program Anti-Kickback Statute constitutes a false or fraudulent claim for purposes of the U.S. False Claims Act. Federal false claims laws, including the U.S. False Claims Act, impose criminal and civil penalties, including through civil whistleblower or qui tam actions, against individuals or entities for knowingly presenting, or causing to be presented, to the federal government, claims for payment that are false or fraudulent or making a false statement to avoid, decrease or conceal an obligation to pay money to the federal government.

Over the past few years, pharmaceutical and other healthcare companies have been prosecuted under these laws for a variety of alleged promotional and marketing activities, such as: allegedly providing free trips, free goods, sham consulting fees and grants and other monetary benefits to prescribers; reporting to pricing services inflated average wholesale prices that were then used by federal programs to set reimbursement rates; engaging in off-label promotion that caused claims to be submitted to Medicare or Medicaid for non-covered, off-label uses; using a charity as an illegal conduit to cover the copays of Medicare patients; and submitting inflated best price information to the Medicaid Drug Rebate Program to reduce liability for Medicaid rebates.

Other restrictions under applicable U.S. federal and state healthcare laws and regulations may include the following:

- the federal Health Insurance Portability and Accountability Act of 1996, or HIPAA, imposes criminal and civil liability for, among other things, knowingly and wilfully executing or attempting

to execute a scheme to defraud any healthcare benefit program or making false statements relating to healthcare matters;

- HIPAA, as amended by the Health Information Technology for Economic and Clinical Health, or HITECH, Act and its implementing regulations, also imposes obligations, including mandatory contractual terms, on certain types of people and entities with respect to safeguarding the privacy, security and transmission of individually identifiable health information;
- the federal Physician Payment Sunshine Act requires applicable manufacturers of covered drugs, devices, biologics, and medical supplies for which payment is available under Medicare, Medicaid, or the Children's Health Insurance Program, with specific exceptions, to report payments and other transfers of value to physicians and teaching hospitals, as well as certain ownership and investment interests held by physicians and their immediate family, which includes annual data collection and reporting obligations; and
- analogous state and foreign laws and regulations, such as state anti-kickback and false claims laws, may apply to sales or marketing arrangements and claims involving healthcare items or services reimbursed by non-governmental third-party payors, including private insurers.

Some state laws require pharmaceutical companies to comply with the pharmaceutical industry's voluntary compliance guidelines and the relevant compliance guidance promulgated by the federal government and may require drug manufacturers to report information related to payments and other transfers of value to physicians and other healthcare providers or marketing expenditures to federal and state agencies. State and foreign laws also govern the privacy and security of health information in some circumstances, many of which differ from each other in significant ways and often are not preempted by HIPAA, thus complicating compliance efforts. Tracking and reporting may be burdensome and require a significant expenditure to comply with applicable requirements.

Our ability to generate product revenues will be diminished if our products sell for inadequate prices or patients are unable to obtain adequate levels of reimbursement.

Our ability to commercialize our products, alone or with collaborators, will depend in part on the extent to which reimbursement will be available from:

- government and health administration authorities;
- private health maintenance organizations and health insurers; and
- other healthcare payers.

Patients generally expect that products such as ours are covered and reimbursed by third-party payors for all or part of the costs and fees associated with their use. If such products are not covered and reimbursed then patients may be responsible for the entire cost of the product, which can be substantial. Therefore, health care providers generally do not prescribe products that are not covered and reimbursed by third-party payors in order to avoid subjecting their patients to such financial liability. The existence of adequate coverage and reimbursement for the products by government and private insurance plans is central to the acceptance of tecarfarin and any future products we provide.

During the past several years, third-party payors have undertaken cost-containment initiatives including different payment methods, monitoring health care expenditures, and anti-fraud initiatives. For some governmental programs, such as Medicaid, coverage and reimbursement differ from state to state, and some state Medicaid programs may not pay an adequate amount for tecarfarin or any of our other products or may make no payment at all. Furthermore, the health care industry in the United States has experienced a trend toward cost containment as government and private insurers seek to control health care costs by imposing lower payment rates and negotiating reduced contract rates with service providers. Therefore, we cannot be certain that our services will be reimbursed at a level that is sufficient to meet our costs.

Obtaining coverage and reimbursement approval of a product from a government or other third-party payor is a time-consuming and costly process that could require us to provide to the payor supporting scientific, clinical and cost-effectiveness data for the use of our products. Even if we obtain coverage for a given product, the resulting reimbursement payment rates might not be adequate for us to achieve or sustain profitability or may require co-payments that patients find unacceptably high. Patients are unlikely to use tecarfarin or any future product candidates unless coverage is provided and reimbursement is adequate to cover a significant portion of the cost of tecarfarin or any future product candidates.

We intend to seek approval to market tecarfarin and future product candidates in both the United States and in selected foreign jurisdictions. If we obtain approval in one or more foreign jurisdictions for tecarfarin or any future product candidates, we will be subject to rules and regulations in those jurisdictions. In some foreign countries, particularly those in the European Union, the pricing of drugs is subject to governmental control. In these countries, pricing negotiations with governmental authorities can take considerable time after obtaining marketing approval of a product candidate. In addition, market acceptance and sales of product candidates will depend significantly on the availability of adequate coverage and reimbursement from third-party payors for product candidates and may be affected by existing and future health care reform measures.

Third-party payors, whether domestic or foreign, or governmental or commercial, are developing increasingly sophisticated methods of controlling healthcare costs. In both the United States and certain foreign jurisdictions, there have been a number of legislative and regulatory changes to the health care system that could impact our ability to sell our products profitably. In particular, in the Affordable Care Act, as amended by the Health Care and Education Affordability Reconciliation Act, among other things, revised the methodology by which rebates owed by manufacturers to the state and federal government for covered outpatient drugs, including product candidates, under the Medicaid Drug Rebate Program are calculated, increased the minimum Medicaid rebates owed by most manufacturers under the Medicaid Drug Rebate Program, extended the Medicaid Drug Rebate program to utilization of prescriptions of individuals enrolled in Medicaid managed care organizations, subjected manufacturers to new annual fees and taxes for certain branded prescription drugs, and provided incentives to programs that increase the federal government's comparative effectiveness research.

There have been, and likely will continue to be, legislative and regulatory proposals at the foreign, federal and state levels directed at broadening the availability of healthcare and containing or lowering the cost of healthcare. We cannot predict the initiatives that may be adopted in the future, particularly in light of the new presidential administration in the United States, and any proposed changes to healthcare laws that could potentially affect our clinical development or regulatory strategy. The continuing efforts of the government, insurance companies, managed care organizations and other payors of healthcare services to contain or reduce costs of healthcare and/or impose price controls may adversely affect:

- the demand for tecarfarin, or future product candidates, if we obtain regulatory approval;
- our ability to set a price that we believe is fair for our products;
- our ability to generate revenue and achieve or maintain profitability;
- the level of taxes that we are required to pay; and
- the availability of capital.

Any reduction in reimbursement from Medicare, Medicaid or other government programs may result in a similar reduction in payments from private payors, which may adversely affect our future profitability.

We will rely on third parties and consultants to conduct all of our clinical trials. If these third parties or consultants do not successfully carry out their contractual duties, comply with regulatory requirements, or meet expected deadlines, we may be unable to obtain regulatory approval for any future product candidates.

We will rely on medical institutions, clinical investigators, contract laboratories, collaborative partners and other third parties, such as CROs, to conduct clinical trials on our product candidates. The third parties with whom we may contract for execution of any of our future clinical trials may play a significant role in the conduct of these trials and the subsequent collection and analysis of data. These third parties would not be our employees, and except for contractual duties and obligations, we would have limited ability to control the amount or timing of resources that they devote to any of our future programs. Although we may rely on these third parties to conduct our clinical trials, we would remain responsible for ensuring that each of our preclinical trials and clinical trials is conducted in accordance with applicable legal requirements, the investigational plan and the protocol. Moreover, whether we conduct trials ourselves or hire third parties to do so, the FDA and other similar regulatory authorities require us to comply with GCPs when we conduct, monitor, record and report the results of clinical trials to ensure that the data and results are scientifically credible and accurate, and that the trial subjects are adequately informed of the potential risks of participating in clinical trials.

In addition, the execution of clinical trials, and the subsequent compilation and analysis of the data produced, requires coordination among various parties. In order for these functions to be carried out effectively and efficiently, it is imperative that these parties communicate and coordinate with one another. Moreover, these third parties may also have relationships with other commercial entities, some of which may compete with us. If the third parties or consultants conducting our clinical trials do not perform their contractual duties or obligations, experience work stoppages, do not meet expected deadlines, terminate their agreements with us or need to be replaced, or if the quality or accuracy of the clinical data they obtain is compromised due to the failure to adhere to our clinical trial protocols or GCPs, or for any other reason, we may need to conduct additional clinical trials or enter into new arrangements with alternative third parties, which could be difficult, costly or impossible, and our clinical trials may be extended, delayed or terminated or may need to be repeated. If any of the foregoing were to occur, we may not be able to obtain, or may be delayed in obtaining, regulatory approval for and will not be able to, or may be delayed in our efforts to, successfully commercialize any future product candidates being tested in such trials.

We currently have limited distribution, marketing, support and sales capabilities and plan to rely on third-party distribution partners for the distribution, marketing, support and sales of our products which could delay or limit our ability to generate revenue.

We plan to utilize third-party service providers for the distribution and marketing and sales of our product candidates, if approved. Upon launch, we intend to promote utilizing third party collaborations in addition to building our own commercial infrastructure in anticipation of the approval of tecarfarin. Reliance on third-party service providers may prevent our direct control of key aspects of those critical functions including regulatory compliance, import and export operations, supply chain security, warehousing and inventory management, distribution, contract administration, invoicing, sales deductions administration, accounts receivable management and call center management. Any future distribution partners may hold significant control over important aspects of the commercialization of our products, including market identification, regulatory compliance, marketing methods, pricing, composition of sales force and promotional activities.

We may not be able to control the amount and timing of resources that any future third-party distribution partners may devote to our products, or prevent any third-party from pursuing the development of alternative technologies or products that compete with our products, except to the extent our contractual arrangements protect us against such activities. Also, we may not be able to prevent any other third-party from withdrawing its support of our products.

If third-party service providers fail to comply with applicable laws and regulations, fail to meet expected deadlines, encounter natural or other disasters at their facilities or otherwise fail to perform their services to us in a satisfactory or predicted manner, or at all, our ability to deliver product to meet commercial demand could be significantly impaired. In addition, we may use third parties to perform various other services for us relating to sample accountability and regulatory monitoring, including adverse event reporting, safety database management and other product maintenance services. If the quality or accuracy of the data maintained by these service providers is insufficient, our ability to continue to market our products could be jeopardized or we could be subject to regulatory sanctions, and any indemnity we may receive from such third-party service providers could be limited by such provider's ability to pay and otherwise might not be sufficient to cover all losses we may experience.

Our employees, independent contractors, consultants, commercial partners and vendors may engage in misconduct or other improper activities, including noncompliance with regulatory standards and requirements.

We are exposed to the risk of employee fraud or other illegal activity by our employees, independent contractors, consultants, commercial partners and vendors. Misconduct by these parties could include intentional, reckless and/or negligent conduct that fails to: (i) comply with the laws of the FDA and other similar foreign regulatory bodies; (ii) provide true, complete and accurate information to the FDA and other similar foreign regulatory bodies; (iii) comply with manufacturing standards we have established; (iv) comply with healthcare fraud and abuse laws in the United States and similar foreign fraudulent misconduct laws; or (v) report financial information or data accurately or to disclose unauthorized activities to us. Any such misconduct or noncompliance could negatively affect the FDA's review of our regulatory submission, including delaying approval or disallowance of certain information to support the submission, and/or delay a federal or state healthcare programs or a commercial insurer's determination regarding the availability of future

reimbursement for product candidates. If we obtain FDA approval of any product candidates and begin commercializing those products in the United States, our potential exposure under such laws will increase significantly, and our costs associated with compliance with such laws are also likely to increase. These laws may

impact, among other things, our current activities with principal investigators and research patients, as well as proposed and future sales, marketing and education programs. In particular, the promotion, sales and marketing of healthcare items and services, as well as certain business arrangements in the healthcare industry, are subject to extensive laws designed to prevent fraud, kickbacks, self-dealing and other abusive practices. These laws and regulations may restrict or prohibit a wide range of pricing, discounting, marketing and promotion, structuring and commission(s), certain customer incentive programs and other business arrangements generally. Activities subject to these laws also involve the improper use of information obtained in the course of patient recruitment for clinical trials.

It is not always possible to identify and deter employee misconduct, and the precautions we take to detect and prevent inappropriate conduct may not be effective in controlling unknown or unmanaged risks or losses or in protecting us from governmental investigations or other actions or lawsuits stemming from a failure to be in compliance with such laws or regulations. Efforts to ensure that our business arrangements will comply with applicable healthcare laws may involve substantial costs. It is possible that governmental and enforcement authorities will conclude that our business practices may not comply with current or future statutes, regulations or case law interpreting applicable fraud and abuse or other healthcare laws and regulations. If any such actions are instituted against us, and we are not successful in defending ourselves or asserting our rights, those actions could have a significant impact on our business, including the imposition of civil, criminal and administrative penalties, damages, disgorgement, monetary fines, possible exclusion from participation in Medicare, Medicaid and other federal healthcare programs, contractual damages, reputational harm, diminished profits and future earnings, and curtailment of our operations, any of which could adversely affect our ability to operate our business and our results of operations. In addition, the approval and commercialization of any product candidates outside the United States will also likely subject us to foreign equivalents of the healthcare laws mentioned above, among other foreign laws.

We intend to establish a sales force to market our product candidates. If we are not successful in doing so, our ability to generate sales and profits will be limited.

Although certain of our employees have commercialization experience, as a company we do not have an internal sales force and we currently have only limited commercial capabilities. We intend to establish an internal specialty cardiorenal sales force for the promotion and sale of tecarfarin, if approved. Establishing a pharmaceutical sales force is a difficult undertaking. Experienced and competent sales representatives and sales managers must be recruited, hired, trained, assigned appropriate territories, managed and compensated in such a way that they can achieve success in selling products to a sophisticated audience of healthcare professionals who frequently have little or no time to spend with sales personnel. In addition, our prospective sales force must compete against the sales forces of some of the largest and most successful pharmaceutical companies in the world, who will be promoting competing products. If we fail to hire and field a high-quality sales force, we may be unable to generate expected revenues and profits.

In addition, there are significant expenses and risks involved with establishing our own sales and marketing capabilities, including our ability to hire, retain and appropriately incentivize qualified individuals, generate sufficient sales leads, provide adequate training to sales and marketing personnel, and effectively manage a geographically dispersed sales and marketing team. Any failure or delay in the development of our internal sales, marketing and distribution capabilities could delay any product launch, which would adversely impact the commercialization of our product candidates. For example, if we recruit any sales representatives or establish marketing capabilities prior to commercial launch and the commercial launch is delayed or does not occur for any reason, we would have prematurely or unnecessarily incurred these commercialization expenses. This may be costly, and our investment would be lost if we cannot retain or reposition our sales and marketing personnel.

In addition to our own internal sales force, we may choose to collaborate with third parties that have direct sales forces and established distribution systems, either to augment our own sales force and distribution systems or in lieu of our own sales force and distribution systems. If we are unable to enter into such arrangements on acceptable terms or at all, we may not be able to successfully commercialize our product candidates. To the extent we commercialize our product candidates by entering into agreements with third-party collaborators, we may have limited or no control over the sales, marketing and distribution activities of these third parties, in which case our future revenues would depend heavily on the success of the efforts of these third parties. If we are not successful in commercializing tecarfarin or any future product candidates, either on our own or through collaborations with one or more third parties, our future product revenue will suffer and we could incur significant additional losses.

We plan to rely on collaborations and license arrangements with third parties to commercialize, market and promote our marketed products which may limit our ability to generate revenue and adversely affect our profitability.

We plan to rely on collaboration and other agreements with third parties with third parties with respect to our product candidates and future marketed products. Our current or any future collaborations or license arrangements may not be successful. With respect to the product candidates we have licensed, including our rights to tecarfarin in China, we depend upon collaborations with third parties to develop these product candidates in the licensed territories and we will depend substantially upon third parties to commercialize these product candidates. If we are unable to maintain current collaborations or enter into additional collaborations with established pharmaceutical or pharmaceutical service companies to provide the services we need, we may not be able to successfully commercialize our products.

Our future growth depends, in part, on our ability to penetrate foreign markets, where we would be subject to additional regulatory burdens and other risks and uncertainties.

Although our focus as this time is primarily on the U.S. market, our future profitability will depend, in part, on our ability to commercialize tecarfarin in foreign markets for which we intend to rely on collaborations with third parties. If we commercialize tecarfarin in foreign markets, we would be subject to additional risks and uncertainties, including:

- our customers' ability to obtain reimbursement in foreign markets;
- our inability to directly control commercial activities because we are relying on third parties;
- the burden of complying with complex and changing foreign regulatory, tax, accounting and legal requirements;
- different medical practices and customs in foreign countries affecting acceptance in the marketplace;
- import, export and foreign licensing requirements;
- different packaging and labeling requirements;
- longer accounts receivable collection times;
- longer lead times for shipping;
- language barriers for technical training;
- differing and/or reduced protection of intellectual property rights in some foreign countries;
- foreign currency exchange rate fluctuations; and
- the interpretation of contractual provisions governed by foreign laws in the event of a contract dispute.

Foreign sales of tecarfarin could also be adversely affected by the imposition of governmental controls, political and economic instability, trade restrictions and changes in tariffs, any of which may adversely affect our results of operations.

The ongoing COVID-19 global health crisis may impact our planned operations, including our pivotal Phase 3 clinical trial

In January 2020, the World Health Organization declared a global pandemic for the novel strain of coronavirus, COVID-19. Since then, the COVID-19 coronavirus has spread to multiple countries, including throughout the United States. We may experience disruptions as a result of the pandemic if the pandemic continues or increases in severity, including:

- unwillingness of potential study participants to enroll in our pivotal Phase 3 clinical trial and/or visit healthcare facilities;

- postponement of enrollment in our pivotal Phase 3 clinical trial
- postponement of the initiation of our pivotal Phase 3 clinical trial;

- diversion of healthcare resources away from the conduct of clinical trials, including the diversion of hospitals serving as our clinical trial sites and hospital staff supporting the conduct of our clinical trial;
- interruption of key clinical trial activities, such as clinical site visits by study participants and clinical trial site monitoring, due to limitations on travel imposed or recommended by federal or state governments, employers and others;
- limitations in employee resources that would otherwise be focused on the conduct of our pivotal Phase 3 clinical trial, including because of sickness of employees or their families or the desire of employees to avoid contact with large groups of people;
- delays in receiving approval from local regulatory authorities to initiate our pivotal Phase 3 clinical trial;
- delays in clinical sites receiving the supplies and materials needed to conduct our pivotal Phase 3 clinical trial;
- interruption in global shipping that may affect the manufacture and transport of clinical trial materials, such as investigational drug product used in our clinical trial;
- changes in local regulations as part of a response to the COVID-19 coronavirus outbreak which may require us to change the ways in which our pivotal Phase 3 clinical trial is conducted, which may result in unexpected costs, or to discontinue the clinical trial altogether;
- delays in necessary interactions with local regulators, ethics committees and other important agencies and contractors due to limitations in employee resources or forced furlough of government employees; and
- delay in the timing of interactions with the FDA due to absenteeism by federal employees or by the diversion of their efforts and attention to approval of other therapeutics or other activities related to COVID-19.

Our business and the business of the suppliers of our clinical product candidate is expected to be materially and adversely affected by the pandemic. While we are currently not experiencing material delays, such events could result in the delay or complete or partial closure of clinical trial sites or one or more manufacturing facilities which could impact our supply of our clinical product candidate. In addition, it could impact economies and financial markets, resulting in an economic downturn that could impact our ability to raise capital or slow down potential partnering relationships.

The effects of the governmental orders may negatively impact productivity, disrupt our business and delay our pivotal Phase 3 clinical trial program and timelines, the magnitude of which will depend, in part, on the length and severity of the restrictions and other limitations on our ability to conduct our business in the ordinary course.

In addition, the COVID-19 outbreak could disrupt our operations due to absenteeism by infected or ill members of management or other employees, or absenteeism by members of management and other employees who elect not to come to work due to the illness affecting others in our office, or due to quarantines. The COVID-19 illness could also impact members of our Board of Directors resulting in absenteeism from meetings of the directors or committees of directors, and making it more difficult to convene the quorums of the full Board of Directors or its committees needed to conduct meetings for the management of our affairs.

The global outbreak of the virus continues to rapidly evolve. The extent to which the virus may continue to impact our business and clinical trials will depend on future developments, which are highly uncertain and cannot be predicted with confidence, such as the ultimate geographic spread of the disease, the duration of the outbreak, travel restrictions and social distancing in the United States, business closures or business disruptions and the effectiveness of actions taken in the United States and other countries to contain and treat the disease. We do not yet know the full extent of potential delays or impacts on our business, operations, or the global economy as a whole. While the spread of COVID-19 may eventually be contained or mitigated, there is no guarantee that a future outbreak of this or any other widespread epidemics will not occur, or that the global economy will recover, either of which could seriously harm our business.

While we are currently not experiencing any delays, we may in the future experience delays. These delays may result in the need for trials to be redesigned and may impact whether they will be completed on schedule, if at all. Clinical trials can be delayed for a variety of reasons, including the COVID-19 pandemic, delays in obtaining regulatory approval to commence a clinical trial, in securing clinical trial agreements with prospective sites with acceptable terms, in obtaining institutional review board approval to conduct a clinical trial at a prospective site, in recruiting patients to participate

in a clinical trial or in obtaining sufficient supplies of clinical trial materials. Manufacturing considerations for clinical development candidates may include an expected several month lead time following a decision to commence any clinical trial(s) and capacity considerations of our third-party contract manufacturers to provide clinical supply of our product candidates could cause delays in clinical trials. Furthermore, due to the COVID-19 pandemic, many manufacturers have been prioritizing the manufacture of COVD-19 related products, increasing the manufacturing lead times for non-COVID-19 related products. Many factors affect patient enrollment, including the size of the patient population, the proximity of patients to clinical sites, the eligibility criteria for the clinical trial, competing clinical trials and new drugs approved for the conditions we are investigating. Clinical investigators will need to decide whether to offer their patients enrollment in clinical trials of our product candidates versus treating these patients with commercially available drugs that have established safety and efficacy profiles. Any delays in completing our clinical trials will increase our costs, slow down our product development and timeliness and approval process and delay our ability to generate revenue.

Compliance with governmental regulations regarding the treatment of animals used in research could increase our operating costs, which would adversely affect the commercialization of our products.

The Animal Welfare Act, or AWA, is the federal law that covers the treatment of certain animals used in research. Currently, the AWA imposes a wide variety of specific regulations that govern the humane handling, care, treatment and transportation of certain animals by producers and users of research animals, most notably relating to personnel, facilities, sanitation, cage size, and feeding, watering and shipping conditions. Third parties with whom we contract are subject to registration, inspections and reporting requirements under the AWA. Furthermore, some states have their own regulations, including general anti-cruelty legislation, which establish certain standards in handling animals. Comparable rules, regulations, and or obligations exist in many foreign jurisdictions. If we or our contractors fail to comply with regulations concerning the treatment of animals used in research, we may be subject to fines and penalties and adverse publicity, and our operations could be adversely affected.

General Company-Related Risks

If we fail to attract and keep senior management and key scientific personnel, we may be unable to successfully develop tecarfarin or any future product candidates, conduct our clinical trials and commercialize our product candidates or any future products we develop.

Our management team has expertise in many different aspects of fundraising, drug development and commercialization. We believe that our future success is highly dependent upon the contributions of our senior management, particularly Quang Pham, our Chief Executive Officer. We do not have an insurance policy on the life of our Chief Executive Officer and we do not have “key person” life insurance policies for any of our other officers or advisors. The loss of services of any of these individuals could delay or prevent the successful development of our product pipeline, completion of our planned clinical trials or the commercialization of tecarfarin or any other future products we develop, which could adversely affect our operating results.

We will need to hire additional personnel, including experienced marketing and sales representatives, as we expand our clinical development and commercial activities. We could experience difficulties attracting and retaining qualified employees in the future. For example, competition for qualified personnel in the pharmaceuticals field is intense due to the limited number of individuals who possess the skills and experience required by our industry. Other pharmaceutical companies with which we compete for qualified personnel have greater financial and other resources, different risk profiles, and a longer history in the industry than we do. They also may provide more diverse opportunities and better chances for career advancement. Some of these characteristics may be more appealing to high-quality candidates than what we have to offer. We may not be able to attract and retain quality personnel on acceptable terms, or at all. In addition, to the extent we hire personnel from competitors, we may be subject to allegations that they have been improperly solicited or that they have divulged proprietary or other confidential information or that their former employers own their research output. If we are unable to continue to attract and retain high-quality personnel, the rate and success at which we can develop and commercialize product candidates could be limited.

We will need to increase the size of our organization, and we may experience difficulties in managing this growth.

As of January 17, 2023, we had one employee who works full time. We will need to continue to expand our managerial, operational, finance and other resources to manage our operations, commercialize tecarfarin or any other product candidates, if approved, and continue our development activities. Our management and personnel systems and facilities currently in place may not be adequate to support this future growth. Our need to effectively execute our growth strategy requires that we:

- manage any of our future clinical trials effectively;
- identify, recruit, retain, incentivize and integrate additional employees;
- manage our internal development efforts effectively while carrying out our contractual obligations to third parties; and
- continue to improve our operational, financial and management controls, reporting systems and procedures.

Due to our limited financial resources and our limited experience in managing a company with such anticipated growth, we may not be able to effectively manage the expansion of our operations or recruit and train additional qualified personnel. The physical expansion of our operations may lead to significant costs and may divert our management and business development resources. Any inability to manage growth could delay the execution of our development and strategic objectives or disrupt our operations.

If product liability lawsuits are brought against us, we may incur substantial liabilities and may be required to limit commercialization of any future products we develop.

We face an inherent risk of product liability as a result of the clinical testing of tecarfarin and any of our future product candidates. We will face further risk if we commercialize tecarfarin or any of our product candidates. For example, we may be sued if any product we sell or any product we develop allegedly causes injury or is found to be otherwise unsuitable during product testing, manufacturing, marketing or sale. Any such product liability claims may include allegations of defects in manufacturing, defects in design, a failure to warn of dangers inherent in the product, negligence, strict liability and a breach of warranties. Claims could also be asserted under state consumer protection acts. If we cannot successfully defend ourselves against product liability claims, we may incur substantial losses or be required to limit commercialization of our products. Even a successful defense would require significant financial and management resources. Regardless of the merits or eventual outcome, liability claims may result in:

- decreased demand for our product candidates or products we develop, including tecarfarin;
- termination of clinical trial sites or entire trial programs;
- injury to our reputation and significant negative media attention;
- withdrawal of clinical trial participants or cancellation of clinical trials;
- significant costs to defend the related litigation;
- a diversion of management's time and our resources;
- substantial monetary awards to trial participants or patients;
- regulatory investigations, product recalls, withdrawals or labeling, marketing or promotional restrictions;
- loss of revenue;
- the inability to commercialize any products we develop; and
- a decline in our share price.

Our inability to obtain and maintain sufficient product liability insurance at an acceptable cost and scope of coverage to protect against potential product liability claims could prevent or inhibit the commercialization of tecarfarin or any future products that we develop. We currently carry product liability insurance covering our marketed products and our clinical trials. Although we maintain such insurance, any claim that may be brought against us could result in a court

judgment or settlement in an amount that is not covered, in whole or in part, by our insurance or that is in excess of the limits of our insurance coverage. Our insurance policies also have various exclusions and deductibles, and we may be subject to a product liability claim for which we have no coverage. We will have to pay any amounts awarded by a court or negotiated in a settlement that exceed our coverage limitations or that are not covered by our insurance, and we may not have, or be able to obtain, sufficient capital to pay such amounts. Moreover, in the future, we may not be able to maintain insurance coverage at a reasonable cost or in sufficient amounts to protect us against losses. If and when we obtain approval for marketing tecarfarin, we intend to expand our insurance coverage to include the sale of tecarfarin, however, we may be unable to obtain this liability insurance on commercially reasonable terms.

Our business and operations would suffer in the event of computer system failures.

Despite the implementation of security measures, our internal computer systems, and those of third parties on which we rely, are vulnerable to damage from computer viruses, malware, natural disasters, terrorism, war, telecommunication and electrical failures, cyber-attacks or cyber-intrusions over the internet, attachments to emails, persons inside our organization, or persons with access to systems inside our organization. The risk of a security breach or disruption, particularly through cyber-attacks or cyber-intrusions, including by computer hackers, foreign governments, and cyber-terrorists, has generally increased as the number, intensity and sophistication of attempted attacks and intrusions from around the world have increased. If such an event were to occur and cause interruptions in our operations, it could result in a material disruption of our current or future product development programs. For example, the loss of clinical trial data from completed or any future ongoing or planned clinical trials could result in delays in our regulatory approval efforts and significantly increase our costs to recover or reproduce the data. To the extent that any disruption or security breach was to result in a loss of or damage to our data or applications, or inappropriate disclosure of confidential or proprietary information, we could incur material legal claims and liability, damage to our reputation, and the further development of our product candidates could be delayed.

Any failure to maintain the security of information relating to our patients, customers, employees and suppliers, whether as a result of cybersecurity attacks or otherwise, could expose us to litigation, government enforcement actions and costly response measures, and could disrupt our operations and harm our reputation.

In connection with the pre-clinical and clinical development, sales and marketing of our products and services, we may from time to time transmit confidential information. We also have access to, collect or maintain private or confidential information regarding our clinical trials and the patients enrolled therein, employees, and suppliers, as well as our business. Cyberattacks are rapidly evolving and becoming increasingly sophisticated. It is possible that computer hackers and others might compromise our security measures, or security measures of those parties that we do business with now or in the future, and obtain the personal information of patients in our clinical trials, vendors, employees and suppliers or our business information. A security breach of any kind, including physical or electronic break-ins, computer viruses and attacks by hackers, employees or others, could expose us to risks of data loss, litigation, government enforcement actions, regulatory penalties and costly response measures, and could seriously disrupt our operations. Any resulting negative publicity could significantly harm our reputation, which could cause us to lose market share and have an adverse effect on our results of operations.

We may acquire other businesses or form joint ventures or make investments in other companies or technologies that could harm our operating results, dilute our stockholders' ownership, increase our debt or cause us to incur significant expense.

As part of our business strategy, we may pursue acquisitions of businesses and assets. We also may pursue strategic alliances and joint ventures that leverage our technology and industry experience to expand our offerings or other capabilities. Though certain company personnel have business development and corporate transaction experience, including with licensing, mergers and acquisitions, and strategic partnering, as a company we have no experience with acquiring other companies and limited experience with forming strategic alliances and joint ventures. We may not be able to find suitable partners or acquisition candidates, and we may not be able to complete such transactions on favorable terms, if at all. If we make any acquisitions, we may not be able to integrate these acquisitions successfully into our existing business, and we could assume unknown or contingent liabilities. Any future acquisitions also could result in significant write-offs or the incurrence of debt

and contingent liabilities, any of which could have a material adverse effect on our financial condition, results of operations and cash flows. Integration of an acquired company also may disrupt ongoing operations and require management resources that would otherwise focus on developing our existing business. We may experience losses related

to investments in other companies, which could have a material negative effect on our results of operations. We may not identify or complete these transactions in a timely manner, on a cost-effective basis, or at all, and we may not realize the anticipated benefits of any acquisition, technology license, strategic alliance or joint venture.

To finance any acquisitions or joint ventures, we may choose to issue shares of our common stock as consideration, which would dilute the ownership of our stockholders. If the price of our common stock is low or volatile, we may not be able to acquire other companies or fund a joint venture project using our stock as consideration. Alternatively, it may be necessary for us to raise additional funds for acquisitions through public or private financings. Additional funds may not be available on terms that are favorable to us, or at all.

Declining general economic or business conditions may have a negative impact on our business.

Continuing concerns over U.S. health care reform legislation and energy costs, geopolitical issues, the availability and cost of credit and government stimulus programs in the United States and other countries have contributed to increased volatility and diminished expectations for the global economy. These factors, combined with low business and consumer confidence, could precipitate an economic slowdown and recession. Additionally, political changes in the U.S. and elsewhere in the world have created a level of uncertainty in the markets. If the economic climate does not improve or deteriorate, our business, as well as the financial condition of our suppliers and our third-party payors, could be adversely affected, resulting in a negative impact on our business, financial condition and results of operations.

Risks Related to Our Intellectual Property

If we are unable to obtain and maintain sufficient exclusivity and/or patent protection for our product candidates, or if the scope of the exclusivity or patent protection is not sufficiently broad, our competitors could develop and commercialize products similar or identical to ours, and our ability to commercialize our product candidates successfully may be adversely affected.

Our success depends in large part on our ability to obtain and maintain exclusivity for our proprietary product candidates through market and data exclusivity granted by regulatory agencies in the United States and other countries with respect to our proprietary product candidates as well as through patent protection. If we do not adequately protect our intellectual property, competitors may be able to erode or negate any competitive advantage we may have, which could harm our business and ability to achieve profitability. To protect our proprietary position, we file patent applications in the United States and abroad related to our novel product candidates that are important to our business. The patent application and approval process are expensive and time-consuming. We may not be able to file and prosecute all necessary or desirable patent applications at a reasonable cost or in a timely manner.

The patent position of pharmaceutical companies generally is highly uncertain. No consistent policy regarding the breadth of claims allowed in pharmaceutical patents has emerged to date in the United States or in many foreign jurisdictions. In addition, the determination of patent rights with respect to pharmaceutical compounds commonly involves complex legal and factual questions, which has in recent years been the subject of much litigation. As a result, the issuance, scope, validity, enforceability and commercial value of our patent rights are highly uncertain.

Assuming the other requirements for patentability are met, currently, the first to file a patent application is generally entitled to the patent. However, prior to March 16, 2013, in the United States, the first to invent was entitled to the patent. Publications of discoveries in the scientific literature often lag behind the actual discoveries, and patent applications in the United States and other jurisdictions are typically not published until 18 months after filing, or in some cases not at all. Therefore, we cannot be certain that we were the first to make the inventions claimed in our patents, or that we were the first to file for patent protection of such inventions.

Moreover, because the issuance of a patent is not conclusive as to its inventorship, scope, validity or enforceability, our patents may be challenged in the courts or patent offices in the United States and abroad. For example, we may be subject to a third-party preissuance submission of prior art to the U.S. Patent and Trademark Office, or USPTO, or become involved in post-grant review procedures, oppositions, derivations, reexaminations, *inter partes* review or interference proceedings, in the United States or elsewhere, challenging our patent rights or the patent rights of others. An adverse determination in any such challenges may result in

loss of exclusivity or in patent claims being narrowed, invalidated or held unenforceable, in whole or in part, which could limit our ability to stop others from using or commercializing similar or identical technology and products, or limit the duration of the patent protection

of our technology and products. In addition, given the amount of time required for the development, testing and regulatory review of new product candidates, patents protecting such candidates might expire before or shortly after such candidates are commercialized.

Our future patent applications may not result in patents being issued which protect our product candidates, in whole or in part, or which effectively prevent others from commercializing competitive products. Changes in either the patent laws or interpretation of the patent laws in the United States and other countries may diminish the value of our patents or narrow the scope of our patent protection. In addition, the laws of foreign countries may not protect our rights to the same extent or in the same manner as the laws of the United States. For example, European patent law restricts the patentability of methods of treatment of the human body more than United States law does.

Even if our patent applications issue as patents, they may not issue in a form that will provide us with any meaningful protection, prevent competitors from competing with us or otherwise provide us with any competitive advantage. Our competitors may be able to circumvent our patents by developing similar or alternative technologies or products in a non-infringing manner. Our competitors may also seek approval to market their own products similar to or otherwise competitive with our products. Alternatively, our competitors may seek to market generic versions of any approved products by submitting ANDAs to the FDA in which they claim that patents owned or licensed by us are invalid, unenforceable or not infringed. In these circumstances, we may need to defend or assert our patents, or both, including by filing lawsuits alleging patent infringement. In any of these types of proceedings, a court or other agency with jurisdiction may find our patents invalid or unenforceable, or that our competitors are competing in a non-infringing manner. Thus, even if we have valid and enforceable patents, these patents still may not provide protection against competing products or processes sufficient to achieve our business objectives.

Depending upon the timing, duration and conditions of FDA marketing approval of our product candidates, one or more of our U.S. patents may be eligible for limited patent term extension under the Hatch-Waxman Act. The Hatch-Waxman Act permits a patent term extension of up to five years for a patent covering an approved product as compensation for effective patent term lost during product development and the FDA regulatory review process. However, we may not receive an extension if we fail to apply within applicable deadlines, fail to apply prior to expiration of relevant patents or otherwise fail to satisfy applicable requirements. Moreover, the length of the extension could be less than we request. If we are unable to obtain patent term extension or the term of any such extension is less than we request, the period during which we can enforce our patent rights for that product will be shortened compared to expectations and our competitors may obtain approval to market competing products sooner. As a result, our revenue from applicable products could be reduced, possibly materially. Further, if this occurs, our competitors may take advantage of our investment in development and trials by referencing our clinical and preclinical data and launch their product earlier than might otherwise be the case.

To date, we have not engaged intellectual property counsel to conduct a freedom to operate analysis.

While we have engaged intellectual property counsel to assist in protecting our patent ownership rights, to date, we have not had intellectual property counsel conduct a freedom to operate analysis regarding our tecarfarin product. As a result, we cannot be certain that we will not be exposed to third party legal claims, liabilities and/or litigation actions when we seek to develop, make and market products using our tecarfarin technology.

We may become involved in lawsuits to protect or enforce our patents or other intellectual property, which could be expensive, time consuming and unsuccessful.

Competitors may infringe our patents, trademarks, copyrights or other intellectual property. To counter infringement or unauthorized use, we may be required to file infringement claims, which can be expensive and time consuming and divert the time and attention of our management and scientific personnel. Any claims we assert against perceived infringers could provoke these parties to assert counterclaims against us alleging that we infringe their patents, in addition to counterclaims asserting that our patents are invalid or unenforceable, or both. In any patent infringement proceeding, there is a risk that a court will decide that a patent of ours is invalid or unenforceable, in whole or in part, and that we do not have the right to stop the other party from using the invention at issue. There is also a risk that, even if the validity of such patents is upheld, the court will construe the patent's claims narrowly or decide that we do not have the right to stop the other party from using the invention at issue on the grounds that our patent claims do not cover the invention. An adverse outcome in a

litigation or proceeding involving our patents could limit our ability to assert our patents against those parties or other competitors, and may curtail or preclude our ability to exclude third parties from making and selling similar or competitive products. Any of these occurrences could adversely affect our competitive business position, business prospects and financial condition.

Similarly, if we assert trademark infringement claims, a court may determine that the marks we have asserted are invalid or unenforceable, or that the party against whom we have asserted trademark infringement has superior rights to the marks in question. In this case, we could ultimately be forced to cease use of such trademarks.

Even if we establish infringement, the court may decide not to grant an injunction against further infringing activity and instead award only monetary damages, which may or may not be an adequate remedy. Furthermore, because of the substantial amount of discovery required in connection with intellectual property litigation, there is a risk that some of our confidential information could be compromised by disclosure during litigation. There could also be public announcements of the results of hearings, motions or other interim proceedings or developments. If securities analysts or investors perceive these results to be negative, it could have a material adverse effect on the price of shares of our common stock. Moreover, there can be no assurance that we will have sufficient financial or other resources to file and pursue such infringement claims, which typically last for years before they are concluded. Even if we ultimately prevail in such claims, the monetary cost of such litigation and the diversion of the attention of our management and scientific personnel could outweigh any benefit we receive as a result of the proceedings.

If we are sued for infringing intellectual property rights of third parties, such litigation could be costly and time consuming and could prevent or delay us from developing or commercializing our product candidates.

Our commercial success depends, in part, on our ability to develop, manufacture, market and sell our product candidates without infringing the intellectual property and other proprietary rights of third parties. If any third-party patents or patent applications are found to cover our product candidates or their methods of use, we may not be free to manufacture or market our product candidates as planned without obtaining a license, which may not be available on commercially reasonable terms, or at all.

There is a substantial amount of intellectual property litigation in the pharmaceutical industry, and we may become party to, or threatened with, litigation or other adversarial proceedings regarding intellectual property rights with respect to our product candidates, including interference and other administrative proceedings before the USPTO. Third parties may assert infringement claims against us based on existing or future intellectual property rights. The outcome of intellectual property litigation is subject to uncertainties that cannot be adequately quantified in advance. The pharmaceutical industry has produced a significant number of patents, and it may not always be clear to industry participants, including us, which patents cover various types of products or methods of use. The coverage of patents is subject to interpretation by the courts, and the interpretation is not always uniform. If we were sued for patent infringement, we would need to demonstrate that our product candidates, products or methods either do not infringe the patent claims of the relevant patent or that the patent claims are invalid or unenforceable, and we may not be able to do this. In the United States, proving invalidity (except in proceedings before the USPTO) requires a showing of clear and convincing evidence to overcome the presumption of validity enjoyed by issued patents. Even if we are successful in these proceedings, we may incur substantial costs and the time and attention of our management and scientific personnel could be diverted in pursuing these proceedings, which could significantly harm our business and operating results. In addition, we may not have sufficient resources to bring these actions to a successful conclusion.

If we are found to infringe a third-party's intellectual property rights, we could be forced, including by court order, to cease developing, manufacturing or commercializing the infringing product candidate or product. Alternatively, we may be required to obtain a license from such third-party in order to use the infringing technology and continue developing, manufacturing or marketing the infringing product candidate. However, we may not be able to obtain any required license on commercially reasonable terms or at all. Even if we were able to obtain a license, it could be non-exclusive, thereby giving our competitors access to the same technologies licensed to us. In addition, we could be found liable for monetary damages, including treble damages and attorney" fees if we are found to have willfully infringed a patent. A finding of infringement could prevent us from commercializing our product candidates or force us to cease some of our business operations, which could materially harm our business. Claims that we have misappropriated the confidential information or trade secrets of third parties could have a similar negative impact on our business.

Changes to the patent law in the United States and other jurisdictions could diminish the value of patents in general, thereby impairing our ability to protect our products.

As is the case with other pharmaceutical companies, our success is heavily dependent on intellectual property, particularly patents. Obtaining and enforcing patents in the pharmaceutical industry involves both technological and legal complexity and is therefore costly, time consuming and inherently uncertain. Recent patent reform legislation in the United States and other countries, including the Leahy-Smith America Invents Act, or the Leahy-Smith Act,

signed into law in September 2011, could increase those uncertainties and costs. The Leahy-Smith Act includes a number of significant changes to U.S. patent law. These include provisions that affect the way patent applications are prosecuted, redefine prior art and provide more efficient and cost-effective avenues for competitors to challenge the validity of patents. In addition, the Leahy-Smith Act has transformed the U.S. patent system into a "first to file" system. The first-to-file provisions, however, only became effective in March 2013. Accordingly, it is not yet clear what, if any, impact the Leahy-Smith Act will have on the operation of our business. However, the Leahy-Smith Act and its implementation could make it more difficult to obtain patent protection for our inventions and increase the uncertainties and costs surrounding the enforcement or defense of our or our collaboration partner" issued patents, all of which could harm our business, results of operations and financial condition.

The U.S. Supreme Court has ruled on several patent cases in recent years, either narrowing the scope of patent protection available in certain circumstances or weakening the rights of patent owners in certain situations. Additionally, there have been recent proposals for additional changes to the patent laws of the United States and other countries that, if adopted, could impact our ability to enforce our proprietary technology. Depending on future actions by the U.S. Congress, the U.S. courts, the USPTO and the relevant law-making bodies in other countries, the laws and regulations governing patents could change in unpredictable ways that would weaken our ability to obtain new patents or to enforce our existing patents and patents that we might obtain in the future.

Obtaining and maintaining our patent protection depends on compliance with various procedural, document submissions, fee payment and other requirements imposed by governmental patent agencies, and our patent protection could be reduced or eliminated for noncompliance with these requirements.

Periodic maintenance fees on any issued patent are due to be paid to the USPTO and foreign patent agencies in several stages over the lifetime of the patent. The USPTO and various foreign governmental patent agencies require compliance with a number of procedural, documentary, fee payment and other similar provisions during the patent application process. While an inadvertent lapse can, in many cases, be cured by payment of a late fee or by other means in accordance with the applicable rules, there are situations in which noncompliance can result in abandonment or lapse of the patent or patent application, resulting in partial or complete loss of patent rights in the relevant jurisdiction. Noncompliance events that could result in abandonment or lapse of a patent or patent application include, but are not limited to, failure to respond to official actions within prescribed time limits, non-payment of fees and failure to properly legalize and submit formal documents. If we fail to maintain the patents and patent applications covering our product candidates, our competitive position would be adversely affected.

We may not be able to enforce our intellectual property rights throughout the world.

Filing, prosecuting and defending patents on our product candidates in all countries throughout the world would be prohibitively expensive. The requirements for patentability may differ in certain countries, particularly in developing countries. Competitors may use our technologies in jurisdictions where we have not obtained patent protection to develop their own products and, further, may export otherwise infringing products to territories where we may obtain patent protection, but where patent enforcement is not as strong as that in the United States. These products may compete with our products in jurisdictions where we do not have any issued or licensed patents or where any future patent claims or other intellectual property rights may not be effective or sufficient to prevent them from competing with us.

Moreover, our ability to protect and enforce our intellectual property rights may be adversely affected by unforeseen changes in foreign intellectual property laws. Additionally, laws of some countries outside of the United States and Europe do not afford intellectual property protection to the same extent as the laws of the United States and Europe. Many companies have encountered significant problems in protecting and defending intellectual property rights in certain foreign jurisdictions. This could make it difficult for us to stop the infringement of our patents or the misappropriation of our other intellectual property rights. For example, many foreign countries have compulsory licensing laws under which a patent owner must grant licenses to third parties. Consequently, we may not be able to prevent third parties from practicing our inventions in certain countries outside the United States and Europe. Competitors may use our technologies in jurisdictions where we have not obtained patent protection to develop their own products and, further, may export otherwise infringing

products to territories where we have patent protection, if our ability to enforce our patents to stop infringing activities is inadequate. These products may compete with our products, and our patents or other intellectual property rights may not be effective or sufficient to prevent them from competing.

Proceedings to enforce our patent rights in foreign jurisdictions, whether or not successful, could result in substantial costs and divert our efforts and resources from other aspects of our business. Furthermore, while we intend to protect our intellectual property rights in major markets for our products, we cannot ensure that we will be able to initiate or maintain similar efforts in all jurisdictions in which we may wish to market our products. Accordingly, our efforts to protect our intellectual property rights in such countries may be inadequate.

Patent terms may be inadequate to protect our competitive position on our products for an adequate amount of time.

Given the amount of time required for the development, testing and regulatory review of new product candidates, patents protecting such candidates might expire before or shortly after such candidates are commercialized. With respect to tecarfarin, we have two issued U.S. patents directed to tecarfarin. The expiration dates of the patents are 2024 for both a composition of matter patent and a method of treatment patent. We expect to seek extensions of patent terms in the United States and, if available, in other countries where we are prosecuting patents. In the United States, the Drug Price Competition and Patent Term Restoration Act of 1984 permits a patent term extension of up to five years beyond the normal expiration of the patent, which is limited to the approved indication (or any additional indications approved during the period of extension). However, the applicable authorities, including the FDA and the USPTO in the United States, and any equivalent regulatory authority in other countries, may not agree with our assessment of whether such extensions are available, and may refuse to grant extensions to our patents, or may grant more limited extensions than we request. If this occurs, our competitors may be able to take advantage of our investment in development and clinical trials by referencing our clinical and preclinical data and launch their product earlier than might otherwise be the case.

We may be subject to claims by third parties asserting that our employees or we have misappropriated their intellectual property, or claiming ownership of what we regard as our own intellectual property.

We anticipate that many of the people that we expect to hire as employees, including our one current employee, were previously employed at other pharmaceutical companies. Some of these employees may have executed proprietary rights, non-disclosure and non-competition agreements, or similar agreements, in connection with such previous employment. Although we try to ensure that our employees do not use the proprietary information or know-how of others in their work for us, we may be subject to claims that we or these employees have used or disclosed intellectual property, including trade secrets or other proprietary information, of any such third-party. Litigation may be necessary to defend against such claims. If we fail in defending any such claims, in addition to paying monetary damages, we may lose valuable intellectual property rights or personnel or sustain damages. Such intellectual property rights could be awarded to a third-party, and we could be required to obtain a license from such third-party to commercialize our technology or products. Such a license may not be available on commercially reasonable terms or at all. Even if we are successful in defending against such claims, litigation could result in substantial costs and be a distraction to management.

In addition, while we typically require our employees, consultants and contractors who may be involved in the development of intellectual property to execute agreements assigning such intellectual property to us, we may be unsuccessful in executing such an agreement with each party who in fact develops intellectual property that we regard as our own, which may result in claims by or against us related to the ownership of such intellectual property. If we fail in prosecuting or defending any such claims, in addition to paying monetary damages, we may lose valuable intellectual property rights. Even if we are successful in prosecuting or defending against such claims, litigation could result in substantial costs and be a distraction to our senior management and scientific personnel.

Risks Related to Ownership of Our Common Stock

An active public trading market for our common stock may not develop or be sustained.

Prior to this offering, there has been no public market or active private market for trading shares of our common stock. We will list our common stock on the Nasdaq Capital Market, in connection with this offering; however, an active trading market may not develop following the completion of this offering or, if developed, may not be sustained. The lack of an active market may impair your ability to sell your shares at the time you wish to sell them or at a price that you consider reasonable. The lack of an active market may also reduce the price of shares

of common stock. An inactive market may impair our ability to raise capital by selling shares and our ability to use our capital stock to acquire other companies or technologies. We cannot predict the prices at which our common stock will trade. The initial public offering price of our common stock may not bear any relationship to the market price at which our common stock will trade after this offering.

We cannot be assured that we will be able to maintain our listing on the Nasdaq Capital Market.

We anticipate that our securities will be listed on The Nasdaq Capital Market, a national securities exchange, upon consummation of this offering. We cannot be assured that we will continue to comply with the rules, regulations or requirements governing the listing of our common stock on Nasdaq or that our securities will continue to be listed on Nasdaq in the future. If Nasdaq should determine after initial listing that we fail to meet Nasdaq requirements, we may be subject to a delisting action by Nasdaq. Nonetheless, this offering is contingent upon our obtaining Nasdaq listing and, in the event such listing is not approved, this offering will be terminated.

Even if we are successful in completing this offering, if Nasdaq delists our securities from trading on its exchange at some future date, we could face significant material adverse consequences, including:

- a limited availability of market quotations for our securities;
- reduced liquidity with respect to our securities;
- a determination that our common stock is a "penny stock" which will require brokers trading in our ordinary shares to adhere to more stringent rules, possibly resulting in a reduced level of trading activity in the secondary trading market for our ordinary shares;
- a limited amount of news and analyst coverage for our company; and
- a decreased ability to issue additional securities or obtain additional financing in the future.

Our stock price may be extremely volatile, and your investment in our common stock could suffer a decline in value.

You should consider an investment in our common stock risky and invest only if you can withstand a significant loss and wide fluctuations in the market value of your investment. Investors who purchase our common stock may not be able to sell their shares at or above the purchase price. Security market prices for securities of pharmaceutical companies have been highly volatile. In addition, the volatility of pharmaceutical company stocks often does not correlate to the operating performance of the companies represented by such stocks. Some of the factors that may cause the market price of our common stock to fluctuate include:

- adverse results or delays in our clinical trials;
- the timing or delay of achievement of our clinical, regulatory, partnering and other milestones, such as the commencement of clinical development, the completion of a clinical trial, the receipt of regulatory approval or the establishment or termination of a commercial partnership for one or more of our product candidates;
- announcement of FDA approval or non-approval of our product candidates or delays in the FDA review process;
- actions taken by regulatory agencies with respect to our product candidates, our clinical trials or our sales and marketing activities;
- the commercial success of any product approved by the FDA or its foreign counterparts;
- regulatory developments in the United States and foreign countries;
- changes in the structure of healthcare payment systems;
- any intellectual property infringement lawsuit involving us;
- announcements of technological innovations or new products by us or our competitors;
- market conditions for the biotechnology or pharmaceutical industries in general;
- changes in financial estimates or recommendations by securities analysts;
- sales of large blocks of our common stock;

- sales of our common stock by our executive officers, directors and significant stockholders;

- direct sales of our common stock through financing arrangements;
- restatements of our financial results and/or material weaknesses in our internal controls;
- the loss of any of our key scientific or management personnel; and
- announcements regarding the ongoing exploration of the strategic options available to us.

The stock markets in general, and the markets for biotechnology stocks in particular, have experienced extreme volatility and price and volume fluctuations that have often been unrelated or disproportionate to the operating performance of particular companies. These broad market and industry factors may seriously harm the market price of our common stock, regardless of our operating performance. In the past, class action litigation has often been instituted against companies whose securities have experienced periods of volatility in market price. Any such litigation brought against us could result in substantial costs, which would hurt our financial condition and results of operations, divert management's attention and resources, and possibly delay our clinical trials or commercialization efforts.

Stock prices in recent initial public offerings have been extremely volatile.

Certain recent initial public offerings of companies with public floats comparable to our anticipated public float have experienced instances of extreme price run-ups followed by rapid price declines and stock volatility seemingly unrelated to the underlying performance of the respective company. Our common stock may experience similar rapid and substantial price volatility following our initial public offering, and any such volatility, including any stock-run up, may be unrelated to our actual or expected operating performance and financial condition or prospects, which could make it difficult for prospective investors to assess the rapidly changing value of our stock. Although the specific cause of such volatility is unclear, our anticipated public float may amplify the impact of actions taken by a few stockholders have on the price of our shares, which may cause our share price to deviate, potentially significantly, from a price that better reflects the underlying performance of our business. Should our securities experience run-ups and declines that are seemingly unrelated to our actual or expected operating performance and financial condition or prospects, prospective investors may have difficulty assessing the rapidly changing value of our securities. In addition, investors of our securities may experience losses, which may be material, if the price of our common stock declines after this offering or if such investors purchase shares of our common stock prior to any price decline.

If financial or industry analysts do not publish research or reports about our business or if they issue inaccurate or unfavorable commentary or downgrade our common stock, our stock price and trading volume could decline.

The trading market for our common stock will be influenced by the research and reports that industry or financial analysts publish about us or our business. We do not control these analysts or the content and opinions included in their reports. As a new public company, we may be slow to attract research coverage, and the analysts who publish information about our common stock will have had relatively little experience with our company, which could affect their ability to accurately forecast our results and make it more likely that we fail to meet their estimates. In the event we obtain industry or financial analyst coverage, if any of the analysts who cover us issue an inaccurate or unfavorable opinion regarding our stock price, our stock price would likely decline. In addition, the stock prices of many companies in the biopharmaceutical industry have declined significantly after those companies have failed to meet, or often times failed to exceed, the financial guidance publicly announced by the companies or the expectations of analysts. If our financial results fail to meet, or significantly exceed, our announced guidance or the expectations of analysts or public investors, analysts could downgrade our common stock or publish unfavorable research about us. If one or more of these analysts cease coverage of our company or fail to publish reports on us regularly, we could lose visibility in the financial markets, which in turn could cause our stock price or trading volume to decline.

After this offering, our officers, directors, and principal stockholders will continue to exercise significant control over our Company, and may be able to control our management and operations, acting in their best interests and not necessarily those of other stockholders.

When this offering is completed, our executive officers, directors and principal stockholders who beneficially own more than 5% or more of our outstanding common stock before this offering will, in the aggregate, beneficially own shares representing approximately 69.02% of our outstanding capital stock immediately after

this offering. When this offering is completed, Quang Pham, our Chief Executive Officer, will beneficially own 53.93% of our outstanding capital stock immediately after this offering. As a result, Mr. Pham alone and together with these other stockholders,

acting together, may be able to significantly influence any matters requiring approval by our stockholders, including the election of directors, the approval of mergers or other business combination transactions. The interests of this group of stockholders may not always coincide with our interests or the interests of other stockholders. The significant concentration of stock ownership may adversely affect the trading price of our common stock due to the perception that conflicts of interest may exist or arise. Therefore, you should not invest in reliance on your ability to have any control over our company.

As a "controlled company" under the rules of the Nasdaq Capital Market, we may choose to exempt our company from certain corporate governance requirements that could have an adverse effect on our public stockholders.

Upon the completion of this offering, we will be a "controlled company" as defined under the Nasdaq Stock Market Rules, because Quang Pham, our Chief Executive Officer and principal stockholder, will beneficially own 52.98% of our common shares and will be able to exercise 52.98% of the total voting power of our issued and outstanding shares, assuming the full exercise of the over-allotment option by the underwriter.

As long as Mr. Pham individually owns at least 50% of the voting power of our company, we are a "controlled company" as defined under Nasdaq Listing Rules. For so long as we are a controlled company under that definition, we are permitted to elect to rely, and may rely, on certain exemptions from corporate governance rules, including:

- an exemption from the rule that a majority of our board of directors must be independent directors;
- an exemption from the rule that the compensation of our CEO must be determined or recommended solely by independent directors; and
- an exemption from the rule that our director nominees must be selected or recommended solely by independent directors.

As a result, you may not have the same protection afforded to stockholders of companies that are subject to these corporate governance requirements.

Although we do not intend to rely on the "controlled company" exemption under the Nasdaq listing rules, we could elect to rely on this exemption in the future. If we elect to rely on the "controlled company" exemption, a majority of the members of our board of directors might not be independent directors and our nominating and corporate governance and compensation committees might not consist entirely of independent directors.

Future sales of common stock by our officers and directors and principal stockholders or others of our common stock, or the perception that such sales may occur, could depress the market price of our common stock.

Sales of a substantial number of shares of our common stock after this offering, particularly sales by our directors, executive officers and principal stockholders could adversely affect the market price of our common stock and may make it more difficult to sell common stock at a time and price that you deem appropriate. Based on the total number of outstanding shares of our common stock as of January 17, 2023, upon completion of this offering, the issuance of common stock pursuant to the terms of the Amendment to Asset Purchase Agreement to HESP LLC, conversion of our outstanding convertible notes into common stock, the issuance subsequent to September 30, 2022 of 67,000 shares of common stock to consultants for the provision of services, the issuance of 50,000 shares of common stock to be issued to an executive officer upon consummation of this offering and the issuance of 250,000 shares (the "Warrant Shares") of common stock upon exercise of the November Warrants at an exercise price of $1.00 per share, with such warrants to be exercised upon consummation of the offering (the Warrant Shares are being registered in the Resale Prospectus), we will have an aggregate of 11,634,575 shares of common stock outstanding, assuming no exercise of our outstanding stock options or other outstanding warrants, and assuming the underwriters do not exercise their option to purchase additional shares.

All of the shares of common stock sold in this offering and the shares of our common stock by the selling stockholders in the resale offering will be freely tradable without restrictions or further registration under the Securities Act of 1933, as amended, or the Securities Act, except for any shares held by our affiliates as defined in Rule 144 under the Securities Act.

A substantial majority our outstanding shares of common stock are currently restricted from resale as a result of market standoff and “lock-up” agreements, as more fully described in “Shares Eligible for Future Sale.” These shares will become available to be sold 181 days after the date of this prospectus. Shares held by directors, executive officers and other affiliates will be subject to volume limitations under Rule 144 under the Securities Act and various vesting agreements. In addition,

the underwriters may, in their sole discretion, release all or some portion of the shares subject to market standoff or lock-up agreements prior to the expiration of the lock-up period. See the section entitled "Shares Eligible for Future Sale" for more information. Sales of a substantial number of such shares upon expiration of the market standoff and lock-up agreements, or the perception that such sales may occur, or early release of these agreements, could cause our market price to fall or make it more difficult for you to sell your common stock at a time and price that you deem appropriate.

We intend to register the offer and sale of an aggregate of approximately 3,010,000 shares of common stock that have been issued or reserved for future issuance under our equity compensation plans on a Form S-8 registration statement. Once we register the offer and sale of these shares, they can be freely sold in the public market upon issuance, subject to the market standoff or lock-up agreements or unless they are held by "affiliates," as that term is defined in Rule 144 of the Securities Act. If the holders of these shares choose to sell a large number of shares, they could adversely affect the market price for our common stock.

We may also issue shares of our common stock or securities convertible into shares of our common stock from time to time in connection with a financing, acquisition, investment or otherwise. Any such issuance could result in substantial dilution to our existing stockholders and cause the trading price of our common stock to decline.

The offering price of the shares and the other terms of the initial public offering have been determined through negotiations between the Company and the underwriter.

The initial public offering price of our common stock and other terms of the initial public offering have been determined through negotiations between the Company and the underwriter. This determined price may not reflect the price at which investors in the market will be willing to buy and sell our shares following this offering. You may be unable to sell your shares of common stock at or above the initial offering price. The market price of shares of our common stock following this offering could be subject to wide fluctuations in response to many risk factors listed in this section, and others beyond our control.

The offering price of the primary offering and resale offering could differ.

The offering price of shares of our common stock in the initial public offering has been determined by negotiations between the Company and the underwriter. The selling stockholders may sell the resale shares at prevailing market prices or privately negotiated prices after close of the initial public offering and listing of our common stock on Nasdaq. Therefore, the offering prices of the initial public offering and resale offering could differ. As a result, the purchasers in the resale offering could pay more or less than the offering price in the primary offering.

The resale by the selling stockholders in our resale offering may cause the market price of our common stock to decline.

The resale of shares of our common stock by the selling stockholders in the resale offering, as well as the issuance of common stock in the initial public offering, could result in resales of our common stock by our current stockholders concerned about the potential dilution of their holdings. In addition, the resale by other unregistered stockholders after expiration of the lock-up period could have the effect of depressing the market price for our common stock.

Our initial public offering price is substantially higher than the pro forma as adjusted net tangible book value per share of our outstanding common stock, and new investors will experience immediate and substantial dilution.

Our initial public offering price is substantially higher than the pro forma as adjusted net tangible book value per share of our common stock based on the expected total value of our total assets, less our goodwill and other intangible assets, less our total liabilities immediately following this offering. If you purchase shares of our common stock in this offering, you will experience immediate and substantial dilution of $4.50 per share in the price you pay for our common stock as compared to the pro forma as adjusted net tangible book value as of September 30, 2022, after giving effect to the issuance of shares of our common stock in this offering at the initial public offering price of $5.00 per share. Furthermore, if the underwriters exercise their option to purchase additional shares, if outstanding options are exercised, if we issue awards to our employees under our equity

incentive plans, or if we otherwise issue additional shares of our common stock, you could experience further dilution. For a further description of the dilution that you will experience immediately after this offering, see the section entitled “Dilution.”

Anti-takeover provisions in our charter documents, and under Delaware law, could make an acquisition of our company more difficult, limit attempts by our stockholders to replace or remove our current management and limit the market price of our common stock.

Provisions in our amended and restated certificate of incorporation and amended and restated bylaws may have the effect of delaying or preventing a change of control or changes in our management. Our amended and restated certificate of incorporation and amended and restated bylaws include provisions that:

- authorize our board of directors to issue, without further action by the stockholders, shares of undesignated preferred stock with terms, rights, and preferences determined by our board of directors that may be senior to our common stock;
- require that any action to be taken by our stockholders be effected at a duly called annual or special meeting and not by written consent;
- specify that special meetings of our stockholders can be called only by our board of directors, the chairperson of our board of directors, or our Chief Executive Officer;
- establish an advance notice procedure for stockholder proposals to be brought before an annual meeting, including proposed nominations of persons for election to our board of directors;
- prohibit cumulative voting in the election of directors;
- establish that our board of directors is divided into three classes — Class I, Class II, and Class III — with each class serving staggered three-year terms;
- provide that, so long as our board of directors is classified, directors may only be removed for cause;
- provide that vacancies on our board of directors may be filled only by a majority of directors then in office, even though less than a quorum; and
- require the approval of our board of directors or the holders of two-thirds of our outstanding shares of voting stock to amend our bylaws and certain provisions of our certificate of incorporation.

These provisions may frustrate or prevent any attempts by our stockholders to replace or remove our current management by making it more difficult for stockholders to replace members of our board of directors, which is responsible for appointing the members of our management. In addition, we are governed by the provisions of Section 203 of the Delaware General Corporation Law, which generally, subject to certain exceptions, prohibits a Delaware corporation from engaging in any of a broad range of business combinations with any “interested” stockholder for a period of three years following the date on which the stockholder became an “interested” stockholder. Any of the foregoing provisions could limit the price that investors might be willing to pay in the future for shares of our common stock, and they could deter potential acquirers of our company, thereby reducing the likelihood that you would receive a premium for your shares of our common stock in an acquisition.

Our amended and restated certificate of incorporation and amended and restated bylaws provide that the Court of Chancery of the State of Delaware or the federal district court for the District of Delaware is the exclusive forum for certain disputes between us and our stockholders, which could result in increased costs for our stockholders to bring a claim and could limit our stockholders’ ability to obtain a favorable judicial forum for disputes with us or our directors, officers or employees.

Our amended and restated certificate of incorporation and amended and restated bylaws provide that, unless the Company consents in writing to the selection of an alternative forum, the Court of Chancery of the State of Delaware (or, in the event that the Court of Chancery does not have jurisdiction, the federal district court for the District of Delaware or other state courts of the State of Delaware) is the exclusive forum for (i) any derivative action or proceeding brought on our behalf; (ii) any action asserting a claim of breach of fiduciary duty owed by, any director, officer, employee or agent of the Company to the Company or our stockholders; (iii) any action asserting a claim arising pursuant to any provision of the Delaware General Corporation Law, our amended and restated certificate of incorporation or our amended and restated bylaws; or (iv) any action asserting a claim against us or any director, officer of employee that is governed by the internal affairs doctrine of the law of the

State of Delaware; provided that, if and only if the Court of Chancery of the State of Delaware dismisses any such action for lack of subject matter jurisdiction, or the Company consents in writing to the selection of an alternative forum, such action may be brought

in another state or federal court sitting in the State of Delaware. Our amended and restated certificate of incorporation and amended and restated bylaws also provide that the federal district courts of the United States of America is the exclusive forum for the resolution of any complaint asserting a cause of action arising under the Securities Act. Notwithstanding the foregoing, the exclusive forum provision will not apply to claims brought to enforce any liability or duty created by the Exchange Act. Nothing in our amended and restated certificate of incorporation or amended and restated bylaws preclude stockholders that assert claims under the Exchange Act from bringing such claims in state or federal court, subject to applicable law.

We believe these provisions may benefit us by providing increased consistency in the application of Delaware law and federal securities laws by chancellors and judges, as applicable, particularly experienced in resolving corporate disputes, efficient administration of cases on a more expedited schedule relative to other forums and protection against the burdens of multi-forum litigation. However, this choice of forum provision could result in increased costs for our stockholders to bring a claim and could may limit a stockholder's ability to bring a claim in a judicial forum that it finds favorable for disputes with us or any of our directors, officers, other employees or stockholders, which may discourage lawsuits with respect to such claims, although our stockholders will not be deemed to have waived our compliance with federal securities laws and the rules and regulations thereunder. Furthermore, the enforceability of similar choice of forum provisions in other companies' certificates of incorporation has been challenged in legal proceedings, and it is possible that a court could find these types of provisions to be inapplicable or unenforceable. While the Delaware courts have determined that such choice of forum provisions are facially valid, a stockholder may nevertheless seek to bring a claim in a venue other than those designated in the exclusive forum provisions, and there can be no assurance that such provisions will be enforced by a court in those other jurisdictions. If a court were to find the choice of forum provision that is contained in our amended and restated certificate of incorporation and amended and restated bylaws to be inapplicable or unenforceable in an action, we may incur additional costs associated with resolving such action in other jurisdictions, which could adversely affect our business and financial condition.

Our management has broad discretion in the use of the net proceeds from this offering, and our use of the net proceeds may not enhance our operating results or the price of our common stock.

We intend to use the net proceeds we receive from this offering to fund ongoing development, primarily CMC preparation, research and development, and other trial preparation expenses needed in order to initiate the pivotal Phase 3 clinical trial of our product candidate and for working capital and general corporate purposes. We may use a portion of the net proceeds to acquire complementary businesses or products. While we do not have agreements or commitments for any specific acquisitions at this time, we continually evaluate potential acquisition candidates to enhance our product offerings. Accordingly, our management will have considerable discretion over the specific use of the net proceeds that we receive in this offering and might not be able to obtain a significant return, if any, on investment of these net proceeds. You will not have the opportunity as part of your investment decision to assess whether the net proceeds are being used appropriately. Investors in this offering will need to rely upon the judgment of our management with respect to the use of proceeds. Until the net proceeds are used, they may be placed in investments that do not produce significant income, may be held in demand deposit accounts, or in investments intended to be highly liquid that may nevertheless lose value. If we do not use the net proceeds that we receive in this offering effectively, our business and prospects could be harmed, and the market price of our common stock could decline.

Claims for indemnification by our directors and officers may reduce our available funds to satisfy successful third-party claims against us and may reduce the amount of money available to us.

Our amended and restated bylaws provide that we will indemnify our directors and executive officers, in each case to the fullest extent permitted by Delaware law.

In addition, as permitted by Section 145 of the DGCL, our amended and restated bylaws and the indemnification agreements that we have entered into with each of our current executive officers and intend to enter into with our directors and certain other officers, among other things provide that:

- We will indemnify our directors and executive officers for serving us in those capacities, or for serving as a director, officer, employee or agent of other business enterprises at our request, to the fullest extent permitted by Delaware law. Delaware law provides that we may indemnify such

person if such person acted in good faith and in a manner such person reasonably believed to be in or not opposed to our best interest and, with respect to any criminal proceeding, had no reasonable cause to believe such person's conduct was unlawful.

- We may, in our discretion, indemnify employees and agents in those circumstances where indemnification is permitted by applicable law.
- We will be required to advance expenses, as incurred, to our directors and officers in connection with defending a proceeding, except that such directors or officers shall undertake to repay such advances if it is ultimately determined that such person is not entitled to indemnification.
- The rights conferred in our bylaws will not be exclusive. We may not retroactively amend our bylaw provisions to reduce our indemnification obligations to directors, officers, employees and agents.

As a result, claims for indemnification by our directors and officers may reduce our available funds to satisfy successful third-party claims against us and may reduce the amount of money available to us.

We do not intend to pay dividends in the foreseeable future. As a result, your ability to achieve a return on your investment will depend on appreciation in the price of our common stock.

We have never declared or paid any cash dividends on our common stock. We currently intend to retain all available funds and any future earnings for use in the operation of our business and do not anticipate paying any dividends on our common stock in the foreseeable future. Any determination to pay dividends in the future will be at the discretion of our board of directors. Consequently, your only opportunity to achieve a return on your investment in our company will be if the market price of our common stock appreciates and you sell your shares at a profit. There is no guarantee that the price of our common stock that will prevail in the market after this offering will ever exceed the price that you pay. For additional information about our dividend policy, see the section entitled “Dividend policy” elsewhere in this prospectus.

Certain members of our management team have limited experience managing a public company.

Some of the members of our management team have limited experience managing a publicly traded company, interacting with public company investors, and complying with laws pertaining to public companies. Our management team may not successfully or efficiently manage our transition to being a public company subject to significant regulatory oversight and reporting obligations under the federal securities laws and the continuous scrutiny of securities analysts and investors. These new obligations and constituents will require significant attention from our senior management and could divert their attention away from the day-to-day management of our business, which could adversely affect our business, financial condition, and operating results.

We will incur significant increased costs as a result of operating as a public company, and our management will be required to devote substantial time to new compliance initiatives.

As a public company, we will incur legal, accounting and other expenses that we did not incur as a private company. We will be subject to the reporting requirements of the Securities Exchange Act of 1934, as amended, or the Exchange Act, and will be required to comply with the applicable requirements of the Sarbanes-Oxley Act of 2002, or SOX, and the Dodd-Frank Wall Street Reform and Consumer Protection Act, or Dodd-Frank. The listing requirements of the Nasdaq Stock Market, and the rules of the Securities and Exchange Commission, or SEC, require that we satisfy certain corporate governance requirements. Our management and other personnel will need to devote a substantial amount of time to ensure that we comply with all of these requirements. Moreover, the reporting requirements, rules and regulations will increase our legal and financial compliance costs and will make some activities more time-consuming and costly. Any changes we make to comply with these obligations may not be sufficient to allow us to satisfy our obligations as a public company on a timely basis, or at all. These reporting requirements, rules and regulations, coupled with the increase in potential litigation exposure associated with being a public company, could also make it more difficult for us to attract and retain qualified persons to serve on our board of directors or board committees or to serve as executive officers, or to obtain certain types of insurance, including directors’ and officers’ insurance, on acceptable terms.

After this offering, we will be subject to Section 404 of the Sarbanes-Oxley Act, or Section 404, and the related rules of the SEC, which generally require our management and independent registered public accounting firm to report on the effectiveness of our internal control over financial reporting. In order to maintain effective internal controls, we will need additional financial personnel, systems and resources. Beginning with the second annual report on Form 10-K that we will be required to file with the SEC, Section 404 requires an annual management

assessment of the effectiveness of our internal control over financial reporting. The rules governing the standards that must be met for management to assess our internal control over financial reporting are complex and require significant documentation, testing, and

possible remediation. To comply with the requirements of being a reporting company under the Exchange Act, we will need to upgrade our systems including information technology; implement additional financial and management controls, reporting systems, and procedures; and hire additional accounting and finance staff.

To date, we have never conducted a review of our internal controls for the purpose of providing the reports required by these rules. During the course of our review and testing, we may identify additional deficiencies and be unable to remediate them before we must provide the required reports. Furthermore, if we have a material weakness in our internal control over financial reporting, we may not detect errors on a timely basis and our financial statements may be materially misstated. We or our independent registered public accounting firm may not be able to conclude on an ongoing basis that we have effective internal control over financial reporting, which could harm our operating results, cause investors to lose confidence in our reported financial information and cause the trading price of our stock to fall. In addition, as a public company we will be required to file accurate and timely quarterly and annual reports with the SEC under the Exchange Act. Any failure to report our financial results on an accurate and timely basis could result in sanctions, lawsuits, delisting of our shares from Nasdaq or other adverse consequences that would materially harm our business and reputation.

For so long as we remain an emerging growth company as defined in the JOBS Act, we intend to take advantage of certain exemptions from various reporting requirements that are applicable to public companies that are not emerging growth companies, including, but not limited to, not being required to comply with the auditor attestation requirements of Section 404. We will remain an emerging growth company until the earlier of (1) the last day of the fiscal year (i) following the fifth anniversary of the completion of this offering, (ii) in which we have total annual gross revenue of at least $1.235 billion, or (iii) in which we are deemed to be a large accelerated filer, which means the market value of our common stock that is held by non-affiliates exceeds $700.0 million as of the prior June 30th, and (2) the date on which we have issued more than $1.0 billion in non-convertible debt during the prior three-year period.

We are an emerging growth company and we cannot be certain if (i) the reduced disclosure requirements or (ii) extended transition periods for complying with new or revised accounting standards applicable to emerging growth companies will make our common stock less attractive to investors. In addition, as a smaller reporting company we will also have reduced disclosure requirements.

We qualify as an emerging growth company. Under the JOBS Act, emerging growth companies can delay adopting new or revised accounting standards until such time as those standards apply to private companies. Under the JOBS Act, emerging growth companies can delay adopting new or revised accounting standards issued subsequent to the enactment of the JOBS Act until such time as those standards apply to private companies. We have elected to use this extended transition period for complying with new or revised accounting standards that have different effective dates for public and private companies until the earlier of the date we (i) are no longer an emerging growth company, or (ii) affirmatively and irrevocably opt out of the extended transition period provided in the JOBS Act. As a result, our financial statements may not be comparable to companies that comply with new or revised accounting pronouncements as of public company effective dates.

In addition, for as long as we continue to be an emerging growth company, we intend to take advantage of certain exemptions from various reporting requirements that are applicable to other public companies including, but not limited to, reduced disclosure obligations regarding executive compensation in our periodic reports and proxy statements and exemptions from the requirements of holding a nonbinding advisory vote on executive compensation and stockholder approval of any golden parachute payments not previously approved. We cannot predict if investors will find our common stock less attractive because we will rely on these exemptions. If some investors find our common stock less attractive as a result, there may be a less active trading market for our common stock and our stock price may be more volatile.

We are also a “smaller reporting company” as defined in the Securities Exchange Act of 1934, as amended, or the Exchange Act, and have elected to take advantage of certain of the scaled disclosures available to smaller reporting companies. To the extent that we continue to qualify as a “smaller reporting company” as such term is defined in Rule 12b-2 under the Exchange Act, after we cease to qualify as an emerging growth company, certain of the exemptions available to us as an “emerging growth company” may continue to be available to us as a “smaller reporting company,” including exemption from compliance with the auditor attestation requirements pursuant to SOX and reduced disclosure about our executive compensation arrangements. We will continue to be a “smaller reporting company” until we have $250 million or more in public float (based on our

Common Stock) measured as of the last business day of our most recently completed second fiscal quarter or, in the event we have no public float (based on our Common Stock) or a public float (based on our Common Stock) that is less than $700 million, annual revenues of $100 million or more during the most recently completed fiscal year.

SPECIAL NOTE REGARDING FORWARD-LOOKING STATEMENTS

This prospectus contains “forward-looking statements.” We use words such as “could,” “may,” “might,” “will,” “expect,” “likely,” “believe,” “continue,” “anticipate,” “estimate,” “intend,” “plan,” “project,” and other similar expressions to identify some forward-looking statements, but not all forward-looking statements include these words. All of our forward-looking statements involve estimates and uncertainties that could cause actual results to differ materially from those expressed in the forward-looking statements. Accordingly, any such statements are qualified in their entirety by reference to the information described under the caption “Risk Factors” and elsewhere in this prospectus.

The forward-looking statements contained in this prospectus are based on assumptions that we have made in light of our industry experience and our perceptions of historical trends, current conditions, expected future developments, and other factors we believe are appropriate under the circumstances. As you read and consider this prospectus, you should understand that these statements are not guarantees of performance or results. They involve risks, uncertainties (many of which are beyond our control), and assumptions. Although we believe that these forward-looking statements are based on reasonable assumptions, you should be aware that many factors could affect our actual operating and financial performance and cause our performance to differ materially from the performance anticipated in the forward-looking statements. We believe these factors include, but are not limited to, those described under “Risk Factors” and “Management’s Discussion and Analysis of Financial Condition and Results of Operations.” Should one or more of these risks or uncertainties materialize, or should any of these assumptions prove incorrect, our actual operating and financial performance may vary in material respects from the performance projected in these forward-looking statements.

Further, any forward-looking statement speaks only as of the date on which it is made, and except as required by law, we undertake no obligation to update any forward-looking statement contained in this prospectus to reflect events or circumstances after the date on which it is made or to reflect the occurrence of anticipated or unanticipated events or circumstances. New factors that could cause our business not to develop as we expect emerge from time to time, and it is not possible for us to predict all of them. Further, we cannot assess the impact of each currently known or new factor on our results of operations or the extent to which any factor, or combination of factors, may cause actual results to differ materially from those contained in any forward-looking statements.

MARKET, INDUSTRY AND OTHER DATA

This prospectus includes market and industry data and forecasts that we have derived from independent consultant reports, publicly available information, various industry publications, and our internal data and estimates.

Our internal data and estimates are based upon information obtained from trade and business organizations and other contacts in the markets in which we operate and our management's understanding of industry conditions. Although we believe that such information is reliable, we have not had this information verified by any independent sources.

USE OF PROCEEDS

We estimate that the net proceeds from the sale of shares of our common stock in this offering will be approximately $5.7 million, after deducting underwriting discounts and commissions and estimated offering expenses payable by us, based on the initial public offering price of $5.00 per share. If the underwriters exercise in full their option to purchase additional shares of common stock, we estimate that the net proceeds to us from this offering will be approximately $6.7 million, after deducting underwriting discounts and commissions and estimated offering expenses payable by us, based on the initial public offering price of $5.00 per share.

We expect to use the net proceeds from this offering, together with our existing cash as follows:

- $3.3 million will be used for CMC preparation, research and development, and other trial preparation expenses necessary for initiation of our planned Phase 3 pivotal trial; and
- the remainder of $2.4 million for working capital and general corporate purposes.

We will require a total of $45 million to complete our pivotal Phase 3 trial. Therefore, we will require additional funding of approximately $42 million to enroll patients and complete our pivotal Phase 3 clinical trial.

The expected use of the net proceeds from this offering and our existing cash represent our intentions based upon our current plans and business conditions. The amounts and timing of our actual expenditures may vary significantly depending on a number of factors, including the progress of our development and commercialization efforts, the status of and results from our clinical trials, and any collaborations that we may enter into with third parties for our products or product candidates, as well as any unforeseen cash needs. As a result, our management will retain broad discretion over the allocation of the net proceeds from this offering. We have no current understandings, agreements or commitments for any material acquisitions or licenses of any products or product candidates, businesses or technologies.

Pending our application of the net proceeds from this offering, we plan to invest the net proceeds in a variety of capital preservation investments, including short and intermediate-term, interest-bearing obligations, investment-grade instruments, certificates of deposit or direct or guaranteed obligations of the U.S. government.

DIVIDEND POLICY

We have never declared or paid, and we do not anticipate declaring or paying in the foreseeable future, any cash dividends on our capital stock. We intend to reinvest any earnings in developing and expanding our business. Any future determination relating to the declaration and payment of dividends will be at the discretion of our board of directors and will depend on then-existing conditions, including our operating results, financial conditions, contractual restrictions, capital requirements, business prospects and other factors our board of directors may deem relevant.

CAPITALIZATION

The following table sets forth our cash and equivalents and capitalization as of September 30, 2022:

- on an actual basis;
- on a pro forma basis to give effect to (i) the conversion upon the closing of this offering of the outstanding Private Placement Notes, in the principal amount of $575,000 into an aggregate of 575,000 shares of common stock based upon an as adjusted conversion price of $1.00 per share; (ii) the conversion upon the closing of this offering of the outstanding June 2022 Note in the principal amount of $50,000, plus accrued interest through September 30, 2022 of $908, into an 50,908 shares of common stock based upon an as adjusted conversion price of $1.00 per share, (iii) the conversion upon the closing of this offering of the outstanding March 2022 Note in the principal amount of $500,000, plus accrued interest through September 30, 2022 of $14,792, into 514,792 shares of common stock based upon an as adjusted conversion price of $1.00 per share, (iv) elimination of the derivative liability related to the notes, (v) the issuance of 600,000 shares of common stock, based upon a price of $3.00 per share, which is sixty percent (60%) of the initial public offering price of $5.00 per share, on the effective date of this offering pursuant to the terms of the Amendment to Asset Purchase Agreement; (vi) the issuance of 67,000 shares of our common stock to consultants subsequent to September 30, 2022 for the provision of services; (vii) the issuance of 50,000 shares of common stock to an executive officer upon consummation of this offering; and (viii) the receipt of $250,000 in proceeds from the issuance of the November Notes and the November Warrants and the automatic exercise of the November Warrants upon consummation of this offering, which will result in the issuance of 250,000 shares of common stock and the repayment of the principal amount of the November Notes, with any interest owed on the November Notes to be payable in cash; and
- on a pro forma as adjusted basis to additionally give effect to the pro forma adjustments set forth above and the sale of 1,400,000 shares of common stock in this offering, at the initial public offering price of $5.00 per share of common stock (the price set forth on the cover page of this prospectus), after deducting estimated underwriting discounts and commissions and estimated offering expenses payable by us.

You should read the information in this table together with our audited financial statements and related notes and unaudited interim condensed financial statements and related notes and "Management's Discussion and Analysis of Financial Condition and Results of Operations" appearing elsewhere in this prospectus.

	As of September 30, 2022		
	Actual	**Pro Forma**	**Pro Forma, as adjusted**
		(unaudited)	***(unaudited)***
Cash	$ 103,082	$ 353,082	$ 6,093,082
Convertible notes, net of debt discount	528,472	—	—
Stockholders' equity:			
Common stock (10,000,000 shares authorized, 8,126,875 shares issued and outstanding as of September 30, 2022, at par value $0.001)	8,127	10,235	11,635
Additional paid-in capital	516,952	5,490,544	11,229,144
Accumulated deficit	(1,704,902)	(5,289,902)	(5,289,902)
Total stockholders' equity (deficit)	$ (1,179,823)	210,877	5,950,877

The number of shares of our common stock to be outstanding upon completion of this offering is based on 8,126,875 shares of our common stock outstanding as of September 30, 2022, and excludes:

- 84,000 shares of common stock issuable upon exercise of the Representative's Warrant;
- 1,100,000 shares of common stock issuable upon exercise of stock options, at a weighted average exercise price of $0.84 per share;
- 26,500 shares of common stock issuable upon exercise of warrants issued in the Private Placement and the November Private Placement to Boustead Securities, LLC, as placement agent; and
- 760,000 shares of our common stock reserved for future issuance under our 2022 Plan, which 2022 Plan will become effective upon the effective date of the registration statement of which this prospectus forms a part.

DILUTION

If you purchase shares of our common stock in this offering, your interest will be diluted immediately to the extent of the difference between the public offering price of $5.00 per share and the pro forma as adjusted net tangible book value per share of our common stock immediately upon the consummation of this offering. As of September 30, 2022, we had a historical net tangible book value of $(1,179,823), or $(0.15) per share of common stock. Our historical net tangible book value per share represents total tangible assets less total liabilities, divided by the number of shares of our common stock outstanding as of September 30, 2022.

Our pro forma net tangible book value as of September 30, 2022 was $122,960, or $0.01 per share of our common stock. Pro forma net tangible book value represents the amount of our total tangible assets less our total liabilities, after giving effect to (i) the conversion upon the closing of this offering of the Private Placement Notes in the principal amount of $575,000 into an aggregate of 575,000 shares of common stock based upon an as adjusted conversion price of $1.00 per share; (ii) the conversion of the June 2022 Note in the principal amount of $50,000, plus accrued interest through September 30, 2022 of $908, into 50,908 shares of common stock based upon an as adjusted conversion price of $1.00 per share; (iii) the conversion upon the closing of this offering of the outstanding of the March 2022 Note in the principal amount of $500,000, plus accrued interest through September 30, 2022 of $14,792, into 514,792 shares of common stock based upon an as adjusted conversion price of $1.00 per share; (iv) elimination of the derivative liability related to the notes; (v) the issuance of 600,000 shares of common stock, based upon a price of $3.00 per share, which is sixty percent (60%) of the initial public offering price of $5.00 per share, on the effective date of this offering pursuant to the terms of the Amendment to Asset Purchase Agreement; (vi) the issuance of 67,000 shares of our common stock to consultants subsequent to September 30, 2022 for the provision of services; (vii) the issuance of 50,000 shares of common stock to an executive officer upon consummation of this offering; and (viii) the receipt of $250,000 in proceeds from the issuance of the November Notes and the November Warrants and the automatic exercise of the November Warrants upon consummation of this offering, which will result in the issuance of 250,000 shares of common stock and the repayment of the principal amount of the November Notes, with any interest owed on the November Notes to be payable in cash.

After giving effect to the pro forma adjustments set forth above and our sale of 1,400,000 shares of common stock in this offering at the public offering price of $5.00 per share of common stock, and after deducting underwriters' commissions and estimated offering expenses, but assuming no exercise of the warrants issued to the Representative of underwriters, our pro forma as adjusted net tangible book value as of September 30, 2022 would have been $5,612,960 or $0.48 per share of common stock. This represents an immediate increase in net tangible book value of $0.49 per share of common stock to existing stockholders and an immediate dilution in net tangible book value of $4.52 per share to purchasers of Common Stock in this offering.

The following table illustrates this dilution on a per share of common stock basis assuming the underwriters do not exercise their option to purchase additional shares of common stock:

Public offering price per share			$	5.00
Net tangible book value per share as of September 30, 2022	$	(0.15)		
Pro forma net tangible book value per share	$	0.01		
Pro forma increase in net tangible book value per share attributable to new investors	$	0.49		
Pro forma as adjusted net tangible book value per share as of September 30, 2022, after giving effect to the offering	$			0.50
Dilution per share to new investors in the offering			$	4.50

The dilution information discussed above is illustrative only and may change based on the actual initial public offering price and other terms of this offering.

If the underwriters exercise their option in full to purchase 210,000 additional shares of common stock in this offering at the offering price of $5.00 per share, the pro forma net tangible book value per share after this offering would be $0.58 per share of common stock and the dilution to new investors purchasing securities in this offering would be $4.42 per share of common stock.

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The following charts illustrate our pro forma proportionate ownership, upon completion of this offering by present stockholders and investors in this offering, compared to the relative amounts paid by each. The charts reflect payment by present stockholders as of the date the consideration was received and by investors in this offering at the public offering price. The charts further assume no changes in net tangible book value other than those resulting from the offering.

	Shares Purchased		Total Consideration		Average Price Per Share
	Amount (#)	Percent (%)	Amount ($)	Percent (%)	($)
Existing stockholders	10,234,575	88%	$ 5,401,150	44%	$ 0.53
New investors	1,400,000	12%	$ 7,000,000	56%	$ 5.00
Total	11,634,575	100%	$ 12,401,150	100%	$ 1.07

The number of shares of our common stock outstanding before and after this offering reflected in the tables and discussion above are based on: (i) the conversion upon the closing of this offering of the Private Placement Notes, in the principal amount of $575,000 into an aggregate of 575,000 shares of common stock based upon an as adjusted conversion price of $1.00 per share; (ii) the conversion upon the closing of this offering of the June 2022 Note in the principal amount of $50,000, plus accrued interest through September 30, 2022 of $908, into 50,908 shares of common stock based upon an as adjusted conversion price of $1.00 per share, (iii) the conversion upon the closing of this offering of the outstanding March 2022 Note in the principal amount of $500,000, plus accrued interest through September 30, 2022 of $14,792, into an aggregate of 514,792 shares of common stock based upon an as adjusted conversion price of $1.00 per share, (iv) the issuance of 600,000 shares of common stock, based upon a price of $3.00 per share which is sixty percent (60%) of the initial public offering price of $5.00 per share, on the effective date of this offering pursuant to the terms of the Amendment to Asset Purchase Agreement; (v) the issuance of 67,000 shares of our common stock to consultants subsequent to September 30, 2022 for the provision of services; (vi) the issuance of 50,000 shares of common stock to an executive officer upon consummation of this offering; and (vii) the issuance of 250,000 shares of common stock upon the exercise of the November Warrants upon consummation of this offering. Excluded, as of that date, are the following:

- 84,000 shares of common stock issuable upon exercise of the Representative's Warrant;
- 1,100,000 shares of common stock issuable upon exercise of stock options, at a weighted average exercise price of $0.84 per share;
- 26,500 shares of common stock issuable upon exercise of warrants issued in the Private Placement and the November Private Placement to Boustead Securities, LLC, as placement agent; and
- 760,000 shares of our common stock reserved for future issuance under our 2022 Plan, which 2022 Plan will become effective upon the effective date of the registration statement of which this prospectus forms a part.

The table below assumes the underwriters' exercise their over-allotment option in full:

	Shares Purchased		Total Consideration		Average Price Per Share
	Amount (#)	Percent (%)	Amount ($)	Percent (%)	($)
Existing stockholders	10,234,575	86%	5,401,150	40%	$ 0.53
New investors	1,610,000	14%	$ 8,050,000	60%	$ 5.00
Total	11,844,575	100%	13,451,150	100%	$ 1.14

The number of shares of our common stock outstanding before and after this offering reflected in the tables and discussion above are based on: (i) the conversion upon the closing of this offering of the Private Placement Notes, in the principal amount of $575,000 into an aggregate of 575,000 shares of common stock based upon an as adjusted conversion price of $1.00 per share; (ii) the conversion upon the closing of this offering of the June 2022 Note in the principal amount of $50,000, plus accrued interest through September 30, 2022 of $908, into 50,908 shares of common stock based upon an as adjusted conversion price of $1.00 per share, (iii) the

conversion upon the closing of this offering of the outstanding of the March 2022 Note in the principal amount of $500,000, plus accrued interest through September 30, 2022 of $14,792, into 514,792 shares of common stock based upon an as adjusted conversion price of $1.00 per share, (iv) the issuance, upon the effective date of this offering pursuant to the terms of the Amendment to Asset Purchase

Agreement of 600,000 shares of common stock, based upon a price of $3.00 per share, which is sixty percent (60%) of the initial public offering price of $5.00 per share; (v) the issuance of 67,000 shares of our common stock to consultants subsequent to September 30, 2022 for the provision of services; (vi) the issuance of 50,000 shares of common stock to an executive officer upon consummation of this offering; and (vii) the issuance of 250,000 shares of common stock upon the exercise of the November Warrants, such warrants to be automatically exercised upon consummation of this offering. Excluded, as of that date, are the following:

- 84,000 shares of common stock issuable upon exercise of the Representative's Warrant;
- 1,100,000 shares of common stock issuable upon exercise of stock options, at a weighted average exercise price of $0.84 per share;
- 26,500 shares of common stock issuable upon exercise of warrants issued in the Private Placement and the November Private Placement to Boustead Securities, LLC, as placement agent; and
- 760,000 shares of our common stock reserved for future issuance under our 2022 Plan, which 2022 Plan will become effective upon the effective date of the registration statement of which this prospectus forms a part.

MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

You should read the following discussion and analysis of our financial condition and results of operations together with our financial statements and the related notes to those statements included elsewhere in this prospectus. This discussion and analysis and other parts of this prospectus contain forward-looking statements based upon current beliefs, plans and expectations related to future events and our future financial performance that involve risks, uncertainties and assumptions, such as statements regarding our intentions, plans, objectives, expectations, forecasts and projections. Our actual results and the timing of selected events could differ materially from those anticipated in these forward-looking statements as a result of several factors, including those set forth under the section titled "Risk Factors" and elsewhere in this prospectus. You should carefully read the "Risk Factors" to gain an understanding of the important factors that could cause actual results to differ materially from our forward-looking statements. Please also see the section titled "Special Note Regarding Forward-Looking Statements."

Overview

We are focused on developing tecarfarin, a novel therapy with orphan drug indication, designed for the prevention of systemic thromboembolism (blood clots) of cardiac origin in patients with end-stage renal disease and atrial fibrillation (irregular heartbeat) or AFib. We secured the rights to tecarfarin on April 1, 2022 via an asset purchase agreement from HESP LLC, a wholly owned subsidiary of Horizon Technology Finance Corporation. HESP LLC acquired the assets of Espero BioPharma, Inc., or Espero, including tecarfarin, in an assignment for the benefit of creditors in which the creditor, Horizon Technology Finance Corporation and Horizon Credit II LLC (collectively, Horizon), a secured lender of Espero, designated HESP LLC as the assignee of Espero's assets.

Tecarfarin is an anticoagulant that uses a drug design process which targets a different pathway than the most commonly prescribed drugs used in the treatment of thrombosis and AFib. Tecarfarin has been evaluated in 11 human clinical trials conducted by third parties in over 1,003 individuals (269 patients were treated for at least six months and 129 patients were treated for one year or more). In Phase 1, Phase 2 and Phase 2/3 clinical trials, tecarfarin has generally been well-tolerated in both healthy adult subjects and patients with CKD. In the Phase 2/3 trial, EMBRACE-AC, the largest tecarfarin trial with 607 patients having completed it, only 1.6% of the blinded tecarfarin subjects suffered from major bleeding and there were no thrombotic events. Five patients died during the trial, but only one death due to intracerebral hemorrhage was considered to be possibly related to the tecarfarin.

In 2019, the FDA provided input on the Phase 3 trial design for tecarfarin, which was submitted by Espero, submitted by the previous owner of tecarfarin. We intend to submit our Phase 3 trial design to the FDA using the same protocol that was submitted by Espero. Assuming the FDA accepts our Phase 3 trial design, we intend to commence the Phase 3 pivotal trial in the second half of 2023. However, there can be no assurance that the trial design will be accepted by the FDA. We are pursuing regulatory approval of tecarfarin as an individual treatment, although we might evaluate, in consultation with the FDA, other potential uses in the future.

In March 2019, the FDA granted orphan drug designation, or ODD, for tecarfarin for the prevention of systemic thromboembolism of cardiac origin in patients with ESRD and AFib. The FDA grants ODD status to drugs that are intended for the treatment, diagnosis, or prevention of rare diseases or conditions, which are defined as a disease or condition that affects fewer than 200,000 people in the U.S. The ODD program provides a drug developer with certain benefits and incentives, including a seven-year period of U.S. marketing exclusivity from the date of marketing authorization, waiver of FDA user fees, and tax credits for clinical research. The granting of an orphan drug designation does not alter the FDA's regulatory requirements to establish safety and effectiveness of a drug through adequate and well-controlled studies to support approval and commercialization. Furthermore, orphan drug designation does not indicate or guarantee FDA approval of the NDA and we might not receive exclusivity.

Tecarfarin was developed by researchers using a retrometabolic drug design process which targets a different metabolic pathway than the most commonly prescribed drugs for the treatment of thrombosis and AFib. "Drug metabolism" refers to the process by which a drug is inactivated by the body and rendered easier to eliminate or to be cleared by the body. Most approved drugs, including warfarin, the only FDA-approved Vitamin K antagonists, or VKAs, which is a prescribed drug for the treatment of thrombosis, are metabolized in the liver

through a pathway known as the Cytochrome CYP450 system, or CYP450, by the enzymes known as CYP2C9 and CYP3A4. By using a different metabolic pathway, tecarfarin eliminates or minimizes the CYP450 metabolism in the liver. Patients taking multiple medications that interact with CYP2C9, or CYP3A4 or those with impaired kidney function, can experience an

overload in the pathway, creating a bottleneck that often leads to insufficient clearance, which results in toxic build-up of one or more drugs. In some instances, patients taking multiple medications metabolized by the same CYP450 pathway may experience decreased efficacy of one or more of the medications due to rapid metabolism or increased drug effect and/or toxicity due to enzyme induction. Patient-specific genetic differences can also hinder drug clearance in the CYP450 pathway. Our product candidate, tecarfarin, was designed using a retrometabolic drug design process so that tecarfarin follows a metabolic pathway distinct from the CYP450 pathway and is metabolized by both CYP450 and non-CYP450 pathways. We believe this may allow elimination by large capacity and non-saturable tissue esterase pathways that exist throughout the body rather than just in the liver.

Tecarfarin is an orphan designated, vitamin K antagonist, oral, once-daily anticoagulant in the same drug class as warfarin designed for use in patients requiring chronic VKA anticoagulation, to prevent systemic thromboembolism of cardiac origin in patients with ESRD and AFib. The prevailing treatment for thrombosis is with an oral anticoagulant, either a VKA, like warfarin, or non-vitamin K oral anticoagulant ("NOAC"). VKAs block the production of vitamin K-dependent blood clotting factors, such that the blood is "thinned," preventing clots, while NOACs directly block the activity of certain of these clotting factors. Tecarfarin, like warfarin, is a VKA.

Vitamin K epoxide Reductase Complex subunit 1 (VKORC1) is a significant enzyme for effective clotting. VKORC1 reduces vitamin K epoxide to its active form (Vitamin K), which is the rate-limiting step in the physiological process of vitamin K recycling. Vitamin K serves as a cofactor for normal function of several clotting/anticoagulation factors including Factors II, VII, IX and X and Proteins C, S, and Z. VKORC1 genetic deficiencies result in increased sensitivity to VKAs, which results in an increase in the risk of significant hemorrhaging. We believe tecarfarin has similar potency for VKORC1 inhibition as warfarin, but it is an investigational new drug, and we must demonstrate it is safe and effective for its proposed indication.

AFib is the most common arrhythmia, with its incidence and prevalence increasing over the last twenty (20) years. AFib is associated with an approximate five-fold increased risk of stroke. The risk of developing AFib increases in patients with CKD. According to 2021 estimates by the Centers for Disease Control and Prevention, or CDC, approximately 15% of the U.S. adult population, or 37 million people, have CKD. An estimated 0.4% of people in the U.S. suffer from Stage 4 CKD and 0.1% of people in the U.S. have ESRD.

Patients with ESRD and AFib represent a spectrum of disorders involving both the heart and kidneys (known as cardiorenal syndrome or CRS) in which acute or chronic dysfunction in one organ may induce acute or chronic dysfunction in the other organ. These patients have typically been excluded from randomized clinical trials because the approved therapies for AFib have metabolic profiles that may increase drug exposures thereby increasing the known risks and challenges in managing these patients. The presence of either chronic kidney disease, or CKD, or AFib, increases the risk of serious thromboembolic adverse clinical outcomes, such as stroke and death. Antithrombotic therapy is typically recommended to decrease this risk in AFib patients, but there are no approved treatment options for patients with ESRD and AFib. Warfarin may cause substantial harm in these patients. Low-dose apixaban (Eliquis) was approved by the FDA for use in ESRD patients on hemodialysis based upon limited pharmacokinetic data by 8 subjects, despite that randomized trials to date of apixaban versus warfarin for AFib excluded patients with severe and end-stage kidney disease. The RENAL-AF (Trial to Evaluate Anticoagulation Therapy in Hemodialysis Patients With Atrial Fibrillation) was terminated early in 2019 by its sponsor.

There are more than 809,000 Americans with ESRD, with approximately 70% on dialysis, according to the United States Renal Data System. Approximately 150,000 ESRD patients also have AFib. AFib nearly doubles the anticipated mortality and increases the stroke risk by approximately five-fold in these patients. There is evidence that AFib is an independent risk factor for developing ESRD in CKD patients. Both diseases share common risk factors including hypertension, diabetes, vascular disease, and advancing age. Cardiovascular disease contributes to more than half of all deaths among patients with ESRD. According to the Annual Data Report published by the United States Renal Data System, total Medicare spending for patients with ESRD reached $51 billion in 2019, accounting for approximately 7% of the Medicare paid claims costs.

We have licensed out the rights to tecarfarin for several Asian markets, including China, to Lee's Pharmaceutical Holdings Limited, an integrated research-driven and market-oriented biopharmaceutical publicly listed company based in Hong Kong with over 25 years' experience in the pharmaceutical industry in

China. Lee's Pharmaceutical Holdings Limited is developing tecarfarin as an anti-thrombotic for patients with mechanical heart valves. In 2020 and 2021, Lee's Pharmaceutical Holdings Limited completed two Phase 1 studies in China and Hong Kong and is currently preparing for its Phase 2 trial.

Recent Developments

On July 16, 2022, we closed the first tranche of the Private Placement pursuant to which we issued First Tranche Private Placement Notes in the principal amount of $450,000 and received gross proceeds of $450,000. The private placement offering was for up to $2,000,000 of convertible promissory notes and was to terminate upon the sale of all $2,000,000 of convertible promissory notes or July 31, 2022 subject to extension until August 31, 2022, in the sole discretion of the placement agent and us. We, together with the placement agent, determined to extend the Private Placement to August 31, 2022. On August 22, 2022, we issued a convertible promissory note pursuant to the Private Placement in the principal amount of $50,000 (the "Second Tranche Private Placement Note") and received gross proceeds of $50,000. The Private Placement was further extended to September 30, 2022, after obtaining the consent of the investors in the Private Placement to date. On September 13, 2022, we issued convertible promissory notes in the aggregate principal amount of $75,000 (together with the First Tranche Private Placement Notes and the Second Tranche Private Placement Note, the "Private Placement Notes) and received gross proceeds of $75,000. The Private Placement Notes bear interest at 6% and mature on the third anniversary of the date of the last closing of the Private Placement Notes, i.e. September 13, 2025. The principal amount due under the Private Placement Notes (and, at our option, any accrued but unpaid interest under the Private Placement Notes) will be automatically converted into shares of our common stock upon the consummation of this offering. The Private Placement Notes were initially convertible into shares of common stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the Private Placement Notes plus (if applicable) any accrued but unpaid interest under the Private Placement Notes by (ii) sixty percent (60%) of the initial offering price per share subject to adjustment in the event of a subsequent issuance of securities by us at a price per share less than the conversion price. Upon the issuance of the November Notes and November Warrants the conversion price was reduced to $1.00 per share. The holders of the Private Placement Notes entered into an Investor Rights Agreement that provides for "drag along" and "tag along" rights and permits the holders to participate in subsequent securities offerings, including this offering, in a percentage amount of such securities offering equal to the percentage of the amount invested by such holder in the Private Placement Notes as compared to the amount invested by all holders in the Private Placement Notes; provided, however, that in no event shall the dollar amount of the participation right of any holder exceed 100% of original principal amount of the Private Placement Notes purchased by such holder.

Boustead Securities, LLC acted as the placement agent for the Private Placement and received a commission equal to 7% of the gross proceeds received by us from the sale of the Private Placement Notes, a non-accountable expense allowance equal to 1% of the gross proceeds received by us from the sale of the Private Placement Notes and five-year warrants, as amended, to purchase an aggregate of 11,500 shares of our common stock at an exercise price equal to 60% of the initial public offering price per share.

On November 30, 2022, we completed a $250,000 note offering pursuant to which we sold to two accredited investors for cash proceeds of $250,000 units consisting of: (i) November Notes in the aggregate principal amount of $250,000, which bear interest at the rate of 10%, repayable at the time of this offering, and (ii) November Warrants to purchase up to an aggregate of 250,000 shares of common stock exercisable at $1.00 per share. The November Warrants are automatically exercisable upon the consummation of this offering, at which time the investors will become the holders of 250,000 shares of our common stock and the proceeds of such warrant exercise will be used to repay the principal amount of the November Notes, with any interest owed on the November Notes to be payable in cash. The shares to be issued upon exercise of the November Warrants are being registered in the Resale Prospectus.

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Results of Operations

The following table summarizes our results of operations for the period January 25, 2022 (inception) to June 30, 2022.

	January 25, 2022 (Inception) to June 30, 2022
Operating expenses:	
General and administrative expenses	$ 432,772
Research and development expenses	220,095
Total operating expenses	652,867
Loss from operations	(652,867)
Other expense:	
Interest expense	8,527
Interest expense, amortization of debt discount	10,832
Change in in fair value of derivative liability	6,562
Total other expenses	25,921
Net loss and comprehensive loss	$ (678,788)

General and administrative expenses

General and administrative expenses were $432,772 for the period January 25, 2022 (inception) to June 30, 2022. General and administrative expenses included $115,000 of consulting services, $191,483 of personnel related expenses, $28,683 of patent renewal fees, $80,546 of professional fees, and $17,060 of other expenses.

Research and development expenses

Research and development expenses were $220,095 for the period January 25, 2022 (inception) to June 30, 2022. Research and development expenses included $200,000 of acquired intangible assets and $20,095 of transaction costs which are expensed as in-process research and development.

The following table summarizes our results of operations for the period January 25, 2022 (inception) to September 30, 2022

	January 25, 2022 (inception) through September 30, 2022
Operating expenses:	
General and administrative expenses	$ 1,348,670
Research and development expenses	276,482
Depreciation and amortization expense	506
Total operating expenses	1,625,658
Loss from operations	(1,625,658)
Other expense:	
Interest expense	22,068
Interest expense, amortization of debt discount	29,302

Change in fair value of derivative liabilities	27,874
Total other expenses	79,244
Net loss and comprehensive loss	$ (1,704,902)

General and administrative expenses

General and administrative expenses were $1,348,670 for the period January 25, 2022 (inception) to September 30, 2022. General and administrative expenses included $435,642 of stock-based compensation, $160,000 of consulting services, $560,527 of personnel related expenses, $30,871 of patent renewal fees, $109,361 of professional fees, and $52,269 of other expenses.

Research and development expenses

Research and development expenses were $276,482 for the period January 25, 2022 (inception) to September 30, 2022. Research and development expenses included $200,000 of acquired intangible assets, $20,095 of transaction costs which are expensed as in-process research and development, and $43,042 of stock-based compensation.

Liquidity and Capital Resources

Since inception, we have incurred losses and negative cash flows from operations. To date, we have funded our operations from the proceeds of the sale of convertible notes, the November Notes, and the November Warrants. We had a net loss of $1,704,902 for the period January 25, 2022 (inception) to September 30, 2022 with cash used in operations of $914,554 for the period then ended. As of September 30, 2022, we had cash of $103,082, negative working capital of $131,007, and an accumulated deficit of $1,704,902. Subsequent to September 30, 2022, we raised an additional $250,000 in gross proceeds from the issuance of promissory notes and warrants to purchase 250,000 shares of common stock. The principal amount of the November Notes will be repaid upon the consummation of this offering from the proceeds of the automatic exercise of the November Warrants, with any interest owed on the November Notes to be payable in cash. Our current cash is not sufficient to fund our planned operations for at least the next twelve months, however, we believe that our existing cash together with the proceeds of this offering will be sufficient to fund our operations for at least the next twelve months. We expect to require additional liquidity as we continue to execute our business strategy including commencing and completing our planned Phase 3 clinical trial and submitting our NDA, the cost of which we anticipate will be approximately $45 million. In addition, we are projecting that our operating losses and expected capital needs will exceed our existing cash balances and cash expected to be generated from operations for the foreseeable future. In order to meet our expected obligations and to fund completion of our Phase 3 clinical trial, we intend to raise additional funds through equity and debt financings. However, there can be no assurance that we will be able to complete any additional equity or debt financings on terms acceptable to the Company or at all. If we are unable to raise additional funding to meet our working capital needs in the future, we will be forced to delay or reduce the scope of our research programs and/or limit or cease our operations. As a result, due to the uncertainty in our ability to meet our current operating and capital expenses, there is substantial doubt about our ability to continue as a going concern.

We estimate that we will require a total of $45 million for the completion of our planned pivotal Phase 3 clinical trial and other expenditures that we will need to incur in order to file our NDA. We intend to use approximately $3.3 million of the proceeds from this offering to fund CMC preparation, research and development and other trial preparation expenses necessary for commencement of our planned pivotal Phase 3 clinical trial, therefore, we will require at least $42 million of additional funding to enroll patients and complete our pivotal Phase 3 clinical trial. In addition, we estimate that we will require $15 million for general and administrative expenses anticipated to be incurred over the next three years.

Cash Flows

The following table summarizes our cash flows for the period presented:

	January 25, 2022 (Inception) to September 30, 2022
Cash used in operating activities	$ (914,554)
Cash used in investing activities	(2,278)

Cash provided by financing activities		1,019,914
Net increase in cash		103,082
Cash, beginning of period		—
Cash, end of period	$	103,082

Operating activities

During the period January 25, 2022 (inception) to September 30, 2022, cash used in operating activities was $914,554 primarily resulting from our net loss of $1,704,902, offset by a $308,221 increase in accounts payable and a $364,141 increase in accrued liabilities.

Financing activities

During the period January 25, 2022 (inception) to September 30, 2022, net cash provided by financing activities was $1,019,914, primarily consisting of the $1,011,964 of net proceeds from the issuance of convertible notes.

Critical Accounting Estimates

This discussion and analysis of our financial condition and results of operations is based on our financial statements, which have been prepared in accordance with generally accepted accounting principles in the United States, or GAAP. The preparation of these financial statements requires us to make estimates and assumptions that affect the reported amounts of assets and liabilities and the disclosure of contingent assets and liabilities at the date of the financial statements, as well as the reported expenses incurred during the reporting periods. Our estimates are based on our historical experience and on various other factors that we believe are reasonable under the circumstances, that results of which form the basis for making judgments about the carrying value of assets and liabilities that are not readily apparent from other sources. Actual results may differ from these estimated under different assumption or conditions. While our significant accounting policies are described in more detail in the notes to our financial statements included elsewhere in the prospectus, we believe that the following accounting policies are critical to understanding our historical and future performance, as these policies relate to the more significant areas involving managements judgements and estimates.

Acquisitions

We evaluate acquisitions of assets and other similar transactions to assess whether or not the transaction should be accounted for as a business combination or asset acquisition by first applying a screen test to determine whether substantially all of the fair value of the gross assets acquired is concentrated in a single identifiable asset or group of similar identifiable assets. If so, the transaction is accounted for as an asset acquisition. If not, further determination is required as to whether or not we have acquired inputs and processes that have the ability to create outputs, which would meet the definition of a business. Significant judgment is required in the application of the screen test to determine whether an acquisition is a business combination or an acquisition of assets.

Acquisitions meeting the definition of business combinations are accounted for using the acquisition method of accounting, which requires that the purchase price be allocated to the net assets acquired at their respective fair values. In a business combination, any excess of the purchase price over the estimated fair values of the net assets acquired is recorded as goodwill.

For asset acquisitions, a cost accumulation model is used to determine the cost of an asset acquisition. Direct transaction costs are recognized as part of the cost of an asset acquisition. We also evaluate which elements of a transaction should be accounted for as a part of an asset acquisition and which should be accounted for separately. The cost of an asset acquisition, including transaction costs, is allocated to identifiable assets acquired and liabilities assumed based on a relative fair value basis. Goodwill is not recognized in an asset acquisition. Any difference between the cost of an asset acquisition and the fair value of the net assets acquired is allocated to the non-monetary identifiable assets based on their relative fair values. When a transaction accounted for as an asset acquisition includes an in-process research and development ("IPR&D") asset, the IPR&D asset is only capitalized if it has an alternative future use other than in a particular research and development project. For an IPR&D asset to have an alternative future use: (a) we must reasonably expect that we will use the asset acquired in the alternative manner and anticipate economic benefit from that alternative use, and (b) our use of the asset acquired is not contingent on further development of the asset subsequent to the acquisition date (that is, the asset can be used in the alternative manner in the condition in which it existed at the acquisition date). Otherwise, amounts allocated to IPR&D that have no alternative use are expensed to research and development. Asset acquisitions may include contingent consideration arrangements that encompass obligations to make future payments to sellers contingent upon the achievement of future financial targets.

Contingent consideration is not recognized until all contingencies are resolved and the consideration is paid or probable of payment, at which point the consideration is allocated to the assets acquired on a relative fair value basis.

Research and Development Expenses

Research and development costs are expensed as incurred and consist of fees paid to other entities that conduct certain research and development activities on the Company's behalf. Acquired intangible assets are expensed as research and development costs if, at the time of payment, the technology is under development; is not approved by the FDA or other regulatory agencies for marketing; has not reached technical feasibility; or otherwise has no foreseeable alternative future use. Nonrefundable advance payments for goods or services to be received in the future for use in research and development activities are capitalized and then expensed as the related goods are delivered or the services are performed.

For the period January 25, 2022 (inception) to September 30, 2022, our research and development expenses were comprised primarily of the cost of the tecarfarin asset and related transaction costs.

Derivative Financial Instruments

We evaluate all of our agreements to determine if such instruments have derivatives or contain features that qualify as embedded derivatives. We account for certain redemption features that are associated with convertible notes as liabilities at fair value and adjust the instruments to their fair value at the end of each reporting period. Derivative financial liabilities are initially recorded at fair value, with gains and losses arising from changes in the fair value recognized in other income (expense) in the accompanying statements of operations and comprehensive loss for each reporting period while such instruments are outstanding. The embedded derivative liability is valued using a probability-weighted expected return model. If we repay the note holders or if, during the next round of financing, the note holders convert the debt into equity, the derivative financial liability will be de-recognized on that date. Derivative instrument liabilities are classified in the balance sheet as current or non-current based on whether or not net-cash settlement of the derivative instrument could be required within 12 months of the balance sheet date.

Stock-Based Compensation

We measure our stock-based awards granted to employees, consultants and directors based on the estimated fair values of the awards and recognize the compensation over the requisite service period. We use the Black-Scholes option-pricing model to estimate the fair value of our stock option awards. Stock-based compensation is recognized using the straight-line method. As the stock compensation expense is based on awards ultimately expected to vest, it is reduced by forfeitures. We account for forfeitures as they occur.

Our equity awards granted through the date of this filing have an estimated grant date fair value of approximately $2.2 million which will be expensed over the requisite service period of each respective stock-based award.

Determination of the Fair Value of Common Stock

The fair value of our common stock for purposes of stock-based awards has been determined by our board of directors. Because there has been no public market for our common stock, and in the absence of arm's-length transactions in our common stock with independent third parties, the board of directors has determined the fair value of our common stock with the assistance of an independent third-party valuation prepared in accordance with the guidelines outlined in the American Institute of Certified Public Accountants Practice Aid, Valuation of Privately-Held-Company Equity Securities Issued as Compensation. In assessing the fair value of our common stock on each grant date, we considered numerous objective and subjective factors, including, but not limited to, the following:

- independent third-party valuation as of May 31, 2022;
- our current and expected operating and financial performance, including our level of available capital resources;
- the value of our tangible and intangible assets;
- market conditions affecting comparable public companies, as reflected in comparable companies' market trading multiples, initial public offering valuations, comparable sales or merger transactions, and other metrics;

- the illiquidity of our common stock by virtue of being a private company, and the resulting discount for lack of marketability;
- the business risks we face;

- the likelihood of achieving a particular liquidity event, such as a sale, a merger, or an initial public offering, given prevailing pharmaceutical and biotechnology industry and capital markets conditions; and
- the experience of management and the board of directors

Use of Estimates

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities as of the date of the financial statements and the reported amounts of expenses during the reporting period. Significant estimates and assumptions made in the accompanying financial statements include but are not limited to the fair value of financial instruments, deferred tax assets and valuation allowance, income tax uncertainties, and certain accruals. We evaluate our estimates and assumptions on an ongoing basis using historical experience and other factors and adjusts those estimates and assumptions when facts and circumstances change. Actual results could differ from those estimates.

JOBS Act

The JOBS Act permits an emerging growth company such as us to take advantage of an extended transition period to comply with new or revised accounting standards applicable to public companies until those standards would otherwise apply to private companies. We have elected to avail ourselves of the extended transition period for complying with new or revised financial accounting standards.

We will remain an emerging growth company until the earliest of (i) the last day of our first fiscal year in which we have total annual gross revenues of $1.235 billion or more; (ii) the date on which we are deemed to be a "large accelerated filer" under the rules of the SEC with at least $700.0 million of outstanding equity securities held by non-affiliates; (iii) the date on which we have issued more than $1.0 billion in non-convertible debt securities during the previous three years; or (iv) the last day of our fiscal year following the fifth anniversary of the date of the completion of this offering.

Quantitative and Qualitative Disclosures About Market Risk

As a smaller reporting company, we are not required to provide disclosure regarding quantitative and qualitative market risk.

BUSINESS

Overview

We are focused on developing tecarfarin, a novel therapy with orphan drug indication, designed for the prevention of systemic thromboembolism (blood clots) of cardiac origin in patients with end-stage renal disease and atrial fibrillation (irregular heartbeat) or AFib. Tecarfarin is an anticoagulant that uses a drug design process which targets a different pathway than the most commonly prescribed drugs used in the treatment of thrombosis and AFib. We secured the rights to tecarfarin on April 1, 2022 via an asset purchase agreement from HESP LLC, a wholly owned subsidiary of Horizon Technology Finance Corporation. HESP LLC acquired the assets of Espero BioPharma, Inc., or Espero, including tecarfarin, in an assignment for the benefit of creditors in which the creditor, Horizon Technology Finance Corporation and Horizon Credit II LLC (collectively, Horizon), a secured lender of Espero, designated HESP LLC as the assignee of Espero's assets.

Tecarfarin has been evaluated in 11 human clinical trials conducted by third parties in over 1,003 individuals (269 patients were treated for at least six months and 129 patients were treated for one year or more). In Phase 1, Phase 2 and Phase 2/3 clinical trials, tecarfarin has generally been well-tolerated in both healthy adult subjects and patients with CKD. In the Phase 2/3 trial, EMBRACE-AC, the largest tecarfarin trial with 607 patients having completed it, only 1.6% of the blinded tecarfarin subjects suffered from major bleeding and there were no thrombotic events. Five patients died during the trial, but only one death due to intracerebral hemorrhage was considered to be possibly related to the tecarfarin.

In 2019, the FDA provided input on the Phase 3 trial design for tecarfarin, which was submitted by Espero, the previous owner of tecarfarin. We intend to submit our Phase 3 trial design to the FDA using the same protocol that was submitted by Espero. Assuming the FDA accepts our Phase 3 trial design, we intend to commence the Phase 3 pivotal trial in the second half of 2023. However, there can be no assurance that the trial design will be accepted by the FDA. We are pursuing regulatory approval of tecarfarin as an individual treatment, although we might evaluate, in consultation with FDA, other potential uses in the future.

In March 2019, the FDA granted orphan drug designation, or ODD, for tecarfarin for the prevention of systemic thromboembolism of cardiac origin in patients with ESRD and AFib. The FDA grants ODD status to drugs that are intended for the treatment, diagnosis, or prevention of rare diseases or conditions, which are defined as a disease or condition that affects fewer than 200,000 people in the U.S. The ODD program provides a drug developer with certain benefits and incentives, including a seven-year period of U.S. marketing exclusivity from the date of marketing authorization, waiver of FDA user fees, and tax credits for clinical research. The granting of an orphan drug designation does not alter the FDA's regulatory requirements to establish safety and effectiveness of a drug through adequate and well-controlled studies to support approval and commercialization. Furthermore, orphan drug designation does not indicate or guarantee FDA approval of the NDA, and we might not receive exclusivity.

Tecarfarin was developed by researchers using a retrometabolic drug design process which targets a different metabolic pathway than the most commonly prescribed drugs for the treatment of thrombosis and AFib. "Drug metabolism" refers to the process by which a drug is inactivated by the body and rendered easier to eliminate or to be cleared by the body. Most approved drugs, including warfarin, the only FDA-approved Vitamin K antagonists, or VKAs, which is a prescribed drug for the treatment of thrombosis, are metabolized in the liver through a pathway known as the Cytochrome CYP450 system, or CYP450, by the enzymes known as CYP2C9 and CYP3A4. By using a different metabolic pathway, tecarfarin eliminates or minimizes the CYP450 metabolism in the liver. Patients taking multiple medications that interact with CYP2C9, or CYP3A4 or those with impaired kidney function, can experience an overload in the pathway, creating a bottleneck that often leads to insufficient clearance, which results in a toxic build-up of one or more drugs. In some instances, patients taking multiple medications metabolized by the same CYP450 pathway may experience decreased efficacy of one or more of the medications due to rapid metabolism or increased drug effect and/or toxicity due to enzyme induction. Patient-specific genetic differences can also hinder drug clearance in the CYP450 pathway. Our product candidate, tecarfarin, was designed using a retrometabolic drug design process so that tecarfarin follows a metabolic pathway distinct from the CYP450 pathway and is metabolized by both CYP450 and non-CYP450 pathways. We believe this may allow elimination by large capacity and non-saturable tissue esterase pathways that exist throughout the body rather than just in the liver.

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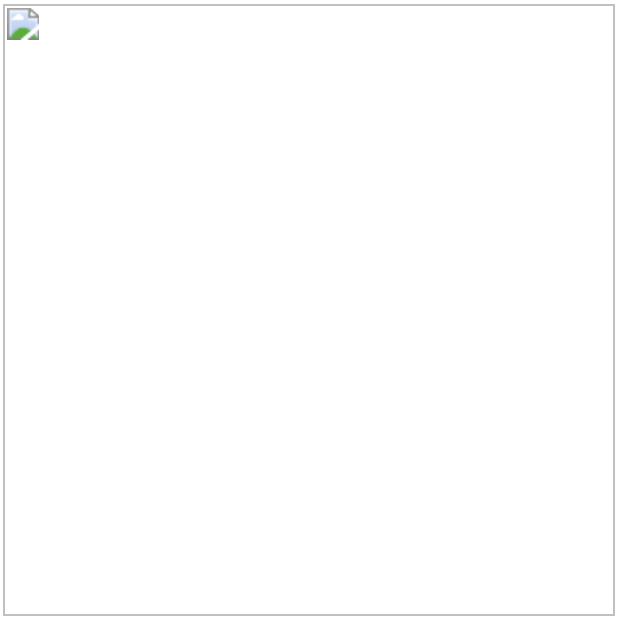
Tecarfarin is an orphan designated, vitamin K antagonist, oral, once-daily anticoagulant in the same drug class as warfarin designed for use in patients requiring chronic VKA anticoagulation, to prevent systemic thromboembolism of cardiac origin in patients with ESRD and AFib. The prevailing treatment for thrombosis is with an oral anticoagulant, either a VKA, like warfarin, or non-vitamin K oral anticoagulant ("NOAC"). VKAs block the production of vitamin K-dependent blood clotting factors, such that the blood is "thinned," preventing clots, while NOACs directly block the activity of certain of these clotting factors. Tecarfarin, like warfarin, is a VKA.

Vitamin K epoxide Reductase Complex subunit 1 (VKORC1) is a significant enzyme for effective clotting. VKORC1 reduces vitamin K epoxide to its active form (Vitamin K), which is the rate-limiting step in the physiological process of vitamin K recycling. Vitamin K serves as a cofactor for normal function of several clotting/anticoagulation factors including Factors II, VII, IX and X and Proteins C, S, and Z. VKORC1 genetic deficiencies result in increased sensitivity to VKAs, which results in an increase in the risk of significant hemorrhaging. We believe tecarfarin has similar potency for VKORC1 inhibition as warfarin, but it is an investigational new drug, and we must demonstrate it is safe and effective for its proposed indication.



AFib is the most common arrhythmia, with its incidence and prevalence increasing over the last twenty (20) years. AFib is associated with an approximate five-fold increased risk of stroke. The risk of developing AFib increases in patients with CKD. According to 2021 estimates by the Centers for Disease Control and Prevention, or CDC, approximately 15% of the U.S. adult population, or 37 million people, have CKD. An estimated 0.4% of people in the U.S. suffer from Stage 4 CKD and 0.1% of people in the U.S. have ESRD.

Patients with ESRD and AFib represent a spectrum of disorders involving both the heart and kidneys (known as cardiorenal syndrome or CRS) in which acute or chronic dysfunction in one organ may induce acute or chronic dysfunction in the other organ. These patients have typically been excluded from randomized clinical trials because the approved therapies for AFib have metabolic profiles that may increase drug exposures thereby increasing the known risks and challenges in managing these patients. The presence of either CKD or AFib, increases the risk of serious thromboembolic adverse clinical outcomes, such as stroke and death. Antithrombotic therapy is typically recommended to decrease this risk in AFib patients, but there are no approved treatment options for patients with ESRD and AFib. Warfarin may cause substantial harm in these patients. Low-dose apixaban (Eliquis) was approved by the FDA for use in ESRD patients on hemodialysis based upon limited pharmacokinetic data by 8 subjects, despite that randomized trials to date of apixaban versus warfarin for AFib excluded patients with severe and end-stage kidney disease. The RENAL-AF (Trial to Evaluate Anticoagulation Therapy in Hemodialysis Patients With Atrial Fibrillation) was terminated early in 2019 by its sponsor.

There are more than 809,000 Americans with ESRD, with approximately 70% on dialysis, according to the United States Renal Data System. Approximately 150,000 ESRD patients also have AFib. AFib nearly doubles

the anticipated mortality and increases the stroke risk by approximately five-fold in these patients. There is evidence that AFib is an independent risk factor for developing ESRD in CKD patients. Both diseases share common risk factors including

hypertension, diabetes, vascular disease, and advancing age. Cardiovascular disease contributes to more than half of all deaths among patients with ESRD. According to the Annual Data Report published by the United States Renal Data System, total Medicare spending for patients with ESRD reached $51 billion in 2019, accounting for approximately 7% of the Medicare paid claims costs.

We have licensed out the rights to tecarfarin for several Asian markets, including China, to Lee's Pharmaceutical Holdings Limited, an integrated research-driven and market-oriented biopharmaceutical publicly listed company based in Hong Kong with over 25 years' experience in the pharmaceutical industry in China. Lee's Pharmaceutical Holdings Limited is developing tecarfarin as an anti-thrombotic for patients with mechanical heart valves. In 2020 and 2021, Lee's Pharmaceutical Holdings Limited completed two Phase 1 studies in China and Hong Kong and is currently preparing for its Phase 2 trial.

* The rights to tecarfarin for several Asian markets including China have been out licensed to Lee's Pharmaceutical, a publicly listed company based in Hong Kong, who is developing tecarfarin as an anti-thrombotic for patients with mechanical heart valves.

Members of our management team have extensive experience in drug discovery, development and commercialization and have held senior leadership positions at a number of pharmaceutical and biotechnology companies. We also benefit from our broad network of established relationships with leaders in the industry and medical community.

As more fully set forth in our risk factors, we are a clinical development biopharmaceutical company with a limited operating history. We have a history of operating losses and expect to continue to incur substantial losses for the foreseeable future. Our independent registered public accounting firm has expressed substantial doubt about our ability to continue as a going concern. Our cash and the proceeds of this offering will only fund our operations for a limited time. The proceeds from this offering will be insufficient to allow us to fully fund our planned pivotal Phase 3 clinical trial. We will need to raise additional capital for the initiation of enrollment of patients and completion of the planned pivotal Phase 3 trial.

With respect to tecarfarin, we have two issued U.S. patents directed to tecarfarin. While the patents currently expire in 2024, we expect to seek extensions of patent terms. In the United States, the Drug Price Competition and Patent Term Restoration Act of 1984 permits a patent term extension of up to five years beyond the normal expiration of the patent, which is limited to the approved indication (or any additional indications approved during the period of extension). We also intend to seek exclusivity for our proprietary product candidates through market and data exclusivity granted by regulatory agencies in the United States and other countries. Further, as discussed above, the ODD program provides a drug developer with certain benefits and incentives, including a seven-year period of U.S. marketing exclusivity from the date of marketing authorization.

While we have engaged intellectual property counsel to assist in protecting our patent ownership rights, to date, we have not had intellectual property counsel conduct a freedom to operate analysis regarding our tecarfarin product. As a result, we cannot be certain that we will not be exposed to third party legal claims, liabilities and/or litigation actions when we seek to develop, make and market products using our tecarfarin technology.

Clinical Trials

Tecarfarin has been evaluated in eleven (11) human clinical trials in over 1,003 individuals in eight Phase 1 trials, two Phase 2 trials and one Phase 2/3 trial evaluating the efficacy and safety of tecarfarin.

In a phase 2/3 randomized and blinded trial sponsored by ARYx Therapeutics, Inc. in 2008, 607 patients with indications for chronic anticoagulation were treated with either tecarfarin or warfarin, the only FDA-approved VKA and a prescribed drug for the treatment of thrombosis. The Time in Therapeutic Range, or TTR, with investigational tecarfarin was similar to that with well-controlled warfarin, and tecarfarin appeared to have a favorable safety profile and be well tolerated with only 1.6% of the blinded tecarfarin subjects suffering from major bleeding and no thrombotic events. When thrombotic and major bleeding events during the blinded period were combined, a numerical imbalance favoring tecarfarin over warfarin was seen (warfarin 11 subjects, 3.6%; tecarfarin 5 subjects, 1.6%). The trial however did not meet its primary endpoint as superiority of tecarfarin over warfarin as measured by TTR was not demonstrated.

In a subsequent phase 1 study with 23 patients with CKD sponsored by Armetheon, Inc. in 2016, the metabolism of warfarin was inhibited, but not tecarfarin. The safety of repeated dosing of tecarfarin in CKD patients remained unknown. However, if the pharmacokinetic findings of this single-dose study are present with repeated dosing, tecarfarin may lead to dosing that is more predictable than warfarin in CKD patients who require anticoagulation therapy.

Recent Events

On April 1, 2022, we entered into an asset purchase agreement with HESP LLC, the assignee of the tecarfarin and related assets (the "Asset Purchase Agreement"). Pursuant to the terms of the Asset Purchase Agreement, we acquired all of the assets of HESP LLC, including all intellectual property and other rights related to tecarfarin, the tecarfarin IND 77041, all rights under the license, development and commercialization agreement dated as of September 16, 2015 by and between Armetheon (which was later assigned to Espero BioPharma, Inc., or Espero) and China Cardiovascular Focus Ltd., an affiliate of Lee's Pharmaceutical Holdings Limited, relating to tecarfarin and related trademarks. In consideration of the purchase of the assets we paid HESP LLC $100,000 on the closing date and agreed to pay an additional $100,000 60 days after the closing date. On June 1, 2022, we made the second $100,000 payment. In addition, we agreed to pay HESP LLC the following development payments.

Development Milestones		Milestone Payments
Completion of enrollment of Lee's Pharmaceutical Phase 3 clinical trial	$	250,000
First MAA submitted in the People's Republic of China	$	350,000
First Commercial Sale to a Third Party	$	1,200,000

As additional consideration, the Company agreed to pay the following amounts, up to $2,000,000, upon each financing milestone as follows (i) 35% of any proceeds received from any licensing or partnering revenue; and (ii) IPO proceeds. The aggregate payments under the development milestone payments and financing milestone payments shall not exceed $2,000,000.

On August 18, 2022, we entered into an Amendment to the Asset Purchase Agreement, whereby in lieu of the $1,800,000 in cash payments that would have been due to HESP LLC pursuant to the Asset Purchase Agreement HESP LLC agreed to accept shares of our common stock to be calculated based upon a 40% discount to the price of our common stock sold in this offering, resulting in 600,000 shares which will be issued to HESP LLC on the effective date of this offering, which shares are being registered in the Resale Prospectus accompanying this registration statement.

Amended and Restated Certificate of Incorporation and Amended and Restated Bylaws

On December 5, 2022, our Board of Directors approved an amendment to our certificate of incorporation to increase our authorized capital stock to 82,500,000 shares, consisting of 75,000,000 shares of common stock, par value $0.001 per share, and 7,500,000 shares of preferred stock, par value $0.001 per share, which was also approved by a majority of our shareholders on the same day. On December 5, 2022, we filed an Amended and

Restated Certificate of Incorporation with the Secretary of State of the State of Delaware, which became effective on December 5, 2022, at which time our authorized capital stock consisted of 75,000,000 shares of common stock, par value $0.001 per share, and 7,500,000 shares of preferred stock, par value $0.001 per share.

On December 5, 2022, our Board of Directors approved the Amended and Restated Bylaws, which became effective as of December 5, 2022 upon the filing and effectiveness of our Amended and Restated Certificate of Incorporation.

Fast Track Designation

On January 13, 2023, the FDA designated as a Fast Track development program the investigation of tecarfarin for the prevention of systemic thromboembolism of cardiac origin in patients with ESRD and AFib. Fast Track is a process designed to facilitate the development and expedite the review of drugs to treat serious conditions and fill an unmet medical need.

Our Strategy

Our goal is to build a pharmaceutical company with a foundation of product candidates that significantly advance patient care in cardiorenal diseases. Key elements of our strategy are as follows:

- Complete the clinical development of and seek FDA approval for tecarfarin. We intend to initiate our pivotal Phase 3 clinical trial in the fourth quarter of 2023, subject to completion of this offering and receipt of additional funding, which we believe will be our remaining pivotal trial based upon the latest feedback that the prior owner of tecarfarin had with the FDA in 2019. ACTOR AF: Anti-Coagulation with Tecarfarin on Outcomes in Renal disease and Atrial Fibrillation is designed as a Phase 3, 492-patient, Randomized, Double-Blind, Placebo-Controlled Outcomes Study of Tecarfarin vs. Placebo in Subjects with End-Stage Renal Disease and Atrial Fibrillation not Currently Treated with Chronic Oral Anticoagulation. If we are able to complete the Phase 3 clinical trial and we are able to obtain FDA approval of our NDA, we believe tecarfarin can be an alternative treatment for patients who are currently without an approved treatment.

- If we obtain FDA approval of our NDA for our first indication, we intend to seek to expand the label for tecarfarin through a supplemental NDA. We intend to explore the full potential of tecarfarin in additional indications, including the treatment of patients with Left Ventricular Assist Devices (LVADs) and Mechanical Heart Valves (MHVs) who require chronic anticoagulation. An LVAD is an implantable mechanical pump attached to the heart, connecting the apex of the left ventricle to the ascending aorta. LVADs are intended to treat patients suffering from advanced heart failure. MHVs are a type of prosthetic heart valve constructed of durable materials such as titanium or carbon that can potentially last a patient's lifetime. Despite their high durability, MHVs are associated with an increased risk of developing blood clots that may interrupt blood flow and travel to various organs (thromboembolism). Oral anticoagulation therapy with a vitamin K antagonist (warfarin) continues to be the recommended treatment option for patients implanted with an MHV based on the 2014 American Heart Association/American College of Cardiology Guidelines.

- We intend to partner and/or in-license and/or acquire clinical and pre-clinical stage cardiorenal products to augment our current pipeline which consists of one investigational product.

- Create a commercial infrastructure for our product candidates. If tecarfarin is approved by the FDA, we intend to expand our commercial infrastructure and hire and train a focused and dedicated specialty cardiorenal salesforce which we believe can efficiently cover the top prescribing physicians and approximately 3,000 anticoagulation clinics in the U.S., which presently monitors patients on warfarin. If approved, we intend to initially target our tecarfarin commercialization efforts at the physicians treating patients at the two largest dialysis providers in the U.S., DaVita Kidney Care and Fresenius Medical Care. DaVita maintains more than 2,800 brick-and-mortar dialysis centers in the U.S., while Fresenius' North America subsidiary owns more than 2,500 on the continent — together accounting for more than 70% of the entire U.S. market.

Retrometabolic Drug Design Process

We utilize a retrometabolic drug design process to design product candidates that follow a metabolic pathway that we believe will confer significant clinical advantages over existing drugs metabolized by the CYP450 pathway. "Drug metabolism" refers to the process by which a drug is inactivated by the body and rendered

easier to eliminate or to be cleared by the body. Most approved drugs are metabolized in the liver through the CYP450 pathway by the enzymes known as CYP2C9 and CYP3A4. The CYP450 metabolic pathway has limited capacity, and patients taking multiple medications that interact with

CYP2C9, or those with impaired kidney function, can experience an overload in the pathway, creating a bottleneck causing insufficient clearance, which results in a toxic build-up of one or more drugs. In some instances, patients taking multiple medications that interact with CYP2C9 may also experience that a drug is eliminated too quickly from the body, reducing the efficacy of the drug. Patient-specific genetic differences can also hinder drug clearance in the CYP450 pathway. Our product candidates were designed so that they follow a metabolic pathway distinct from or in addition to the CYP450 pathway, eliminating or minimizing the CYP450 metabolism by the liver, and are instead or additionally eliminated by large capacity and non-saturable tissue esterase pathways that exist throughout the body rather than just in the liver. We believe that the use of these alternative pathways can minimize the impact of drug-to-drug interactions, impaired kidney function and genetic variability, on the metabolism of our drugs, thereby ultimately minimizing clearance-related safety issues.

As part of the retrometabolic design process, we first design a metabolite, which we refer to as the "ideal metabolite," that is non-toxic, pharmacologically inactive, water soluble and rapidly eliminated from the body through a pathway distinct from the CYP450 pathway. This ideal metabolite becomes the foundation upon which we develop a limited series of potential molecular drug candidates that have the same pharmacology and function as the original drug. We then test the potential drug candidates we have created preclinically to assess whether the candidates have the same activity as the original drug and to ensure the candidates break down properly to the ideal metabolite. This process allows us to identify the drug candidates that would be best suited for further clinical development.

By designing drugs that break down to the ideal metabolite, and accordingly, are not cleared through the CYP450 pathway, we create product candidates that we believe would reduce many of the safety risks and complications that patients experience with drugs that are cleared through the CYP450 pathway. As a result, we believe there may be better compliance with a tecarfarin treatment regimen, if approved by FDA, which may also result in increased efficacy for the patients taking our drug product candidates.

Our Investigational Product Candidate

Tecarfarin for Use in Patients with ESRD and AFib

ESRD and AFib Current Treatment Landscape

Thrombosis is the formation or presence of a blood clot (a thrombus) within a blood vessel that blocks normal blood flow. A formed thrombus can detach from the vessel or heart atrium wall, resulting in a thromboembolism that causes a blockage of the blood flow to vital organs, such as the brain, heart and lungs. According to the CDC, each year, approximately 800,000 people in the U.S. experience a new or recurrent stroke, of which approximately 87% are ischemic strokes, which are caused by either a thrombotic event or embolism, in which blood flow to the brain is blocked. In addition, the CDC estimates that as many as 900,000 people in the U.S. could be affected by venous thromboembolism (blood clotting in the veins) or a pulmonary embolism (blood clotting in the lungs) each year.

The American Heart Association estimates that 5 million Americans suffer from heart-valve disease, which forces the heart to work harder to pump blood and can lead to heart failure and sudden death. The disease can be present at birth or result from infections, heart attacks or other heart conditions. Further, more than 182,000 heart valve replacements are performed every year in the U.S. Mechanical heart valves also create a risk of thrombotic events.

The prevailing treatment for patients at risk of thrombosis is an oral anticoagulant, or OAC, of which there are two common types: vitamin K antagonists, or VKAs, and non-vitamin K oral anticoagulants, or NOACs.

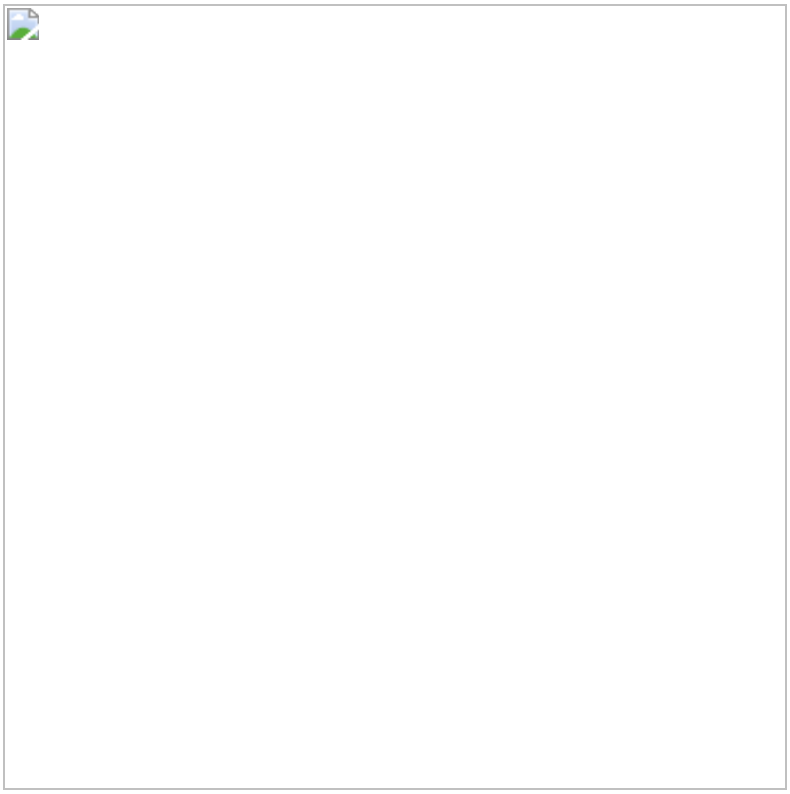
Vitamin K Antagonists: Warfarin

Vitamin K antagonists, or VKAs, are substances that block the production of vitamin K-dependent blood clotting factors such that the blood is "thinned," preventing clots. VKAs are used as anticoagulants in the treatment of thrombosis. For patients treated with a VKA, the international normalized ratio, or INR, a system established by the World Health Organization and the International Committee on Thrombosis and Hemostasis, is a commonly available, inexpensive measure of the body's coagulation status. Each VKA patient's dose must be individualized, based on a target range for his or her INR test. The percentage of time that a patient's INR is

maintained within his or her target range is known as the time in therapeutic range, or TTR. TTR is a well-established FDA metric used to evaluate anticoagulation control (safety and efficacy) of a VKA based on prothrombin time and the INR. When used as a therapy, VKAs are titrated to a patient's individual INR range and that patient is expected to visit a clinic for regular INR monitoring. A higher TTR reflects better anticoagulation control and is related to improved clinical

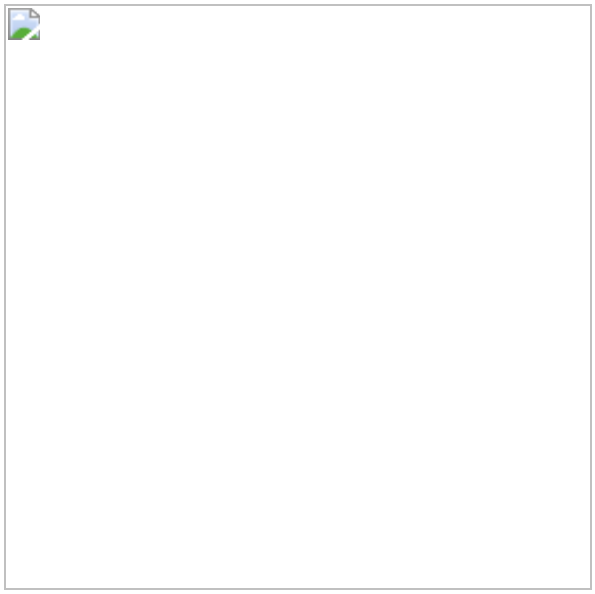
outcomes, including rates of death, bleeding, myocardial infarction, stroke and systemic embolism, and a TTR measure of ≥ 70% is generally accepted as the goal for stable anticoagulation with a VKA. When patients are above their individual INR range, they are at higher risk for bleeding, due to reduced clotting ability, while patients are below their target INR range are at higher risk for thrombotic events. Potential benefits of monitoring INR include ascertaining patient compliance with their drug treatment regimen, the ability to detect when dose adjustments are needed and maintaining safety and efficacy of the drug treatment. TTR is predictive of adverse events, including mortality, stroke and myocardial infarction.

As depicted in the chart below, higher TTR is generally correlated with higher survival rates.



Source: Currie et al. Heart 2006 (92) 196-200

Higher TTR levels are also associated with better kidney functioning, as measured by estimated glomerular filtration rate (eGFR), as reported by the Journal of the American Heart Association in 2017.



VKAs are reversible, meaning that in cases of over-anticoagulation, vitamin K or fresh frozen plasma, or a combination, can be administered to bring patients back down into their INR range.

Warfarin is currently one VKA treatment option for thrombosis in the U.S. and has been in use since the 1950s. However, as reported by the National Center for Biotechnology Information, there are many adverse events associated with warfarin, including bleeding, skin necrosis and hair loss, and warfarin has been reported as number three on the list of drugs implicated in adverse effects causing hospital admission due to its many drug-to-drug interactions. Due to these side effects and the increasing use of NOACs, the use of warfarin has decreased during the last decade.

Limitations of Warfarin Treatment

Warfarin has significant safety risks stemming from its metabolic process, including its elimination pathway. Other drawbacks of warfarin are widely recognized such as narrow therapeutic range, slow onset and offset of action causing difficulty to manage during peri-invasive procedures, and multiple drug and food interactions.

Warfarin's efficacy and safety profile are affected by its metabolism and elimination characteristics and various interactions with other drugs. Warfarin is metabolized through the CYP450 pathway, primarily by the CYP2C9 enzyme, and approximately 15% of clinically used drugs are metabolized by the same enzyme, including certain anticoagulants, antiplatelets and non-steroidal anti-inflammatory drugs, or NSAIDS. Patients taking warfarin and on CYP2C9 interacting drugs may experience either or both of, warfarin being eliminated by the body too quickly, thereby decreasing its anticoagulation effect, or warfarin being eliminated by the body too slowly, resulting in excessive and dangerous thinning of the blood. In both of these situations, increased monitoring is required and dose adjustments are often necessary. Patients who take both warfarin and these CYP2C9 interacting drugs also have an increased risk of being outside their individual INR target range and experiencing lower or higher TTR. For these patients, their CYP2C9 interacting drugs must be used with caution, or at times, their use must cease.

Warfarin's efficacy and safety profile are also affected by genetic mutations that lead to a lower activity of the CYP2C9 enzyme, the primary enzyme used to eliminate warfarin from the body. Clinical studies have shown that these persons require lower dosages of warfarin and are at an increased risk of anticoagulation.

Currently, warfarin is commonly used in patients with non-valvular AFib and in patients with valvular heart diseases (VHD) with AFib. However, as reported by an article published by the Egyptian Heart Journal on March 28, 2022, an analysis of 6,454 patients with AFib taking warfarin showed that almost 50% of the time the INR was outside the target range of 2–3, leading to a higher risk of bleeding and thrombotic complications. The major adverse effect associated with warfarin is bleeding. Major and fatal bleeding events occur at rates of 7.2 and 1.3 per 100 patient-years, respectively, according to a meta-analysis of 33 studies.

INR should be more frequently monitored in patients with impaired kidney function. Patients with impaired kidney function have a decreased ability to metabolize drugs through the CYP450 pathway, and accordingly have an increased risk of being outside their individual INR target range and experiencing lower TTR. Further, according to an article published by Frontiers in Medicine in January 2021, warfarin was associated with an increase in the risk of major bleeding without reduction in stroke/thromboembolism or mortality in patients with end-stage CKD requiring dialysis.

As a result of some or all of the above, and other factors, trials have shown that patients treated with warfarin often experience TTRs lower than 70%, the generally accepted TTR threshold representing stable anticoagulation. In a 2019 study conducted to evaluate the TTR of 300 patients on long-term warfarin for non-valvular AFib, as reported by an article published in Health and Quality of Life Outcomes on October 20, 2020, 75.5% of patients had a poor TTR with a mean of only 39.5%, with the mean TTR of all patients in the study being 47%. In another study of 406 AFib patients conducted in Lithuania to evaluate the quality of warfarin as anticoagulation therapy, more than half (57.3%) of INR values were outside of the target range and the median TTR was only 40%, with only 20% of patients having a TTR greater than or equal to 65%.

Non-Vitamin K Oral Anticoagulants (NOACs) or Direct Acting Oral Anticoagulants (DOACs)

Non-vitamin K oral anticoagulants, or NOACs, or DOACs, are a form of OAC treatment that inhibits certain blood clotting factors. While VKAs block the synthesis of vitamin K-dependent blood clotting factors, NOACs block the activity of these clotting factors. There are two classes of NOACs, oral direct thrombin inhibitors and oral direct factor Xa inhibitors. Currently, there are only four NOACs approved by the FDA for use outside of a hospital setting: apixaban (the generic name for Eliquis), dabigatran (the generic name for Pradaxa), rivaroxaban (the generic name for Xarelto) and edoxaban (the generic name for Savaysa). NOACs are generally more rapid in onset and offset of action than VKAs, have few strong drug-to-drug interactions and do not require INR monitoring.

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Limitations of NOAC Treatment for Patients with ESRD and AFib

NOACs have been approved in the U.S. for the treatment of specific oral anticoagulation indications; however, there are anticoagulation indications for which NOACs are warned against use or are not recommended for use, including for anticoagulation treatment in patients with mechanical heart valves. NOACs do not have the same broad label indication as warfarin and are only indicated for some thrombosis indications.

Our Proposed Solution: Tecarfarin for Treatment of ESRD and AFib

Tecarfarin, our lead investigational product candidate, is a VKA, once-daily OAC designed for use in patients with ESRD and AFib. Tecarfarin was designed using a retrometabolic drug design which targets a different metabolic pathway than the most commonly prescribed drugs for the treatment of thrombosis and AFib. Like warfarin, tecarfarin will also require INR monitoring. Due to its retrometabolic design, tecarfarin is eliminated by large capacity and non-saturable tissue esterase pathways that exist throughout the body, rather than just in the liver. This is a metabolic pathway that is distinct from the CYP450 pathway and infrequently used by other medications, which could potentially reduce the risk for drug-to-drug interactions. Moreover, unlike warfarin, we do not believe tecarfarin's metabolism is affected by CYP2C9 genetic variant alleles or by kidney function.

Given the metabolic process and related safety issues with warfarin and the limited treatment indications for which NOACs are approved, we believe there is a significant thrombosis patient population in need of an alternative anticoagulation treatment. The lack of stable and predictable anticoagulation control is particularly problematic in large underserved patient subpopulations with risk factors such as:

- Patients treated with CYP2C9 interacting drugs;
- Patients with severely impaired kidney function;
- Patients with genetic variant alleles for CYP2C9; and
- Patients with mechanical heart valve implants.

Tecarfarin Clinical Program

Tecarfarin has been evaluated in eleven clinical trials: eight Phase 1 trials, two Phase 2 trials and one Phase 2/3 trial evaluating the efficacy and safety of tecarfarin. We are currently planning to commence what we believe to be our remaining pivotal Phase 3 trial in the second half of 2023, subject to completion of this offering. A readout of the two-year animal carcinogenicity study is expected to be completed in August 2023. We will also conduct any further trials as may be required by the FDA.

A summary of the clinical trials conducted to date with tecarfarin is shown below.

Study Number	Study Description	Study Population	Number Exposed	Date started	Date completed	Sponsor
ZK-TEK-201905	**Multiple-dose tolerance and PK-PD study of tecarfarin**	**Healthy Chinese Volunteers**	**40**	**2020**	**January 2021**	**Zhaoke Pharmaceutical (Guangzhou) Co., Ltd**
LP-HK-001	**Phase 1, Sequential Cohort, Single-dose escalation study**	**Healthy Chinese Volunteers**	**40**	**June 2018**	**July 2019**	**Lee's Pharmaceutical (Hong Kong) Limited**
CLN-512	**Phase 1 pharmacokinetic study in chronic kidney disease subjects**	**Chronic kidney disease subjects; healthy volunteers**	**23**	**November 2015**	**May 2016**	**Armetheon, Inc.**

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Study Number	Study Description	Study Population	Number Exposed	Date started	Date completed	Sponsor
CLN-505	**Phase 2/3 randomized, blinded head-to-head anticoagulation in broad indications (EMBRACE-AC)**	**Patients (all indications)**	**609**	**June 2008**	**December 2009**	**ARYx Therapeutics, Inc.**
CLN-509	**Pilot Phase 2 study for trial methodology to be used in CLN-505**	**Patients (all indications)**	**50**	**January 2008**	**August 2008**	**ARYx Therapeutics, Inc.**
CLN-504	**Phase 2a open-label anticoagulation in AFib**	**Patients Requiring Oral Anticoagulation**	**66**	**December 2006**	**October 2007**	**ARYx Therapeutics, Inc.**
CLN-508	**Phase 1 DDI study with amiodarone**	**Healthy volunteers**	**19**	**July 2007**	**September 2007**	**ARYx Therapeutics, Inc.**
CLN-507	**Phase 1 DDI study with fluconazole**	**Healthy volunteers**	**20**	**June 2007**	**August 2007**	**ARYx Therapeutics, Inc.**
CLN-503	**Phase 1 Dose titration to target**	**Healthy volunteers**	**28**	**October 2006**	**January 2007**	**ARYx Therapeutics, Inc.**
CLN-502	**Phase 1 Effect on INR in multiple dose response**	**Healthy volunteers**	**42**	**November 2005**	**July 2006**	**ARYx Therapeutics, Inc.**
CLN-501 & CLN-501.X	**Phase 1 Safety and human pharmacokinetics**	**Healthy volunteers**	**66**	**August 2005**	**April 2006**	**ARYx Therapeutics, Inc.**

Phase 2 Trials

CLN-504: Trial CLN-504 was an open-label study in which 66 patients with AFib were treated with tecarfarin for a period of six weeks, with the option of continuing treatment for an additional six weeks. The trial, which was conducted by our predecessor company that owned the rights to tecarfarin, was primarily designed to determine an optimal dosing regimen and monitoring schedule and to describe the efficacy and explore the quality of anticoagulation as measured by TTR for INR. Before the trial, warfarin-treated patients had a mean TTR of 59.4%. After the initial three weeks of dose titration, the tecarfarin-treated patients were within the target INR range 71.4% of the time ($p<0.001$). The most commonly reported treatment-related adverse events, or TEAEs, were mild hemorrhagic complications of anticoagulation, such as bruising and nosebleed.

There were two deaths after trial drug treatment was completed: one patient died due to idiopathic pulmonary fibrosis and pneumonia two weeks following his last dose of tecarfarin, and one patient died due to bronchial carcinoma two weeks following his last dose of tecarfarin. These deaths were not attributed to tecarfarin.

CLN-509: The CLN-509 trial was a pilot Phase 2 study to assess clinical trial methodology to be used in CLN-505, in patients having a variety of clinical indications requiring chronic oral anticoagulation, as measured by INR. Fifty patients, including patients with AFib, some of whom were already taking warfarin and some of

whom had not, received daily doses of tecarfarin ranging from 1 mg to 60 mg to maintain their INR value (the INR varied based upon the patient's condition). The objectives of evaluating safety and INR control in patients with a variety of clinical indications for chronic oral anticoagulation and assessing the feasibility of tecarfarin treatment in multiple dose strengths were met. INR control was shown for patients with atrial fibrillation, venous thromboembolic disease, prosthetic heart valves, and cardiomyopathy. There were no off-target adverse events due to tecarfarin and there were no clinically important safety signals in other measures of safety. The results of this trial resulted in the development of the clinical trial methodology for the Phase 2/3 trial (EMBRACE-AC).

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CLN-505 (EMBRACE-AC): The Phase 2/3 CLN-505 trial, referred to as the EMBRACE-AC trial, was a multi-center, randomized, stratified, double-blind, parallel group, active control trial for a minimum period of six months and up to one year designed to compare the quality of anticoagulation of tecarfarin and warfarin as determined by TTR. Dosing of study drugs was managed by a centralized dose control center. In total, 609 patients were enrolled and of those, 607 patients completed the trial and of these, 304 patients received warfarin and 303 patients received tecarfarin. The EMBRACE-AC trial did not achieve statistical significance on its primary endpoint and the results of the primary analysis showed that tecarfarin was not superior to warfarin as measured by TTR. However, the TTR observed in patients taking tecarfarin (72.3%) was numerically similar to patients taking warfarin (71.5%) (difference of 0.8%; p=0.51).

As part of its original design, the EMBRACE-AC trial included analyses of INR measurements while patients were temporarily off their trial drug due to other medical reasons. Subsequently a post-hoc analysis was conducted in which we excluded INR values collected during these periods and showed that the percentage of TTR was higher on tecarfarin (68.8%) than on warfarin (66.4%) (difference of 2.3%; p<0.04).

Post-hoc analyses were conducted in other patient subgroups in our EMBRACE-AC trial. The following chart depicts the findings of the analysis of the study and the degree to which TTR% on tecarfarin was higher than on warfarin:

- In the 179 patients taking CYP2C9 interacting drugs, the TTR of patients taking tecarfarin was similar to that of patients taking warfarin (72.2% and 69.9%, respectively; p=0.15).
- In the 55 patients taking CYP2C9 interacting drugs who also had a CYP2C9 genetic variant allele, the TTR of patients taking tecarfarin was similar to that of patients taking warfarin (76.5% and 69.5%, respectively; p=0.09).

- In the 84 patients with mechanical heart valve implants, the TTR of patients taking tecarfarin was similar to that of patients taking warfarin (68.4% and 66.3%, respectively; p=0.51).

The potential benefit of tecarfarin over warfarin as measured by TTR was not demonstrated in EMBRACE-AC. However, the TTR observed in patients taking tecarfarin in the trial, and in the subpopulations described above, were numerically similar to the TTR observed in the patients taking warfarin, and the TTR observed in patients taking tecarfarin demonstrated stable anticoagulation. The TTR in patients treated with warfarin exceeded previously reported TTR rates observed in patients taking warfarin, which are typically in the 50% to 65% range, which we believe was due to the use of dose control centers in the administration of the trial. When dose control centers are used in the administration of warfarin, large teams of medical professionals are able to closely monitor patients and mitigate many of the drug-to-drug and genetic variant-related limitations of the drug that are not easily managed in real world settings.

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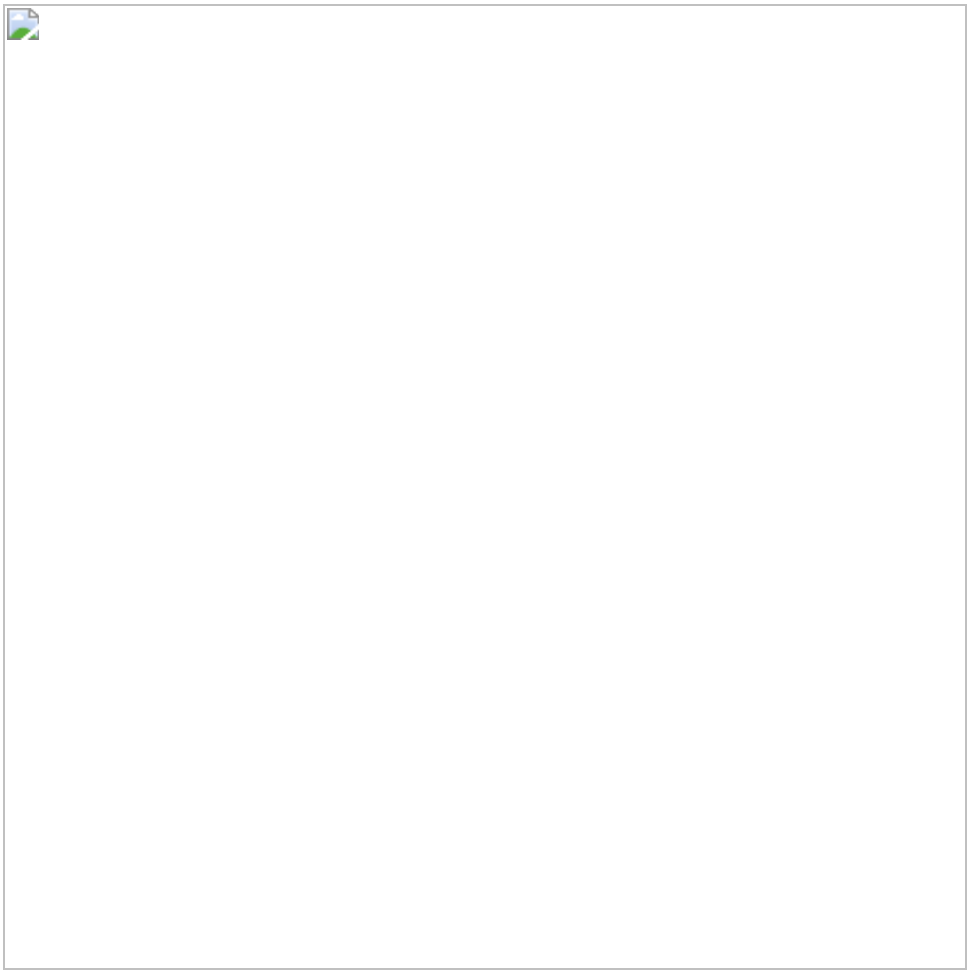
EMBRACE-AC also provided information for dosing and dose adjustments for tecarfarin. The average daily doses required of both tecarfarin and warfarin was analyzed in patients who had poor, intermediate or extensive metabolism capacity of the CYP2C9 enzyme. Patients with poor metabolism capacity of the CYP2C9 enzyme who were treated with warfarin required a significantly lower average daily dose compared to those patients with extensive CY2C9 enzyme metabolism capacity. In contrast, as depicted below, the required dosage of tecarfarin did not vary significantly based on the patient's CYP2C9 activity level.



Tecarfarin appeared to be well tolerated with only 1.6% of the blinded tecarfarin subjects suffering from major bleeding and no thrombotic events. When thrombotic and major bleeding events during the blinded period were combined, a numerical imbalance favoring tecarfarin over warfarin was seen (warfarin 11 subjects, 3.6%; tecarfarin 5 subjects, 1.6%). The safety data from EMBRACE-AC showed comparable rates of adverse events between the two treatment groups. TEAEs were reported for 93.2% of patients who received tecarfarin and 90.5% of patients who received warfarin. TEAEs reported by ≥10% of patients in either treatment group were nasopharyngitis (18.6% and 19.3%, blinded tecarfarin and warfarin, respectively), contusion (15.6% and 14.8%, respectively), epistaxis (8.1% and 11.1%, respectively), upper respiratory tract infection (10.7% and 10.8%, respectively), diarrhea (10.1% and 9.2%, respectively) and headache (10.7% and 8.9%, respectively). Most TEAEs were mild (32.2%, tecarfarin and 30.2%, warfarin) or moderate (45.0% and 46.6%, respectively) in severity.

The trial had some limitations. The TTR with warfarin achieved in Embrace-AC was much higher than that typically seen in clinical trials and as compared to "real-world practice," exceeding 71% on an interpolated basis in both treatment arms. The most likely reason stemmed from the dosing of study drugs which was managed by a centralized dose control center, which had access to genotyping.

Five patients died during the trial, with four deaths occurring during the double-blind period: one patient (tecarfarin; off drug) died due to mantle cell lymphoma, pneumonia and sepsis; one patient (tecarfarin; on drug) died due to cardiorespiratory arrest and myocardial infarction; one patient (warfarin; off drug) died due to

metastatic colon cancer; one patient (warfarin; off drug) died due to lung cancer; and one patient (not randomized) died due to intracerebral hemorrhage. The patient who died due to intracerebral hemorrhage was considered to be possibly related to the study drug, but the remaining four deaths were not attributed to the drug.

During the blinded period of the trial, five patients on tecarfarin and six patients on warfarin experienced major bleeding events. The occurrence of major bleeding events for both tecarfarin and warfarin was lower when compared to prior anticoagulation trials. Among warfarin-treated patients, there were five thrombotic events (two ischemic strokes, two deep vein thromboses and one pulmonary embolism), while there were no such events among tecarfarin-treated patients.

Phase 1 Trials

CLN-501: Trials CLN-501 and CLN-501.X evaluated the safety tolerability of tecarfarin in a total of 64 healthy volunteers. The studies were sufficiently similar in their requirements and study populations to be combined and analyzed together. The primary differences between the studies were the study drug formulation (CLN-501 used a solution formulation while study 501.X used tablets) and the range of single doses studied. In CLN-501, cohorts of 6 eligible subjects were randomly assigned to receive tecarfarin at one of eight ascending dose levels between 0.2 and 10.0 mg or placebo. In CLN-501.X, similar cohorts received tecarfarin at one of three ascending dose levels

(20.0, 30.0, or 40.0 mg) or placebo. The studies demonstrated that there were no apparent differences between pharmacokinetics parameters after tecarfarin was administered at single doses from 0.2 to 10.0 mg as an oral solution or at single doses from 20.0 to 40.0 mg as an oral solid tablet formulation and that tecarfarin was well-tolerated.

CLN-502: Trial CLN-502 evaluated tecarfarin pharmacokinetics, dose range, and duration of dosing that would attain a stead steady state INR of 1.7 to 2.0 and would give stead-state plasma concentrations of tecarfarin. Forty-two healthy volunteer subjects were randomized and received either 1, 3, 6, 10, 20, 30, or 40 mg of tecarfarin or placebo (the 3 and 6 mg cohorts were discontinued after one week due to lack of pharmacodynamic effect). The study successfully determined the active dose of tecarfarin and provided the pharmacokinetic and pharmacodynamic basis for subsequent multidose trials. Doses of 20 mg and above brought subjects into the target INR range of 1.7 to 2.0, with the 40 mg dose bringing all subjects into the target range within one week of dosing. The trial demonstrated that tecarfarin was well-tolerated at all doses studied as assessed by adverse events, vital signs, electrocardiography, and laboratory testing, and that a loading dose of 40 mg could be appropriate for initiating anticoagulation in Phase II trials.

CLN-503: Trial CLN-503 evaluated the safety and tolerability of tecarfarin versus warfarin when administered alone and in combination with amiodarone as measured by INR in 28 healthy subjects. During the first phase, subjects were administered tecarfarin or warfarin for 10 days, with doses titrated daily to achieve a target INR range of 1.5 to 2.0. Subjects who remained within the target INR range without requiring a dose change continued to the next phase with amiodarone, with all subjects receiving 200 mg amiodarone twice daily in addition to tecarfarin or warfarin. The primary objectives of the trial were completely met. Both tecarfarin and warfarin were well-tolerated, both alone and in the presence of amiodarone. There were no safety signals as ascertained by adverse event reports, clinical laboratory testing, vital sign measurement, and by electrocardiography. The quality of anticoagulation was good for both cohorts and a target INR was reached and maintained during the 3-day maintenance period. The results of the trial suggested the use of the same INR therapeutic range for tecarfarin as is recommended for warfarin.

CLN-507: Trial CLN-507 evaluated the effects of co-administration of fluconazole, a drug that blocks the activity of the CYP450 enzyme, with either 50 mg tecarfarin or 17.5 mg warfarin in 20 healthy volunteers. The trial demonstrated that co-administration of fluconazole did not affect the metabolism or elimination of tecarfarin. In contrast, the co-administration of fluconazole prolonged the half-life of warfarin.

CLN-512: Trial CLN-512 evaluated the effects of severe chronic kidney dysfunction on the metabolism and elimination of tecarfarin and warfarin. Thirteen patients with severe kidney dysfunction (stage 4 chronic kidney disease, or CKD) and 10 healthy volunteers (matched for age, weight, gender and CYP2C9 genotype) were administered 30 mg tecarfarin and 10 mg warfarin in a randomized crossover design. The trial demonstrated that tecarfarin's elimination from the body was not affected by severe kidney dysfunction: the half-life and the amount of drug in the body were similar in people with CKD and healthy patients. In contrast, the plasma concentration and half-life of warfarin was increased in patients with CKD, with warfarin's exposure increasing 44% in these patients. These effects were exaggerated in patients with CYP2C9 genetic variant alleles and in those who required concomitant CYP2C9 interacting drugs. At the conclusion of the trial, the safety of repeated dosing of tecarfarin in CKD patients remained unknown. However, overall, the results of this study suggest that no adjustment in the dose of tecarfarin is needed for patients with CKD.

CLN-508: Trial CLN-508 evaluated the effects of co-administration of 400 mg amiodarone with either 50 mg tecarfarin or 17.5 mg warfarin in 19 healthy volunteers, nine on tecarfarin and 10 on warfarin. Amiodarone, a drug used to treat irregular heartbeat, is a moderately potent inhibitor of CYP2C9 metabolism and is frequently used as a treatment for AFib in combination with warfarin. The effects of amiodarone on the pharmacokinetics of warfarin and tecarfarin showed that the exposure was increased to about the same extent for both drugs. The exposure of R-warfarin increased by 27% and the exposure for S-warfarin increased by 38%. The exposure of tecarfarin increased by approximately 31%. These changes in exposure did not result in any changes in INR in either the tecarfarin or the warfarin cohorts and demonstrated that tecarfarin behaved similarly to warfarin when administered in combination with amiodarone.

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An Open-label, Phase 1, Sequential Cohort, Single-Dose Escalation Study to Assess the Safety and Tolerability of Tecarfarin (ATI-5923) in Healthy Chinese Volunteers

Study Protocol: LP-HK-001 completed in July 2019

This was an open-label, phase 1, sequential cohort, single-dose escalation study conducted in China to assess the safety and tolerability of tecarfarin (ATI-5923) in healthy Chinese volunteers. The study site enrolled up to a total of 40 subjects. Ten (10) healthy Chinese subjects received tecarfarin (ATI-5923) at each dose level (i.e., 10 mg, 20 mg, 30 mg and 40 mg).

The safety assessment results of this study were consistent with the results of CLN-501 study, and there was no safety risk after single dose administration of 10mg~40mg tecarfarin. AEs with higher incidence rate included headache and dizziness, and such AEs were graded as mild in severity. Tecarfarin showed a promising safety and tolerability profile in Chinese subjects.

Based on the results in this open-labelled, single-dose escalation, phase 1 study of tecarfarin, the following conclusions were made:

Single dose administration of tecarfarin in dose level ranging from 10 mg to 40 mg had no clinically significant effect on coagulation function. However, a slightly increasing trend in INR and PT values were observed with dose escalation. A slightly decreasing trend in coagulation factors II, VII and X were observed with dose escalation. Tecarfarin showed a promising safety and tolerability profile in Chinese subjects. The results of this study warrant further multiple-dose pharmacokinetic studies in the Chinese population. We do not believe we can extrapolate this data to other populations, including the United States, but other trials were performed in the U.S.

A Multiple-Dose, Safety and Tolerability PK/PD Study of Tecarfarin in Healthy Chinese Volunteers

Study Protocol: ZK-TEK-201905 completed in January 2021

This was a multiple-dose phase 1 pharmacokinetic-pharmacodynamic study conducted in China to assess the safety and tolerability of tecarfarin in healthy Chinese volunteers. The study site enrolled up to a total of 40 subjects. Ten (10) healthy Chinese subjects received tecarfarin once-daily on fasting every morning for 14 days at each dose level (i.e., 10 mg, 20 mg, 30 mg and 40 mg).

Tecarfarin was well tolerated in Chinese volunteers without serious adverse events in both single ascending dose and multiple ascending dose ("MAD") studies. There was only one treatment related adverse event (hematochezia) that resulted in early withdrawal in the MAD 40mg cohort. Exposure levels of tecarfarin were generally dose proportional.

Summary of Tecarfarin Clinical Trials

Clinical and preclinical trials of tecarfarin have demonstrated lack of drug-to-drug interactions with tecarfarin, predictable clearance that is independent of CYP450 blood clotting factors and any genetic variation in these factors, and the lack of impact of kidney function on clearance of tecarfarin. In the largest and longest of the clinical trials, EMBRACE-AC, tecarfarin and warfarin were found to have similar major and overall bleeding risks. In EMBRACE-AC, warfarin-treated patients had five thrombotic events, while there were no such events among tecarfarin-treated patients. When thrombotic and major bleeding events were combined, a trend favoring tecarfarin over warfarin was seen (five tecarfarin patients (1.6%) compared to 11 warfarin patients (3.6%)). We will conduct further studies and intend to submit this data to FDA in the NDA.

Upcoming Pivotal Phase 3 Trial: CLN-515 (ACTOR AF)

In the second half of 2023, we intend to commence our Phase 3, randomized, double-blind, placebo-controlled study of tecarfarin in subjects with ESRD and AFib not currently treated with chronic oral anticoagulation. The study will assess the safety and efficacy of evaluate the efficacy and safety of tecarfarin (target INR 2.0-3.0) in subjects with ESRD (stage 5 — eGFR < 15 mL/min/1.73 mm2) and AFib. Subjects must have chronic paroxysmal, persistent or permanent AFib documented. All subjects will undergo genetic testing for VKORC1

prior to randomization, which will also be used for stratification at the time of randomization. Subjects will be randomly assigned to receive either blinded tecarfarin or placebo in a 1:1 ratio. Approximately 540 subjects (270 per arm) will be enrolled in the study.

An enrollment period of 15 months is anticipated. A 10% dropout rate (48 subjects) is anticipated resulting in 492 evaluable subjects. All subjects enrolled in the study will remain on study drug until the last subject enrolled completes a minimum of 12 months of therapy or until the required number of adjudicated major adverse cardiovascular events (death, ischemic stroke, pulmonary embolus, and/or myocardial infarction) have been obtained, whichever is later. The primary efficacy assessment is time to first major adverse cardiovascular event, or MACE. There will be approximately 125 study sites in the United States and Canada, with other trial sites to be determined. Based upon internal statistical projections, assuming the Phase 3 clinical trial is powered at 80%, the study is expected to demonstrate a treatment effect of 25%.

Sub-License

Lee's Pharmaceutical Holdings Limited License

In September 2015, China Cardiovascular Focus Ltd., a wholly owned subsidiary of Lee's Pharmaceutical Holdings Limited, or LPH, entered into an agreement (the "LPH License") with Armetheon for the license, development and commercialization of our tecarfarin compound in China, Hong Kong, Macau, Taiwan and Thailand (the "Territory"). In October 2017, Armetheon merged with Espero BioPharma, Inc., or Espero. The assets owned by Espero were assigned to HESP LLC in a court-approved assignment for the benefit of creditors. On April 1, 2022 we acquired from HESP LLC, pursuant to an asset purchase agreement, the assets related to tecarfarin, including the LPH License. Under the terms of the LPH License, LPH provided a non-refundable up-front payment of $1 million and agreed, during the term of the agreement, not to develop, manufacture or commercialize a competitive product in the Territory. Conversely, we agreed not to develop, manufacture or commercialize a competitive product in the Territory. If all potential development, regulatory and commercial milestones under the LPH License are met, we are entitled to receive payments of approximately $52.0 million. In addition, we are also entitled to receive royalties between 9% to 15% of the net sales of tecarfarin in certain specified markets. The LPH License expires on a country-by-country basis within

the Territory, upon the latest of the expiration of the last intellectual property covering the tecarfarin compound in such country of the Territory, or the twelfth anniversary of the first commercial sale of tecarfarin in such country of the Territory.

Manufacturing

We do not have a manufacturing infrastructure and do not intend to develop one. We intend to contract with third parties for the production and packaging of our products and product candidates. With respect to tecarfarin, we are negotiating a contract with a third-party contract pharmaceutical manufacturer to perform the work necessary to develop a validated manufacturing process and to scale up for commercial production. However, we have not entered into any long-term supply agreements or commercialization partnership with these vendors. We anticipate that certain of the manufacturing sites for our products and product candidates may be in locations outside of the U.S.

While the drug substances used in our product candidate are manufactured by more than one supplier, the number of manufacturers is limited. In the event it is necessary or advisable to acquire supplies from an alternative supplier, we might not be able to obtain them on commercially reasonable terms, if at all. It could also require significant time and expense to redesign our manufacturing processes to work with another company. If approved by the FDA, we anticipate that we will be able to enter into agreements with suppliers to formulate and distribute tecarfarin on commercially reasonable terms.

Sales and Marketing

If any of our product candidates are approved by the FDA or other regulatory authorities, we intend to commercialize our products by leveraging our existing commercial infrastructure and hiring and training a small and dedicated cardiorenal salesforce to commercialize our products in the U.S., and possibly other major markets. In addition, we anticipate entering into a variety of distribution agreements and commercial partnerships in those territories where we do not establish an internal sales force, including if we expand outside of the U.S. We expect that our specialized commercial cardiovascular team would be comprised of experienced marketing and sales management professionals.

Market Opportunity

Based upon data of untreated patients with ESRD and AFib (40,823) and orphan drug pricing of $65 per day (or $23,400 annually) derived from a 2019 study commissioned by us (adjusted for inflationary increases in the per day pricing), and assuming that we receive FDA approval of tecarfarin, we estimate that the annual U.S. market revenue potential for tecarfarin is approximately $1 billion.

Competition

There have been several randomized trials to definitively assess the treatment effects of apixaban compared with VKAs in the population dependent on dialysis. The RENAL-AF (Trial to Evaluate Anticoagulation Therapy in Hemodialysis Patients With Atrial Fibrillation) was terminated early in 2019 by its sponsor. In addition, the AXADIA study (Compare Apixaban and Vitamin-K Antagonists in Patients With Atrial Fibrillation and End-Stage Kidney Disease), which is currently recruiting patients, will randomize patients to apixaban 2.5 mg twice daily versus phenprocoumon. The randomization of study drug and blinded event adjudication in these trials will help to minimize bias and confounding, and will better elucidate the risks and benefits of standard versus low-dose apixaban. Neither of these trials are adequately powered to address the important questions relating to intracerebral hemorrhage.

The development and commercialization of new drugs is highly competitive. We face competition with respect to developing our current product candidate, and we will face competition with respect to any products that we may seek to develop or commercialize in the future, from major pharmaceutical companies, specialty pharmaceutical companies and biotechnology companies worldwide. We are seeking to develop tecarfarin as a marketable VKA, once-daily OAC for chronic anticoagulation, and we are seeking to develop as a marketable oral drug for the treatment and prevention of refractory AFib. If we succeed in developing the lead indication or additional indications, we will face substantial competition. Existing anticoagulant treatments for thrombosis include warfarin and NOACs such as Pradaxa (dabigatran), Xarelto (rivaroxaban), Eliquis (apixaban) and Savaysa (edoxaban) for specific indications. The entry of the first generic NOACs, starting with Boehringer Ingelheim's loss of U.S., Japanese and Canadian patent protection for Pradaxa (dabigatran) in November 2018 and the remaining NOACs by 2024, could increase competition and reduce the total dollars spent on the treatment of thrombosis, as a result of lower generic drug pricing. The next generation of anticoagulants in development, Factor XI inhibitors, are currently in Phase 2 studies.

Many of these named products are marketed by some of the largest and most successful pharmaceutical companies worldwide. The companies that market these products have substantially more resources than we do and substantially more experience developing and marketing pharmaceuticals. We may not be able to successfully compete with these existing products. Potential competitors also include academic institutions, government agencies and other public and private research organizations that conduct research, seek patent protection and establish collaborative arrangements for research, development, manufacturing and commercialization of competing drugs and potentially competing drugs. Our competitors are developing or may be attempting to develop therapeutics for our target indications.

Factors affecting competition in these markets include the financial, research and development, testing, and marketing strengths of individual competitors, trends in industry consolidation, consumers' product options, product quality, price, technology, reputation, customer service capabilities and access to market partners and customers. Eliquis is manufactured and distributed by Bristol Myers Squibb, amiodarone is manufactured and distributed by several companies, including Sanofi, Baxter, and Pfizer, Pradaxa is manufactured and distributed by Boehringer Ingelheim, Xarelto is manufactured and distributed by Janssen Pharmaceuticals, and Savaysa is manufactured and distributed by Daiichi Sankyo. Each of these organizations has a long operating history,

extensive resources, strong brand recognition and large customer base. As a result, we expect they will be able to devote greater resources than we can to the manufacture, promotion and sale of their products, receive greater resources and support than we will from market partners and independent distributors, initiate and withstand substantial price competition, and take advantage more

readily than we could of acquisition and other strategic market opportunities. In addition, these or other organizations could succeed in developing new products that perform better or more cost-effectively than our products and product candidates in their respective markets. Moreover, changes in health trends, diet or other factors could substantially reduce the commercial attractiveness or viability of anti-anginal, anticoagulant, anti-arrhythmic and anti-platelet products.

The high level of competition in these markets could result in pricing pressure, reduced margins, the inability of our product candidates to achieve market acceptance and other impediments to commercial success. As a result, there can be no assurance that we will be able to complete the development of competitive products and commercialize them on a competitive basis.

Mergers and acquisitions in the pharmaceutical, biotechnology and diagnostic industries may result in even more resources being concentrated among a smaller number of our competitors. Smaller or early-stage companies may also prove to be significant competitors, particularly through collaborative arrangements with large and established companies. These companies compete with us in recruiting and retaining qualified scientific and management personnel, establishing clinical trial sites and patient registration for clinical trials, as well as in acquiring technologies complementary to, or necessary for, our programs.

Intellectual Property

Our success will significantly depend upon our ability to obtain and maintain patent and other intellectual property and proprietary protection for our drug candidates, including market and data exclusivity granted by regulatory agencies and composition-of-matter, dosage and formulation patents, as well as patent and other intellectual property and proprietary protection for our novel biological discoveries and other important technology inventions and know-how. In addition to patents, we rely upon unpatented trade secrets, know-how, and continuing technological innovation to develop and maintain our competitive position. We protect our proprietary information, in part, using confidentiality agreements with our commercial partners, collaborators, employees and consultants and invention assignment agreements with our employees. We also have confidentiality agreements or invention assignment agreements with our commercial partners and selected consultants. Despite these measures, any of our intellectual property and proprietary rights could be challenged, invalidated, circumvented, infringed or misappropriated, or such intellectual property and proprietary rights may not be sufficient to permit us to take advantage of current market trends or otherwise to provide competitive advantages. For more information, please see “Risk Factors — Risks Related to Our Intellectual Property.”

We have two issued U.S. patents directed to tecarfarin. The expiration dates of the patents are 2024 for both our composition of matter patent and our method of treatment patent, not including any possible patent term extension. Foreign patents corresponding to the tecarfarin patents expire in 2025. If our patents expire, we may not be able to adequately protect our intellectual property, and competitors may be able to erode or negate any competitive advantage we may have, which could harm our business and ability to achieve profitability.

However, in the U.S., the term of a patent covering an FDA-approved drug may be eligible for a patent term extension under the Hatch-Waxman Act as compensation for the loss of patent term during the FDA regulatory review process. The period of extension may be up to five years beyond the expiration of the patent, but cannot extend the remaining term of a patent beyond a total of 14 years from the date of product approval. Only one patent among those eligible for an extension may be extended. For patents that might expire during the application phase, the patent owner may request an interim patent extension. An interim patent extension increases the patent term by one year and may be renewed up to four times. For each interim patent extension granted, the post-approval patent extension is reduced by one year. The director of the United States Patent and Trademark Office must determine that approval of the drug covered by the patent for which a patent extension is being sought is likely. Interim patent extensions are not available for a drug for which an NDA has not been submitted. Provisions are available in certain other jurisdictions to extend the term of a patent that covers an approved drug or to provide data exclusivity. For example, data exclusivity in the EU may be available for 10 years from approval and in Japan for eight years from approval. It is possible that issued U.S. patents covering tecarfarin may be entitled to patent term extensions. If our product candidates receive FDA approval, we intend to apply for patent term extensions, if available, to extend the term of patents that cover the approved product candidates. We also intend to seek patent term extensions in any jurisdictions where they are available, however, there is no guarantee that the applicable authorities, including the FDA, will agree with our assessment of whether such extensions should be granted, and even if granted, the length of such extensions.

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The following is a list of our U.S. and foreign patents:

Tecarfarin Patents

Country	Patent No. (Application No.)	Grant Date	Type of Patent Protection	Expiration Date
U.S.A.	7666902	02/23/2010	Method of treatment	April 8, 2024
U.S.A.	7253208	08/07/2007	Composition of matter	April 8, 2024
U.S.A.	7285671	10/23/2007	Composition of matter (chloro derivative)[1]	April 8, 2024
Australia	2005233614	7/12/2012	Composition of matter	April 8, 2025
Austria	1735296	12/9/2009	Composition of matter	April 8, 2025
Belgium	1735296	12/9/2009	Composition of matter	April 8, 2025
Belgium	2161261	8/28/2013	Use	April 8, 2025
Brazil	PI0508392-3		Composition of matter	April 8, 2025
Canada	2559568	5/28/2013	Composition of matter	April 8, 2025
China	1950353-B (200580012074.6)	6/1/2011	Composition of matter	April 8, 2025
Cyprus	1735296	12/9/2009	Composition of matter	April 8, 2025
Denmark	1735296	12/9/2009	Composition of matter	April 8, 2025
Denmark	2161261	8/28/2013	Use	April 8, 2025
Europe	1735296 (05733799.0)	12/9/2009	Composition of matter	April 8, 2025
Europe	2161261 (09175606.4)	8/28/2013	Use	April 8, 2025
Finland	1735296	12/9/2009	Composition of matter	April 8, 2025
Finland	2161261	8/28/2013	Use	April 8, 2025
France	1735296	12/9/2009	Composition of matter	April 8, 2025
France	2161261	8/28/2013	Use	April 8, 2025
Germany	602005018181.4	12/9/2009	Composition of matter	April 8, 2025
Germany	602005041073.2	8/28/2013	Use	April 8, 2025
Great Britain	1735296	12/9/2009	Composition of matter	April 8, 2025
Great Britain	2161261	8/28/2013	Use	April 8, 2025
Greece	3071104	12/9/2009	Composition of matter	April 8, 2025
Hong Kong	1105200	3/9/2012	Composition of matter	April 8, 2025
Hong Kong	1138265	5/23/2014	Use	April 8, 2025
India	250594 (2793/KOLNP/2006)	1/11/2012	Composition of matter	April 8, 2025
Ireland	1735296	12/9/2009	Composition of matter	April 8, 2025
Ireland	2161261	8/28/2013	Use	April 8, 2025
Israel	178122	5/4/2013	Composition of Matter	April 8, 2025
Italy	502010901815570	12/9/2009	Composition of matter	April 8, 2025
Italy	502013902212109	8/28/2013	Use	April 8, 2025
Japan	5036532	7/13/2012	Composition of matter	April 8, 2025
Luxembourg	1735296	12/9/2009	Composition of matter	April 8, 2025
Luxembourg	2161261	8/28/2013	Use	April 8, 2025

Mexico	274321 (PA/a/2006/011637)	3/3/2010	Composition of matter	April 8, 2025
Monaco	1735296	12/9/2009	Composition of matter	April 8, 2025
Monaco	2161261	8/28/2013	Use	April 8, 2025

(1) This patent is not directed to tecarfarin, but rather is a derivative of tecarfarin.

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Country	Patent No. (Application No.)	Grant Date	Type of Patent Protection	Expiration Date
Netherlands	1735296	12/9/2009	Composition of matter	April 8, 2025
Netherlands	2161261	8/28/2013	Use	April 8, 2025
Norway	338837	10/24/2016	Composition of matter	April 8, 2025
Philippines	1-2006-501866	11/19/2010	Composition of matter	April 8, 2025
Portugal	1735296	12/9/2009	Composition of matter	April 8, 2025
Russia	2335501-C2	10/10/2008	Composition of matter	April 8, 2025
Russia	2495034-C2	10/10/2013	Method of Treatment	April 8, 2025
South Africa	2006/07667	11/28/2007	Composition of matter	April 8, 2025
South Korea	10-1203124	11/14/2012	Composition of matter	April 8, 2025
Spain	1735296	12/9/2009	Composition of matter	April 8, 2025
Spain	09175606.4	8/28/2013	Use	April 8, 2025
Sweden	1735296	12/9/2009	Composition of matter	April 8, 2025
Sweden	2161261	8/28/2013	Use	April 8, 2025
Switzerland	1735296	12/9/2009	Composition of matter	April 8, 2025
Switzerland	2161261	8/28/2013	Use	April 8, 2025

Data Exclusivity

If tecarfarin is approved by the FDA, we expect to receive five years of data exclusivity, often referred to as new chemical entity exclusivity, for our tecarfarin NDA, so long as FDA has not approved a drug containing the same active moiety as tecarfarin. It is possible that the FDA may disagree with our position and not approve tecarfarin or grant new chemical exclusivity to our NDA for tecarfarin. Assuming the FDA approves tecarfarin and new chemical entity exclusivity is granted, during the five-year period, no generic applicant can file an abbreviated drug application referencing our NDA for tecarfarin, unless the generic applicant challenges a patent listed in the FDA Orange Book for the referenced NDA, in which case the generic applicant can file after four years. If the patent is asserted against the generic applicant within 45 days of receipt of a required notice letter by the generic applicant, the generic abbreviated drug application cannot be approved by FDA for up to thirty months.

Government Regulation

The process of obtaining regulatory approvals and the subsequent compliance with appropriate federal, state, local and foreign statutes and regulations require the expenditure of substantial time and financial resources. Failure to comply with the applicable requirements at any time during the product development process, approval process or after approval, may subject an applicant to administrative or judicial sanctions. These sanctions could include the FDA's refusal to approve pending applications, withdrawal of an approval, a clinical hold, warning or untitled letters, product recalls or withdrawals from the market, product seizures, total or partial suspension of production or distribution, injunctions, fines, refusals of government contracts, restitution, disgorgement, or civil or criminal penalties.

Product development and marketing activities are subject to extensive regulation by various government authorities, including the FDA, other federal, state and local agencies and comparable regulatory authorities in other countries, which regulate the design, research, clinical and non-clinical development, testing, manufacturing, storage, distribution, import, export, labeling, advertising and marketing of pharmaceutical products and devices. Generally, before a new drug can be sold, considerable data demonstrating its quality, safety and efficacy must be obtained, organized into a format specific to each regulatory authority, submitted for review and approved by the regulatory authority. The data are often generated in two distinct development states: pre-clinical and clinical.

Among other matters, U.S. and foreign anti-corruption, anti-money laundering, export control, sanctions, and other trade laws and regulations, which are collectively referred to as Trade Laws, prohibit companies and their employees, agents, clinical research organizations, legal counsel, accountants, consultants, contractors, and

other partners from authorizing, promising, offering, providing, soliciting, or receiving directly or indirectly, corrupt or improper payments or anything else of value to or from recipients in the public or private sector. Violations of Trade Laws can result in substantial criminal fines and civil penalties, imprisonment, the loss of trade privileges, debarment, tax reassessments, breach of contract and fraud litigation, reputational harm, and other consequences. We have direct or indirect interactions with officials and employees of government agencies or government-affiliated hospitals, universities,

and other organizations. We also expect our non-U.S. activities to increase in time. We plan to engage third parties for clinical trials and/or to obtain necessary permits, licenses, patent registrations, and other regulatory approvals and we can be held liable for the corrupt or other illegal activities of our personnel, agents, or partners, even if we do not explicitly authorize or have prior knowledge of such activities.

Development of Drugs in the United States

Pharmaceutical products must be approved by the FDA before they may be legally marketed in the United States. Pharmaceutical product development for a new product or certain changes to an approved product in the U.S. typically involves pre-clinical laboratory and animal tests, the submission to the FDA of an investigational new drug application, or IND, which must become effective before clinical testing may commence, and adequate and well-controlled clinical trials to establish the safety and effectiveness of the drug for each indication for which FDA approval is sought. Satisfaction of FDA pre-market approval requirements typically takes many years and the actual time required may vary substantially based upon the type, complexity and novelty of the product or disease.

The pre-clinical development stage generally involves synthesizing the active component, developing the formulation and determining the manufacturing process, as well as carrying out non-human toxicology, pharmacology and drug metabolism trials that support subsequent clinical testing. These pre-clinical laboratory and animal tests must comply with federal regulations and requirements, including the FDA's good laboratory practices regulations. A drug's sponsor must submit the result of the pre-clinical tests, together with manufacturing information, analytical data and any available clinical data or literature and a proposed clinical protocol to the FDA as part of an IND application. A 30-day waiting period after the submission of each IND is required prior to the commencement of clinical testing in humans. If the FDA has neither commented on nor questioned the IND within this 30-day period, the clinical trial proposed in the IND may begin.

Clinical trials involve the administration of the investigational new drug to healthy volunteers or patients under the supervision of a qualified investigator. Clinical trials must be conducted (i) in compliance with federal regulations, including good clinical practices, or GCPs, an international standard meant to protect the rights and health of patients and to define the roles of clinical trial sponsors, administrators and monitors; and (ii) under protocols detailing the objectives of the trial, the parameters to be used in monitoring safety, and the effectiveness criteria to be evaluated. Each protocol involving testing on U.S. patients and subsequent protocol amendments must be submitted to the FDA as part of the IND.

Clinical trials to support NDAs for marketing approval can generally be divided into three sequential phases that may overlap, Phase 1, Phase 2 and Phase 3 clinical trials. In Phase 1, generally, small numbers of healthy volunteers are initially exposed to single escalating doses and then multiple escalating doses of the product candidate. The primary purpose of these trials is to assess the metabolism, pharmacologic action and general safety of the drug. Phase 2 trials typically involve trials in disease-affected patients to determine the dose required to produce the desired benefits, common short-term side effects and risks. Phase 2 trials are typically well-controlled, closely monitored, and conducted in a relatively small number of patients, usually involving no more than several hundred patients. Phase 3 trials are intended to gather the additional information about effectiveness and safety in a larger number of patients, typically at geographically dispersed clinical trial sites, that is needed to evaluate the overall benefit-risk relationship of the drug and to provide an adequate basis for physician labeling. Phase 3 trials usually include from several hundred to several thousand patients and are closely controlled and monitored. In many cases, the FDA requires two adequate and well-controlled Phase 3 clinical trials to demonstrate the efficacy of the drug. A single Phase 3 trial with other confirmatory evidence may be sufficient in some instances. In addition to these Phase 1-3 trials, other trials may be conducted to gather additional safety, pharmacokinetic and pharmacodynamic information. Pharmaceutical products with active ingredients that are the same as or similar to those already approved by the FDA may have more streamlined development programs than new chemical entities.

The FDA may order the temporary, or permanent, discontinuation of a clinical trial at any time, or impose other sanctions, if it believes that the clinical trial either is not being conducted in accordance with FDA requirements or presents an unacceptable risk to the clinical trial patients. Trials must be conducted in accordance with GCPs and reporting of study progress and any adverse experiences is required. The study protocol and informed consent information for patients in clinical trials must also be submitted to an institutional review board, or IRB, responsible for overseeing trials at particular sites and protecting human research trial patients. An independent institutional review

board may also suspend or terminate a trial once initiated, for failure to comply with the IRB's requirements, or may impose other conditions. Accordingly, we cannot be sure that submission of an IND, will result in the FDA allowing clinical trials to begin, or that once begun, issues will not arise that could cause the trial to be suspended or terminated.

Post-approval trials, sometimes referred to as Phase 4 clinical trials, may be conducted after initial marketing approval. Sometimes, these trials are used to gain additional experience from the treatment of patients in the intended therapeutic condition. In certain instances, the FDA may mandate the performance of Phase 4 trials. In other situations, post-approval trials aim to gain additional indications for a medication.

Changes to some of the conditions established in an approved application, including changes in indications, labeling, or manufacturing processes or facilities, require submission and FDA approval of a new NDA or NDA supplement before the change can be implemented. An NDA supplement for a new indication typically requires clinical data similar to that in the original application, and the FDA uses the same procedures and actions in reviewing NDA supplements as it does in reviewing NDAs.

Review and Approval in the United States

Following Phase 3 trial completion, data are analyzed to determine safety and efficacy, with any final such determination to be made by the FDA. Data are then submitted to the FDA in an NDA, along with proposed labeling for the product and information about the manufacturing and testing processes and facilities that will be used to ensure product quality. The cost of preparing and submitting an NDA is substantial. Manufacturers may be assessed up to five program fees for a fiscal year for prescription drug products identified in a single approved NDA. These fees are typically increased annually. In the United States, FDA approval of an NDA must be obtained before marketing a new drug.

The FDA has 60 days from its receipt of an NDA to determine whether the application will be accepted for filing based on the agency's threshold determination that the application is sufficiently complete to permit substantive review. Once the submission is accepted for filing, the FDA begins an in-depth review. The FDA has agreed to certain performance goals in the review of NDAs. Most applications for standard review drug products are reviewed within 10 to 12 months; most applications for priority review drugs are reviewed in six to eight months. Priority review can be applied to drugs that the FDA determines offer major advances in treatment, or provide a treatment where no adequate therapy exists. The review process for both standard and priority review may be extended by the FDA for three additional months to consider certain late-submitted information, or information intended to clarify information already provided in the submission.

The FDA may also refer applications for novel drug products, or drug products that present difficult questions of safety or efficacy, to an advisory committee — typically a panel that includes clinicians and other experts — for review, evaluation, and a recommendation as to whether the application should be approved. The FDA is not bound by the recommendations of advisory committees, but it generally follows such recommendations. Before approving an NDA, the FDA will typically inspect one or more clinical sites to assure compliance with GCP. Additionally, the FDA will inspect the facility or the facilities at which the drug is manufactured.

The FDA may conduct a pre-approval inspection of the manufacturing facilities for the new product to determine whether they comply with current good manufacturing practice requirements. The FDA will not approve the product unless compliance with current good manufacturing practices, or GMPs, is satisfactory and the NDA contains data that provide substantial evidence that the drug is safe and effective in the indication studied.

After the FDA evaluates the NDA and the manufacturing facilities, it issues either an approval letter or a complete response letter. A complete response letter generally outlines the deficiencies in the submission and may require substantial additional testing, or information, in order for the FDA to reconsider the application. If, or when, those deficiencies have been addressed to the FDA's satisfaction in a resubmission of the NDA, the FDA will issue an approval letter. The FDA has committed to reviewing such resubmissions in two to six months depending on the type of information included.

An approval letter authorizes commercial marketing of the drug with specific prescribing information for specific indications. As a condition of NDA approval, the FDA may require a risk evaluation and mitigation strategy, or REMS, to help ensure that the benefits of the drug outweigh the potential risks. REMS can include

medication guides, communication plans for healthcare professionals, and elements to assure safe use, or ETASU. ETASU can include, but are not limited to, special training or certification for prescribing or dispensing, dispensing only under certain

circumstances, special monitoring, and the use of patient registries. The requirement for a REMS can materially affect the potential market and profitability of the drug. Moreover, product approval may require substantial post-approval testing and surveillance to monitor the drug's safety or efficacy. Once granted, product approvals may be withdrawn if compliance with regulatory standards is not maintained or problems are identified following initial marketing.

Pediatric Information

Under the Pediatric Research Equity Act, or PREA, NDAs or supplements to NDAs must contain data to assess the safety and effectiveness of the drug for the claimed indications in all relevant pediatric subpopulations and to support dosing and administration for each pediatric subpopulation for which the drug is safe and effective. The FDA may grant full or partial waivers, or deferrals, for submission of data. Unless otherwise required by regulation, PREA does not apply to any drug for an indication for which orphan designation has been granted.

The Best Pharmaceuticals for Children Act, or BPCA, provides NDA holders a six-month extension of any exclusivity — patent or non-patent — for a drug if certain conditions are met. Conditions for exclusivity include the FDA's determination that information relating to the use of a new drug in the pediatric population may produce health benefits in that population, the FDA's written request for pediatric studies, and the applicant's agreeing to perform, and reporting on, the requested studies within the statutory timeframe. Applications under the BPCA are treated as priority applications, with all of the benefits that designation confers.

Orphan Drug Designation

Under the Orphan Drug Act, the FDA may grant orphan drug designation to a drug intended to treat a rare disease or condition, which is generally a disease or condition that affects fewer than 200,000 individuals in the United States. Orphan product designation must be requested before submitting an NDA. After the FDA grants orphan drug designation, the identity of the therapeutic agent and its potential orphan use are disclosed publicly by the FDA. Orphan product designation does not convey any advantage in or shorten the duration of regulatory review and approval process. It also does not suggest FDA approval or exclusivity. The first NDA applicant to receive FDA approval for a particular active ingredient to treat a particular disease with FDA orphan drug designation is entitled to a seven-year exclusive marketing period in the U.S. for that product, for that indication. In addition to the potential period of exclusivity, orphan designation makes a company eligible for grant funding of up to $500,000 per year for four years to defray costs of clinical trial expenses, tax credits for clinical research expenses and potential exemption from the FDA application user fee.

Orphan drug exclusivity means the FDA may not approve any other applications to market the same drug for the same indication for seven years, except in limited circumstances, such as (i) the drug's orphan designation is revoked; (ii) its marketing approval is withdrawn; (iii) the orphan exclusivity holder consents to the approval of another applicant's product; (iv) the orphan exclusivity holder is unable to assure the availability of a sufficient quantity of drug; or (v) a showing of clinical superiority to the product with orphan exclusivity by a competitor product. Orphan drug exclusivity does not prevent the FDA from approving a different drug for the same disease or condition, or the same drug for a different disease or condition. If a drug designated as an orphan product receives marketing approval for an indication broader than what is designated, it may not be entitled to orphan drug exclusivity. There has been recent litigation concerning FDA's interpretation of the orphan drug exclusivity provisions.

Accelerated Approval

There are a variety of pathways under which applicants may seek expedited approval from FDA, including Fast Track, breakthrough therapy, priority review and accelerated approval. Fast Track is a process designed to facilitate the development and expedite the review of investigational drugs to treat serious conditions and fill an unmet medical need. Drugs that receive Fast Track designation may be eligible for more frequent communications and meetings with the FDA to discuss the drug's development plan, including the design of the proposed clinical trials, use of biomarkers and the extent of data needed to support approval. Drugs with Fast Track designation may also qualify for accelerated approval and priority review of new drug applications if relevant criteria are met. However, Fast Track designation may be withdrawn by the FDA if the FDA believes that the designation is no longer supported by data emerging in the clinical trial process.

The FDA accelerated approval program provides for early approval of drugs based on a drug on a clinical trial(s) showing that the drug meets a surrogate or an intermediate clinical endpoint rather than a clinical benefit endpoint. Accelerated approval is possible for drugs for serious conditions that fill an unmet medical need. Under priority review, the FDA reviews an application in six months rather than ten months after it is accepted for filing.

A surrogate endpoint used for accelerated approval is a marker, such as a laboratory measurement, that is thought to predict clinical benefit, but is not itself a measure of clinical benefit. Likewise, an intermediate clinical endpoint is a measure of a therapeutic effect that is considered reasonably likely to predict the clinical benefit of a drug, such as an effect on irreversible morbidity and mortality. Because it sometimes can take many years for a drug trial to show a clinical benefit, the use of a surrogate endpoint or an intermediate clinical endpoint can significantly shorten the time required to complete clinical trials and obtain FDA approval.

If a drug receives an accelerated approval, the company that sponsored the application must conduct a post-approval trial to confirm the anticipated clinical benefit. These trials are known as Phase 4 or post-approval confirmatory trials. If the confirmatory trial shows that the drug actually provides a clinical benefit, then the FDA grants traditional approval for the drug. Failure to conduct required post-approval studies, or confirm a clinical benefit during post-marketing studies, will allow the FDA to withdraw the drug from the market on an expedited basis. All promotional materials for drug candidates approved under accelerated regulations are subject to prior review by the FDA. If the confirmatory trial does not show that the drug provides clinical benefit, FDA has regulatory procedures in place that could lead to removing the drug from the market.

Drug Development in Europe

In the European Union, our future products may also be subject to extensive regulatory requirements. Similar to the United States, the marketing of medicinal products is subject to the granting of marketing authorizations by regulatory agencies. Also, as in the United States, the various phases of pre-clinical and clinical research in the European Union are subject to significant regulatory controls.

Review and Approval in the European Union

In the European Union, approval of new medicinal products can be obtained through one of three processes: the mutual recognition procedure, the centralized procedure and the decentralized procedure. We intend to determine which process we will follow, if any, in the future.

Mutual Recognition Procedure: An applicant submits an application in one European Union member state, known as the reference member state. Once the reference member state has granted the marketing authorization, the applicant may choose to submit applications in other concerned member states, requesting them to mutually recognize the marketing authorizations already granted. Under this mutual recognition process, authorities in other concerned member states have 55 days to raise objections, which must then be resolved by discussion among the concerned member states, the reference member state and the applicant within 90 days of the commencement of the mutual recognition procedure. If any disagreement remains, all considerations by authorities in the concerned member states are suspended and the disagreement is resolved through an arbitration process. The mutual recognition procedure results in separate national marketing authorizations in the reference member state.

Centralized Procedure: This procedure is currently mandatory for products developed by means of a biotechnological process and optional for new active substances and other “innovative medicinal products with novel characteristics.” Under this procedure, an application is submitted to the European Agency for the Evaluation of Medical Products. Two European Union member states are appointed to conduct an initial evaluation of each application. These countries each prepare an assessment report that is then used as the basis of a scientific opinion of the Committee on Proprietary Medical Products. If this opinion is favorable, it is sent to the European Commission, which drafts a decision. After consulting with the member states, the European Commission adopts a decision and grants a marketing authorization, which is valid throughout the European Union and confers the same rights and obligations in each of the member states as a marketing authorization granted by that member state.

Decentralized Procedure: The most recently introduced of the three processes for obtaining approval of new medicinal processes in the European Union, the decentralized procedure is similar to the mutual recognition procedure described above, but with differences in the timing that key documents are provided to concerned member states by the reference member state, the overall timing of the procedure and the possibility of, among other things, "clock stops" during the procedure.

Post-Marketing Requirements

Following approval of a new product, a pharmaceutical company and the approved product are subject to continuing regulation by the FDA and other regulatory authorities, including, among other things, monitoring and recordkeeping activities, reporting to applicable regulatory authorities of adverse experiences with the product, providing the regulatory authorities with updated safety and efficacy information, product sampling and distribution requirements, and complying with promotion and advertising requirements, which include, among others, standards for direct-to-consumer advertising, restrictions on promoting drugs for uses or in patient populations not described in the drug's approved labeling (known as "off-label use"), and limitations on industry-sponsored scientific and educational activities. Although physicians may prescribe legally available drugs for off-label uses, drugs may be marketed only for the approved indications and in accordance with the provisions of the approved labeling. Modifications or enhancements to the products or labeling or changes of site of manufacture are often subject to the approval of the FDA and other regulators, which may or may not be received or may result in a lengthy review process. The FDA regulations require the products be manufactured in specific approved facilities and in accordance with current good manufacturing practices, and NDA holders must list their products and register their manufacturing establishments with the FDA. These regulations also impose certain organizational, procedural and documentation requirements with respect to manufacturing and quality assurance activities. Drug manufacturers and other entities involved in the manufacture and distribution of approved drugs are subject to periodic unannounced inspections by the FDA and certain state agencies for compliance with current good manufacturing practice and other laws. NDA holders using contract manufacturers, laboratories or packagers are responsible for the selection and monitoring of qualified firms. These firms are subject to inspections by the FDA at any time, and the discovery of violative conditions could result in enforcement actions that interrupt the operation of any such facilities or the ability to distribute products manufactured, processed or tested by them.

Other Regulatory Matters

Manufacturing, sales, promotion and other activities following product approval are also subject to regulation by numerous regulatory authorities in addition to the FDA, including, in the United States, the Centers for Medicare & Medicaid Services, or CMS, other divisions of the Department of Health and Human Services, the Drug Enforcement Administration, the Consumer Product Safety Commission, the Federal Trade Commission, the Occupational Safety & Health Administration, the Environmental Protection Agency, and state and local governments. These laws and regulations include:

- The federal healthcare program anti-kickback law which prohibits, among other things, persons from soliciting, receiving or providing remuneration, directly or indirectly, to induce either the referral of an individual, for an item or service or the purchasing or ordering of a good or service, for which payment may be made under federal healthcare programs such as the Medicare and Medicaid programs;

- Federal false claims laws which prohibit, among other things, individuals or entities from knowingly presenting, or causing to be presented, claims for payment from Medicare, Medicaid, or other government reimbursement programs that are false or fraudulent. The government may assert that a claim including items or services resulting from a violation of the federal healthcare program anti-kickback law or related to off-label promotion constitutes a false or fraudulent claim for purposes of the federal false claims laws;

- The Federal Physician Payments Sunshine Act within the Affordable Care Act, or the ACA, and its implementing regulations, require that certain manufacturers of drugs, devices, biological and medical supplies for which payment is available under Medicare, Medicaid or the Children's Health Insurance Program (with certain exceptions) to report on an annual basis information related to certain payments or other transfers of value made or distributed to physicians and teaching hospitals, or to entities or individuals at the request of, or designated on behalf of, the physicians

and teaching hospitals and certain ownership and investment interests held by physicians and their immediate family members, with the information made publicly available on a searchable website; and

- The Health Insurance Portability and Accountability Act, or HIPAA, as amended by the Health Information Technology for Economic and Clinical Health Act, or HITECH, and its implementing regulations, imposes certain requirements relating to the privacy, security and transmission of individually identifiable health information. Among other things, HITECH makes HIPAA's privacy and security standards directly applicable to "business associates" — independent contractors or agents of covered entities that receive or obtain protected health information in connection with providing a service on behalf of a covered entity. HITECH also created four new tiers of civil monetary penalties, amended HIPAA to make civil and criminal penalties directly applicable to business associates and possibly other persons, and gave state attorneys general new authority to file civil actions for damages or injunctions in federal courts to enforce the federal HIPAA laws and seek attorneys' fees and costs associated with pursuing federal civil actions.

- Applicable child-resistant packaging requirements under the U.S. Poison Prevention Packaging Act.

- The Lanham Act and federal antitrust laws.

- State law equivalents of each of the above federal laws, such as anti-kickback and false claims laws, which may apply to items or services reimbursed by any third-party payer, including commercial insurers, and state laws governing the privacy and security of health information in certain circumstances, many of which differ from each other in significant ways and often are not preempted by federal laws, thus complicating compliance efforts. In addition, several states now require prescription drug companies to report expenses relating to the marketing and promotion of drug products and to report gifts and payments to individual physicians in these states. Other states prohibit various other marketing-related activities, and still other states require the posting of information relating to clinical studies and their outcomes. In addition, California, Connecticut, Massachusetts and Nevada require pharmaceutical companies to implement compliance programs and/or marketing codes. Several additional states are considering similar proposals. Compliance with these laws is difficult and time consuming, and companies that do not comply with these state laws face civil penalties.

Distribution of pharmaceutical products is subject to additional requirements and regulations, including extensive record-keeping, licensing, traceability, and storage and security requirements intended to prevent the unauthorized sale of pharmaceutical products.

Third-Party Payer Coverage and Reimbursement

Significant uncertainty exists as to the coverage and reimbursement status of any of our drug candidates that ultimately may obtain regulatory approval. In both the United States and foreign markets, our ability to commercialize our product candidates successfully, and to attract commercialization partners for our product candidates, depends in significant part on the availability of adequate financial coverage and reimbursement from third-party payers, including, in the United States, governmental payers such as the Medicare and Medicaid programs, managed care organizations, and private health insurers. Medicare is a federally funded program managed by the CMS, through local fiscal intermediaries and carriers that administer coverage and reimbursement for certain healthcare items and services furnished to the elderly and disabled. Medicaid is an insurance program for certain categories of patients whose income and assets fall below state defined levels and who are otherwise uninsured that is both federally and state funded and managed by each state. The federal government sets general guidelines for Medicaid and each state creates specific regulations that govern its individual program. Each payer has its own process and standards for determining whether it will cover and reimburse a procedure or particular product. Private payers often rely on the lead of the governmental payers in rendering coverage and reimbursement determinations. Therefore, achieving favorable CMS coverage and reimbursement is usually a significant gating issue for successful introduction of a new product. The competitive position of some of our products will depend, in part, upon the extent of coverage and adequate reimbursement for such products and for the procedures in which such products are used. Prices at which we or our customers seek reimbursement for our products can be subject to challenge, reduction or denial by the government and other payers.

The United States Congress and state legislatures may, from time to time, propose and adopt initiatives aimed at cost containment, which could impact our ability to sell our products and product candidates profitably. For example, in the first quarter of 2018, President Trump signed a law requiring pharmaceutical companies to pay for a substantially larger percentage of the coverage gap, or the so-called "donut hole," between regular and catastrophic Medicare Part D prescription drug coverage, a change that is estimated to have a multi-billion-dollar effect on brand-name drug companies. Additional changes could be made in the future to governmental healthcare programs and many other laws that could significantly impact the success of our products.

The cost of pharmaceuticals continues to generate substantial governmental and third-party payer interest. We expect that the pharmaceutical industry will experience pricing pressures due to the trend toward managed healthcare, the increasing influence of managed care organizations and additional legislative proposals. Our results of operations could be adversely affected by current and future healthcare reforms.

Some third-party payers also require pre-approval of coverage for new or innovative devices or drugs before they will reimburse healthcare providers that use such drugs. While we cannot predict whether any proposed cost-containment measures will be adopted or otherwise implemented in the future, the announcement or adoption of these proposals could have a material adverse effect on our ability to obtain adequate prices for our products and product candidates and operate profitably.

In addition, in some foreign countries, the proposed pricing for a drug must be approved before it may be lawfully marketed. The requirements governing drug pricing vary widely from country to country. For example, the European Union provides options for its member states to restrict the range of medicinal products for which their national health insurance systems provide reimbursement and to control the prices of medicinal products for human use. A member state may approve a specific price for the medicinal product or it may instead adopt a system of direct or indirect controls on the profitability of the company placing the medicinal product on the market. There can be no assurance that any country that has price controls or reimbursement limitations for pharmaceutical products will allow favorable reimbursement and pricing arrangements for any of our products. Historically, products launched in the European Union do not follow price structures of the United States and generally tend to be significantly lower.

Trade Laws

Among other matters, U.S. and foreign anti-corruption, anti-money laundering, export control, sanctions, and other trade laws and regulations, which are collectively referred to as Trade Laws, prohibit companies and their employees, agents, clinical research organizations, legal counsel, accountants, consultants, contractors, and other partners from authorizing, promising, offering, providing, soliciting, or receiving directly or indirectly, corrupt or improper payments or anything else of value to or from recipients in the public or private sector. Violations of Trade Laws can result in substantial criminal fines and civil penalties, imprisonment, the loss of trade privileges, debarment, tax reassessments, breach of contract and fraud litigation, reputational harm, and other consequences. We have direct or indirect interactions with officials and employees of government agencies or government-affiliated hospitals, universities, and other organizations. We also expect our non-U.S. activities to increase in time. We plan to engage third parties for clinical trials and/or to obtain necessary permits, licenses, patent registrations, and other regulatory approvals and we can be held liable for the corrupt or other illegal activities of our personnel, agents, or partners, even if we do not explicitly authorize or have prior knowledge of such activities.

Human Capital-Employees

As of January 17, 2023, we had one full-time employee, two full-time contractors, and one part-time contractor. Of the team, one is engaged in research and development and three are engaged in general administration, business development and marketing. Our employees are not represented by labor unions or covered by collective bargaining agreements. We consider our relationship with our employees to be good.

Corporate Information

We were incorporated as a Delaware corporation in January 2022 and secured the rights to tecarfarin on April 1, 2022 via an asset purchase agreement (the "Asset Purchase Agreement") from HESP LLC. HESP LLC acquired the assets of Espero BioPharma, Inc., or Espero, from Horizon Technology Finance Corporation and Horizon Credit II LLC (collectively, Horizon), a secured lender of Espero, including tecarfarin and the other assets of Espero in an assignment for the benefit of creditors. Quang Pham, our Chief Executive Officer, was the Chief Executive Officer of Espero, and served as a consultant to HESP LLC from July 2020 until December 2021.

Our principal executive offices are located at 822 A1A North, Suite 306, Ponte Vedra, Florida 32082, and our telephone number is (904) 300-0701. Our website address is *www.cadrenal.com*. The information contained on, or that can be accessed through, our website is not incorporated by reference into this prospectus, and you should not consider any information contained on, or that can be accessed through, our website as part of this prospectus or in deciding whether to purchase our common stock.

Facilities

Our corporate headquarters are located at 822 A1A North, Suite 306, Ponte Vedra, Florida 32082, which are leased pursuant to a Lease Agreement dated October 15, 2022 with Veranda III Partners, Ltd. (the "Lease Agreement"). The Lease Agreement has an initial term of 24 months commencing on November 1, 2022. The monthly rent is $2,167. We believe that these headquarters are adequate for our current operations and needs.

Legal Proceedings

We are not currently a party to any material legal proceedings. We may, however, in the ordinary course of business face various claims brought by third parties, and we may, from time to time, make claims or take legal actions to assert our rights, including intellectual property rights as well as claims relating to employment matters and the safety or efficacy of our products. Any of these claims could subject us to costly litigation. If this were to happen, the payment of any such awards could have a material adverse effect on our business, financial condition and results of operations. Additionally, any such claims, whether or not successful, could damage our reputation and business.

MANAGEMENT

Executive Officers and Directors

The following table sets forth information as of January 17, 2023, for individuals who are expected to serve as executive officers, directors and director nominees following the completion of this offering.

Name	Age	Position
Executive Officers and Directors		
Quang Pham	58	Chairman and Chief Executive Officer
Matthew Szot	48	Acting Chief Financial Officer (Chief Financial Officer upon completion of this offering)
Douglas Losordo	64	Acting Chief Medical Officer (Chief Medical Officer upon completion of this offering)
Non-Employee Directors		
John R. Murphy	72	Director Nominee*
Steven Zelenkofske	64	Director Nominee*
Glynn Wilson	75	Director Nominee*

* Each director nominee will become a member of our board of directors upon the effectiveness of the registration statement of which this prospectus forms a part.

Quang Pham, Chairman and Chief Executive Officer

Quang Pham has served as our Chief Executive Officer since he formed the Company on January 25, 2022. He previously served as Chief Executive Officer, Chairman of the Board of Directors and founder of Espero BioPharma, Inc. ("Espero"), the previous sponsor of the tecarfarin IND, since its formation in March 2015 until July 2020, at which time a petition for assignment for the benefit of creditors was filed in the Delaware Chancery Court, seeking an assignment of Espero's assets. He then served as a consultant to HESP LLC, the assignee of Espero, from July 2020 until December 2021. From February 2012 to August 2015, Mr. Pham was a partner with D+R LATHIAN, LLC, a life sciences multichannel marketing agency. Prior to joining D+R LATHIAN, he founded and served as Chairman and Chief Executive Officer of Lathian Systems, Inc., a digital and database marketing company serving the pharmaceutical industry from 2000 until 2003 and from 2008 until 2012 when the company was acquired by D&R Communications, LLC in February 2012. He has a Bachelor of Arts in Economics from UCLA, and served as a U.S. Marine Corps Officer. We believe Mr. Pham is qualified to serve on our Board of Directors because of his significant business, mergers and acquisitions, and fundraising experience, numerous interactions with the FDA, continuous five-year history with tecarfarin development, and his extensive knowledge of the pharmaceutical industry and our competitors.

Matthew Szot, Acting Chief Financial Officer

Matthew Szot has served as our Acting Chief Financial Officer since May 2022. Upon the completion of this offering, he will become our full-time Chief Financial Officer. From March 2010 to November 2021, Mr. Szot served as Executive Vice President and Chief Financial Officer of S&W Seed Company, a Nasdaq-listed agricultural seed biotechnology company. Since September 2020, Mr. Szot has served on the Board of Directors and as Chairman of the Audit and Compensation Committees of INVO Bioscience, Inc., a Nasdaq-listed commercial-stage fertility company. He also serves on the Board of Directors and serves as Vice-Chairman of the Board and as Chairman of the Audit Committee of SenesTech, Inc., a Nasdaq-listed life science company with next-generation technologies for managing animal pest populations through fertility control. From June 2018 to August 2019, Mr. Szot served on the Board of Directors and as Chairman of the Audit Committee of Eastside Distilling, a Nasdaq-listed craft spirits company. From 2007 until 2011, Mr. Szot served as the Chief Financial Officer for Cardiff Partners, LLC, a strategic consulting company that provided executive financial services to various publicly traded and privately held companies. From 2003 to 2006, he served as Chief Financial Officer of Rip Curl, Inc., a market leader in wetsuit and action sports apparel products. From 1996 to

2003, Mr. Szot was a Certified Public Accountant with KPMG in the San Diego and Chicago offices and served as an Audit Manager for various publicly traded companies. Mr. Szot graduated from the University of Illinois, Champaign-Urbana with a BS in Agricultural Economics/Accountancy. He is a Certified Public Accountant in the State of California. Mr. Szot brings a wealth of knowledge in mergers and acquisitions, corporate strategy, equity and

debt financings, corporate governance, SEC reporting and compliance, and developing and implementing financial and operational workflows and process improvements. He also has extensive experience in international operations, joint ventures, and technology license agreements.

Douglas Losordo, M.D., Acting Chief Medical Officer

Douglas Losordo has served as our Acting Chief Medical Officer since August 8, 2022. Upon completion of this Offering he will become our Chief Medical Officer. Dr. Losordo has worked in the biotech industry developing cell-based therapies for over twenty years. Since February 2021, he has served on the Board of Directors of Longeveron Inc., a clinical-stage biotechnology company developing cellular therapies for aging-related and life-threatening conditions. Dr. Losordo also served as Global Head Clinical Development and Operations of American Regent, Inc., a clinical development pharmaceutical company from June 2021 until August 2022. Prior thereto he served as Chief Medical Officer of KBP Biosciences Co., Ltd., a biotechnology research and development company, from November 2020 until June 2021 and as Executive Vice President, Global Head of Research and Development, Chief Medical Officer of Caladrius Biosciences, a clinical-stage biopharmaceutical company dedicated to the development of cellular therapies designed to reverse chronic disease, from August 2013 until November 2020. Dr. Losordo has extensive knowledge of clinical, regulatory, manufacturing, supply chain and commercial factors unique to cellular therapy technologies as a result of his prior industry experience. Dr. Losordo also previously served as a Professor of Medicine at NYU Langone Medical Center and Northwestern University's Feinberg School of Medicine. He received his MD from the University of Vermont College of Medicine, and his B.A. in Zoology from the University of Vermont.

Steven Zelenkofske, D.O.

Dr. Steven Zelenkofske will serve on our Board of Directors upon effectiveness of the registration statement of which this prospectus is a part. Dr. Zelenkofske has served on the Board of Directors of Dinaqor AG since May 2020. He has served as Chief Medical Officer of SwanBio Therapeutics since June 1, 2020. Dr. Zelenkofske is also an advisor to Veralox Therapeutics, Inc., as Chair of the Scientific Advisory Board, a position he has held since March 2020. Previously, he served as Executive Vice President and Chief Medical Officer of Achillion Pharmaceuticals, Inc. from August 2018 until April 2020. Dr. Zelenkofske also served as Chief Medical Officer of uniQure N.V., from June 2017 to August 2018. Prior to joining uniQure, N.V., Dr. Zelenkofske was Vice President and Therapeutic Head of Cardiovascular/Metabolism for AstraZeneca, a biopharmaceutical company, from November 2014 to June 2017. From January 2009 to November 2014, Dr. Zelenkofske was Senior Vice President Clinical and Medical Affairs and Chief Medical Officer of Regado Biosciences, Inc., a biotechnology company. Dr. Zelenkofske has held leadership positions at Sanofi-Aventis, Boston Scientific, a medical device company, and Novartis Pharmaceuticals, a global healthcare company. Dr. Zelenkofske holds Bachelor of Science and Master of Science degrees from Emory University and a Doctor of Osteopathic Medicine degree from the Philadelphia College of Osteopathic Medicine. He conducted his graduate medical education at the Philadelphia College of Osteopathic Medicine and is board-certified in internal medicine, cardiology and cardiac electrophysiology. We believe that Dr. Zelenkofske's knowledge and experience working in the biotech and pharmaceutical space will assist us as we work to complete our drug development and commercialization activities.

John R. Murphy

John R. Murphy will serve on our Board of Directors upon effectiveness of the registration statement of which this prospectus is a part. Since 2003, John R. Murphy has served on the Board of Directors of O'Reilly Automotive, Inc., where he served as Chairman of the Audit Committee from 2003 until 2019. Currently, he serves on the Audit Committee and Human Capital and Compensation Committee (Chair). Mr. Murphy has also served on the Board of Directors of Summit Materials, Inc. since 2012, where he is currently the Chair of the Audit Committee. Previously he served as a Director, Audit Committee Chairman, and Member of the Nominating and Governance Committee of Apria, Inc. ("Apria") from August 2019 until April 2022. He also served on the Board of Directors of Alight Solutions LLC and Audit Committee Chairman from February 2020 until May 2022 and DJO Global, Inc. from 2012 until March 2019. Mr. Murphy also previously served on the Board of Directors of Graham Packaging, Inc. and Accuride Corporation, Inc. He previously served as Interim Chief Financial Officer of Summit Materials, Inc. in 2013, Senior Vice President and Chief Financial Officer of Smurfit-Stone Container Corporation from 2009 to 2010, and Chief Financial Officer, then President and Chief

Operating Officer, then President and Chief Executive Officer with Accuride Corporation, Inc. from 1998 to 2008. Mr. Murphy holds a Bachelor of Science in Accounting from Pennsylvania State University and a Master of Business Administration from the University of Colorado, and

is a Certified Public Accountant. We believe Mr. Murphy's substantial experience guiding public company boards and knowledge and experience as chief financial officer will assist us in navigating the public reporting process when we become a public reporting company.

Glynn Wilson, Ph.D.

Dr. Glynn Wilson will serve on our Board of Directors upon effectiveness of the registration statement of which this prospectus is a part. He has served on the Board of Directors of Jupiter Wellness, Inc. ("Jupiter") since November 2018, serving as Chairman since October 2019. Dr. Wilson also serves as Jupiter's Chief Scientific Officer since April 2021 and served as its Head of Research and Development from October 2019 to July 2021. Dr. Wilson previously served as a Director of TapImmune, Inc. from February 2005 until October, 2018 and as Chief Executive Officer from July 2009 through September 2017. Dr. Wilson also served as President of Auriga Laboratories, Inc. from June 1, 2005 through March 13, 2006, and as Chief Scientific Officer from March 13, 2016 through August 25, 2006. He was the Chief Scientific Officer at Tacora Corporation from 1994 to 1997 and was the Vice-President, R&D, at Access Pharmaceuticals from 1997 to 1998. Dr. Wilson was Research Area Head, Cell and Molecular Biology in Advanced Drug Delivery at Ciba-Geigy Pharmaceuticals from 1984 – 1989 and Worldwide Head of Drug Delivery at SmithKline Beecham from 1989 to 1994. He was a faculty member at Rockefeller University, New York, in the laboratory of the Nobel Laureates, Sanford Moore and William Stein, from 1974 to 1979. Dr. Wilson is a recognized leader in the development of drug delivery systems and has been involved in taking lead products & technologies from concept to commercialization.

Dr. Wilson has a Ph. D. in Biochemistry and conducted medical research at The Rockefeller University, New York. We believe that Dr. Wilson's extensive background of success in corporate management and product development, with tenures in both multinational and start-up biotech organizations, will assist us as we work to complete our drug development and commercialization activities.

Selection of Officers

Our executive officers serve at the discretion of our Board of Directors. There are no familial relationships among our directors and executive officers.

Board Composition

Pursuant to our amended and restated certificate of incorporation, our Board of Directors is divided into three classes — Class I, Class II, and Class III — with each class serving staggered three-year terms. Following the effectiveness of the registration statement of which this prospectus is a part, and subject to the terms of our amended and restated certificate of incorporation and amended and restated bylaws, our Board of Directors will consist of four members, as a classified board of directors. As a result, only one class of directors will be elected at each annual meeting of our stockholders, with the other classes continuing for the remainder of their respective three-year terms. The terms of the directors will expire upon the election and qualification of successor directors at the annual meeting of stockholders to be held during 2023 for the Class I directors, 2024 for the Class II directors and 2025 for the Class III directors. Our directors will be divided among the three classes as follows:

- the Class I directors will be Quang Pham and Glynn Wilson and their terms will expire at the annual meeting of stockholders to be held in 2023;
- the Class II director will be John Murphy and his term will expire at the annual meeting of stockholders to be held in 2024; and
- the Class III director will be Steven Zelenkofske and his term will expire at the annual meeting of stockholders to be held in 2025.

Upon expiration of the term of a class of directors, new directors for that class will be elected for three-year terms at the annual meeting of stockholders during the year in which that term expires. Each director's term shall continue until the election and qualification of his or her successor, or the director's earlier death, resignation or removal. Any additional directorships resulting from an increase in the number of authorized directors will be distributed among the three classes so that, as nearly as possible, each class will consist of one-third of the directors.

The classification of our Board of Directors may have the effect of delaying or preventing a change of our management, a change of control or other corporate actions. Under Delaware law, our amended and restated

certificate of incorporation our directors may be removed only for cause.

Principal Investigator

Sean Pokorney, M.D.

Sean Pokorney has been designated as our Principal Investigator. Dr. Pokorney is a cardiologist, electrophysiologist and researcher specializing in patients with ESRD and AFib. He is an Assistant Professor of Medicine at Duke University and is a member of the Duke Clinical Research Institute. We entered into a Scientific Advisory Board and Consulting Agreement with Dr. Pokorney ("Pokorney Agreement"), dated June 15, 2022, which provides that he shall be granted stock options, pursuant to the Cadrenal Therapeutics, Inc. 2022 Equity Incentive Plan, to purchase 100,000 shares of our Common Stock at an exercise price per share equal to the fair market value per share of the Company's Common Stock on the date of the grant, which stock options will vest over a three-year vesting schedule. The Pokorney Agreement further provides that we shall pay Dr. Pokorney at the rate of $650 per hour for services that exceed the scope of work of general members of the Scientific Advisory Board.

Scientific Advisory Board (SAB)

We intend that our Scientific Advisory Board will work with our management team in planning, developing and executing further scientific, clinical, and research and development initiatives and strategies. We anticipate that our scientific advisory board will be comprised of the following individuals, who have significant experience in the field of cardiovascular medicine:

Christopher Granger, MD

- Professor of Medicine in the Division of Cardiology at Duke University
- Director of Cardiac Care Unit for Duke University Medical Center
- Member, Duke Clinical Research Institute (DCRI)

C. Michael Gibson, MS, MD

- CEO of not-for profit Baim/PERFUSE Research Institutes
- Harvard University Professor
- Cardiologist at Beth Israel Deaconess Medical Center of Boston
- Founder and Chairman WikiDoc.org

Richard Whitlock, MD, PhD

- Cardiac Surgeon and Associate Professor at Population Health Research Institute, McMaster University Medical Center
- Investigator, Population Health Research Institute

A. Michael Lincoff, MD

- Vice Chairman, Department of Cardiovascular Medicine, Cleveland Clinic
- Director of Clinical Research, Lerner Research Institute

Wolfgang C. Winkelmayer, MD, MPH, ScD

- Chief, Section of Nephrology and Professor of Medicine, Baylor College of Medicine
- Director, Selzman Institute for Kidney Health at Baylor College of Medicine

Elaine Hylek, MD, MPH

- Professor of Medicine at the Boston University School of Medicine
- Director, Thrombosis and Anticoagulation Service at Boston Medical Center
- Researcher focused on anticoagulation and stroke prevention in AFib

Director Independence

Under the rules of the Nasdaq Stock Market, independent directors must comprise a majority of our Board of Directors. The rules of the Nasdaq Stock Market, as well as those of the SEC, impose several requirements with respect to the independence of our directors. Our Board of Directors has conducted a review of its proposed composition, the composition of its proposed committees and the independence of each director in accordance with these rules. Based upon information requested from and provided by each director concerning his or her background, employment and affiliations, including family relationships, our Board of Directors has determined that John R. Murphy, Dr. Steven Zelenkofske, and Dr. Glynn Wilson do not have relationships that would interfere with the exercise of independent judgment in carrying out the responsibilities of a director and that each of these directors is "independent" as that term is defined under the rules of the Nasdaq Stock Market and the SEC. In making this determination, our Board of Directors considered relationships that each director has with the Company, including the transactions described under the section entitled "Certain Relationships and Related Party Transactions."

Controlled Company

After the completion of this offering, Quang Pham, our Chief Executive Officer, will continue to beneficially own shares representing more than 50% of the voting power of our shares eligible to vote in the election of directors. As a result, we will be a "controlled company" within the meaning of the corporate governance standards of the Nasdaq. Under these corporate governance standards, a company of which more than 50% of the voting power is held by an individual, group or another company is a "controlled company" and may elect not to comply with certain corporate governance standards, including the requirements (1) that a majority of our board of directors consist of independent directors, (2) that our board of directors have a compensation committee that is comprised entirely of independent directors with a written charter addressing the committee's purpose and responsibilities and (3) that our board of directors have a nominating and governance committee that is comprised entirely of independent directors with a written charter addressing the committee's purpose and responsibilities. At this time we do not intend to utilize any of these exemptions; however they will be available to us should we choose to avail ourselves of any of the exemptions.

Committees of the Board of Directors

Our Board of Directors intends to establish an audit committee, a compensation committee and a nominating and corporate governance committee in connection with this offering, each of which will have the composition and responsibilities described below. Each committee will operate under a written charter, to be effective prior to the completion of this offering, that satisfies the applicable rules of the SEC and the listing standards of the Nasdaq Stock Market. Members will serve on these committees until their resignation or until otherwise determined by our Board of Directors. From time to time, our Board of Directors may establish other committees to facilitate the management of our business as it sees fit and in accordance with applicable law and our corporate governance documents.

Audit Committee. After the completion of this offering, our Audit Committee will consist of John Murphy, Dr. Steven Zelenkofske and Dr. Glynn Wilson, with John Murphy serving as the Chair of the Audit Committee. Our Board of Directors has determined that all of the directors who will serve on our Audit Committee are independent within the meaning of the rules and regulations of the Nasdaq Stock Market and Rule 10A-3 under the Exchange Act. In addition, our Board of Directors has determined that John Murphy qualifies as an audit committee financial expert within the meaning of SEC regulations and meets the financial sophistication requirements of the Nasdaq Stock Market. The primary purpose of the audit committee is to oversee the quality and integrity of our accounting and financial reporting processes and the audit of our financial statements. Specifically, the audit committee will:

- select and hire the independent registered public accounting firm to audit our financial statements;
- help to ensure the independence and performance of the independent registered public accounting firm;
- approve audit and non-audit services and fees;
- review financial statements and discuss with management and the independent registered public accounting firm our annual audited and quarterly financial statements, the results of the independent

audit and the quarterly reviews and the reports and certifications regarding internal controls over financial reporting and disclosure controls;

- prepare the audit committee report that the SEC requires to be included in our annual proxy statement;
- review reports and communications from the independent registered public accounting firm;

- review the adequacy and effectiveness of our internal controls and disclosure controls and procedure;
- review our policies on risk assessment and risk management;
- review and approve related party transactions; and
- establish and oversee procedures for the receipt, retention and treatment of accounting related complaints and the confidential submission by our employees of concerns regarding questionable accounting or auditing matters.

Compensation Committee. After the completion of this offering, our Compensation Committee will consist of Dr. Steven Zelenkofske and John Murphy, with Dr. Steven Zelenkofske serving as the Chair of the Compensation Committee. Our Board of Directors has determined that all of the directors who will serve on our compensation committee are independent under the listing standards, are "non-employee directors" as defined in rule 16b-3 promulgated under the Exchange Act and are "outside directors" as that term is defined in Section 162(m) of the Internal Revenue Code of 1986, as amended, or the Code. Our compensation committee oversees our compensation policies, plans and benefits programs. The compensation committee will also:

- oversee our overall compensation philosophy and compensation policies, plans and benefit programs;
- review and recommend to our board of directors for approval compensation for our executive officers and directors;
- prepare the compensation committee report that the SEC would require to be included in our annual proxy statement if we were no longer deemed to be an emerging growth company or a smaller reporting company; and
- administer our equity compensation plans.

Nominating and Corporate Governance Committee. After the completion of this offering, our Nominating and Corporate Governance Committee will consist of Dr. Glynn Wilson and Dr. Steven Zelenkofske, with Dr. Glynn Wilson serving as the Chair of the Nominating and Corporate Governance Committee. The Nominating and Corporate Governance Committee will oversee and assist our Board of Directors in reviewing and recommending nominees for election as directors. All members who will serve on the Nominating and Corporate Governance Committee are independent directors as defined under the listing standards of the Nasdaq Stock Market. Specifically, the corporate governance and nominating committee will:

- identify, evaluate and make recommendations to our board of directors regarding nominees for election to our board of directors and its committees, including consideration of recommendations for election to the Board of Directors by stockholders if submitted in a timely manner in accordance with the procedures set forth in our bylaws;
- consider and make recommendations to our board of directors regarding the composition of our board of directors and its committees;
- review developments in corporate governance practices;
- evaluate the adequacy of our corporate governance practices and reporting; and
- evaluate the performance of our board of directors and of individual directors.

Compensation Committee Interlocks and Insider Participation

No member of our Compensation Committee will be serving, or will have ever served, as an officer or employee of ours. None of our executive officers currently serves, or has served during the last completed year, as a member of the Board of Directors, Compensation Committee or other board committee performing equivalent functions of any entity that has one or more executive officers who served as a member of our Board of Directors during the last completed year.

Code of Business Conduct and Ethics

Prior to the completion of this offering, we will adopt a code of business conduct and ethics that is applicable to all of our employees, officers and directors. The full text of our code of business conduct and ethics will be available on our website at *www.cadrenal.com*. If we amend or grant any waiver from a provision of our code of ethics that applies to our directors or executive officers, we will publicly disclose such amendment or waiver on our website and as required by applicable law, including by filing a Current Report on Form 8-K.

Limitation of Liability and Indemnification

Our amended and restated bylaws provide indemnification for our directors and executive officers to the fullest extent permitted by the Delaware General Corporation Law. The indemnification agreements that we have entered into with each of our current executive officers and that we intend to enter into with each of our directors and executive officers may, in some cases, be broader than the specific indemnification provisions contained under Delaware law.

In addition, as permitted by Delaware law, our amended and restated certificate of incorporation includes provisions that eliminate the personal liability of our directors and officers for monetary damages resulting from breaches of certain fiduciary duties as a director or officer, as applicable, except to the extent such an exemption from liability thereof is not permitted under the Delaware General Corporation Law. The effect of these provisions is to restrict our rights and the rights of our stockholders in derivative suits to recover monetary damages against a director or officer for breach of fiduciary duties as a director or officer, subject to certain exceptions in which case the director or officer would be personally liable. An officer may not be exculpated for any action brought by or in the right of the corporation. A director may not be exculpated for improper distributions to stockholders. Further, pursuant to Delaware law a director or officer may not be exculpated for:

- any breach of his or her duty of loyalty to us or to our stockholders;
- acts or omissions not in good faith or that involve intentional misconduct or a knowing violation of law; and
- any transaction from which the director or officer derived an improper personal benefit.

If Delaware law is amended to authorize corporate action further eliminating or limiting the personal liability of directors and officers, then the liability of our directors and officers will be eliminated or limited to the fullest extent permitted by Delaware law, as so amended. Our certificate of incorporation does not eliminate the duty of care owed by our directors and officers and, in appropriate circumstances, equitable remedies, such as injunctive or other forms of non-monetary relief, remain available under Delaware law. This provision also does not affect the responsibilities of directors and officers under any other laws, such as the federal securities laws or other state or federal laws. Under our amended and restated bylaws, we will also be empowered to purchase insurance on behalf of any person whom we are required or permitted to indemnify.

In the case of an action or proceeding by or in the right of our company or any of our subsidiaries, no indemnification will be provided for any claim where a court determines that the indemnified party is prohibited from receiving indemnification. We believe that these charter and bylaw provisions are necessary to attract and retain qualified persons as directors and officers.

The limitation of liability and indemnification provisions in our amended and restated certificate of incorporation and amended and restated bylaws may discourage stockholders from bringing a lawsuit against directors and officers for breach of their fiduciary duties. They may also reduce the likelihood of derivative litigation against directors and officers, even though an action, if successful, might benefit us and our stockholders. Moreover, a stockholder's investment may be harmed to the extent we pay the costs of settlement and damage awards against directors and officers pursuant to these indemnification provisions.

Insofar as indemnification for liabilities arising under the Securities Act may be permitted to our directors, officers and controlling persons pursuant to the foregoing provisions, or otherwise, we have been advised that, in the opinion of the SEC, such indemnification is against public policy as expressed in the Securities Act, and is, therefore, unenforceable. There is no pending litigation or proceeding naming any of our directors or officers as to which indemnification is being sought, nor are we aware of any pending or threatened litigation that may result in claims for indemnification by any director or officer.

We have entered into separate indemnification agreements with each of our current executive officers and intend to enter, into separate indemnification agreements with each of our directors and executive officers, in addition to the indemnification that will be provided for in our amended and restated bylaws. The indemnification agreements and our amended and restated bylaws require us to indemnify our directors and executive officers to the fullest extent permitted by Delaware law. See the section titled “Description of Securities — Limitations on Liability and Indemnification of Officers and Directors” for additional information.

Director Compensation

Upon completion of this offering, our non-employee directors will receive annual cash compensation in the amount of $35,000 and the Chair of the Audit Committee will receive additional annual cash compensation of $15,000 per year. We have also granted stock options to our non-employee director nominees (see section entitled “Certain Relationships and Related Party Transactions”). From time to time, we may grant additional stock options to certain of our non-employee directors as compensation for their services as directors.

EXECUTIVE COMPENSATION

Our named executive officers for the year ended December 31, 2022, which consist of our principal executive officer and the next most highly compensated executive officers, are:

- Quang Pham, Chairman and Chief Executive Officer
- Matthew Szot, Acting Chief Financial Officer (Chief Financial Officer upon completion of this offering)
- Douglas Losordo, Acting Chief Medical Officer (Chief Medical Officer upon completion of this offering)

SUMMARY COMPENSATION TABLE

The following table shows compensation awarded to or earned by our named executive officers, for the period from January 25, 2022 (inception) to December 31, 2022.

Name and Principal Position	Salary ($)	Bonus ($)	Stock Award ($)[1]	Option Awards ($)[1]	Non-Equity Incentive Plan Compensation ($)	All Other Compensation ($)	Total ($)
Quang Pham							
Chief Executive Officer[5]	$ 350,000	—	—	$ —	$ —	$ 128,723[2]	$ 478,723
Matthew Szot							
Chief Financial Officer[6]	$ —	—	—	$ —	$ —	$ 172,105[3]	$ 172,105
Douglas Losordo							
Chief Medical Officer[7]	$ —	—	—	$ 625,200	$ —	$ 23,360[4]	$ 648,560

(1) In accordance with SEC rules, this column reflects the aggregate fair value of the stock and option awards granted as of their respective grant dates in accordance with Financial Accounting Standard Board Accounting Standards Codification Topic 718 for stock-based compensation transactions (ASC 718). The valuation assumptions used in determining such amounts are described in Note 1 and Note 10 to our unaudited financial statements included elsewhere in this registration statement on Form S-1. These amounts do not correspond to the actual value that may be realized by the Named Executive Officers upon vesting or exercise of such awards.

(2) All other compensation for Mr. Pham includes $115,000 of accrued but unpaid consulting fees due to Phamace, LLC, a consulting firm of which Quang Pham, our Chief Executive Officer, is the sole member. Other compensation also includes $13,723 for the cost of his monthly insurance premiums for health insurance.

(3) Amount represents Mr. Szot's monthly consulting fee of $22,500 from May 17, 2022 through December 31, 2022 plus $2,836 of insurance premiums for health insurance.

(4) Amount represents Mr. Losordo's consulting fees from August 8, 2022 through December 31, 2022.

(5) Mr. Pham became our Chief Executive Officer on January 25, 2022 (inception).

(6) Mr. Szot became our Acting Chief Financial Officer on May 17, 2022.

(7) Mr. Losordo became our Acting Chief Medical Officer on August 8, 2022.

Agreements with Our Named Executive Officers

Quang Pham Employment Agreement

We entered into an employment agreement with Quang Pham, our Chief Executive Officer, on March 1, 2022. Mr. Pham's employment is at-will. Mr. Pham's current annual base salary is $420,000, which will increase to a minimum of $675,000 upon the completion of a public listing of our stock on a national stock exchange. Mr. Pham is eligible for an annual target bonus of up to 50% of his base salary, with the actual amount of the bonus, if any, based upon the achievement by Mr. Pham and us of the applicable performance targets and goals as set by our board of directors or our compensation committee, with individual performance targets determined in consultation with Mr. Pham.

Pursuant to Mr. Pham's employment agreement, we will need to provide 90 days' written notice to terminate his employment without cause. If Mr. Pham resigns for Good Reason, as such term is defined in the employment agreement, or is terminated without cause (as such terms are defined below), he is entitled to (i) a lump sum payment equal to 24 months of his base salary, (ii) a lump sum payment equal to his target bonus for the calendar year in which his termination date occurs, (iii) full acceleration of any outstanding equity or equity-

based awards that he has with respect to us or any of our affiliates as of his termination date, (iv) extension of exercisability for the full term of any stock option, and (v) payment of his full COBRA premiums for 24 months following his termination date, if applicable conditions are met.

Mr. Pham is required to provide us 90 days' written notice of the condition that qualifies as a Good Reason for his resignation and we will have 30 days from receipt of such notice to remedy such condition. If Mr. Pham fails to provide the required notice such that we have the opportunity to cure the condition prior to his resignation, or if he resigns more than nine months after the initial existence of the condition, his resignation shall not be deemed for Good Reason.

If we terminate Mr. Pham's employment for Cause, as such term is defined in the employment agreement, or if Mr. Pham voluntarily terminates his employment without Good Reason upon 30 days written notice to us, Mr. Pham shall be entitled to receive Accrued Obligations, as such term is defined in the employment agreement and which includes earned, but unpaid, base salary, accrued, but unused, vacation, and vested benefits, as of the date of termination.

Pursuant to Mr. Pham's employment agreement, if his employment is terminated due to his death or disability (as defined in the employment agreement), he is entitled to (i) a lump sum payment equal to twelve months of his base salary, (ii) full acceleration of any outstanding equity or equity-based awards that he has with respect to us or any of our affiliates as of his termination date, and (iii) Accrued Obligations.

Matthew Szot Consulting Agreement

We entered into a consulting agreement with Matthew Szot, our Acting Chief Financial Officer, on May 17, 2022. Mr. Szot's engagement as a consultant is at-will. Mr. Szot's current monthly consulting fee is $22,500, which will increase to a minimum of $375,000 annual salary upon the completion of a public listing of our stock on a national stock exchange. Mr. Szot is eligible for an annual target bonus of up to 50% of his base salary, with the actual amount of the bonus, if any, based upon the achievement of Mr. Szot and us of the applicable performance targets and goals as set by our board of directors.

Pursuant to Mr. Szot's employment agreement (which will become effective upon completion of this Offering), we will need to provide 90 days' written notice to terminate his employment without Cause, as such term is defined in the employment agreement. If Mr. Szot resigns for Good Reason, as such term is defined in the employment agreement, or is terminated without Cause, unrelated to a Change of Control, as such term is defined in the employment agreement, he is entitled to (i) continuation of his base salary in effect immediately prior to termination for a period of 12 months, (ii) a lump sum payment equal to his target bonus for the calendar year in which his termination date occurs, (iii) full acceleration of any outstanding equity or equity-based awards as of his termination date, (iv) extension of exercisability for the full term of any stock option, and (v) payment of his full COBRA premiums for 12 months following his termination date, if applicable conditions are met.

Mr. Szot will be required to provide us 90 days' written notice of the condition that qualifies as a Good Reason for his resignation and we will have 30 days from receipt of such notice to remedy such condition. If Mr. Szot fails to provide the required notice such that we do not have the opportunity to cure the condition prior to his resignation, or if he resigns more than nine months after the initial existence of the condition, his resignation shall not be deemed for Good Reason.

If at any time during a Change of Control Period, as such term is defined in the employment agreement, Mr. Szot's employment is terminated without Cause or Mr. Szot resigns for Good Reason, he is entitled to: (i) a lump sum payment equal to 12 months of his base salary in effect immediately prior to termination plus his target bonus for the fiscal year in which his termination date occurs; (ii) full acceleration of any outstanding equity or equity-based awards as of his termination date; (iii) extension of exercisability for the full term of any stock option; and payment of his full COBRA premiums for 12 months following his termination date, if applicable conditions are met.

If we terminate Mr. Szot's employment for Cause, or if Mr. Szot voluntarily terminates his employment without Good Reason upon 30 days written notice to us, Mr. Szot shall be entitled to receive Accrued Obligations, as such term is defined in the employment agreement, as of the date of termination.

Pursuant to Mr. Szot's employment agreement, if his employment is terminated due to his death or Disability (as defined in the employment agreement), he is entitled to (i) a lump sum payment equal to twelve months of his base salary, (ii) full acceleration of any outstanding equity or equity-based awards that he has with respect to us or any of our affiliates as of his termination date; and (iii) Accrued Obligations.

Douglas Losordo Letter Agreement

We entered into a letter agreement with Douglas Losordo, our Acting Chief Medical Officer, effective as of August 8, 2022. Dr. Losordo was engaged to perform Chief Medical Officer services at a rate of $160/hour as a contractor. Pursuant to the letter agreement, Dr. Losordo received a stock option grant to purchase 300,000 shares of our common stock on August 18, 2022 and will be granted an additional equity award of 50,000 stock options upon the initiation of a pivotal clinical trial's first patient in (FPI).

Effective immediately upon completion of this Offering, Dr. Losordo will become an employee and will receive an annual base salary of $425,000, with an annual targeted cash bonus of 40% of his base salary.

Pursuant to Dr. Losordo's employment agreement (which will become effective upon completion of this Offering), we will need to provide 90 days' written notice to terminate his employment without Cause, as such term is defined in the employment agreement. If Dr. Losordo resigns for Good Reason, as such term is defined in the employment agreement, or is terminated without Cause, unrelated to a Change of Control, as such term is defined in the employment agreement, he is entitled to (i) continuation of his base salary in effect immediately prior to termination for a period of 6 months, (ii) a lump sum payment equal to 50% of his target bonus for the calendar year in which his termination date occurs, (iii) full acceleration of any outstanding equity or equity-based awards as of his termination date, (iv) extension of exercisability for the full term of any stock option, and (v) payment of his full COBRA premiums for 6 months following his termination date, if applicable conditions are met.

Dr. Losordo will be required to provide us 90 days' written notice of the condition that qualifies as a Good Reason for his resignation and we will have 30 days from receipt of such notice to remedy such condition. If Dr. Losordo fails to provide the required notice such that we do not have the opportunity to cure the condition prior to his resignation, or if he resigns more than nine months after the initial existence of the condition, his resignation shall not be deemed for Good Reason.

If at any time during a Change of Control Period, as such term is defined in the employment agreement, Dr. Losordo's employment is terminated without Cause or Dr. Losordo resigns for Good Reason, he is entitled to: (i) a lump sum payment equal to 12 months of his base salary in effect immediately prior to termination plus his target bonus for the fiscal year in which his termination date occurs; (ii) full acceleration of any outstanding equity or equity-based awards as of his termination date, (iii) extension of exercisability for the full term of any stock option; and (iv) payment of his full COBRA premiums for 12 months following his termination date, if applicable conditions are met.

If we terminate Dr. Losordo's employment for Cause, or if Dr. Losordo voluntarily terminates his employment without Good Reason upon 30 days written notice to us, Dr. Losordo shall be entitled to receive Accrued Obligations, as such term is defined in the employment agreement, as of the date of termination.

Outstanding Equity Awards at Fiscal Year End

The following table sets forth information concerning the number of shares of common stock underlying outstanding equity incentive awards for each of our executive officers as of December 31, 2022:

	Option Awards					Stock Awards	
Name	**Grant Date**	**Number of Securities Underlying Unexercised Options Exercisable (#)**	**Number of Securities Underlying Unexercised Options Unexercisable (#)**	**Option Exercise Price ($)**	**Option Expiration Date**	**Number of Shares or Units of Stock not yet Vested (#))**	**Market Value of Shares or Units not yet Vested ($)**
Douglas Losordo	08/18/2022[(1)]	25,000	275,000	0.64	08/17/2032	—	—

(1) These options vest quarterly over three years commencing October 1, 2022.

Equity Incentive Plans

2022 Initial Equity Incentive Plan

We adopted the Cadrenal Therapeutics, Inc. 2022 Equity Incentive Plan, or the Initial Plan, on July 11, 2022, which was later amended and restated on October 16, 2022, for purposes of clarifying the application of certain of the rules of the Initial Plan to awards approved before such amendment and restatement of the Initial Plan and to facilitate the transition to the Cadrenal Therapeutics, Inc. 2022 Successor Equity Incentive Plan (the "Successor Plan") for the issuance and approval of awards after consummation of this offering. The principal provisions of the Initial Plan are

summarized below. On October 16, 2022, the Board adopted and our stockholders approved the Cadrenal Therapeutics, Inc. 2022 Successor Equity Incentive Plan, or the 2022 Plan, which will be a successor to and continuation of the Initial Plan and will become effective upon the effective date of the registration statement of which this prospectus forms a part. Upon the effectiveness of the 2022 Plan, it will replace the Initial Plan, except with respect to awards outstanding under the Initial Plan, and no further awards will be available for grant under the Initial Plan.

Administration

The Initial Plan vests broad powers in a committee to administer and interpret the Initial Plan. Our board of directors has initially been designated to administer the Initial Plan. Except when limited by the terms of the Initial Plan, our board of directors has the authority to, among other things: select the persons to be granted awards; determine the type, size and term of awards; establish performance objectives and conditions for earning awards; determine whether such performance objectives and conditions have been met; and accelerate the vesting or exercisability of an award. In its discretion, board of directors may delegate all or part of its authority and duties with respect to granting awards to one or more of our officers, subject to certain limitations and provided applicable law so permits.

Our board of directors may amend, alter or discontinue the Initial Plan and is able to amend any outstanding award at any time; provided, however, that no such amendment or termination may adversely affect awards then outstanding without the holder's permission. In addition, any amendments seeking to increase the total number of shares reserved for issuance under the Initial Plan or modifying the classes of participants eligible to receive awards under the Initial Plan requires ratification by our stockholders in accordance with applicable law. Additionally, as described more fully below, neither the board of directors nor any committee designated to administer the Initial Plan, or the Administrator, is permitted to reprice outstanding options or stock appreciation rights without shareholder consent.

Eligibility

Any of our employees, directors, consultants, and other service providers, or those of our affiliates, are eligible to participate in the Initial Plan and may be selected by the Administrator to receive an award.

Vesting

The Administrator determines the vesting conditions for awards. These conditions may include the continued employment or service of the participant, the attainment of specific individual or corporate performance goals, or other factors as determined in the Administrator's discretion (collectively, "Vesting Conditions").

Shares of Stock Available for Issuance

Subject to certain adjustments, the maximum number of shares of common stock that could have been issued under the Initial Plan in connection with awards was 2,000,000 shares, of which 810,000 remained available for issuance as of the date of this prospectus. Once the 2022 Plan becomes effective all outstanding awards under the Initial Plan will remain outstanding but no further grants will be made under the Initial Plan. All available shares may be utilized toward the grant of any type of award under the Initial Plan. The Initial Plan imposes a $100,000 limitation on the total grant date fair value with respect to which incentive stock options are exercisable for the first time by an individual optionee during any single calendar year.

In the event of any merger, consolidation, reorganization, recapitalization, stock split, reverse stock split, split up, spin-off, combination of shares, exchange of shares, stock dividend, dividend in kind, or other like change in capital structure (other than ordinary cash dividends), or other similar corporate event or transaction that affects our common stock, the Administrator shall make adjustments to the number and kind of shares authorized by the Initial Plan and covered under outstanding Initial Plan awards as it determines appropriate and equitable.

Shares subject to Initial Plan awards that expire without being fully exercised or that are otherwise forfeited, cancelled or terminated may again be made available for issuance under the Initial Plan. In addition, shares withheld in settlement of a tax withholding obligation, or in satisfaction of the exercise price payable upon exercise of an option, will again become available for issuance under the Initial Plan.

Types of Awards

The following types of awards may be granted to participants under the Initial Plan: (i) incentive stock options, or ISOs; (ii) nonqualified stock options, or NQOs and together with ISOs, options, (iii) stock appreciation rights, (iv) restricted stock, or (v) restricted stock units.

Stock Options. An option entitles the holder to purchase from us a stated number of shares of common stock. An ISO may only be granted to an employee of ours or our eligible affiliates. The Administrator will specify the number of shares of common stock subject to each option and the exercise price for such option, provided that the exercise price may not be less than the fair market value of a share of common stock on the date the option is granted. Notwithstanding the foregoing, if ISOs are granted to any 10% stockholder, the exercise price shall not be less than 110% of the fair market value of common stock on the date the option is granted.

Generally, options may be exercised in whole or in part through a cash payment. The Administrator may, in its sole discretion, permit payment of the exercise price of an option in the form of previously acquired shares based on the fair market value of the shares on the date the option is exercised, through means of "net settlement," which involves the cancellation of a portion of the option to cover the cost of exercising the balance of the option or by such other means as it deems acceptable.

All options shall be or become exercisable in accordance with the terms of the applicable award agreement. The maximum term of an option shall be determined by the Administrator on the date of grant but shall not exceed 10 years (5 years in the case of ISOs granted to any 10% stockholder). In the case of ISOs, the aggregate fair market value (determined as of the date of grant) of common stock with respect to which such ISOs become exercisable for the first time during any calendar year cannot exceed $100,000. ISOs granted in excess of this limitation will be treated as non-qualified stock options.

Stock Appreciation Rights. A stock appreciation right represents the right to receive, upon exercise, any appreciation in a share of common stock over a particular time period. The base price of a stock appreciation right shall not be less than the fair market value of a share of common stock on the date the stock appreciation right is granted. This award is intended to mirror the benefit the participant would have received if the Administrator had granted the participant an option. The maximum term of a stock appreciation right shall be determined by the Administrator on the date of grant but shall not exceed 10 years. Distributions with respect to stock appreciation rights may be made in cash, shares of common stock, or a combination of both, at the board of director's discretion.

Unless otherwise provided in an award agreement or determined by the Administrator, if a participant terminates employment with us (or our affiliates) due to death or disability, the participant's unexercised options and stock appreciation rights may be exercised, to the extent they were exercisable on the termination date, for a period of twelve months from the termination date or until the expiration of the original award term, whichever period is shorter. Unless otherwise provided in an award agreement or determined by the Administrator, if a participant terminates employment with us (or our affiliates) due to retirement from active employment with the us or any affiliate on or after age 65, the participant's unexercised NQOs may be exercised, to the extent they were exercisable on the termination date, for a period of twelve months from the termination date or until the expiration of the original award term, whichever period is shorter. If the participant terminates employment with us (or our affiliates) for cause, all unexercised options and stock appreciation rights (whether vested or unvested) shall terminate and be forfeited on the termination date. If the participant's employment terminates for any other reason, any vested but unexercised options and stock appreciation rights may be exercised by the participant, to the extent exercisable at the time of termination, for a period of three months from the termination date (or such time as specified by the Administrator at or after grant) or until the expiration of the original option or stock appreciation right term, whichever period is shorter. Unless otherwise provided by the Administrator, any options and stock appreciation rights that are not exercisable at the time of termination of employment shall terminate and be forfeited on the termination date.

Restricted Stock. A restricted stock award is a grant of shares of common stock, which are subject to forfeiture restrictions during a restriction period. The Administrator will determine the price, if any, to be paid by the participant for each share of common stock subject to a restricted stock award. The restricted stock may be subject to Vesting Conditions. If the specified Vesting Conditions are not attained, the participant will forfeit the portion of the restricted stock award with respect to which those conditions are not attained, and the underlying common stock will be forfeited to us. At the end of the restriction period, if the Vesting Conditions have been satisfied, the restrictions imposed will lapse with respect to the applicable number of shares. Unless otherwise provided in an award agreement or determined by the Administrator, upon termination a participant will forfeit all restricted stock that then remains subject to forfeiture restrictions.

Restricted Stock Units. Restricted stock units are granted in reference to a specified number of shares of common stock and entitle the holder to receive, on the achievement of applicable Vesting Conditions, shares of common stock. Unless otherwise provided in an award agreement or determined by the Administrator, upon termination a participant will forfeit all restricted stock units that then remain subject to forfeiture.

Change of Control

In the event of a change of control, unless otherwise provided in a grant agreement, employment agreement or other agreement between the Company and the participant, and unless otherwise determined by an affirmative vote of a majority of the board of directors prior to the occurrence of such change of control: (i) all outstanding stock options and stock appreciation rights which have been outstanding for at least six months shall become exercisable in full, whether or not otherwise exercisable at such time, and any such stock option and stock appreciation right shall remain exercisable in full thereafter until it expires pursuant to its terms; and (ii) all restrictions and deferral limitations contained in restricted stock and restricted stock unit awards granted under the Initial Plan shall lapse and the shares of stock subject to such awards shall be distributed to the participant within thirty (30) days of the change of control to the extent permitted under Section 409A of the Code. In addition, in the event of a change of control, unless otherwise specifically prohibited under applicable laws or by the rules and regulations of any governing governmental agencies or national securities exchanges, or unless the board of directors or the Administrator shall specify otherwise in the applicable award agreement, the Administrator is authorized (but not obligated) to make any of the following adjustments (or any combination thereof) in the terms and conditions of outstanding awards: (a) continuation or assumption of such outstanding awards under the Initial Plan by the Company (if it is the surviving company or corporation) or by the surviving company or corporation or its parent; (b) substitution by the surviving company or corporation or its parent of equity, equity-based and/or cash awards with substantially the same terms for outstanding awards (to the extent permitted under Section 409A of the Code), including, in the case of options, substitution by the surviving company or corporation or its parent of restricted stock or other equity in an amount equal to the intrinsic value of such options; (c) accelerated exercisability, vesting and/or lapse of restrictions under outstanding awards immediately prior to the occurrence of such event; (d) upon written notice, provide that any outstanding awards must be exercised, to the extent then exercisable, during a reasonable period of time immediately prior to the scheduled consummation of the event or such other period as determined by the Administrator (contingent upon the consummation of the event), and at the end of such period, such awards shall terminate to the extent not so exercised within the relevant period; and (e) cancellation of all or any portion of outstanding awards for fair value (in the form of cash, stock, other property or any combination thereof) as determined in the sole discretion of the Administrator and which value (for example, in the case of options that are not in the money) may be zero.

Repricing

The Administrator cannot without obtaining prior approval of our stockholders, reduce the exercise price in effect for outstanding options under the Initial Plan.

Miscellaneous

Generally, awards granted under the Initial Plan shall be nontransferable except by will or by the laws of descent and distribution. No participant shall have any rights as a stockholder with respect to shares covered by options or restricted stock units, unless and until such awards are settled in shares of common stock. The Company's obligation to issue shares or to otherwise make payments in respect of Initial Plan awards will be conditioned on the Company's ability to do so in compliance with all applicable laws and exchange listing requirements. The awards will be subject to our recoupment and stock ownership policies, as may be in effect from time to time.

2022 Successor Equity Incentive Plan

In October 2022, the Board adopted and our stockholders approved the 2022 Successor Equity Incentive Plan, as a successor to and continuation of the Initial Plan, which will become effective upon the effective date of the registration statement of which this prospectus forms a part. Upon the effectiveness of the 2022 Plan, all outstanding awards under the Initial Plan will remain outstanding but no further grants will be made under the Initial Plan. The shares of common stock underlying any awards under the 2022 Plan and the Initial Plan that,

after the effective date of the 2022 Plan, are forfeited, canceled or otherwise terminated, other than by exercise, will be added back to the shares of common stock available for issuance under the 2022 Plan. In addition, if any shares subject to an award under the 2022 Plan and

the Initial Plan are tendered or withheld by the Company to satisfy any exercise price or tax withholding obligation once the 2022 Plan becomes effective, such tendered or withheld shares will be added back to the shares of common stock available for issuance under the 2022 Plan. Shares of common stock repurchased on the open market will not be added back to the shares of Common Stock available for issuance under the 2020 Plan. The principal purpose of the 2022 Plan is to attract, retain and incentivize the Company's employees and other service providers through the granting of certain stock-based awards, including performance-based awards. The material terms of the 2022 Plan, as it is currently contemplated, are summarized below.

Administration

The 2022 Plan vests broad powers in a committee to administer and interpret the 2022 Plan. Our board of directors has initially designated the compensation committee to administer the 2022 Plan. Except when limited by the terms of the 2022 Plan, the compensation committee has the authority to, among other things: select the persons to be granted awards; determine the type, size and term of awards; establish performance objectives and conditions for earning awards; determine whether such performance objectives and conditions have been met; and accelerate the vesting or exercisability of an award. In its discretion, the compensation committee may delegate all or part of its authority and duties with respect to granting awards to one or more of our officers, subject to certain limitations and provided applicable law so permits.

Our board of directors may amend, alter or discontinue the 2022 Plan and the compensation committee is able to amend any outstanding award at any time; provided, however, that no such amendment or termination may adversely affect awards then outstanding without the holder's permission. In addition, any amendments seeking to increase the total number of shares reserved for issuance under the 2022 Plan or modifying the classes of participants eligible to receive awards under the 2022 Plan requires ratification by our stockholders in accordance with applicable law. Additionally, as described more fully below, neither the compensation committee nor the board of directors is permitted to reprice outstanding options or stock appreciation rights without shareholder consent.

Eligibility

Any of our employees, directors, and consultants, or those of our affiliates, are eligible to participate in the 2022 Plan and may be selected by the compensation committee to receive an award.

Vesting

The compensation committee determines the vesting conditions for awards. These conditions may include the continued employment or service of the participant, the attainment of specific individual or corporate performance goals, or other factors as determined in the compensation committee's discretion (collectively, "Vesting Conditions").

Shares of Stock Available for Issuance

Subject to certain adjustments, as of the date of this prospectus, the maximum number of shares of common stock that may be issued under the 2022 Plan in connection with awards is 1,910,000 shares, consisting of (i) the 810,000 shares of common stock reserved and available for issuance pursuant to the grant of new awards under our Initial Plan, and (ii) the 1,100,000 shares subject to outstanding stock options or other awards granted under our Initial Plan that on or after the 2022 Plan becomes effective, terminate or expire prior to exercise or settlement; are not issued because the award is settled in cash; are forfeited because of the failure to vest; or are reacquired or withheld (or not issued) to satisfy a tax withholding obligation or the purchase or exercise price, if any, as such shares become available from time to time. However, the number of shares reserved for future issuance under the 2022 Plan will be 760,000 shares as a result of the 50,000 shares that will be issued to an executive officer upon consummation of the offering. In addition, the maximum number of shares of common stock that may be issued under the 2022 Plan will automatically increase on January 1 of each calendar year for a period of ten years commencing on January 1, 2023 and ending on (and including) January 1, 2032, to a number of shares of common stock equal to 20% of the total number of shares of common stock outstanding on December 31 of the preceding calendar year; provided, however that the board of directors, or the compensation committee, may act prior to January 1 of a given calendar year to provide that the increase for such year will be a lesser number of shares of common stock. All available shares may be utilized toward the grant of any type of

award under the 2022 Plan. The 2022 Plan imposes a $100,000 limitation on the total grant date fair value with respect to which incentive stock options are exercisable for the first time by an individual optionee during any single calendar year.

In the event of any merger, consolidation, sale or disposition of all or substantially all our assets, sale or disposition of at least 50% of our outstanding securities, or other similar corporate transaction that affects our common stock, the board of directors or compensation committee shall make adjustments to the number and kind of shares authorized by the 2022 Plan and covered under outstanding 2022 Plan awards as it determines appropriate and equitable.

Shares subject to 2022 Plan awards that expire without being fully exercised or that are otherwise forfeited, cancelled or terminated may again be made available for issuance under the 2022 Plan. In addition, shares withheld in settlement of a tax withholding obligation, or in satisfaction of the exercise price payable upon exercise of an option, will again become available for issuance under the 2022 Plan.

Types of Awards

The following types of awards may be granted to participants under the 2022 Plan: (i) incentive stock options, or ISOs; (ii) nonqualified stock options, or NQOs and together with ISOs, options, (iii) stock appreciation rights, (iv) restricted stock, or (v) restricted stock units.

Stock Options. An option entitles the holder to purchase from us a stated number of shares of common stock. An ISO may only be granted to an employee of ours or our eligible affiliates. The compensation committee will specify the number of shares of common stock subject to each option and the exercise price for such option, provided that the exercise price may not be less than the fair market value of a share of common stock on the date the option is granted. Notwithstanding the foregoing, if ISOs are granted to any 10% stockholder, the exercise price shall not be less than 110% of the fair market value of common stock on the date the option is granted.

Generally, options may be exercised in whole or in part through a cash payment. The compensation committee may, in its sole discretion, permit payment of the exercise price of an option pursuant to a “cashless exercise,” in the form of previously owned shares of common stock based on the fair market value of the shares on the date the option is exercised, or through means of “net settlement,” which involves the cancellation of a portion of the option to cover the cost of exercising the balance of the option or by such other means as it deems acceptable.

All options shall be or become exercisable in accordance with the terms of the applicable award agreement. The maximum term of an option shall be determined by the compensation committee on the date of grant but shall not exceed 10 years (5 years in the case of ISOs granted to any 10% stockholder). In the case of ISOs, the aggregate fair market value (determined as of the date of grant) of common stock with respect to which such ISOs become exercisable for the first time during any calendar year cannot exceed $100,000. ISOs granted in excess of this limitation will be treated as non-qualified stock options.

Stock Appreciation Rights. A stock appreciation right represents the right to receive, upon exercise, any appreciation in a share of common stock over a particular time period. The base price of a stock appreciation right shall not be less than the fair market value of a share of common stock on the date the stock appreciation right is granted. This award is intended to mirror the benefit the participant would have received if the compensation committee had granted the participant an option. The maximum term of a stock appreciation right shall be determined by the compensation committee on the date of grant but shall not exceed 10 years. Distributions with respect to stock appreciation rights may be made in cash, shares of common stock, or a combination of both, at the board of director’s discretion.

Unless otherwise provided in an award agreement or determined by the compensation committee, if a participant terminates employment with us (or our affiliates) for any reason other than cause, the participant may exercise his or her unexercised options and stock appreciation rights, to the extent they were exercisable on the termination date, within the following period of time, provided however that in no event may any award be exercised after termination of its maximum term: (i) three months following the date of such termination if such termination is a termination without cause (other than any termination due to the participant’s disability or death); (ii) 12 months following the date of such termination if such termination is due to the participant’s disability; (iii) 18 months following the date of such termination if such termination is due to the participant’s death; or (iv) 18 months following the date of the participant’s death if such death occurs following the date of such termination but during the period such Award is otherwise exercisable. If the participant terminates employment with us (or our affiliates) for cause, all unexercised options and stock appreciation rights (whether

vested or unvested) shall terminate and be forfeited on the termination date. Unless otherwise provided by the compensation committee, any options and stock appreciation rights that are not exercisable at the time of termination of employment shall terminate and be forfeited on the termination date.

Restricted Stock. A restricted stock award is a grant of shares of common stock, which are subject to forfeiture restrictions during a restriction period. The compensation committee will determine the price, if any, to be paid by the participant for each share of common stock subject to a restricted stock award. The restricted stock may be subject to Vesting Conditions. If the specified Vesting Conditions are not attained, the participant will forfeit the portion of the restricted stock award with respect to which those conditions are not attained, and the underlying common stock will be forfeited to us. At the end of the restriction period, if the Vesting Conditions have been satisfied, the restrictions imposed will lapse with respect to the applicable number of shares. Unless otherwise provided in an award agreement or determined by the compensation committee, upon termination a participant will forfeit all restricted stock that then remains subject to forfeiture restrictions.

Restricted Stock Units. Restricted stock units are granted in reference to a specified number of shares of common stock and entitle the holder to receive, on the achievement of applicable Vesting Conditions, shares of common stock. Unless otherwise provided in an award agreement or determined by the compensation committee, upon termination a participant will forfeit all restricted stock units that then remain subject to forfeiture.

Change of Control

In the event of a change of control, unless otherwise provided in a grant agreement, employment agreement or other agreement between the Company and the participant, and unless otherwise determined by an affirmative vote of a majority of the board of directors prior to the occurrence of such change of control: (i) the vesting and settlement of all outstanding awards to non-employee directors will be automatically accelerated and the shares immediately issued to the participant (or the Board may direct the payment of a cash settlement equal to the fair market value of the shares that would otherwise be issued to the participant); (ii) the settlement of vested awards to employees and consultants will automatically be accelerated and the shares immediately issued to the participant; and (iii) unvested awards to employees and consultants shall be terminated and forfeited unless the acquiring entity assumes, continues or substitutes any such awards. Notwithstanding the foregoing, to the extent permitted and in compliance with the requirements of Section 409A of the Code, the Board may in its discretion determine to elect to accelerate the vesting and settlement of the unvested awards to employees and consultants upon a Change of Control, or direct the payment of a cash settlement equal to the fair market value of the shares that would otherwise be issued to the participant).

Repricing

Neither our board of directors nor the compensation committee may reduce the exercise price in effect for outstanding options under the 2022 Plan without obtaining the consent of any participant whose award would be materially impaired by such action.

Miscellaneous

Generally, awards granted under the 2022 Plan shall be nontransferable except by will or by the laws of descent and distribution. No participant shall have any rights as a stockholder with respect to shares covered by options or restricted stock units, unless and until such awards are settled in shares of common stock. Our obligation to issue shares or to otherwise make payments in respect of 2022 Plan awards will be conditioned on our ability to do so in compliance with all applicable laws and exchange listing requirements. The awards will be subject to our recoupment and stock ownership policies, as may be in effect from time to time. The 2022 Plan expires 10 years after it becomes effective.

CERTAIN RELATIONSHIPS AND RELATED PARTY TRANSACTIONS

Each of the related party transactions described below was negotiated on an arm's length basis. We believe that the terms of such agreements are as favorable as those we could have obtained from parties not related to us. The following are summaries of certain provisions of our related party agreements and are qualified in their entirety by reference to all of the provisions of such agreements. Because these descriptions are only summaries of the applicable agreements, they do not necessarily contain all of the information that you may find useful. We therefore urge you to review the agreements in their entirety. Copies of the forms of the agreements have been filed as exhibits to the registration statement of which this prospectus is a part and are available electronically on the website of the SEC at *www.sec.gov*.

In addition to the compensation arrangements, including employment, termination of employment and change in control arrangements, with our directors and executive officers, including those discussed in the sections titled "Management" and "Executive Compensation," the following is a description of each transaction since January 25, 2022 or any currently proposed transaction in which:

- we have been or are to be a party to;
- the amount involved exceeded or exceeds $120,000 or 1% of the average of our total assets as of the end of the last two completed fiscal years; and
- any of our directors, executive officers or holders of more than 5% of our outstanding capital stock, or any immediate family member of, or person sharing the household with, any of these individuals or entities, had or will have a direct or indirect material interest.

For information on our compensation arrangements, including employment, termination of employment and change in control arrangements, with our directors and executive officers, see the sections titled "Management" and "Executive Compensation."

On January 25, 2022, we entered into an agreement with Phamace, LLC, a consulting firm of which Quang Pham, our Chief Executive Officer, is the sole member, for an initial term of January 25, 2022 through February 28, 2022 to provide advisory and administrative services relating to preparing the Company to launch as an operating company. Pursuant to the agreement, the Company shall pay the sum of $115,000 to Phamace, LLC for services rendered, which is due and payable on September 30, 2022.

On January 25, 2022, we issued 7,500,000 shares of common stock, pursuant to a subscription agreement, to Quang Pham, our Chief Executive Officer. Mr. Pham paid a total of $7,500 for such founders shares.

On March 1, 2022, we issued a convertible promissory note in the amount of $500,000 to John Murphy, a director nominee, which bears interest at 5% and matures on March 1, 2025. The note is convertible, as amended in December 2022, into shares of Common Stock at a conversion price equal to $1.00.

On May 17, 2022, we issued 450,000 shares of restricted common stock, pursuant to a restricted stock purchase agreement, to Matthew Szot, our Acting Chief Financial Officer, which shares shall vest quarterly over a period of two (2) years, subject to certain adjustments, as provided in the Restricted Stock Purchase Agreement dated May 17, 2022.

On July 11, 2022, we issued stock options to purchase an aggregate of 150,000 shares of common stock to two (2) director nominees. The options have an at an exercise price of $0.64 per common share and will vest over a period of three (3) years, with 33.33% vesting on the first anniversary of the effective date, and thereafter 1/36 on a monthly basis.

On August 18, 2022, we issued stock options to purchase 300,000 shares of common stock to Douglas Losordo, our Acting Chief Medical Officer. The options have an at an exercise price of $0.64 per common share and will vest quarterly over a period of three (3) years, subject to certain adjustments.

On August 18, 2022, we issued 40,000 shares of common stock to Steven Zelenkofske, a director nominee, for recruiting services performed.

On August 22, 2022, we issued a convertible promissory note in the amount of $50,000 to Glynn Wilson, a director nominee, which bears interest at 6% and matures on September 13, 2025. The note is convertible into shares of Common Stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal

amount of the note plus (if applicable) any accrued but unpaid interest under the Note by (ii) sixty percent (60%) of the price per share of the equity securities issued and sold at the initial closing of our next equity financing.

On September 16, 2022, we issued 50,000 shares of common stock to John Murphy, a director nominee, and agreed to issue 50,000 shares of common stock to Matthew Szot, our Acting Chief Financial Officer, upon consummation of this offering.

On October 3, 2022, the Company issued stock options to purchase 50,000 shares of common stock to Glynn Wilson, a director nominee. The options have an exercise price of $5.00 per common share and will vest over a three-year period, with 33.33% vesting on the first anniversary of the effective date, and thereafter 1/36 on a monthly basis.

Indemnification Agreements

We have entered into separate indemnification agreements with each of our current executive officers and intend to enter into separate indemnification agreements with each of our directors and executive officers, in addition to the indemnification that will be provided for in our amended and restated bylaws. The indemnification agreements and our amended restated bylaws require us to indemnify our directors and executive officers to the fullest extent permitted by Delaware law. See the section titled "Description of Securities — Limitations on Liability and Indemnification of Officers and Directors" for additional information.

Our Policy Regarding Related Party Transactions

Our board of directors recognizes the fact that transactions with related persons present a heightened risk of conflicts of interest and/or improper valuation (or the perception thereof). Prior to the closing of this offering, our board of directors will adopt a written policy on transactions with related persons that is in conformity with the requirements for issuers having publicly held common stock that is listed on the Nasdaq Stock Market. Under the new policy:

- any related person transaction, and any material amendment or modification to a related person transaction, must be reviewed and approved or ratified by the Audit Committee; and
- any employment relationship or transaction involving an executive officer and any related compensation must be approved by the compensation committee of the board of directors or recommended by the compensation committee to the board of directors for its approval.

In connection with the review and approval or ratification of a related person transaction:

- management must disclose to the committee or disinterested directors, as applicable, the name of the related person and the basis on which the person is a related person, the material terms of the related person transaction, including the approximate dollar value of the amount involved in the transaction, and all the material facts as to the related person's direct or indirect interest in, or relationship to, the related person transaction;
- management must advise the committee or disinterested directors, as applicable, as to whether the related person transaction complies with the terms of our agreements governing our material outstanding indebtedness that limit or restrict our ability to enter into a related person transaction;
- management must advise the committee or disinterested directors, as applicable, as to whether the related person transaction will be required to be disclosed in our applicable filings under the Securities Act or the Exchange Act, and related rules, and, to the extent required to be disclosed, management must ensure that the related person transaction is disclosed in accordance with the Securities Act and the Exchange Act and related rules; and
- management must advise the committee or disinterested directors, as applicable, as to whether the related person transaction constitutes a "personal loan" for purposes of Section 402 of the Sarbanes-Oxley Act.

In addition, the related person transaction policy provides that the committee or disinterested directors, as applicable, in connection with any approval or ratification of a related person transaction involving a non-employee director, should consider whether such transaction would compromise the director's status as an "independent," "outside," or "non-employee" director, as applicable, under the rules and regulations of the SEC, the Nasdaq Stock Market, and the Code.

PRINCIPAL STOCKHOLDERS

The following table sets forth the beneficial ownership of our common stock as of January 17, 2023, by:

- each person, or group of affiliated persons, who is known by us to beneficially own more than 5% of our common stock;
- each of the named executive officers;
- each of our directors and director nominees; and
- all of our current executive officers and directors as a group

As of January 17, 2023, we had 8,193,875 shares of common stock outstanding, held by approximately eighteen (18) stockholders of record.

We have determined beneficial ownership in accordance with the rules of the SEC, and thus it represents sole or shared voting or investment power with respect to our securities. Unless otherwise indicated below, to our knowledge, the persons and entities named in the table have sole voting and sole investment power with respect to all shares that they beneficially owned, subject to community property laws where applicable. The information does not necessarily indicate beneficial ownership for any other purpose, including for purposes of Sections 13(d) and 13(g) of the Exchange Act.

We have based our calculation of the percentage of beneficial ownership prior to this offering on 10,234,575 shares of our common stock outstanding as of January 17, 2023, which gives effect to: (i) the issuance of 1,140,700 shares of common stock upon conversion of the Private Placement Notes, the March 2022 Note, the June 2022 Note, (ii) the issuance of 600,000 shares of common stock to HESP LLC pursuant to the terms of the Amendment to Asset Purchase Agreement; (iii) the issuance of 67,000 shares of our common stock to consultants subsequent to September 30, 2022 for the provision of services; and (iv) the issuance of 50,000 shares of common stock to an executive officer upon consummation of this offering and (v) the issuance of 250,000 shares of common stock upon the exercise of the November Warrants. We have based our calculation of the percentage of beneficial ownership after this offering on 11,634,575 shares of our common stock outstanding immediately after the completion of this offering, assuming no exercise by the underwriters of their option to purchase additional shares and based on the initial public offering price of $5.00 per share. We have deemed shares of our common stock subject to securities that are currently convertible or exercisable into shares of common stock, or convertible or exercisable into shares of our common stock within 60 days of January 17, 2023, to be outstanding and to be beneficially owned by the person holding the stock option for the purpose of computing the percentage ownership of that person. We did not deem these shares outstanding, however, for the purpose of computing the percentage ownership of any other person. Unless otherwise indicated, the address of each beneficial owner listed in the table below is c/o Cadrenal Therapeutics, Inc., 822 A1A North, Suite 306, Ponte Vedra, Florida 32082.

	Shares Beneficially Owned Prior to this Offering		Shares Beneficially Owned After this Offering	
Name of Beneficial Owner	**Shares**	**Percentage**	**Shares**	**Percentage**
Named Executive Officers and Continuing Directors and Director Nominees				
Quang Pham	6,275,000[1]	61.31%	6,275,000	53.93%
Matthew Szot	500,000[2]	4.89%	500,000	4.30%
Douglas Losordo	—[3]	—	—	—
John Murphy	564,792[4]	5.52%	564,792	4.85%
Steven Zelenkofske	40,000[5]	*	40,000	*
Glynn Wilson	50,000[6]	*	50,000	*
All executive officers, continuing directors and director nominees as a group (6 persons)	7,429,792	72.60%	7,429,792	63.86%

5% Stockholders other than executive officers and directors				
The PVBQ Living Trust	3,000,000[1]	29.31%	3,000,000	25.79%
HESP LLC	600,000[7]	5.86%	600,000	5.16%

* Represents beneficial ownership of less than one percent.

(1) Includes (i) 3,275,000 shares of common stock owned by Quang Pham; and (ii) 3,000,000 shares of common stock owned by The PVBQ Living Trust. The beneficiary of The PVBQ Living Trust (the "Trust") is Mr. Pham's child and Mr. Pham is the trustee of the Trust and has sole voting and disposition power with respect to the shares owned by the Trust. The address for the Trust is 822 A1A North, Suite 306, Ponte Vedra, Florida 32082.

(2) Consists of 450,000 shares of restricted common stock, which shares shall vest quarterly over a period of two (2) years, subject to certain adjustments and 50,000 shares issuable upon consummation of the offering.

(3) Does not include a grant of an option to purchase 300,000 shares of common stock that was granted to Dr. Losordo on August 18, 2022.

(4) Includes (i) 50,000 shares of common stock; and (ii) 514,792 shares of common stock issuable upon the conversion of a convertible note with the Company held by Mr. Murphy, based on an as adjusted conversion price of $1.00 per share. Does not include a grant of an option to purchase 100,000 shares of common stock that was granted to Mr. Murphy on July 11, 2022.

(5) Does not include a grant of an option to purchase 50,000 shares of common stock that was granted to Dr. Zelenkofske on July 11, 2022.

(6) Dr. Wilson is the holder of a convertible note with the Company that is convertible into 50,000 shares of common stock at an as adjusted conversion price equal to $1.00 per share. Does not include a grant of an option to purchase 50,000 shares of common stock that was granted to Dr. Wilson on October 3, 2022.

(7) On the effective date of this offering, HESP LLC will be issued 600,000 shares of our common stock, based on a $3.00 price per share, which is sixty percent (60%) of the public offering price of $5.00, pursuant to the terms of the Amendment to Asset Purchase Agreement executed on August 18, 2022. Daniel Devorsetz, Chief Operating Officer of Horizon Technology Finance Corporation ("Horizon"), the sole member of HESP LLC, has voting power over the shares that will be issued to HESP LLC and an Investment Committee of Horizon, of which Daniel Devorsetz is one of several members, has investment power over such shares. The address for HESP, LLC is 312 Farmington Avenue, Farmington, CT 06032.

DESCRIPTION OF SECURITIES

The following description of our securities and the provisions of our amended and restated certificate of incorporation and our amended and restated bylaws are summaries and are qualified by reference to the amended and restated certificate of incorporation and the amended and restated bylaws. We have filed copies of these documents with the SEC as exhibits to our registration statement of which this prospectus forms a part.

General

On December 5, 2022, we filed an amended and restated certificate of incorporation with the Secretary of State of the State of Delaware, pursuant to which our authorized capital stock consists of:

- 75,000,000 shares of common stock, par value $0.001 per share; and
- 7,500,000 shares of preferred stock, par value $0.001 per share.

We are selling shares of common stock in this offering based on the initial public offering price of $5.00 per share. All of our common stock outstanding upon consummation of this offering will be fully paid and non-assessable.

The following description of our capital stock and provisions of our amended and restated certificate of incorporation and amended and restated bylaws, both of which became effective on December 5, 2022, are summaries of material terms and provisions and are qualified by reference to our amended and restated certificate of incorporation and amended and restated bylaws, copies of which have been filed with the SEC as exhibits to the registration statement of which this prospectus is a part. The descriptions of our common stock and preferred stock reflect the content of the amended and restated certificate of incorporation and amended and restated bylaws.

Certain provisions of our amended and restated certificate of incorporation and our amended and restated bylaws summarized below, and under Delaware law, may be deemed to have an anti-takeover effect and may delay or prevent a tender offer or takeover attempt that a stockholder might consider in its best interest, including those attempts that might result in a premium over the market price for the shares of common stock.

Common Stock

We are authorized to issue one class of common stock. Holders of our common stock are entitled to one vote for each share of common stock held of record for the election of our directors and all other matters requiring stockholder action, except with respect to amendments to our certificate of incorporation that alter or change the powers, preferences, rights or other terms of any outstanding preferred stock if the holders of such affected series of preferred stock are entitled to vote on such an amendment. Our directors are elected by a plurality of the votes cast by the stockholders entitled to vote at our annual meeting of stockholders, and are not entitled to cumulative voting rights. Holders of common stock are entitled to receive such dividends, if any, as may be declared from time to time by our Board of Directors in its discretion out of funds legally available therefor. The payment of dividends, if any, on shares of our common stock are subject to the prior payment of dividends on any outstanding preferred stock, of which there is none as of the date of this offering. Upon our liquidation or dissolution, the holders of our common stock are entitled to receive a pro rata portion of all assets remaining available for distribution to stockholders after payment of all liabilities and provision for the liquidation of any shares of preferred stock outstanding at that time. The holders of our common stock have no preemptive, subscription or redemption rights, and have no rights to convert their common stock into any other securities. The absence of preemptive rights could result in a dilution of the interest of the existing stockholders should additional shares of our common stock be issued. In addition, the rights of holders of our common stock are subject to, and may be adversely affected by, the rights of holders of shares of any series of preferred stock that we may designate and issue in the future.

Preferred Stock

No shares of preferred stock are outstanding as of the date of this offering. Our Board of Directors is authorized, without action by our stockholders, to designate and issue up to 7,500,000 shares of preferred stock in one or more series. Our Board of Directors has the right to fix the voting rights, if any, designations, powers, preferences, the relative, participating, optional or other special rights, if any, and any qualifications, limitations and restrictions thereof, applicable to the shares of each series of designated preferred stock. Our Board of

Directors is able to, without stockholder approval, issue shares of preferred stock with voting and other rights that could adversely affect the voting power and other rights of the holders of our common stock and could have anti-takeover effects. The ability of our

Board of Directors to issue preferred stock without stockholder approval could have the effect of delaying, deferring or preventing a change of control of us or the removal of existing management. Although we do not currently intend to issue any shares of preferred stock, we cannot assure you that we will not do so in the future.

Convertible Notes

On March 1, 2022, we issued a convertible promissory note in the amount of $500,000 (the "March 2022 Note"). The March 2022 Note bears interest at 5% and matures on March 1, 2025 ("March Note Maturity Date"). The principal amount due under the March 2022 Note (and, at the Company's option, any accrued but unpaid interest under the March 2022 Note) will be automatically converted, on or before the March Note Maturity Date, into equity securities issued and sold at the initial closing of our next equity financing ("Next Equity Securities") in a single transaction or a series of related transactions yielding gross proceeds to us of at least $3,000,000 (the "Next Equity Financing"). The March 2022 Note was initially convertible into shares of common stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the March 2022 Note plus (if applicable) any accrued but unpaid interest under the March 2022 Note by (ii) eighty percent (80%) of the price per share of the Next Equity Securities sold in the Next Equity Financing. In the event of a Change of Control which occurs prior to repayment in full of the March 2022 Note, immediately prior to the Change of Control, the outstanding principal and any accrued but unpaid interest on the March 2022 Note will convert directly into our common equity (or directly into proceeds paid to the holders of our common equity in connection with the Change of Control) at a price per share that is 80% of the price per share of common equity paid at the Change of Control. In December 2022, we amended the conversion price of the March 2022 Note to $1.00 per share.

On June 13, 2022, we issued a convertible promissory note in the amount of $50,000 (the "June 2022 Note"). The June 2022 Note bears interest at 6% and matures on June 13, 2025 ("June Note Maturity Date"). The principal amount due under the June 2022 Note (and, at the Company's option, any accrued but unpaid interest under the June 2022 Note) will be automatically converted, on or before the June Note Maturity Date, into Next Equity Securities in the Next Equity Financing. The June 2022 Note was initially convertible into shares of common stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the June 2022 Note plus (if applicable) any accrued but unpaid interest under the June 2022 Note by (ii) sixty percent (60%) of the price per share of the Next Equity Securities sold in the Next Equity Financing. In the event of a Change of Control which occurs prior to repayment in full of the June 2022 Note, immediately prior to the Change of Control, the outstanding principal and any accrued but unpaid interest on the June 2022 Note will convert directly into our common equity (or directly into proceeds paid to the holders of our common equity in connection with the Change of Control) at a price per share that is 60% of the price per share of common equity paid at the Change of Control. The March 2022 Note and the June 2022 Note (collectively, the "Notes") may be prepaid at any time without penalty. For the purposes of the Notes, the term "Change of Control" means (i) a sale of all or substantially all of our assets other than to an Excluded Entity (as defined), (ii) a merger, consolidation or other capital reorganization or business combination transaction of us with or into another corporation, limited liability company or other entity other than an Excluded Entity, or (iii) the consummation of a transaction, or series of related transactions, in which any "person" (as such term is used in Sections 13(d) and 14(d) of the Exchange Act)) becomes the "beneficial owner" (as defined in Rule 13d-3 of the Exchange Act), directly or indirectly, of all of our then outstanding voting securities. In December 2022, we amended the conversion price of the June 2022 Note to $1.00 per share.

On July 16, 2022, we closed the first tranche of the Private Placement pursuant to which we issued First Tranche Private Placement Notes in the principal amount of $450,000 and received gross proceeds of $450,000. The private placement offering was for up to $2,000,000 of convertible promissory notes and was to terminate upon the sale of all $2,000,000 of convertible promissory notes or July 31, 2022 subject to extension until August 31, 2022, in the sole discretion of the placement agent and us. We, together with the placement agent, determined to extend the Private Placement to August 31, 2022. On August 22, 2022, we issued the Second Tranche Private Placement Note in the principal amount of $50,000 and received gross proceeds of $50,000. The Private Placement was further extended to September 30, 2022, after obtaining the consent of the investors in the Private Placement to date. On September 13, 2022, we issued convertible promissory notes in the aggregate principal amount of $75,000 (together with the First Tranche Private Placement Notes and the Second Tranche Private Placement Note, the Private Placement Notes) and received gross proceeds of $75,000. The Private Placement Notes bear interest at 6% and mature on the third anniversary of the last closing of the Private Placement Notes, i.e. September 13, 2025. The principal amount due under the Private Placement Notes

(and, at our option, any accrued but unpaid interest under the Private Placement Notes) will be automatically converted into shares of our common stock upon the consummation of this offering. The Private Placement Notes were initially convertible into shares of common stock at

a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the Private Placement Notes plus (if applicable) any accrued but unpaid interest under the Private Placement Notes by (ii) sixty percent (60%) of the initial offering price per share subject to adjustment in the event of a subsequent issuance by us of securities at a price per share less than the conversion price. Upon the issuance of the November Notes and November Warrants the conversion price was reduced to $1.00 per share. The holders of the Private Placement Notes entered into an Investor Rights Agreement that provides for "drag along" and "tag along" rights and permits the holders to participate in subsequent securities offerings, including this offering, in a percentage amount of such securities offering equal to the percentage of the amount invested by such holder in the Private Placement Notes as compared to the amount invested by all holders in the Private Placement Notes; provided, however, that in no event shall the dollar amount of the participation right of any holder exceed 100% of original principal amount of the Private Placement Notes purchased by such holder. The holder is prohibited from converting the Private Placement Notes if such conversion would result in the holder owning in excess of 4.99% of our outstanding common stock.

Upon completion of this offering, the holder of the March 2022 Note will convert such Note into 514,792 shares of our common stock, the holder of the June 2022 Note will convert such Note into 50,908 shares of our common stock, and the holders of the Private Placement Notes will convert such Notes into an aggregate of 575,000 shares of our common stock. The holders of the Notes and the Private Placement Notes have agreed not to publicly sell or assign such common stock for a period of six months following completion of this offering.

Boustead Securities, LLC acted as the placement agent for the Private Placement and received a commission equal to 7% of the gross proceeds received by us from the sale of the Private Placement Notes, a non-accountable expense allowance equal to 1% of the gross proceeds received by us from the sale of the Private Placement Notes and five-year warrants, as amended, to purchase an aggregate of 11,500 shares of our common stock at an exercise price equal to 60% of the initial public offering price per share.

Non-Convertible Promissory Notes

On November 30 2022, we closed the November Private Placement pursuant to which we sold to two accredited investors, both of whom had a pre-existing relationship with us and Boustead Securities, LLC, units consisting of (i) November Notes in the principal amount of $250,000, which bear interest at the rate of 10%, repayable at the time of the completion of this offering and (ii) November Warrants to purchase up to 250,000 shares of common stock exercisable at $1.00 per share which are automatically exercisable into shares of our common stock upon the consummation of this offering. The principal amount of the November Notes will be repaid upon the consummation of this offering out of the proceeds of the automatic exercise of the November Warrants, with any interest owed on the November Notes to be payable in cash.

Boustead Securities, LLC acted as the placement agent for the November Private Placement and received a commission equal to 7% of the gross proceeds received by us from the sale of the November Notes, a non-accountable expense allowance equal to 1% of the gross proceeds received by us from the sale of the November Notes and five-year warrants to purchase shares of our common stock in an amount equal to 6% of the shares of common stock underlying the securities issued in the November Private Placement at an exercise price of $1.00 per share.

Representative's Warrants

We have agreed to issue to the Representative warrants to purchase up to 84,000 shares of our common stock as a portion of the underwriting compensation payable to the underwriters in connection with this offering. The warrants will be exercisable, at an exercise price equal to 120% of the initial public offering price, for five years; however such warrants may not be sold, transferred, assigned, pledged or hypothecated for a period commencing 180 days following the commencement of sales in this offering; provided, however, that the warrants (and the underlying shares) may be transferred to the underwriters' officers, partners, registered persons or affiliates as long as the warrants (and the underlying shares) remain subject to the lockup. We have agreed to a one-time demand registration of the shares of common stock underlying the underwriter's warrants for a period of five years from the effective date of the registration statement. The underwriter's warrants also provide for immediate "piggyback" registration rights with respect to the underlying shares of common stock during the three-year period commencing from the effective date of the registration statement related to this offering. Such piggyback rights shall expire on a date which shall be three years from the date of

commencement of sales of the share offered hereby. The warrants are not redeemable by us. The warrants and the shares of common stock issuable upon exercise of the warrants have been included in the registration statement of which this prospectus forms a part.

Forum Selection

Our amended and restated certificate of incorporation and amended and restated bylaws provide that unless we consent in writing to the selection of an alternative forum, the Court of Chancery of the State of Delaware (or, in the event that the Court of Chancery does not have jurisdiction, the federal district court for the District of Delaware or other state courts of the State of Delaware) is the exclusive forum for (i) any derivative action or proceeding brought on our behalf; (ii) any action asserting a claim of breach of fiduciary duty owed by any director, officer, employee or agent of the Company to the Company or our stockholders; (iii) any action asserting a claim arising pursuant to the provisions of the Delaware General Corporation Law, our amended and restated certificate of incorporation or our amended and restated bylaws; or (iv) any action asserting a claim against us that is governed by the internal affairs doctrine of the State of Delaware; provided that, if and only if the Court of Chancery of the State of Delaware dismisses any such action for lack of subject matter jurisdiction, or the Company consents in writing to the selection of an alternative forum, such action may be brought in another state or federal court sitting in the State of Delaware. Our amended and restated certificate of incorporation and amended and restated bylaws also provide that the federal district courts of the United States of America is the exclusive forum for the resolution of any complaint asserting a cause of action against under the Securities Act. Notwithstanding the foregoing, the exclusive forum provision does not apply to suits brought to enforce any liability or duty created by the Exchange Act or any other claim for which the federal courts have exclusive jurisdiction. Nothing in our amended and restated certificate of incorporation or amended and restated bylaws preclude stockholders that assert claims under the Exchange Act from bringing such claims in state or federal court, subject to applicable law.

Anti-Takeover Provisions

Our amended and restated certificate of incorporation and amended and restated bylaws contain provisions that may delay, defer, or discourage another party from acquiring control of us. We expect that these provisions, which are summarized below, will discourage coercive takeover practices or inadequate takeover bids. These provisions are also designed to encourage persons seeking to acquire control of us to first negotiate with our board of directors, which we believe may result in an improvement of the terms of any such acquisition in favor of our stockholders. However, they also give our board of directors the power to discourage acquisitions that some stockholders may favor.

Section 203 of the Delaware General Corporation Law

We are subject to Section 203 of the Delaware General Corporation Law. Subject to certain exceptions, Section 203 prevents a publicly held Delaware corporation from engaging in a “business combination” with any “interested stockholder” for three years following the date that the person became an interested stockholder, unless the interested stockholder attained such status with the approval of our board of directors or unless the business combination is approved in a prescribed manner. A “business combination” includes, among other things, a merger or consolidation involving us and the “interested stockholder” and the sale of more than 10% of our assets. In general, an “interested stockholder” is any entity or person beneficially owning 15% or more of our outstanding voting stock and any entity or person affiliated with or controlling or controlled by such entity or person.

Classified Board of Directors

Our board of directors is divided into three classes serving three-year terms, with one class being elected each year by a plurality of the votes cast by the stockholders entitled to vote on the election.

Proposals of Business and Nominations

Our amended and restated bylaws generally regulates proposals of business and nominations for election of directors by stockholders. In general, Section 5 requires stockholders intending to submit proposals or nominations at a stockholders’ meeting to provide the Company with advance notice thereof, including information regarding the stockholder proposing the business or nomination as well as information regarding the proposed business or nominee. Section 5 also provides a time period during which business or nominations must be provided to the Company that creates a predictable window for the submission of such notices, eliminating the risk that the Company finds a meeting will be contested after printing its proxy materials for an uncontested election and providing the Company with a reasonable opportunity to respond to nominations and proposals by stockholders.

Blank Check Preferred Stock

Our board of directors has the right to issue preferred stock in one or more series and to determine the designations, rights, preferences of such preferred stock without stockholder approval.

Board Vacancies

Our amended and restated bylaws generally provide that only our board of directors (and not the stockholders) may fill vacancies and newly created directorships.

Stockholder Action by Written Consent

Our amended and restated certificate of incorporation and our amended and restated bylaws provide that any action required or permitted to be taken by our stockholders at an annual meeting or special meeting of stockholders may only be taken if it is properly brought before such meeting.

Special Meetings of Stockholders

Our amended and restated certificate of incorporation and amended and restated bylaws also provide that, except as otherwise required by law, special meetings of the stockholders may only be called by our board of directors, Chairman of the board of directors, or our Chief Executive Officer.

Amendment of Certificate of Incorporation or Bylaws

The Delaware General Corporation Law provides generally that the affirmative vote of a majority of the shares entitled to vote on any matter is required to amend a corporation's certificate of incorporation or bylaws, unless a corporation's certificate of incorporation or bylaws, as the case may be, requires a greater percentage. Our bylaws may be amended or repealed by a majority vote of our board of directors or by the affirmative vote of the holders of at least 66 2/3% of the votes which all our stockholders would be eligible to cast in an election of directors.

Limitations on Liability and Indemnification of Officers and Directors

Our amended and restated bylaws provide indemnification for our directors and executive officers to the fullest extent permitted by the Delaware General Corporation Law. The indemnification agreements that we have entered into with each of our current executive officers and that we intend to enter into with each of our directors and executive officers may, in some cases, be broader than the specific indemnification provisions contained under Delaware law. In addition, as permitted by Delaware law, our amended and restated certificate of incorporation includes provisions that eliminate the personal liability of our directors and officers for monetary damages resulting from breaches of certain fiduciary duties as a director or officer, as applicable, except to the extent such an exemption from liability thereof is not permitted under the Delaware General Corporation Law. The effect of these provisions is to restrict our rights and the rights of our stockholders in derivative suits to recover monetary damages against a director or officer for breach of fiduciary duties as a director or officer, subject to certain exceptions in which case the director or officer would be personally liable. An officer may not be exculpated for any action brought by or in the right of the corporation. A director may not be exculpated for improper distributions to stockholders. Further, pursuant to Delaware law a director or officer may not be exculpated for:

- any breach of his or her duty of loyalty to us or our stockholders;
- acts or omissions not in good faith or which involve intentional misconduct or a knowing violation of law; and
- any transaction from which the director or officer derived an improper personal benefit.

These limitations of liability do not apply to liabilities arising under the federal or state securities laws and do not affect the availability of equitable remedies such as injunctive relief or rescission.

Our amended and restated bylaws provide that we will indemnify our directors and executive officers to the fullest extent permitted by law, and may indemnify other officers, employees and other agents. Our amended and restated bylaws also provide that we are obligated to advance expenses incurred by a director or executive officer in advance of the final disposition of any action or proceeding.

We have entered into separate indemnification agreements with each of our current executive officers and plan to enter into separate indemnification agreements with our directors and executive officers. These agreements, among other things, require us to indemnify our directors and officers for any and all expenses (including reasonable attorneys' fees, retainers, court costs, transcript costs, fees of experts, witness fees, travel expenses, duplicating costs, printing and binding costs, telephone charges, postage, delivery service fees) judgments, fines and amounts paid in settlement actually and reasonably incurred by such directors or officers or on his or her behalf in connection with any action or proceeding arising out of their services as one of our directors or officers, or any of our subsidiaries or any other company or enterprise to which the person provides services at our request provided that such person follows the procedures for determining entitlement to indemnification and advancement of expenses set forth in the indemnification agreement. We believe that these bylaw provisions and indemnification agreements are necessary to attract and retain qualified persons as directors and officers.

The limitation of liability and indemnification provisions in our amended and restated certificate of incorporation and amended and restated bylaws may discourage stockholders from bringing a lawsuit against directors for breach of their fiduciary duties. They may also reduce the likelihood of derivative litigation against directors and officers, even though an action, if successful, might provide a benefit to us and our stockholders. Our results of operations and financial condition may be harmed to the extent we pay the costs of settlement and damage awards against directors and officers pursuant to these indemnification provisions.

Insofar as indemnification for liabilities arising under the Securities Act may be permitted to directors, officers or persons controlling us, we have been informed that, in the opinion of the SEC, such indemnification is against public policy as expressed in the Securities Act and is therefore unenforceable.

At present, there is no pending litigation or proceeding involving any of our directors or officers as to which indemnification is required or permitted, and we are not aware of any threatened litigation or proceeding that may result in a claim for indemnification.

Dissenters' Rights of Appraisal and Payment

Under the Delaware General Corporation Law, with certain exceptions, our stockholders will have appraisal rights in connection with a merger or consolidation of Cadrenal Therapeutics, Inc. Pursuant to the Delaware General Corporation Law, stockholders who properly request and perfect appraisal rights in connection with such merger or consolidation will have the right to receive payment of the fair value of their shares as determined by the Delaware Court of Chancery.

Stockholders' Derivative Actions

Under the Delaware General Corporation Law, any of our stockholders may bring an action in our name to procure a judgment in our favor, also known as a derivative action, provided that the stockholder bringing the action is a holder of our shares at the time of the transaction to which the action relates or such stockholder's stock thereafter devolved by operation of law.

Transfer Agent and Registrar

The transfer agent and registrar for our common stock is Transfer Online, Inc., 512 SE Salmon Street, Portland, OR 97214.

Trading Symbol and Market

Our common stock has been approved for listing on The Nasdaq Capital Market under the symbol "CVKD" and will begin trading on January 20, 2023.

SHARES ELIGIBLE FOR FUTURE SALE

If our stockholders sell substantial amounts of our common stock, including shares issued upon the exercise of outstanding options or warrants, in the public market following the offering, the market price of our common stock could decline. These sales also might make it more difficult for us to sell equity or equity related securities in the future at a time and price that we deem appropriate.

Upon completion of the offering, we will have outstanding an aggregate of 11,634,575 shares of our common stock, assuming no exercise of the underwriters' option to purchase additional shares and no exercise of outstanding options based on an initial public offering price of $5.00 per share. Of these shares, all of the shares sold in the offering will be freely tradeable without restriction or further registration under the Securities Act, unless the shares are purchased by "affiliates" as that term is defined in Rule 144 under the Securities Act.

Upon consummation of this offering, our existing stockholders will hold shares of common stock. The shares of common stock other than the shares registered in the resale offering will be "restricted securities" as defined in Rule 144 unless we register such issuances. The shares issued upon conversion of the Private Placement Notes held by an affiliate and the March 2022 Note are subject to a lock-up agreement and will not be distributed or transferred for a period of six months from the date of closing.

Rule 144

In general, under Rule 144 as in effect on the date of this prospectus, beginning 90 days after the completion of this offering, a person (or persons whose shares are required to be aggregated) who is an affiliate and who has beneficially owned our shares for at least six months is entitled to sell in any three-month period a number of shares that does not exceed the greater of:

- 1% of the number of shares of common stock then outstanding, which will equal approximately 116,346 shares of common stock immediately after completion of this offering; or
- the average weekly trading volume in our shares on the applicable stock exchange during the four calendar weeks preceding the filing of a notice on Form 144 with respect to such a sale.

Sales by our affiliates under Rule 144 are also subject to manner of sale provisions and notice requirements and to the availability of current public information about us. An "affiliate" is a person that directly, or indirectly through one or more intermediaries, controls or is controlled by, or is under common control with an issuer.

Under Rule 144, a person (or persons whose shares are aggregated) who is not deemed to have been an affiliate of ours at any time during the 90 days preceding a sale, and who has beneficially owned the shares of common stock proposed to be sold for at least six months (including the holding period of any prior owner other than an affiliate), would be entitled to sell those shares subject only to availability of current public information about us, and after beneficially owning such shares for at least 12 months (including the holding period of any prior owner other than an affiliate), would be entitled to sell an unlimited number of such shares without restriction. To the extent that our affiliates sell their shares, other than pursuant to Rule 144 or a registration statement, the purchaser's holding period for the purpose of affecting a sale under Rule 144 commences on the date of transfer from the affiliate.

Lock-Up Agreements

Pursuant to "lock-up" agreements, we, our executive officers and directors, and our majority stockholder prior to completion of this offering, have agreed, without the prior written consent of the representative not to directly or indirectly, offer to sell, sell, pledge or otherwise transfer or dispose of any of shares of (or enter into any transaction or device that is designed to, or could be expected to, result in the transfer or disposition by any person at any time in the future of) our common stock, enter into any swap or other derivatives transaction that transfers to another, in whole or in part, any of the economic benefits or risks of ownership of shares of our common stock, make any demand for or exercise any right or cause to be filed a registration statement, including any amendments thereto, with respect to the registration of any shares of common stock or securities convertible into or exercisable or exchangeable for common stock or any other securities of ours or publicly disclose the intention to do any of the foregoing, subject to customary exceptions: (i) for a period of one year in the case of our majority stockholder after the date of this prospectus, as well as our directors, executive officers

and affiliates of our directors and executive officers; and (ii) for a period of six months with respect to us and owners of 1% or more of our outstanding common stock, except the selling stockholders as identified in the Resale Prospectus.

Equity Awards

In general, under Rule 701 of the Securities Act as currently in effect, any of our employees, consultants, or advisors who purchase shares of our common stock from us in connection with a compensatory stock or option plan or other written agreement is eligible to resell those shares 90 days after the effective date of the offering in reliance on Rule 144, but without compliance with some of the restrictions, including the holding period, contained in Rule 144.

Following the offering, we intend to file a registration statement on Form S-8 under the Securities Act covering approximately 2,000,000 shares of common stock issued or issuable upon the exercise of stock options, subject to outstanding options or reserved for issuance under our employee and director stock benefit plans. Accordingly, shares registered under the registration statement will, subject to Rule 144 provisions applicable to affiliates, be available for sale in the open market, except to the extent that the shares are subject to vesting restrictions or the contractual restrictions described above.

Selling Stockholder Resale Prospectus

As described in the Explanatory Note to the registration statement of which this prospectus forms a part, the registration statement also contains the Resale Prospectus to be used in connection with the potential resale by certain selling stockholders of our common stock. These shares of common stock have been registered to permit public resale of such shares, and the selling stockholders may offer the shares for resale from time to time pursuant to the Resale Prospectus. The selling stockholders may also sell, transfer or otherwise dispose of all or a portion of their shares in transactions exempt from the registration requirements of the Securities Act or pursuant to another effective registration statement covering those shares. Any shares sold by the selling stockholders until our common stock is listed or quoted on an established public trading market will take place at $5.00 per share. Thereafter, any sales will occur at prevailing market prices or in privately negotiated prices.

MATERIAL U.S. FEDERAL INCOME TAX CONSIDERATIONS FOR NON-U.S. HOLDERS OF OUR COMMON STOCK

The following is a summary of the material U.S. federal income tax consequences of the ownership and disposition of our common stock acquired in this offering by a "non-U.S. holder" (as defined below), but does not purport to be a complete analysis of all the potential tax considerations relating thereto. This summary is based upon the provisions of the United States Internal Revenue Code of 1986, as amended, or the Code, Treasury Regulations promulgated thereunder, administrative rulings and judicial decisions, all as of the date hereof. These authorities may be changed, possibly retroactively, so as to result in U.S. federal income tax consequences different from those set forth below. We have not sought, and do not intend to seek, any ruling from the Internal Revenue Service, or IRS, with respect to the statements made and the conclusions reached in the following summary, and there can be no assurance that the IRS or a court will agree with such statements and conclusions.

This summary also does not address the tax considerations arising under the laws of any state or local or non-U.S. jurisdiction or under U.S. federal gift and estate tax rules, or rising out of other non-income tax rules, except to the limited extent set forth below. In addition, this discussion does not address tax considerations applicable to an investor's particular circumstances or to investors that may be subject to special tax rules, including, without limitation:

- banks, insurance companies, regulated investment companies, real estate investment trusts or other financial institutions;
- persons subject to the alternative minimum tax or the tax on net investment income;
- persons subject to special tax accounting rules as a result of any item of gross income with respect to our common stock being taken into account in an applicable financial statement;
- tax-exempt organizations or governmental organizations;
- pension plans and tax-qualified retirement plans;
- controlled foreign corporations, passive foreign investment companies and corporations that accumulate earnings to avoid U.S. federal income tax;
- partnerships or other entities or arrangements treated as partnership for U.S. federal income tax purposes (and investors therein);
- brokers or dealers in securities or currencies;
- traders in securities that elect to use a mark-to-market method of accounting for their securities holdings;
- persons that own, or are deemed to own, more than five percent of our capital stock (except to the extent specifically set forth below);
- certain former citizens or long-term residents of the United States;
- persons who hold our common stock as a position in a hedging transaction, "straddle," "conversion transaction" or other risk reduction transaction or integrated investment;
- persons who hold or receive our common stock pursuant to the exercise of any option or otherwise as compensation;
- persons who do not hold our common stock as a capital asset within the meaning of Section 1221 of the Code (generally, property held for investment); and
- persons deemed to sell our common stock under the constructive sale provisions of the Code.

In addition, if a partnership, entity or arrangement classified as a partnership or flow-through entity for U.S. federal income tax purposes holds our common stock, the tax treatment of a partner generally will depend on the status of the partner and upon the activities of the partnership or other entity. A partner in a partnership or

other such entity that will hold our common stock should consult his, her or its own tax advisor regarding the tax consequences of the ownership and disposition of our common stock through a partnership or other such entity, as applicable.

Each non-U.S. holder is urged to consult its own tax advisor with respect to the application of the U.S. federal income tax laws to its particular situation, as well as any tax consequences of the purchase, ownership and disposition of our common stock arising under the U.S. federal gift or estate tax rules or under the laws of any state, local, non-U.S. or other taxing jurisdiction or under any applicable tax treaty.

Non-U.S. Holder Defined

For purposes of this discussion, a "non-U.S. holder" is a beneficial owner of our common stock that, for U.S. federal income tax purposes, is neither a "U.S. person" nor an entity (or arrangement) treated as a partnership. A "U.S. person" is any person that, for U.S. federal income tax purposes, is or is treated as any of the following:

- an individual who is a citizen or resident of the United States;
- a corporation or other entity taxable as a corporation created or organized in the United States or under the laws of the United States or any political subdivision thereof, or otherwise treated as such for U.S. federal income tax purposes;
- an estate whose income is subject to U.S. federal income tax regardless of its source; or
- a trust (x) whose administration is subject to the primary supervision of a U.S. court and that has one or more U.S. persons who have the authority to control all substantial decisions of the trust or (y) that has made a valid election under applicable Treasury Regulations to be treated as a U.S. person.

Distributions

As described in the section titled "Dividend Policy," we have never declared or paid cash dividends on our common stock, and we do not anticipate paying any dividends on our common stock following the completion of this offering. However, if we do make distributions of cash or property on our common stock to non-U.S. holders, such distributions will constitute dividends for U.S. federal income tax purposes to the extent paid from our current or accumulated earnings and profits, as determined under U.S. federal income tax principles. To the extent those distributions exceed both our current and our accumulated earnings and profits, the excess will first constitute a return of capital and will reduce each non-U.S. holder's adjusted tax basis in our common stock, but not below zero. Any additional excess will then be treated as capital gain from the sale of stock, as discussed under "Gain on Disposition of Common Stock."

Subject to the discussions below on effectively connected income, and backup withholding and the Foreign Account Tax Compliance Act, or FATCA, withholding, any dividend paid to a non-U.S. holder generally will be subject to U.S. federal withholding tax either at a rate of 30% of the gross amount of the dividend or such lower rate as may be specified by an applicable income tax treaty between the United States and such non-U.S. holder's country of residence. In order to receive a reduced treaty rate, such non-U.S. holder must provide the applicable withholding agent with an IRS Form W-8BEN or W-8BEN-E or other appropriate version of IRS Form W-8 certifying qualification for the reduced treaty rate. A non-U.S. holder of shares of our common stock eligible for a reduced rate of U.S. federal withholding tax pursuant to an income tax treaty may obtain a refund of any excess amounts withheld by filing an appropriate claim for refund with the IRS. If such non-U.S. holder holds our common stock through a financial institution or other agent acting on the non-U.S. holder's behalf, the non-U.S. holder will be required to provide appropriate documentation to such agent, which then will be required to provide certification to the applicable withholding agent, either directly or through other intermediaries. Each non-U.S. holder should consult its own tax advisors regarding their entitlement to benefits under any applicable income tax treaty.

Dividends received by a non-U.S. holder that are treated as effectively connected with such non-U.S. holder's conduct of a trade or business within the United States (and, if an applicable income tax treaty so provides, such non-U.S. holder maintains a permanent establishment or fixed base in the United States to which such dividends are attributable) are generally exempt from the 30% U.S. federal withholding tax, subject to the discussion below on backup withholding and FATCA withholding. To claim this exemption, a non-U.S. holder must provide the applicable withholding agent with a properly executed IRS Form W-8ECI or other applicable IRS Form W-8 properly certifying such exemption. Such effectively connected dividends, although not subject to

U.S. federal withholding tax, are taxed at the same graduated rates applicable to U.S. persons, net of certain deductions and credits, subject to an applicable income tax treaty providing otherwise. In addition, if a non-U.S. holder is a corporation, dividends such non-U.S. holder receives

that are effectively connected with its conduct of a U.S. trade or business may also be subject to a branch profits tax at a rate of 30% or such lower rate as may be specified by an applicable income tax treaty between the United States and such non-U.S. holder's country of residence. Each non-U.S. holder should consult its own tax advisor regarding the tax consequences of the ownership and disposition of our common stock, including any applicable tax treaties that may provide for different rules.

Gain on Disposition of Common Stock

Subject to the discussion below regarding backup withholding and FATCA withholding, a non-U.S. holder generally will not be required to pay U.S. federal income tax on any gain realized upon the sale or other disposition of our common stock unless:

- the gain is effectively connected with such non-U.S. holder's conduct of a U.S. trade or business (and, if an applicable income tax treaty so provides, such non-U.S. holder maintains a permanent establishment or fixed base in the United States to which such gain is attributable);
- such non-U.S. holder is an individual who is present in the United States for an aggregate 183 days or more during the taxable year in which the sale or disposition occurs and certain other conditions are met; or
- our common stock constitutes a United States real property interest, or USRPI, by reason of our status as a "United States real property holding corporation," or USRPHC, for U.S. federal income tax purposes.

We believe that we are not currently and will not become a USRPHC for U.S. federal income tax purposes, and the remainder of this discussion so assumes. However, because the determination of whether we are a USRPHC depends on the fair market value of our U.S. real property interests relative to the fair market value of our U.S. and worldwide real property interests plus our other business assets, there can be no assurance that we will not become a USRPHC in the future. Even if we become a USRPHC, however, as long as our common stock is regularly traded on an established securities market, your common stock will be treated as U.S. real property interests only if you actually (directly or indirectly) or constructively hold more than 5% of such regularly traded common stock at any time during the shorter of the five-year period preceding your disposition of, or your holding period for, our common stock.

A non-U.S. holder described in the first bullet above will be required to pay U.S. federal income tax on the gain derived from the sale (net of certain deductions and credits) under regular graduated U.S. federal income tax rates. Such a non-U.S. holder that is a corporation may be subject to the branch profits tax at a 30% rate on a portion of its effectively connected earnings and profits for the taxable year that are attributable to such gain, as adjusted for certain items. A lower rate may be specified by an applicable income tax treaty.

A non-U.S. holder described in the second bullet above will be subject to tax at 30% (or such lower rate specified by an applicable income tax treaty) on the gain derived from the sale, which gain may be offset by U.S. source capital losses of such non-U.S. holder for the taxable year, provided such non-U.S. holder has timely filed U.S. federal income tax returns with respect to such losses.

Each non-U.S. holder should consult its own tax advisor regarding any applicable income tax or other treaties that may provide for different rules.

Information Reporting and Backup Withholding

Generally, we or an applicable withholding agent must report annually to the IRS the amount of dividends paid to a non-U.S. holder, such non-U.S. holder's name and address, and the amount of tax withheld, if any. A similar report is sent to such non-U.S. holder. Pursuant to any applicable income tax treaty or other agreement, the IRS may make such report available to the tax authority in such non-U.S. holder's country of residence.

Dividends paid by us (or our paying agent) to a non-U.S. holder may also be subject to backup withholding at a current rate of 24%.

Such information reporting and backup withholding requirements may be avoided, however, if such non-U.S. holder establishes an exemption by providing a properly executed, and applicable, IRS Form W-8, or

otherwise establishes an exemption. Generally, such information reporting and backup withholding requirements will not apply to

a non-U.S. holder where the transaction is effected outside the United States, through a non-U.S. office of a non-U.S. broker. Notwithstanding the foregoing, backup withholding and information reporting may apply, however, if the applicable withholding agent has actual knowledge, or reason to know, that such non-U.S. holder is a U.S. person.

Backup withholding is not an additional tax; rather, the U.S. federal income tax liability of persons subject to backup withholding will be reduced by the amount of tax withheld. If withholding results in an overpayment of taxes, a refund or credit may generally be obtained from the IRS, provided that the required information is furnished to the IRS in a timely manner.

Foreign Account Tax Compliance Act (FATCA)

Sections 1471 to 1474 of the Code, Treasury Regulations issued thereunder and related official IRS guidance, commonly referred to as FATCA, generally impose a U.S. federal withholding tax of 30% on dividends on our common stock paid to a "foreign financial institution" (as defined under FATCA, and which may include banks, traditional financial institutions, investment funds, and certain holding companies), unless such institution enters into an agreement with the U.S. Department of the Treasury to, among other things, identify accounts held by certain "specified United States persons" or "United States-owned foreign entities" (each as defined under FATCA), report annually substantial information about such accounts, and withhold on certain payments to non-compliant foreign financial institutions and certain other account holders. FATCA also generally imposes a U.S. federal withholding tax of 30% on dividends on our common stock paid to a "non-financial foreign entity" (as specially defined under FATCA), unless such entity provides identifying information regarding each direct or indirect "substantial United States owners" (as defined under FATCA), certifies that it does not have any substantial United States owners, or otherwise establishes an exemption. Accordingly, the institution or entity through which our common stock is held will affect the determination of whether such withholding is required.

The withholding obligations under FATCA generally apply to dividends on our common stock. Such withholding will apply regardless of whether the beneficial owner of the payment otherwise would be exempt from withholding pursuant to an applicable tax treaty with the United States, the Code, or other exemptions described above. Under certain circumstances, a non-U.S. holder might be eligible for refunds or credits of such taxes.

While withholding under FATCA would have also applied on or after January 1, 2019 to payments of gross proceeds from the sale or other disposition of stock which can proceed U.S. source dividends, such as our common stock, proposed regulations under FATCA have eliminated withholding on payment of gross proceeds entirely. Taxpayers generally may rely on these proposed regulations until final Treasury Regulations are issued.

An intergovernmental agreement between the United States and an applicable foreign country may modify the requirements described in this section. Prospective investors are encouraged to consult with their own tax advisors regarding the application of FATCA withholding to their investment in, and ownership and disposition of, our common stock.

The preceding discussion of U.S. federal tax considerations is not tax advice to investors in their particular circumstances. Each prospective investor should consult its own tax advisor regarding the particular U.S. federal, state and local and non-U.S. tax consequences of purchasing, holding and disposing of our common stock, including the consequences of any proposed change in applicable laws.

UNDERWRITING

In connection with this offering, we have entered into an underwriting agreement with Boustead Securities, LLC to serve as representative of the underwriters in this offering. Subject to the terms and conditions of the underwriting agreement dated as of the date of this prospectus, each underwriter will severally agree to purchase the number of shares of common stock set forth opposite its name below, at the public offering price, less the underwriting discount set forth on the cover page of this prospectus.

Underwriter	Number of Shares Common Stock
Boustead Securities, LLC	1,400,000

Subject to the terms and conditions set forth in the underwriting agreement, the underwriters have agreed to purchase all of the shares offered by this prospectus (other than those covered by the overallotment option described below, if any are purchased).

The underwriters are offering the shares of common stock subject to various conditions and may reject all or part of any order. The representative of the underwriters has advised us that the underwriters propose initially to offer the shares of common stock to the public at the public offering price set forth on the cover page of this prospectus and to dealers at a price less a concession not in excess of $0.35 per share of common stock. After the shares of common stock are released for sale to the public, the representative may change the offering price, the concession, and other selling terms at various times.

The following table provides information regarding the amount of the discounts and commissions to be paid to the underwriters by us, before expenses. The information assumes either no exercise or full exercise of the over-allotment option we granted to the representative of the underwriters:

	Per Share of Common Stock	Total Without Over-Allotment Option	Total With Over-Allotment Option
Public offering price	$ 5.00	$ 7,000,000	$ 8,050,000
Underwriting discounts and commission	$ 0.35	$ 490,000	563,500
Proceeds, before expenses, to us	$ 4.65	$ 6,510,000	7,486,500
Non-accountable expense allowance	$ 0.05	$ 70,000	80,500

We estimate that our total expenses of the offering, excluding the estimated underwriting discounts and commissions and excluding the non-accountable expense allowance, will be approximately $700,000.

We have also agreed to issue to the representative of the underwriters warrants to purchase a number of shares of common stock equal to an aggregate of 6% of the aggregate number of the shares sold in this offering. The warrants will be exercisable on a cashless basis at an exercise price equal to 120% of the offering price of the shares sold in this offering. The warrants are exercisable commencing six months after the date of effectiveness of the registration statement of which this prospectus forms a part, and will be exercisable for a period of five years from the effective date of the registration statement of which this prospectus forms a part. We have agreed to a one-time demand registration of the shares of common stock underlying the underwriter's warrants for a period of five years from the effective date of the registration statement. The underwriter's warrants also provide for immediate "piggyback" registration rights with respect to the underlying shares of common stock during the three-year period commencing from the effective date of the registration statement related to this offering. Such piggyback rights shall expire on a date which shall be three years from the date of commencement of sales of the share offered hereby. The warrants are not redeemable by us. The warrants and the shares of common stock issuable upon exercise of the warrants have been included on the registration statement of which this prospectus forms a part. Pursuant to applicable FINRA rules, and in particular Rule 5110, the warrants (and underlying shares) issued to the underwriters may not be sold, transferred,

assigned, pledged, or hypothecated, or the subject of any hedging, short sale, derivative, put or call transaction that would result in the effective disposition of the securities by any person for a period of 180 days after the effective date of the

registration statement related to this offering; provided, however, that the warrants (and the underlying shares) may be transferred to the underwriters' officers, partners, registered persons or affiliates as long as the warrants (and the underlying shares) remain subject to the lockup.

We have agreed to indemnify the underwriters against certain liabilities, including liabilities under the Securities Act of 1933, as amended.

Pursuant to the underwriting agreement, we will provide the representative of the underwriters the right of first refusal for one year from the date of commencement of sales of this public offering to act as financial advisor or to act as joint financial advisor on at least equal economic terms on any public or private financing (debt or equity), merger, business combination, recapitalization or sale of some or all of the equity or assets of our company.

We have agreed to a 12-month "lock-up" from the closing of this offering, during which, without the prior written consent of Boustead Securities, LLC, we will not issue, sell or register with the SEC (other than on Form S-8 or on any successor form) with respect to any of our equity securities (or any securities convertible into, exercisable for or exchangeable for any of our equity securities), except for (i) the issuance of the shares of common stock offered pursuant to this prospectus; and (ii) the issuance of shares of common stock pursuant to our existing equity incentive or bonus plan as described in the registration statement of which this prospectus forms a part.

Our executive officers, directors and certain of our significant stockholders have also agreed to a 12-month "lock-up," during which, without the prior written consent of Boustead Securities, LLC, they will not, directly or indirectly, (i) offer, pledge, assign, encumber, announce the intention to sell, sell, contract to sell, sell any option or contract to purchase, purchase any option or contract to sell, grant any option, right or warrant to purchase, or otherwise transfer or dispose of, any shares of common stock or any securities convertible into or exercisable or exchangeable for common stock, owned either of record or beneficially (as defined in the Securities Exchange Act of 1934, as amended (the "Exchange Act") by any signatory of the lock-up agreement on the date of the prospectus or thereafter acquired; (ii) enter into any swap or other agreement that transfers, in whole or in part, any of the economic consequences of ownership of the common stock or any securities convertible into or exercisable or exchangeable for common stock, whether any such transaction described in clauses (i) or (ii) above is to be settled by delivery of common stock or such other securities, in cash or otherwise, or publicly announce an intention to do any of the foregoing; and (iii) make any demand for or exercise any right with respect to, the registration of any shares of common stock or any security convertible into or exercisable or exchangeable for common stock. The foregoing shall not apply to (i) common stock to be transferred as a gift or gifts (*provided*, that (a) any donee shall execute and deliver to Boustead Securities, LLC, acting on behalf of the underwriters, not later than one business day prior to such transfer, a lock-up agreement to Boustead Securities, LLC and (b) if the lock-up signatory is required to file a report under Section 16(a) of the Exchange Act, reporting a reduction in beneficial ownership of shares of common stock or beneficially owned shares or any securities convertible into or exercisable or exchangeable for common stock or beneficially owned shares during the 15-month "lock-up," the lock-up signatory shall include a statement in such report to the effect that such transfer is being made as a gift), and (ii) the sale of the shares of common stock to be sold pursuant to this prospectus.

Rules of the SEC may limit the ability of the underwriters to bid for or purchase shares of our common stock before the distribution of the shares is completed. However, the underwriters may engage in the following activities in accordance with the rules:

- Stabilizing transactions — the representative may make bids or purchases for the purpose of pegging, fixing or maintaining the price of the common stock, so long as stabilizing bids do not exceed a specified maximum.

- Over-allotments and syndicate covering transactions — the underwriters may sell more shares of common stock in connection with this offering than the number of shares that they have committed to purchase. This over-allotment creates a short position for the underwriters. This short sales position may involve either "covered" short sales or "naked" short sales. Covered short sales are short sales made in an amount not greater than the underwriter's over-allotment option to purchase additional shares in this offering described above. The underwriters may close out any covered short position either by exercising their over-allotment option or by purchasing shares in the open market.

To determine how they will close the covered short position, the underwriters will consider, among other things, the price of shares available for purchase in the open market, as compared to the price at which they may purchase shares through

the over-allotment option. Naked short sales are short sales in excess of the over-allotment option. The underwriters must close out any naked short position by purchasing shares in the open market. A naked short position is more likely to be created if the underwriters are concerned that, in the open market after pricing, there may be downward pressure on the price of the shares that could adversely affect investors who purchase shares in this offering.

- Penalty bids — if the representative purchases shares of common stock in the open market in a stabilizing transaction or syndicate covering transaction, it may reclaim a selling concession from the underwriters and selling group members who sold those shares of common stock as part of this offering.
- Passive market making — market makers in the common stock who are underwriters or prospective underwriters may make bids for or purchases of shares of common stock, subject to limitations, until the time, if ever, at which a stabilizing bid is made.

Similar to other purchase transactions, the underwriters' purchases to cover the syndicate short sales or to stabilize the market price of our common stock may have the effect of raising or maintaining the market price of our common stock or preventing or mitigating a decline in the market price of our common stock. As a result, the price of our common stock may be higher than the price that might otherwise exist in the open market. The imposition of a penalty bid might also have an effect on the price of the common stock if it discourages resales of our shares of common stock.

Neither we nor the underwriters make any representation or prediction as to the effect that the transactions described above may have on the price of our common stock. These transactions may occur on the Nasdaq Capital Market or otherwise. If such transactions are commenced, they may be discontinued without notice at any time.

In July 2022, we closed the first tranche of the Private Placement pursuant to which we issued First Tranche Private Placement Notes in the aggregate principal amount of $450,000 and received gross proceeds of $450,000, on August 22, 2022, we issued a $50,000 convertible promissory note pursuant to the Private Placement and received gross proceeds of $50,000, and in September 2022 we issued convertible promissory notes in the aggregate principal amount of $75,000 and received gross proceeds of $75,000 (collectively, the "Private Placement Notes"). The Private Placement Notes bear interest at 6% and mature on the third anniversary of the last closing of the Private Placement Notes, i.e. September 13, 2025. The principal amount due under the Private Placement Notes (and, at our option, any accrued but unpaid interest under the Private Placement Notes) will be automatically converted into shares of our common stock upon the consummation of this offering. The Private Placement Notes were initially convertible into shares of common stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the Private Placement Notes by (ii) sixty percent (60%) of the initial offering price per share subject to adjustment in the event of a subsequent issuance of securities by us at a price per share less than the conversion price. Upon the issuance of the November Notes and November Warrants the conversion price was reduced to $1.00 per share. The Private Placement Note offering was completed pursuant to an exemption from registration under Rule 506(b) of the Securities Act. Boustead Securities, LLC acted as the placement agent for the private placement and received cash compensation of $31,500, a one percent non-accountable expense equal to 1% of the offering, or $4,500, and warrants (the "Placement Agent Warrants"), as amended, to purchase an aggregate of 11,500 shares of our common stock at an exercise price equal to 60% of the initial public offering price per share. The Placement Agent Warrants will not be exercisable for a period of more than five years following commencement of sales in this public offering, in compliance with FINRA Rule 5110(g)(8)(A). Further, pursuant to applicable FINRA rules and, in particular, Rule 5110(e)(1), the Placement Agent Warrants may not be sold, transferred, assigned, pledged or hypothecated, or be the subject of any hedging, short sale, derivative, put or call transaction that would result in the effective economic disposition of the securities for a period of 180 days beginning on the date of commencement of sales of this public offering; provided, however, the Placement Agent Warrants may be transferred to the underwriter's officers, partners, registered persons or affiliates as long as the warrants remain subject to the lock-up restriction set forth above.

In November 2022, we closed the November Private Placement pursuant to which we issued the November Notes and the November Warrants and received gross proceeds of $250,000. The November Notes bear interest at 10% and mature on the date of the closing of this offering. Boustead Securities, LLC acted as the placement agent for the private placement and received cash compensation of $15,000, a one percent non-accountable

expense equal to 1% of the offering, or $2,500, and warrants (the "Placement Agent Warrants") to purchase shares of our common stock in an amount equal to 6% of the shares of common stock underlying the securities issued in the November Private

Placement at an exercise price of $1.00 per share. The Placement Agent Warrants will not be exercisable for a period of more than five years following commencement of sales in this public offering, in compliance with FINRA Rule 5110(g)(8)(A). Further, pursuant to applicable FINRA rules and, in particular, Rule 5110(e)(1), the Placement Agent Warrants may not be sold, transferred, assigned, pledged or hypothecated, or be the subject of any hedging, short sale, derivative, put or call transaction that would result in the effective economic disposition of the securities for a period of 180 days beginning on the date of commencement of sales of this public offering; provided, however, the Placement Agent Warrants may be transferred to the underwriter's officers, partners, registered persons or affiliates as long as the warrants remain subject to the lock-up restriction above.

Electronic Delivery of Prospectus: A prospectus in electronic format may be delivered to potential investors by one or more of the underwriters participating in this offering. The prospectus in electronic format will be identical to the paper version of such prospectus. Other than the prospectus in electronic format, the information on any underwriter's website and any information contained in any other website maintained by an underwriter is not part of this prospectus or the registration statement of which this prospectus forms a part.

LEGAL MATTERS

The validity of the securities being offered by this prospectus will be passed upon for us by Blank Rome LLP, New York, New York. Michelman & Robinson, LLP, New York, New York and Los Angeles, California, is acting as counsel for the underwriters.

EXPERTS

The financial statements of Cadrenal Therapeutics, Inc. as of June 30, 2022, and for the period from January 25, 2022 (inception) to June 30, 2022, have been audited by WithumSmith+Brown, PC, independent registered public accounting firm, as stated in their report, which includes an explanatory paragraph as to the Company's ability to continue as a going concern, appearing herein. Such financial statements have been included herein in reliance on the report of such firm given upon their authority as experts in accounting and auditing.

WHERE YOU CAN FIND ADDITIONAL INFORMATION

We have filed with the SEC a registration statement on Form S-1 under the Securities Act with respect to the shares of our common stock offered by this prospectus. This prospectus, which constitutes a part of the registration statement, does not contain all of the information set forth in the registration statement, as permitted by the rules and regulations of the SEC. For further information with respect to us and our common stock, we refer you to the registration statement, including the exhibits filed as a part of the registration statement. Statements contained in this prospectus concerning the contents of any contract or any other document are not necessarily complete. If a contract or document has been filed as an exhibit to the registration statement, please see the copy of the contract or document that has been filed. Each statement in this prospectus relating to a contract or document filed as an exhibit is qualified in all respects by the filed exhibit. The SEC also maintains an Internet website that contains reports, proxy and information statements, and other information regarding issuers that file electronically with the SEC, including the registration statement of which this prospectus forms a part, as well as the exhibits thereto. These documents, along with future reports, proxy statements and other information about us, are available at the SEC's website, *www.sec.gov*.

As a result of this offering, we will become subject to the information and reporting requirements of the Exchange Act, as amended, and, in accordance with this law, will file periodic reports, proxy statements and other information with the SEC. These periodic reports, proxy statements and other information will be available at the SEC's website, *www.sec.gov.* We also maintain a website *www.cadrenal.com*. Upon the completion of this offering, you may access these materials free of charge as soon as reasonably practicable after they are electronically filed with, or furnished to, the SEC. Information contained on our website is not a part of this prospectus, and the inclusion of our website address in this prospectus is an inactive textual reference only.

DISCLOSURE OF COMMISSION POSITION ON INDEMNIFICATION FOR SECURITIES ACT LIABILITIES

Insofar as indemnification for liabilities arising under the Securities Act may be permitted to our directors, officers, and controlling persons, we have been informed that in the opinion of the SEC this indemnification is against public policy as expressed in the Securities Act and is therefore unenforceable.

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REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Stockholders and Board of Directors
Cadrenal Therapeutics, Inc.

Opinion on the Financial Statements

We have audited the accompanying balance sheet of Cadrenal Therapeutics, Inc. (the "Company") as of June 30, 2022, the related statements of operations and comprehensive loss, changes in stockholders' deficit, and cash flows for the period from January 25, 2022 (inception) to June 30, 2022, and the related notes to the financial statements. In our opinion, the financial statements present fairly, in all material respects, the financial position of the Company at June 30, 2022, and the results of its operations and its cash flows for the period from January 25, 2022 (inception) to June 30, 2022, in conformity with accounting principles generally accepted in the United States of America.

Going Concern Uncertainty

The accompanying financial statements have been prepared assuming that the Company will continue as a going concern. As discussed in Note 1, the Company has incurred recurring losses from operations and negative cash flows since inception, and has an accumulated deficit and insufficient cash at June 30, 2022 to fund operations for twelve months from the date of issuance. All of these matters raise substantial doubt about the Company's ability to continue as a going concern. Management's plans in regard to these matters are also described in Note 1. The financial statements do not include any adjustments that might result from the outcome of this uncertainty.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audit. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) ("PCAOB") and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB and in accordance with auditing standards generally accepted in the United States of America. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. The Company is not required to have, nor were we engaged to perform, an audit of its internal control over financial reporting. As part of our audit, we are required to obtain an understanding of internal control over financial reporting, but not for the purpose of expressing an opinion on the effectiveness of the Company's internal control over financial reporting. Accordingly, we express no such opinion.

Our audit included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audit also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audit provides a reasonable basis for our opinion.

/s/ WithumSmith+Brown, PC

We have served as the Company's auditor since 2022.

San Francisco, California
October 7, 2022

CADRENAL THERAPEUTICS, INC.
BALANCE SHEET

	June 30, 2022
Assets:	
Current assets:	
Cash	$ 59,646
Prepaid expenses	27,691
Deferred offering costs	153,814
Total current assets	241,151
Other assets	2,195
Total Assets	$ 243,346
Liabilities:	
Current liabilities:	
Accounts payable	$ 207,505
Accrued liabilities	132,201
Total current liabilities	339,706
Convertible note payable, net of debt discount – related party	404,198
Convertible note payable, net of debt discount	20,759
Derivative liabilities	140,977
Accrued interest – convertible note payable	8,544
Total liabilities	914,184
Stockholders' deficit:	
Common stock, $0.001 par value; 10,000,000 shares authorized, 7,950,000 issued and outstanding as of June 30, 2022	7,950
Additional paid-in capital	—
Accumulated deficit	(678,788)
Total stockholders' deficit	(670,838)
Total liabilities and stockholders' deficit	$ 243,346

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
STATEMENT OF OPERATIONS AND COMPREHENSIVE LOSS

	January 25, 2022 (Inception) through June 30, 2022
Operating expenses:	
General and administrative expenses	$ 432,772
Research and development expenses	220,095
Total operating expenses	652,867
Loss from operations	(652,867)
Other expense:	
Interest expense	8,527
Interest expense, amortization of debt discount	10,832
Change in in fair value of derivative liabilities	6,562
Total other expenses	25,921
Net loss and comprehensive loss	$ (678,788)
Net loss per common share, basic and diluted	$ (0.09)
Weighted average number of common shares used in computing net loss per common share, basic and diluted	7,626,923

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
STATEMENT OF CHANGES IN STOCKHOLDERS' DEFICIT

	Common Stock		Additional Paid-In Capital	Accumulated Deficit	Total Stockholders' Deficit
	Shares	Amount			
Balance, January 25, 2022 (inception)	—	$ —	$ —	$ —	$ —
Issuance of founder shares	7,950,000	7,950	—	—	7,950
Net loss	—	—	—	(678,788)	(678,788)
Balance, June 30, 2022	7,950,000	$ 7,950	$ —	$ (678,788)	$ (670,838)

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
STATEMENT OF CASH FLOWS

	January 25, 2022 (Inception) through June 30, 2022
Cash flows from operating activities:	
Net loss	$ (678,788)
Adjustments to reconcile net loss to net cash used in operating activities:	
Amortization of debt discount	10,832
Change in fair value of derivative liabilities	6,562
Changes in operating assets and liabilities:	
Prepaid expenses	(27,691)
Deferred equity offering costs	(153,814)
Other assets	(2,195)
Accounts payable	207,505
Accrued liabilities	132,201
Accrued interest – convertible notes payable	8,544
Net cash used in operating activities	(496,844)
Cash flows from financing activities:	
Proceeds from issuance of convertible notes, net of debt issuance costs	548,540
Proceeds from sale of common stock	7,950
Net cash provided by financing activities	556,490
Net change in cash	59,646
Cash – beginning of the period	—
Cash – end of the period	$ 59,646

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies

Cadrenal Therapeutics, Inc. (the "Company" or "Cadrenal") was incorporated on January 25, 2022 (inception) in the State of Delaware and is headquartered in Ponte Vedra, Florida. Cadrenal is focused on developing a novel therapy with orphan drug indication, tecarfarin, for the prevention of systemic thromboembolism (blood clots) of cardiac origin in patients with end-stage renal disease (on dialysis) and atrial fibrillation (irregular heartbeat) or AFib. Tecarfarin is an anticoagulant designed using a drug design process which targets a different pathway than most commonly prescribed drugs for the treatment of thrombosis and AFib.

Basis of Presentation

The accompanying financial statements have been prepared in accordance with accounting principles generally accepted in the United States of America (GAAP) and applicable rules and regulations of the U.S. Securities and Exchange Commission ("SEC") for the fair presentation of the Company's financial position for the period from January 25, 2022 (inception) to June 30, 2022. The Company's date of inception was January 25, 2022 and the fiscal year-end is December 31.

Liquidity and Going Concern

The accompanying financial statements have been prepared assuming that the Company will continue as a going concern, which contemplates the realization of assets and settlement of liabilities and commitments in the normal course of business. The financial statements do not reflect any adjustments relating to the recoverability and reclassification of assets and liabilities that might be necessary if the Company is unable to continue as a going concern. Since inception, the Company has incurred operating losses and negative cash flows from operations. For the period from January 25, 2022 (inception) to June 30, 2022, the Company had a net loss of $678,788 with cash used in operations of $496,844. As of June 30, 2022, the Company had cash of $59,646, negative working capital of $98,555, and an accumulated deficit of $678,788. In July 2022, the Company raised $450,000 in gross proceeds from the issuance of a convertible notes. The Company is projecting that its operating losses and expected capital needs will exceed its existing cash balances and cash expected to be generated from operations for the foreseeable future. In order to meet the Company's expected obligations, management intends to raise additional funds through equity and debt financings. However, there can be no assurance that the Company will be able to complete any additional equity or debt financings on terms acceptable to the Company or at all. If the Company is unable to raise additional funding to meet its working capital needs in the future, it will be forced to delay or reduce the scope of its research programs and/or limit or cease its operations. As a result, there is uncertainty in the Company's ability to meet its current operating and capital expenses. These factors, among others, raise substantial doubt about the Company's ability to continue as a going concern for at least one year from the date the accompanying financial statements are issued.

In March 2020, the World Health Organization declared COVID-19 a pandemic. The COVID-19 pandemic has negatively impacted the global economy, disrupted global supply chains and created significant volatility and disruption of financial markets. The Company cannot at this time predict the specific extent, duration, or full impact that the ongoing COVID-19 pandemic will have on its financial condition and operations, including planned clinical trials. The impact of the COVID-19 pandemic on the financial performance of the Company will depend on future developments, including the duration and spread of the pandemic and related governmental advisories and restrictions. These developments and the impact of the COVID-19 pandemic on the financial markets and the overall economy are highly uncertain. If business conditions, financial markets and/or the overall economy are impacted for an extended period, the Company's results may be adversely affected.

Emerging Growth Company Status

From time to time, new accounting pronouncements, or Accounting Standards Updates ("ASU") are issued by the FASB or other standard setting bodies and adopted by the Company as of the specified effective date. Unless otherwise discussed, the impact of recently issued standards that are not yet effective is not expected to have a material impact on the Company's financial position or results of operations upon adoption.

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies (cont.)

As an "emerging growth company" ("EGC") under the Jumpstart Our Business Startups Act ("JOBS Act"), the Company may elect to take advantage of certain forms of relief from various reporting requirements that are applicable to public companies. The relief afforded under the JOBS Act includes an extended transition period for the implementation of new or revised accounting standards. The Company has elected to take advantage of this extended transition period and, as a result, the Company's financial statements may not be comparable to those of companies that implement accounting standards as of the effective dates for public companies. The Company may take advantage of the relief afforded under the JOBS Act up until the last day of the fiscal year following the fifth anniversary of an offering or such earlier time that it is no longer an EGC.

Use of Estimates

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities as of the date of the financial statements and the reported amounts of expenses during the reporting period. Significant estimates and assumptions made in the accompanying financial statements include but are not limited to the fair value of financial instruments, deferred tax assets and valuation allowance, income tax uncertainties, and certain accruals. The Company evaluates its estimates and assumptions on an ongoing basis using historical experience and other factors and adjusts those estimates and assumptions when facts and circumstances change. Actual results could differ from those estimates.

Concentration of Credit and other Risks and Uncertainties

Financial instruments, which potentially subject the Company to significant concentrations of credit risk, consist primarily of cash. Cash is maintained at high credit quality financial institutions and, at times, balances may exceed federally insured limits. All interest-bearing and non-interest-bearing cash balances are insured up to $250,000 per depositor at each financial institution. The Company has never experienced any losses related to these balances.

The Company is subject to a number of risks common for early-stage biopharmaceutical companies including, but not limited to, dependency on the clinical and commercial success of its product candidates, ability to obtain regulatory approval of its product candidates, the need for substantial additional financing to achieve its goals, uncertainty of broad adoption of its approved products, if any, by physicians and patients, significant competition and untested manufacturing capabilities.

Segments

Operating segments are defined as components of an entity for which separate financial information is available and that is regularly reviewed by the Chief Operating Decision Maker (CODM) in deciding how to allocate resources to an individual segment and in assessing performance. The Company's CODM is its Chief Executive Officer. The Company has determined it operates in a single operating segment and has one reportable segment.

Cash and Cash Equivalents

The Company considers all highly liquid investments purchased with original maturities of three months or less from the purchase date to be cash equivalents. The Company did not have any cash equivalents at June 30, 2022.

Derivative Financial Instruments

The Company evaluates all of its agreements to determine if such instruments have derivatives or contain features that qualify as embedded derivatives. The Company accounts for certain redemption features that are associated with convertible notes as liabilities at fair value and adjusts the instruments to their fair value at the end of each reporting period. Derivative financial liabilities are initially recorded at fair value, with gains and

losses arising from changes in the fair value recognized in other income (expense) in the accompanying statements of operations for each reporting period while such instruments are outstanding. The embedded derivative liability is valued using a

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies (cont.)

probability-weighted expected return model. If the Company repays the note holders or if, during the next round of financing, the note holders convert the debt into equity, the derivative financial liability will be de-recognized and reclassified to stockholders' equity (deficit) on that date. Derivative instrument liabilities are classified in the balance sheet as current or non-current based on whether or not net-cash settlement of the derivative instrument could be required within 12 months of the balance sheet date.

Deferred Offering Costs

The Company capitalizes certain legal, professional and other third-party costs that are directly associated with in-process equity financings until such financings are consummated at which time such costs are recorded against the gross proceeds of the offering. Should an in-process equity financing be abandoned, the deferred offering costs will be expensed immediately as a charge to operating expenses in the statements of operations and comprehensive loss. As of June 30, 2022, deferred offering costs were $153,814 and are included on the accompanying balance sheet.

Acquisitions

The Company evaluates acquisitions of assets and other similar transactions to assess whether or not the transaction should be accounted for as a business combination or asset acquisition by first applying a screen test to determine whether substantially all of the fair value of the gross assets acquired is concentrated in a single identifiable asset or group of similar identifiable assets. If so, the transaction is accounted for as an asset acquisition. If not, further determination is required as to whether or not the Company has acquired inputs and processes that have the ability to create outputs, which would meet the definition of a business. Significant judgment is required in the application of the screen test to determine whether an acquisition is a business combination or an acquisition of assets.

Acquisitions meeting the definition of business combinations are accounted for using the acquisition method of accounting, which requires that the purchase price be allocated to the net assets acquired at their respective fair values. In a business combination, any excess of the purchase price over the estimated fair values of the net assets acquired is recorded as goodwill.

For asset acquisitions, a cost accumulation model is used to determine the cost of an asset acquisition. Direct transaction costs are recognized as part of the cost of an asset acquisition. The Company also evaluates which elements of a transaction should be accounted for as a part of an asset acquisition and which should be accounted for separately. The cost of an asset acquisition, including transaction costs, is allocated to identifiable assets acquired and liabilities assumed based on a relative fair value basis. Goodwill is not recognized in an asset acquisition. Any difference between the cost of an asset acquisition and the fair value of the net assets acquired is allocated to the non-monetary identifiable assets based on their relative fair values. When a transaction accounted for as an asset acquisition includes an in-process research and development ("IPR&D") asset, the IPR&D asset is only capitalized if it has an alternative future use other than in a particular research and development project. For an IPR&D asset to have an alternative future use: (a) the Company must reasonably expect that it will use the asset acquired in the alternative manner and anticipate economic benefit from that alternative use, and (b) the Company's use of the asset acquired is not contingent on further development of the asset subsequent to the acquisition date (that is, the asset can be used in the alternative manner in the condition in which it existed at the acquisition date). Otherwise, amounts allocated to IPR&D that have no alternative use are expensed to research and development. Asset acquisitions may include contingent consideration arrangements that encompass obligations to make future payments to sellers contingent upon the achievement of future financial targets. Contingent consideration is not recognized until all contingencies are resolved and the consideration is paid or probable of payment, at which point the consideration is allocated to the assets acquired on a relative fair value basis.

Income Taxes

Income taxes are accounted for under the asset and liability method. Under this method, deferred tax assets and liabilities are determined based on the difference between the financial statement and tax bases of assets and liabilities using enacted tax rates in effect for the year in which the differences are expected to affect taxable income. Management makes an assessment of the likelihood that the resulting deferred tax assets will be realized. A valuation allowance is

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies (cont.)

provided when it is more likely than not that some portion or all of a deferred tax asset will not be realized. Due to the Company's historical operating performance and net losses, the net deferred tax assets have been fully offset by a valuation allowance.

The Company recognizes uncertain income tax positions at the largest amount that is more likely than not to be sustained upon audit by the relevant taxing authority. An uncertain income tax position will not be recognized if it has less than a 50% likelihood of being sustained. Changes in recognition or measurement are reflected in the period in which judgment occurs. The Company's policy is to recognize interest and penalties related to the underpayment of income taxes as a component of the provision for income taxes.

Net Loss Per Common Share

Basic net loss per common share is calculated by dividing the net loss by the weighted-average number of shares of common stock outstanding for the period, without consideration for potential dilutive shares of common stock. Diluted net loss per common share is computed by dividing net loss by the weighted average number of shares of common stock and common stock equivalents of potentially dilutive securities outstanding for the period determined using the treasury stock or if-converted methods. Since the Company was in a loss position for all periods presented, basic net loss per common share is the same as diluted net loss per common share since the effects of potentially dilutive securities are anti-dilutive. Shares of common stock subject to repurchase are excluded from the weighted-average shares.

Comprehensive Loss

Comprehensive loss is defined as the change in equity during a period from transactions and other events or circumstances from non-owner sources. Net loss and comprehensive loss were the same for all periods presented in the accompanying financial statements.

Research and Development Expenses

Research and development costs are expensed as incurred and consist of fees paid to other entities that conduct certain research and development activities on the Company's behalf. Acquired intangible assets are expensed as research and development costs if, at the time of payment, the technology is under development; is not approved by the FDA or other regulatory agencies for marketing; has not reached technical feasibility; or otherwise has no foreseeable alternative future use. Nonrefundable advance payments for goods or services to be received in the future for use in research and development activities are capitalized and then expensed as the related goods are delivered or the services are performed.

For the period from January 25, 2022 (inception) to June 30, 2022, the Company's research and development expenses were comprised primarily of the cost of the tecarfarin asset and related transaction costs.

Patents

Patent costs are comprised primarily of external legal fees, filing fees incurred to file patent applications, and periodic renewal fees to keep the patent in force and are expensed as incurred as a component of general and administrative expense.

Note 2. Recent Accounting Guidance

Recent Accounting Pronouncements Not Yet Adopted

In August 2020, the FASB issued ASU No. 2020-06, Debt — Debt with Conversion and Other Options (Subtopic 470-20) and Derivatives and Hedging — Contracts in Entity's Own Equity (Subtopic 815-40):

Accounting for Convertible Instruments and Contracts in an Entity, which simplifies the accounting for certain financial instruments with characteristics of liabilities and equity, including convertible instruments and contracts on an entity's own equity.

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 2. Recent Accounting Guidance (cont.)

ASU 2020-06 is effective on a prospective basis for annual reporting periods beginning after December 15, 2023 and for interim periods within those periods. Early adoption is permitted. The Company is currently evaluating the impact that this new standard will have on its financial position and results of operations.

Note 3. Fair Value Measurements

Assets and liabilities recorded at fair value on a recurring basis in the balance sheet are categorized based upon the level of judgment associated with the inputs used to measure their fair values. Fair value is defined as the exchange price that would be received for an asset or an exit price that would be paid to transfer a liability in the principal or most advantageous market for the asset or liability in an orderly transaction between market participants on the measurement date. Valuation techniques used to measure fair value must maximize the use of observable inputs and minimize the use of unobservable inputs. The authoritative guidance on fair value measurements establishes a three-tier fair value hierarchy for disclosure of fair value measurements as follows:

- Level 1 — Observable inputs such as unadjusted, quoted prices in active markets for identical assets or liabilities at the measurement date.
- Level 2 — Inputs (other than quoted prices included in Level 1) are either directly or indirectly observable for the asset or liability. These include quoted prices for similar assets or liabilities in active markets and quoted prices for identical or similar assets or liabilities in markets that are not active.
- Level 3 — Unobservable inputs that are supported by little or no market activity and that are significant to the fair value of the assets or liabilities.

The Company classifies its embedded derivative liability as a Level 3 financial instrument and measures and reports its embedded derivatives at fair value.

Financial liabilities subject to fair value measurements on a recurring basis and the level of inputs used in such measurements by major security type as of June 30, 2022 are presented in the following table:

	June 30, 2022			
	Level 1	Level 2	Level 3	Fair Value
Financial Liabilities:				
Derivative liabilities	$ —	$ —	$ 140,977	$ 140,977
Total Financial Liabilities	$ —	$ —	$ 140,977	$ 140,977

	Derivative liabilities
Balance at January 25, 2022	$ —
Fair value of financial instruments at issuance	134,415
Change in fair value	6,562
Balance at June 30, 2022	$ 140,977

The carrying amounts of cash, prepayments, deferred offering costs, accounts payable and accrued liabilities approximate their fair values due to their short-term nature. There were no transfers of liabilities among the fair value measurement categories during the period from January 25, 2022 (inception) to June 30, 2022.

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 4. Accrued Liabilities

Accrued liabilities consist of the following:

	June 30, 2022
Accrued consulting fees	$ 115,000
Other	17,201
Total accrued liabilities	$ 132,201

Note 5. Asset Purchase Agreement

On April 1, 2022, the Company completed an asset purchase agreement with HESP LLC, the assignee of tecarfarin and related assets (the "Asset Purchase Agreement"). Pursuant to the terms of the Asset Purchase Agreement, the Company acquired all of the assets of HESP LLC, including all intellectual property and other rights related to tecarfarin, the tecarfarin IND 77041, all rights under the license, development and commercialization agreement dated as of September 16, 2015 by and between Armetheon, Inc. ("Armetheon") (which was later assigned by Armetheon to Espero BioPharma, Inc. ("Espero"), and China Cardiovascular Focus Ltd, an affiliate of Lee's Pharmaceutical Holdings Limited ("Lee's Pharmaceutical"), relating to tecarfarin and related trademarks. In consideration of the purchase of the assets, the Company paid HESP LLC $100,000 on the closing date and paid an additional $100,000 on June 1, 2022. As additional consideration, the Company agreed to pay HESP LLC the following development milestone payments:

Development Milestones	Milestone Payments
Completion of enrollment of Lee's Pharmaceutical Phase 3 clinical trial	$ 250,000
First MAA submitted in the People's Republic of China	350,000
First Commercial Sale to a Third Party	$ 1,200,000

Financing Milestones

As additional consideration, the Company agreed to pay the following amounts, up to $2,000,000, upon each financing milestone as follows (i) 35% of any proceeds received from any licensing or partnering revenue; and (ii) IPO proceeds. The aggregate payments under the development milestone payments and financing milestone payments shall not exceed $2,000,000.

The Company accounted for the transaction as an asset acquisition as substantially all of the estimated fair value of the gross assets acquired was concentrated in a single identified in-process research and development asset, the tecarfarin asset, thus satisfying the requirements of the screen test in accordance with the criteria under ASC 805-10-55-5C. The assets acquired in the transaction were measured based on the fair value of the consideration paid including the direct transaction costs of $20,095, as the fair value of the consideration paid was more readily determinable than the fair value of the assets acquired. The following table summarizes the purchase price of the assets acquired:

In process research and development	$ 200,000
Transaction costs	20,095
Total	$ 220,095

All costs the Company incurred in connection with this Asset Purchase Agreement were recognized as research and development expenses in the Company's statement of operations and comprehensive loss as these assets had no alternative future use at the time of the acquisition transaction. Due to the nature of the regulatory, sales and financing-based milestones, the contingent consideration was not included in the initial cost of assets purchased as they are contingent upon events that are outside the Company's control. However, upon

achievement or anticipated achievement of each milestone, the Company will recognize the related appropriate payment as additional research and development expense. Contingent consideration will not be recorded until it is probable the milestone events occur. As of June 30, 2022, none of the contingent events have occurred.

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 6. Convertible Debt

March 2022 Note

In March 2022, the Company entered into a convertible promissory note agreement (the "March 2022 Note") and received cash proceeds of $500,000. The March 2022 Note bears interest at a rate equal to simple interest of 5.0% per annum computed on the basis of the 360-day year of twelve (12) 30-day months. The March 2022 Note is due and payable on March 1, 2025, unless earlier converted or repaid.

Pursuant to the March 2022 Note, the principal and accrued but unpaid interest will be automatically converted into equity securities sold in the Next Equity Financing of the Company comprising a single transaction or a series of related transactions in which total proceeds of at least $3.0 million is raised. The principal and unpaid and accrued interest of the March 2022 Note at the date of conversion will be converted into shares at a conversion price equal to 80% of the price per share paid by investors purchasing such shares in the Next Equity Financing. If the Company consummates a Change of Control prior to repayment in full of the March 2022 Note, immediately prior to the Change of Control, the outstanding principal and any unpaid and accrued interest will automatically convert into common equity of the Company (or directly into proceeds paid to the holders of common equity in connection with the Change of Control) at a price per share that is 80% of the price per share of common equity paid at the Change of Control.

The Company evaluated whether the March 2022 Note contain embedded features that meet the definition of derivatives under FASB ASC 815, Derivatives and Hedging. The Company determined that these redemption features contained rights and obligations for conversion contingent upon a potential future financing event or a change in control. Thus, the embedded put options were bifurcated from the face value of the March 2022 Note and accounted for as a derivative liability to be remeasured at the end of each reporting period with the change in the fair value included in other expense, in the accompanying statement of operations and comprehensive loss. The fair value of the put option derivative liability at issuance was $104,883 with the offsetting amount being recorded as a debt discount. Debt issuance costs totaled $1,460. The debt discount and debt issuance costs are being amortized to interest expense using the effective interest method over the expected term of the March 2022 Note. The effective interest rate of the March 2022 Note is 12.1% compared to a stated interest rate of 5.0%.

June 2022 Note

In June 2022, the Company entered into a convertible promissory note agreement (the "June 2022 Note") and received cash proceeds of $50,000. The June 2022 Note bears interest at a rate equal to simple interest of 6.0% per annum computed on the basis of the 360-day year of twelve (12) 30-day months. The June 2022 Note is due and payable on June 13, 2025, unless earlier converted or repaid.

Pursuant to the June 2022 Note, the principal and accrued but unpaid interest will be automatically converted into equity securities sold in the Next Equity Financing of the Company comprising a single transaction or a series of related transactions in which total proceeds of at least $3.0 million is raised. The principal and unpaid and accrued interest of the June 2022 Note at the date of conversion will be converted into shares at a conversion price equal to 60% of the price per share paid by investors purchasing such shares in the Next Equity Financing. If the Company consummates a Change of Control prior to repayment in full of the June 2022 Note, immediately prior to the Change of Control, the outstanding principal and any unpaid and accrued interest will automatically convert into common equity of the Company (or directly into proceeds paid to the holders of common equity in connection with the Change of Control) at a price per share that is 60% of the price per share of common equity paid at the Change of Control.

The Company evaluated whether the June 2022 Note contain embedded features that meet the definition of derivatives under FASB ASC 815, Derivatives and Hedging. The Company determined that these redemption features contained rights and obligations for conversion contingent upon a potential future financing event or a change in control. Thus, the embedded put options were bifurcated from the face value of the June 2022 Note

and accounted for as a derivative liability to be remeasured at the end of each reporting period with the change in the fair value included in other expense, in the accompanying statement of operations and comprehensive loss. The fair

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 6. Convertible Debt (cont.)

value of the put option derivative liability at issuance was $29,532 with the offsetting amount being recorded as a debt discount. The debt discount is being amortized to interest expense using the effective interest method over the expected term of the June 2022 Note. The effective interest rate of the June 2022 Note is 25.7% compared to a stated interest rate of 6.0%.

Note 7. Related Party Transactions

On January 25, 2022, the Company into an agreement with Phamace, LLC, a consulting firm of which Quang Pham, the Company's Chief Executive Officer, is the sole member, for an initial term of January 25, 2022 through February 28, 2022. Pursuant to the agreement, the Company shall pay the sum of $115,000 to Phamace, LLC for advisory and administrative services rendered relating to preparing the Company to launch as an operating company, which is due and payable on September 30, 2022.

On January 25, 2022, the Company issued 7,500,000 shares of common stock, pursuant to a subscription agreement, to Quang Pham, our Chief Executive Officer, of which 4,500,000 were subsequently transferred to related trusts, of which Mr. Pham's child is a beneficiary and Mr. Pham is the trustee with sole voting and disposition power with respect to the shares owned by the Trusts. Mr. Pham paid a total of $7,500 for such founders shares.

On March 1, 2022, the Company issued a convertible promissory note in the amount of $500,000 to John Murphy, a member of the Company's board of directors, which bears interest at 5% and matures on March 1, 2025. The note is convertible into shares of Common Stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the note plus (if applicable) any accrued but unpaid interest under the Note by (ii) eighty percent (80%) of the price per share of the equity securities issued and sold at the initial closing of our next equity financing. See Note 6 for further discussion.

On May 17, 2022, the Company issued 450,000 shares of restricted common stock, pursuant to a restricted stock purchase agreement, to Matthew Szot its Acting Chief Financial Officer, which shares shall vest quarterly over a period of two (2) years, subject to certain adjustments, as provided in the Restricted Stock Purchase Agreement dated May 17, 2022.

Note 8. Commitments and Contingencies

Operating Lease

As of June 30, 2022, the Company has not entered into any long-term operating lease agreements.

Contingencies

In the normal course of business, the Company enters into contracts and agreements that contain a variety of representations and warranties and provide for general indemnifications. The Company's exposure under these agreements is unknown, because it involves claims that may be made against the Company in the future, but have not yet been made. The Company accrues a liability for such matters when it is probable that future expenditures will be made and such expenditures can be reasonably estimated.

Indemnification

In accordance with the Company's certificate of incorporation and bylaws, the Company indemnifies its officers and directors for certain events or occurrences, subject to certain limits, while they are serving in such capacity. There have been no claims to date, and the Company has a directors and officers liability insurance policy that may enable it to recover a portion of any amounts paid for future claims.

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 9. Stockholders' Equity

Common Stock

Pursuant to the Certificate of Incorporation filed on January 25, 2022, the Company is authorized to issue a total of 10,000,000 shares of common stock with a par value of $0.001 per share. As of June 30, 2022, 7,950,000 shares of common stock were issued and outstanding.

Holders of common stock are entitled to one vote for each share of common stock held of record for the election of the Company's directors and all other matters requiring stockholder action. Holders of common stock will be entitled to receive such dividends, if any, as may be declared from time to time by the Company's Board in its discretion out of funds legally available therefor.

Note 10. Income Taxes

The Company's income (loss) before provision (benefit) for income taxes for the period ended June 30, 2022 was generated in the following jurisdictions:

	January 25, 2022 (Inception) to June 30, 2022
Domestic	$ (678,788)
Foreign	—
Loss before income taxes	$ (678,788)

The Company's provision is $0, which is primarily driven by the federal and state statutory income tax rates on current year losses, offset by the Company's full valuation allowance.

The components of income tax expense (benefit) were as follows:

	January 25, 2022 (Inception) to June 30, 2022
Current	
Federal	$ —
State	—
Foreign	—
Total current provision	—
Deferred:	
Federal	—
State	—
Foreign	—
Total deferred provision	—
Provision for income taxes	$ —

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 10. Income Taxes (cont.)

A reconciliation of income tax expense to the amount computed by applying the statutory federal income tax rate to the loss from operations is summarized for the period ended June 30, 2022, as follows:

	January 25, 2022 (Inception) to June 30, 2022
Tax benefit at statutory tax rate	$ (142,546)
State benefit, net of federal benefit	(23,604)
Permanent differences	1,794
Other	3,907
Valuation allowance	160,449
	$ —

Deferred income taxes reflect the net tax effects of temporary differences between the carrying amount of assets and liabilities for financial reporting purposes and the amounts used for income tax purposes. Significant components of the Company's deferred tax assets and liabilities at June 30, 2022 are shown below. The Company has established a full valuation allowance against net deferred tax assets due to the uncertainty that such assets will be realized. The Company periodically evaluates the recoverability of its deferred tax assets. At such time as it is determined that it is more likely than not that the deferred tax asset will be realized, the valuation allowance will be reduced. The increase in the valuation allowances of $160,449 for the period from January 25, 2022 (inception) to June 30, 2022 was primarily due to the Company's net operating losses and basis differences in intangible assets generated during the current year.

The components of deferred tax assets and liabilities consisted of the following at June 30, 2022:

	June 30, 2022
Deferred tax assets:	
Net operating loss carryforwards	$ 107,187
Licenses acquired	53,262
Total deferred tax assets	160,449
Valuation allowance for deferred tax assets	(160,449)
Net deferred tax asset	$ —

As of June 30, 2022, the Company has net operating loss carryforwards of approximately $437,000 and $437,000 available to reduce future taxable income, if any, for federal and state income tax purposes, respectively. The Company's US federal and state net operating loss carryovers can be carried forward indefinitely, but the deduction related to these net operating losses is limited to 80% of taxable income when utilized in future years.

The utilization of net operating loss carryforwards and tax credit carryovers could be subject to annual limitations under Section 382 and 383 of the Internal Revenue Code of 1986, and similar state tax provisions, due to ownership change limitations that may have occurred previously or that could occur in the future. These ownership changes limit the amount of net operating loss carryforwards and other deferred tax assets that can be utilized to offset future taxable income and tax, respectively. In general, an ownership change, as defined by Section 382 and 383, results from transactions increasing ownership of certain stockholders or public groups in

the stock of the corporation by more than 50 percent points over a three-year period. The Company has not conducted an analysis of an ownership change under section 382. To the extent that a study is completed and an ownership change is deemed to occur, the Company's net operating losses and tax credits could be limited.

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 10. Income Taxes (cont.)

The Company applies the two-step approach to recognizing and measuring uncertain tax positions. The first step is to evaluate the tax position for recognition by determining if the weight of available evidence indicates it is more likely than not that the position will be sustained upon audit, including resolution of related appeals or litigation processes, if any. The second step is to measure the tax benefit as the largest amount, which is more than 50% likely of being realized upon ultimate settlement. Income tax positions must meet a more likely than not recognition threshold to be recognized under ASC 740 upon initial measurement and in subsequent periods. ASC 740-10 also provides guidance on measurement, derecognition, classification, interest and penalties, accounting in interim periods, disclosure and transition.

At June 30, 2022, the Company did not have any significant uncertain tax positions. The Company will recognize interest and penalties related to uncertain tax positions in income tax expense. As of June 30, 2022, the Company had no accrued interest or penalties related to uncertain tax positions and no amounts have been recognized in the Company's statement of operations and comprehensive loss. The Company does not anticipate a material change to unrecognized tax benefits in the next twelve months.

All of the Company's tax years will remain open for examination by the federal and state taxing authorities to the extent that the Company's tax attributes are utilized in future years to offset income or income taxes.

Note 11. Net Loss Per Share

The following table sets forth the computation of the basic and diluted net loss per common share:

	January 25, 2022 (Inception) to June 30, 2022
Numerator:	
Net loss	$ (678,788)
Denominator:	
Weighted average common shares outstanding	7,626,923
Net loss per share, basic and diluted	$ (0.09)

Since the Company was in a loss position for the period presented, basic net loss per share is the same as diluted net loss per share as the inclusion of all potential dilutive securities would have been anti-dilutive. For the period from inception to June 30, 2022, there were no potential dilutive securities other than the March 2022 and June 2022 Notes.

Note 12. Subsequent Events

The Company has evaluated subsequent events that occurred after the balance sheet date up to October 7, 2022, the date that the financial statements were issued and determined that there have been no events that have occurred that would require adjustments to our disclosures in the financial statements except for the transaction described below:

On July 11, 2022, the Company issued a restricted stock grant of 46,875 shares of common stock to a consultant. These shares fully vested on October 1, 2022.

On July 11, 2022, the Company granted stock options to purchase 650,000 shares of common stock at an exercise price of $0.64 per common share. The stock option grants were issued to directors, members of the Scientific Advisory Board, and other consultants to the Company and vest in periods ranging over one to three years.

In July 2022, the Company issued convertible promissory notes in the aggregate amount of $450,000 (the "July 2022 Notes"), in August 2022, the Company issued a convertible promissory note in the amount of $50,000 (the "August 2022 Note") and in September 2022, the Company issued convertible promissory notes in the aggregate amount of $75,000 (the "September 2022 Notes"). The July 2022 Notes, the August 2022 Note and the September 2022 Note bear interest at 6% and mature on September 13, 2025 ("Maturity Date").

CADRENAL THERAPEUTICS, INC.
Notes to Financial Statements

Note 12. Subsequent Events (cont.)

The principal amount due under the July 2022 Notes, the August 2022 Note and the September 2022 Notes (and, at the Company's option, any accrued but unpaid interest under the July 2022 Notes, the August 2022 Note and the September 2022 Notes) will be automatically converted, on or before the Maturity Date, into Next Equity Securities in the Next Equity Financing. The July 2022 Notes, the August 2022 Note and September 2022 Notes are convertible into shares of common stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the July 2022 Notes, the August 2022 Note and the September 2022 Notes, plus (if applicable) any accrued but unpaid interest under the July 2022 Notes, the August 2022 Note and the September 2022 Notes by (ii) sixty percent (60%) of the price per share of the Next Equity Securities sold in the Next Equity Financing. In the event of a Change of Control which occurs prior to repayment in full of the July 2022 Notes, the August 2022 Note or the September 2022 Notes, immediately prior to the Change of Control, the outstanding principal and any accrued but unpaid interest on the July 2022 Notes, the August 2022 Note and the September 2022 Notes will convert directly into our common equity (or directly into proceeds paid to the holders of our common equity in connection with the Change of Control) at a price per share that is 60% of the price per share of common equity paid at the Change of Control. Boustead Securities, LLC acted as the placement agent for the private placement and received as part of its compensation and five-year warrants to purchase shares of our common stock at a price equal to the conversion price of the Private Placement Notes in an amount equal to 6% of the shares of common stock underlying the Private Placement Notes.

The Company has determined the redemption features in the July 2022 Notes, August 2022 Note and September 2022 Notes contain rights and obligations for conversion contingent upon a potential future financing event or a change in control, and that such features are required to be bifurcated from the host debt instrument and accounted for as a derivative liability to be remeasured at the end of each reporting period.

On August 18, 2022, the Company entered into an Amendment to Asset Purchase Agreement, whereby in lieu of the $1,800,000 cash payment that would have been due to HESP LLC pursuant to the Asset Purchase Agreement as a result of an initial public offering, HESP LLC agreed to accept shares of the Company's common stock, such number of shares to be calculated based upon a 40% discount to the price of the Company's common stock sold in the initial public offering. If and when it is determined that it is probable this financing-based milestone (initial public offering) will be achieved, the Company will recognize the payment of $3.0 million in research and development expense.

On August 18, 2022, the Company issued 40,000 shares of common stock to a consultant. These shares were fully vested upon the date of grant.

On August 18, 2022, the Company issued 40,000 shares of common stock to Steven Zelenkofske, a director, for recruiting services performed. These shares were fully vested upon the date of grant.

On August 18, 2022, the Company issued stock options to purchase 300,000 shares of common stock to Douglas Losordo, our Acting Chief Medical Officer. The options have an exercise price of $0.64 per common share and will vest quarterly over a period of three (3) years, subject to certain adjustments.

On August 18, 2022, the Company granted stock options to purchase 100,000 shares of common stock at an exercise price of $0.64 per common share. The stock option grants were issued to members of the Scientific Advisory Board and vest over three years.

On September 16, 2022, the Company issued 50,000 shares of common stock to John Murphy, a director nominee. These shares were fully vested upon the date of grant.

On October 3, 2022, the Company issued a stock grant totaling 137,000 shares of common stock to various consultants for services. These shares were fully vested upon the date of grant.

On October 3, 2022, the Company issued stock options to purchase 50,000 shares of common stock to a director nominee. The options have an exercise price of $5.00 per common share and will vest over a three-year period, with 33.33% vesting on the first anniversary of the effective date, and thereafter 1/36 on a monthly basis.

CADRENAL THERAPEUTICS, INC.
BALANCE SHEET

	September 30, 2022 (Unaudited)	June 30, 2022
Assets:		
Current assets:		
Cash	$ 103,082	$ 59,646
Prepaid expenses	24,033	27,691
Deferred offering costs	414,240	153,814
Total current assets	541,355	241,151
Property and equipment	1,772	—
Other assets	2,195	2,195
Total Assets	$ 545,322	$ 243,346
Liabilities:		
Current liabilities:		
Accounts payable	$ 308,221	$ 207,505
Accrued liabilities	364,141	132,201
Total current liabilities	672,362	339,706
Convertible note payable, net of debt discount – related party	432,952	404,198
Convertible note payable, net of debt discount	95,520	20,759
Derivative liabilities	502,223	140,977
Accrued interest – convertible note payable	22,088	8,544
Total liabilities	1,725,145	914,184
Stockholders' deficit:		
Common stock, $0.001 par value; 10,000,000 shares authorized , 8,126,875 and 7,950,000 issued and outstanding as of September 30, 2022 and June 30, 2022, respectively	8,127	7,950
Additional paid-in capital	516,952	—
Accumulated deficit	(1,704,902)	(678,788)
Total stockholders' deficit	(1,179,823)	(670,838)
Total Liabilities and Stockholder's Deficit	$ 545,322	$ 243,346

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
STATEMENTS OF OPERATIONS AND COMPREHENSIVE LOSS
(Unaudited)

	Three Months Ended September 30, 2022	January 25, 2022 (inception) through September 30, 2022
Operating expenses:		
General and administrative expenses	$ 915,897	$ 1,348,670
Research and development expenses	56,388	276,482
Depreciation expense	506	506
Total operating expenses	972,791	1,625,658
Loss from operations	(972,791)	(1,625,658)
Other expense:		
Interest expense	13,541	22,068
Interest expense, amortization of debt discount	18,470	29,302
Change in in fair value of derivative liabilities	21,312	27,874
Total other expenses	53,323	79,244
Net loss and comprehensive loss	$ (1,026,114)	$ (1,704,902)
Net loss per common share, basic and diluted	$ (0.13)	$ (0.22)
Weighted average number of common shares used in computing net loss per common share, basic and diluted	8,037,218	7,778,778

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
STATEMENTS OF CHANGES IN STOCKHOLDERS' DEFICIT
(Unaudited)

	Common Stock		Additional Paid-In Capital	Accumulated Deficit	Total Stockholders' Deficit
	Shares	Amount			
Balance, June 30, 2022	7,950,000	$ 7,950	$ —	$ (678,788)	$ (670,838)
Equity-based compensation – options, restricted stock and RSUs	176,875	177	478,507	—	478,684
Issuance of placement agent warrants	—	—	38,445	—	38,445
Net loss	—	—	—	(1,026,114)	(1,026,114)
Balance, September 30, 2022	8,126,875	$ 8,127	$ 516,952	$ (1,704,902)	$ (1,179,823)
Balance, January 25, 2022 (inception)	—	$ —	$ —	$ —	$ —
Issuance of founder shares	7,950,000	7,950	—	—	7,950
Equity-based compensation – options, restricted stock and RSUs	176,875	177	478,507	—	478,684
Issuance of placement agent warrants	—	—	38,445	—	38,445
Net loss	—	—	—	(1,704,902)	(1,704,902)
Balance, September 30, 2022	8,126,875	$ 8,127	$ 516,952	$ (1,704,902)	$ (1,179,823)

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
STATEMENT OF CASH FLOWS
(Unaudited)

	January 25, 2022 (inception) through September 30, 2022
Cash flows from operating activities:	
Net loss	$ (1,704,902)
Adjustments to reconcile net loss to net cash used in operating activities:	
Depreciation	506
Stock based compensation	478,684
Amortization of debt discount	29,302
Change in fair value of derivative liability	27,874
Changes in operating assets and liabilities:	
Prepaid expenses	(24,033)
Deferred equity offering costs	(414,240)
Other assets	(2,195)
Accounts payable	308,221
Accrued liabilities	364,141
Accrued interest – convertible notes payable	22,088
Net cash used in operating activities	(914,554)
Cash flows used in investing activities:	
Purchase of property and equipment	(2,278)
Net cash used in investing activities	(2,278)
Cash flows from financing activities:	
Proceeds from issuance of convertible notes, net of debt issuance costs	1,011,964
Proceeds from sale of common stock	7,950
Net cash provided by financing activities	1,019,914
Net change in cash	103,082
Cash – beginning of the period	—
Cash – end of the period	$ 103,082

The accompanying notes are an integral part of these financial statements.

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies

Cadrenal Therapeutics, Inc. (the "Company" or "Cadrenal") was incorporated on January 25, 2022 (inception) in the State of Delaware and is headquartered in Ponte Vedra, Florida. Cadrenal is focused on developing a novel therapy with orphan drug indication, tecarfarin, for the prevention of systemic thromboembolism (blood clots) of cardiac origin in patients with end-stage renal disease (on dialysis) and atrial fibrillation (irregular heartbeat) or AFib. Tecarfarin is an anticoagulant designed using a drug design process which targets a different pathway than most commonly prescribed drugs for the treatment of thrombosis and AFib.

Basis of Presentation

The accompanying financial statements have been prepared in accordance with accounting principles generally accepted in the United States of America (GAAP) and applicable rules and regulations of the U.S. Securities and Exchange Commission ("SEC") for the fair presentation of the Company's financial position for the period from January 25, 2022 (inception) through September 30, 2022. The Company's date of inception was January 25, 2022 and the fiscal year-end is December 31.

Unaudited Interim Financial Information

The accompanying interim balance sheet as of September 30, 2022, the interim statements of operations and comprehensive loss, and changes in stockholders' deficit for the three months ended September 30, 2022 and for the period from January 25, 2022 (inception) through September 30, 2022, and cash flows for the period from January 25, 2022 (inception) through September 30, 2022 are unaudited. The unaudited interim financial statements have been prepared on the same basis as the audited financial statements and, in the opinion of management, reflect all adjustments, which include only normal recurring adjustments, necessary to present fairly the balance sheet as of September 30, 2022, the statements of operations and comprehensive loss, and changes in stockholders' deficit for the three months ended September 30, 2022 and for the period from January 25, 2022 (inception) through September 30, 2022, and cash flows for the period from January 25, 2022 (inception) through September 30, 2022. The financial data disclosed in these notes to the financial statements related to the three months ended September 30, 2022 and the period from January 25, 2022 (inception) through September 30, 2022 and as of September 30, 2022 are also unaudited. The results of operations for the are not necessarily indicative of the results to be expected for the full year ending December 31, 2022, or for any other future annual or interim period. The accompanying unaudited interim financial statements should be read in conjuction with the audited financial statements and notes thereto included elsewhere in this prospectus.

Liquidity and Going Concern

The accompanying financial statements have been prepared assuming that the Company will continue as a going concern, which contemplates the realization of assets and settlement of liabilities and commitments in the normal course of business. The financial statements do not reflect any adjustments relating to the recoverability and reclassification of assets and liabilities that might be necessary if the Company is unable to continue as a going concern. Since inception, the Company has incurred operating losses and negative cash flows from operations. For the period from January 25, 2022 (inception) to September 30, 2022, the Company had a net loss of $1,704,902 with cash used in operations of $914,554. As of September 30, 2022, the Company had cash of $103,082, negative working capital of $131,007, and an accumulated deficit of $1,704,902. During the period from January 25, 2022 (inception) through September 30, 2022, the Company raised $1,125,000 in gross proceeds from the issuance of a convertible notes. The Company is projecting that its operating losses and expected capital needs will exceed its existing cash balances and cash expected to be generated from operations for the foreseeable future. In order to meet the Company's expected obligations, management intends to raise additional funds through equity and debt financings. However, there can be no assurance that the Company will be able to complete any additional equity or debt financings on terms acceptable to the Company or at all. If the Company is unable to raise additional funding to meet its working capital needs in the

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies (cont.)

future, it will be forced to delay or reduce the scope of its research programs and/or limit or cease its operations. As a result, there is uncertainty in the Company's ability to meet its current operating and capital expenses. These factors, among others, raise substantial doubt about the Company's ability to continue as a going concern for at least one year from the date the accompanying financial statements are issued.

In March 2020, the World Health Organization declared COVID-19 a pandemic. The COVID-19 pandemic has negatively impacted the global economy, disrupted global supply chains and created significant volatility and disruption of financial markets. The Company cannot at this time predict the specific extent, duration, or full impact that the ongoing COVID-19 pandemic will have on its financial condition and operations, including planned clinical trials. The impact of the COVID-19 pandemic on the financial performance of the Company will depend on future developments, including the duration and spread of the pandemic and related governmental advisories and restrictions. These developments and the impact of the COVID-19 pandemic on the financial markets and the overall economy are highly uncertain. If business conditions, financial markets and/or the overall economy are impacted for an extended period, the Company's results may be adversely affected.

Emerging Growth Company Status

From time to time, new accounting pronouncements, or Accounting Standards Updates ("ASU") are issued by the FASB or other standard setting bodies and adopted by the Company as of the specified effective date. Unless otherwise discussed, the impact of recently issued standards that are not yet effective is not expected to have a material impact on the Company's financial position or results of operations upon adoption.

As an "emerging growth company" ("EGC") under the Jumpstart Our Business Startups Act ("JOBS Act"), the Company may elect to take advantage of certain forms of relief from various reporting requirements that are applicable to public companies. The relief afforded under the JOBS Act includes an extended transition period for the implementation of new or revised accounting standards. The Company has elected to take advantage of this extended transition period and, as a result, the Company's financial statements may not be comparable to those of companies that implement accounting standards as of the effective dates for public companies. The Company may take advantage of the relief afforded under the JOBS Act up until the last day of the fiscal year following the fifth anniversary of an offering or such earlier time that it is no longer an EGC.

Use of Estimates

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and disclosure of contingent assets and liabilities as of the date of the financial statements and the reported amounts of expenses during the reporting period. Significant estimates and assumptions made in the accompanying financial statements include but are not limited to the fair value of financial instruments, fair value of common stock, deferred tax assets and valuation allowance, income tax uncertainties, and certain accruals. The Company evaluates its estimates and assumptions on an ongoing basis using historical experience and other factors and adjusts those estimates and assumptions when facts and circumstances change. Actual results could differ from those estimates.

Concentration of Credit and other Risks and Uncertainties

Financial instruments, which potentially subject the Company to significant concentrations of credit risk, consist primarily of cash. Cash is maintained at high credit quality financial institutions and, at times, balances may exceed federally insured limits. All interest-bearing and non-interest-bearing cash balances are insured up to $250,000 per depositor at each financial institution. The Company has never experienced any losses related to these balances.

The Company is subject to a number of risks common for early-stage biopharmaceutical companies including, but not limited to, dependency on the clinical and commercial success of its product candidates, ability to obtain regulatory

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies (cont.)

approval of its product candidates, the need for substantial additional financing to achieve its goals, uncertainty of broad adoption of its approved products, if any, by physicians and patients, significant competition and untested manufacturing capabilities.

Segments

Operating segments are defined as components of an entity for which separate financial information is available and that is regularly reviewed by the Chief Operating Decision Maker (CODM) in deciding how to allocate resources to an individual segment and in assessing performance. The Company's CODM is its Chief Executive Officer. The Company has determined it operates in a single operating segment and has one reportable segment.

Cash and Cash Equivalents

The Company considers all highly liquid investments purchased with original maturities of three months or less from the purchase date to be cash equivalents. The Company did not have any cash equivalents at September 30, 2022 or June 30, 2022.

Derivative Financial Instruments

The Company evaluates all of its agreements to determine if such instruments have derivatives or contain features that qualify as embedded derivatives. The Company accounts for certain redemption features that are associated with convertible notes as liabilities at fair value and adjusts the instruments to their fair value at the end of each reporting period. Derivative financial liabilities are initially recorded at fair value, with gains and losses arising from changes in the fair value recognized in other income (expense) in the accompanying statements of operations and comprehensive loss for each reporting period while such instruments are outstanding. The embedded derivative liability is valued using a probability-weighted expected return model. If the Company repays the note holders or if, during the next round of financing, the note holders convert the debt into equity, the derivative financial liability will be de-recognized and reclassified to stockholders' equity (deficit) on that date. Derivative instrument liabilities are classified in the balance sheet as current or non-current based on whether or not net-cash settlement of the derivative instrument could be required within 12 months of the balance sheet date.

Stock-Based Compensation

The Company measures its stock-based awards granted to employees, consultants and directors based on the estimated fair values of the awards and recognize the compensation over the requisite service period. The Company uses the Black-Scholes option-pricing model to estimate the fair value of its stock option awards. Stock-based compensation is recognized using the straight-line method. As the stock compensation expense is based on awards ultimately expected to vest, it is reduced by forfeitures. The Company accounts for forfeitures as they occur.

Determination of the Fair Value of Common Stock

The fair value of the Company's common stock for purposes of stock-based awards has been determined by its board of directors. Because there has been no public market for the Company's common stock, and in the absence of arm's-length transactions in the Company's common stock with independent third parties, the board of directors has determined the fair value of its common stock with the assistance of an independent third-party valuation prepared in accordance with the guidelines outlined in the American Institute of Certified Public Accountants Practice Aid, Valuation of Privately-Held-Company Equity Securities Issued as Compensation. In assessing the fair value of the Company's common stock on each grant date, the Company considered numerous objective and subjective factors, including, but not limited to, the following:

- independent third-party valuation as of May 31, 2022;

- the Company's current and expected operating and financial performance, including our level of available capital resources;

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies (cont.)

- the value of the Company's tangible and intangible assets;
- market conditions affecting comparable public companies, as reflected in comparable companies' market trading multiples, initial public offering valuations, comparable sales or merger transactions, and other metrics;
- the illiquidity of the Company's common stock by virtue of being a private company, and the resulting discount for lack of marketability;
- the business risks we face;
- the likelihood of achieving a particular liquidity event, such as a sale, a merger, or an initial public offering, given prevailing pharmaceutical and biotechnology industry and capital markets conditions; and
- the experience of management and the board of directors

Deferred Offering Costs

The Company capitalizes certain legal, professional and other third-party costs that are directly associated with in-process equity financings until such financings are consummated at which time such costs are recorded against the gross proceeds of the offering. Should an in-process equity financing be abandoned, the deferred offering costs will be expensed immediately as a charge to operating expenses in the statements of operations and comprehensive loss. As of September 30, 2022 and June 30, 2022 deferred offering costs were $414,240 and $153,814, respectively, and are included on the accompanying balance sheets.

Acquisitions

The Company evaluates acquisitions of assets and other similar transactions to assess whether or not the transaction should be accounted for as a business combination or asset acquisition by first applying a screen test to determine whether substantially all of the fair value of the gross assets acquired is concentrated in a single identifiable asset or group of similar identifiable assets. If so, the transaction is accounted for as an asset acquisition. If not, further determination is required as to whether or not the Company has acquired inputs and processes that have the ability to create outputs, which would meet the definition of a business. Significant judgment is required in the application of the screen test to determine whether an acquisition is a business combination or an acquisition of assets.

Acquisitions meeting the definition of business combinations are accounted for using the acquisition method of accounting, which requires that the purchase price be allocated to the net assets acquired at their respective fair values. In a business combination, any excess of the purchase price over the estimated fair values of the net assets acquired is recorded as goodwill.

For asset acquisitions, a cost accumulation model is used to determine the cost of an asset acquisition. Direct transaction costs are recognized as part of the cost of an asset acquisition. The Company also evaluates which elements of a transaction should be accounted for as a part of an asset acquisition and which should be accounted for separately. The cost of an asset acquisition, including transaction costs, is allocated to identifiable assets acquired and liabilities assumed based on a relative fair value basis. Goodwill is not recognized in an asset acquisition. Any difference between the cost of an asset acquisition and the fair value of the net assets acquired is allocated to the non-monetary identifiable assets based on their relative fair values. When a transaction accounted for as an asset acquisition includes an in-process research and development ("IPR&D") asset, the IPR&D asset is only capitalized if it has an alternative future use other than in a particular research and development project. For an IPR&D asset to have an alternative future use: (a) the Company must reasonably expect that it will use the asset acquired in the alternative manner and anticipate economic benefit from that alternative use, and (b) the Company's use of the asset acquired is not contingent on further development of the asset subsequent to the acquisition date (that is, the asset can be used in the alternative

manner in the condition in which it existed at the acquisition date). Otherwise, amounts allocated to IPR&D that have no alternative use are expensed to research and development. Asset acquisitions may include contingent consideration arrangements

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 1. Description of Business and Summary of Significant Accounting Policies (cont.)

that encompass obligations to make future payments to sellers contingent upon the achievement of future financial targets. Contingent consideration is not recognized until all contingencies are resolved and the consideration is paid or probable of payment, at which point the consideration is allocated to the assets acquired on a relative fair value basis.

Income Taxes

Income taxes are accounted for under the asset and liability method. Under this method, deferred tax assets and liabilities are determined based on the difference between the financial statement and tax bases of assets and liabilities using enacted tax rates in effect for the year in which the differences are expected to affect taxable income. Management makes an assessment of the likelihood that the resulting deferred tax assets will be realized. A valuation allowance is provided when it is more likely than not that some portion or all of a deferred tax asset will not be realized. Due to the Company's historical operating performance and net losses, the net deferred tax assets have been fully offset by a valuation allowance.

The Company recognizes uncertain income tax positions at the largest amount that is more likely than not to be sustained upon audit by the relevant taxing authority. An uncertain income tax position will not be recognized if it has less than a 50% likelihood of being sustained. Changes in recognition or measurement are reflected in the period in which judgment occurs. The Company's policy is to recognize interest and penalties related to the underpayment of income taxes as a component of the provision for income taxes.

Net Loss Per Common Share

Basic net loss per common share is calculated by dividing the net loss by the weighted-average number of shares of common stock outstanding for the period, without consideration for potential dilutive shares of common stock. Diluted net loss per common share is computed by dividing net loss by the weighted average number of shares of common stock and common stock equivalents of potentially dilutive securities outstanding for the period determined using the treasury stock or if-converted methods. Since the Company was in a loss position for all periods presented, basic net loss per common share is the same as diluted net loss per common share since the effects of potentially dilutive securities are anti-dilutive. Shares of common stock subject to repurchase are excluded from the weighted-average shares.

Comprehensive Loss

Comprehensive loss is defined as the change in equity during a period from transactions and other events or circumstances from non-owner sources. Net loss and comprehensive loss were the same for all periods presented in the accompanying financial statements.

Research and Development Expenses

Research and development costs are expensed as incurred and consist of fees paid to other entities that conduct certain research and development activities on the Company's behalf. Acquired intangible assets are expensed as research and development costs if, at the time of payment, the technology is under development; is not approved by the FDA or other regulatory agencies for marketing; has not reached technical feasibility; or otherwise has no foreseeable alternative future use. Nonrefundable advance payments for goods or services to be received in the future for use in research and development activities are capitalized and then expensed as the related goods are delivered or the services are performed.

For the period from January 25, 2022 (inception) through September 30, 2022, the Company's research and development expenses were comprised primarily of the cost of the acquired tecarfarin asset and related transaction costs.

Patents

Patent costs are comprised primarily of external legal fees, filing fees incurred to file patent applications, and periodic renewal fees to keep the patent in force and are expensed as incurred as a component of general and administrative expense.

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 2. Recent Accounting Guidance

Recent Accounting Pronouncements

In August 2020, the FASB issued ASU No. 2020-06, Debt — Debt with Conversion and Other Options (Subtopic 470-20) and Derivatives and Hedging — Contracts in Entity's Own Equity (Subtopic 815-40): Accounting for Convertible Instruments and Contracts in an Entity, which simplifies the accounting for certain financial instruments with characteristics of liabilities and equity, including convertible instruments and contracts on an entity's own equity.

ASU 2020-06 is effective on a prospective basis for annual reporting periods beginning after December 15, 2023 and for interim periods within those periods. Early adoption is permitted. The Company adopted ASU 2020-06 on July 1, 2022. The adoption of ASU 2020-06 did not have a material impact on the Company's financial position and results of operations.

Note 3. Fair Value Measurements

Assets and liabilities recorded at fair value on a recurring basis in the balance sheet are categorized based upon the level of judgment associated with the inputs used to measure their fair values. Fair value is defined as the exchange price that would be received for an asset or an exit price that would be paid to transfer a liability in the principal or most advantageous market for the asset or liability in an orderly transaction between market participants on the measurement date. Valuation techniques used to measure fair value must maximize the use of observable inputs and minimize the use of unobservable inputs. The authoritative guidance on fair value measurements establishes a three-tier fair value hierarchy for disclosure of fair value measurements as follows:

- Level 1 — Observable inputs such as unadjusted, quoted prices in active markets for identical assets or liabilities at the measurement date.
- Level 2 — Inputs (other than quoted prices included in Level 1) are either directly or indirectly observable for the asset or liability. These include quoted prices for similar assets or liabilities in active markets and quoted prices for identical or similar assets or liabilities in markets that are not active.
- Level 3 — Unobservable inputs that are supported by little or no market activity and that are significant to the fair value of the assets or liabilities.

The Company classifies its embedded derivative liability as a Level 3 financial instrument and measures and reports its embedded derivatives at fair value.

Financial liabilities subject to fair value measurements on a recurring basis and the level of inputs used in such measurements by major security type as of September 30, 2022 and June 30, 2022 are presented in the following table:

	September 30, 2022			
	Level 1	Level 2	Level 3	Fair Value
Financial Liabilities:				
Derivative liabilities	$ —	$ —	$ 502,223	$ 502,223
Total Financial Liabilities	$ —	$ —	$ 502,223	$ 502,223

	June 30, 2022			
	Level 1	Level 2	Level 3	Fair Value
Financial Liabilities:				
Derivative liabilities	$ —	$ —	$ 140,977	$ 140,977
Total Financial Liabilities	$ —	$ —	$ 140,977	$ 140,977

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 3. Fair Value Measurements (cont.)

	Derivative liabilities
Balance at January 25, 2022	$ —
Fair value of financial instruments at issuance	134,415
Change in fair value	6,562
Balance at June 30, 2022	$ 140,977
Fair value of financial instruments at issuance	339,934
Change in fair value	21,312
Balance at September 30, 2022	$ 502,223

The carrying amounts of cash, prepayments, deferred offering costs, accounts payable and accrued liabilities approximate their fair values due to their short-term nature. There were no transfers of liabilities among the fair value measurement categories during any of the periods presented.

Note 4. Accrued Liabilities

Accrued liabilities consist of the following:

	September 30, 2022	June 30, 2022
Accrued consulting fees	$ 160,000	$ 115,000
Accrued bonus	195,721	—
Other	8,420	17,201
Total accrued liabilities	$ 364,141	$ 132,201

Note 5. Asset Purchase Agreement

On April 1, 2022, the Company completed an asset purchase agreement with HESP LLC, the assignee of tecarfarin and related assets (the "Asset Purchase Agreement"). Pursuant to the terms of the Asset Purchase Agreement, the Company acquired all of the assets of HESP LLC, including all intellectual property and other rights related to tecarfarin, the tecarfarin IND 77041, all rights under the license, development and commercialization agreement dated as of September 16, 2015 by and between Armetheon, Inc. ("Armetheon") (which was later assigned by Armetheon to Espero BioPharma, Inc. ("Espero"), and China Cardiovascular Focus Ltd, an affiliate of Lee's Pharmaceutical Holdings Limited ("Lee's Pharmaceutical"), relating to tecarfarin and related trademarks. In consideration of the purchase of the assets, the Company paid HESP LLC $100,000 on the closing date and paid an additional $100,000 on June 1, 2022. As additional consideration, the Company agreed to pay HESP LLC the following development milestone payments:

Development Milestones	Milestone Payments
Completion of enrollment of Lee's Pharmaceutical Phase 3 clinical trial	$ 250,000
First MAA submitted in the People's Republic of China	350,000
First Commercial Sale to a Third Party	$ 1,200,000

Financing Milestones

As additional consideration, the Company agreed to pay the following amounts, up to $2,000,000, upon each financing milestone as follows (i) 35% of any proceeds received from any licensing or partnering revenue; and

(ii) IPO proceeds. The aggregate payments under the development milestone payments and financing milestone payments shall not exceed $2,000,000.

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 5. Asset Purchase Agreement (cont.)

The Company accounted for the transaction as an asset acquisition as substantially all of the estimated fair value of the gross assets acquired was concentrated in a single identified in-process research and development asset, the tecarfarin asset, thus satisfying the requirements of the screen test in accordance with the criteria under ASC 805-10-55-5C. The assets acquired in the transaction were measured based on the fair value of the consideration paid including the direct transaction costs of $20,095, as the fair value of the consideration paid was more readily determinable than the fair value of the assets acquired. The following table summarizes the purchase price of the assets acquired:

In process research and development	$	200,000
Transaction costs		20,095
Total	$	220,095

All costs the Company incurred in connection with this Asset Purchase Agreement were recognized as research and development expenses in the Company's statement of operations and comprehensive loss as these assets had no alternative future use at the time of the acquisition transaction. Due to the nature of the regulatory, sales and financing-based milestones, the contingent consideration was not included in the initial cost of assets purchased as they are contingent upon events that are outside the Company's control.

However, upon achievement or anticipated achievement of each milestone, the Company will recognize the related appropriate payment as additional research and development expense. Contingent consideration will not be recorded until it is probable the milestone events occur.

On August 18, 2022, the Company entered into an Amendment to Asset Purchase Agreement, whereby in lieu of the $1,800,000 cash payment that would have been due to HESP LLC pursuant to the Asset Purchase Agreement as a result of an initial public offering, HESP LLC agreed to accept shares of the Company's common stock, such number of shares to be calculated based upon a 40% discount to the price of the Company's common stock sold in the initial public offering. If and when it is determined that it is probable this financing-based milestone (initial public offering) will be achieved, the Company will recognize the payment of $3.0 million as research and development expense.

As of September 30, 2022, none of the contingent events have occurred.

Note 6. Convertible Debt

March 2022 Note

In March 2022, the Company entered into a convertible promissory note agreement (the "March 2022 Note") and received cash proceeds of $500,000. The March 2022 Note bears interest at a rate equal to simple interest of 5.0% per annum computed on the basis of the 360-day year of twelve (12) 30-day months. The March 2022 Note is due and payable on March 1, 2025, unless earlier converted or repaid.

Pursuant to the March 2022 Note, the principal and accrued but unpaid interest will be automatically converted into equity securities sold in the Next Equity Financing of the Company comprising a single transaction or a series of related transactions in which total proceeds of at least $3.0 million is raised. The principal and unpaid and accrued interest of the March 2022 Note at the date of conversion will be converted into shares at a conversion price equal to 80% of the price per share paid by investors purchasing such shares in the Next Equity Financing. If the Company consummates a Change of Control prior to repayment in full of the March 2022 Note, immediately prior to the Change of Control, the outstanding principal and any unpaid and accrued interest will automatically convert into common equity of the Company (or directly into proceeds paid to the holders of common equity in connection with the Change of Control) at a price per share that is 80% of the price per share of common equity paid at the Change of Control.

The Company evaluated whether the March 2022 Note contain embedded features that meet the definition of derivatives under FASB ASC 815, Derivatives and Hedging. The Company determined that these redemption features contained rights and obligations for conversion contingent upon a potential future financing event or a change in control. Thus, the embedded put options were bifurcated from the face value of the March 2022 Note and accounted for as a derivative liability to be remeasured at the end of each reporting period with the change in the fair value

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 6. Convertible Debt (cont.)

included in other expense, in the accompanying statement of operations and comprehensive loss. The fair value of the put option derivative liability at issuance was $104,883 with the offsetting amount being recorded as a debt discount. Debt issuance costs totaled $1,460. The debt discount and debt issuance costs are being amortized to interest expense using the effective interest method over the expected term of the March 2022 Note. The effective interest rate of the March 2022 Note is 12.1% compared to a stated interest rate of 5.0%.

June 2022 Note

In June 2022, the Company entered into a convertible promissory note agreement (the "June 2022 Note") and received cash proceeds of $50,000. The June 2022 Note bears interest at a rate equal to simple interest of 6.0% per annum computed on the basis of the 360-day year of twelve (12) 30-day months. The June 2022 Note is due and payable on June 13, 2025, unless earlier converted or repaid.

Pursuant to the June 2022 Note, the principal and accrued but unpaid interest will be automatically converted into equity securities sold in the Next Equity Financing of the Company comprising a single transaction or a series of related transactions in which total proceeds of at least $3.0 million is raised. The principal and unpaid and accrued interest of the June 2022 Note at the date of conversion will be converted into shares at a conversion price equal to 60% of the price per share paid by investors purchasing such shares in the Next Equity Financing. If the Company consummates a Change of Control prior to repayment in full of the June 2022 Note, immediately prior to the Change of Control, the outstanding principal and any unpaid and accrued interest will automatically convert into common equity of the Company (or directly into proceeds paid to the holders of common equity in connection with the Change of Control) at a price per share that is 60% of the price per share of common equity paid at the Change of Control.

The Company evaluated whether the June 2022 Note contain embedded features that meet the definition of derivatives under FASB ASC 815, Derivatives and Hedging. The Company determined that these redemption features contained rights and obligations for conversion contingent upon a potential future financing event or a change in control. Thus, the embedded put options were bifurcated from the face value of the June 2022 Note and accounted for as a derivative liability to be remeasured at the end of each reporting period with the change in the fair value included in other expense, in the accompanying statement of operations and comprehensive loss. The fair value of the put option derivative liability at issuance was $29,532 with the offsetting amount being recorded as a debt discount. The debt discount is being amortized to interest expense using the effective interest method over the expected term of the June 2022 Note. The effective interest rate of the June 2022 Note is 25.7% compared to a stated interest rate of 6.0%.

Boustead Private Placement Notes

In July 2022, the Company issued convertible promissory notes in the aggregate amount of $450,000 (the "July 2022 Notes"), in August 2022, the Company issued a convertible promissory note in the amount of $50,000 (the "August 2022 Note") and in September 2022, the Company issued convertible promissory notes in the aggregate amount of $75,000 (the "September 2022 Notes"). The July 2022 Notes, the August 2022 Note and the September 2022 Note bear interest at 6% and mature on September 13, 2025 ("Maturity Date").

The principal amount due under the July 2022 Notes, the August 2022 Note and the September 2022 Notes (and, at the Company's option, any accrued but unpaid interest under the July 2022 Notes, the August 2022 Note and the September 2022 Notes) will be automatically converted, on or before the Maturity Date, into Next Equity Securities in the Next Equity Financing. The July 2022 Notes, the August 2022 Note and September 2022 Notes are convertible into shares of common stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the July 2022 Notes, the August 2022 Note and the September 2022 Notes, plus (if applicable) any accrued but unpaid interest under the July 2022 Notes, the August 2022 Note and the September 2022 Notes by (ii) sixty percent (60%) of the price per share of the Next Equity Securities sold in the Next Equity Financing. In the event of a Change of Control which occurs prior to

repayment in full of the July 2022 Notes, the August 2022 Note or the September 2022 Notes, immediately prior to the Change of Control, the outstanding principal and any accrued but unpaid interest on the July 2022 Notes, the August 2022 Note and the September 2022 Notes will convert directly into our common equity

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 6. Convertible Debt (cont.)

(or directly into proceeds paid to the holders of our common equity in connection with the Change of Control) at a price per share that is 60% of the price per share of common equity paid at the Change of Control. Boustead Securities, LLC acted as the placement agent for the private placement and received as part of its compensation and five-year warrants to purchase shares of our common stock at a price equal to the conversion price of the Private Placement Notes in an amount equal to 6% of the shares of common stock underlying the Private Placement Notes. The Company has determined that the warrants issued to the placement agent receive equity treatment. The warrants were recorded at fair value at issuance and recorded as debt issuance costs associated with the Boustead Private Placement Notes.

The Company has determined the redemption features in the July 2022 Notes, August 2022 Note and September 2022 Notes contain rights and obligations for conversion contingent upon a potential future financing event or a change in control, and that such features are required to be bifurcated from the host debt instrument and accounted for as a derivative liability to be remeasured at the end of each reporting period.

The Company evaluated whether the July, August and September 2022 Notes contain embedded features that meet the definition of derivatives under FASB ASC 815, Derivatives and Hedging. The Company determined that these redemption features contained rights and obligations for conversion contingent upon a potential future financing event or a change in control. Thus, the embedded put options were bifurcated from the face value of the July, August and September Notes and accounted for as a derivative liabilities to be remeasured at the end of each reporting period with the change in the fair value included in other expense, in the accompanying statements of operations and comprehensive loss. The fair value of the put option derivative liabilities at issuance was $339,934 with the offsetting amount being recorded as a debt discount. The debt discount is being amortized to interest expense using the effective interest method over the expected term of the July, August and September 2022 Notes. The effective interest rates of the July, August and September 2022 Notes were 36.3%, 26.1%, and 28.8%, respectively, compared to a stated interest rate of 6.0%.

Note 7. Related Party Transactions

On January 25, 2022, the Company into an agreement with Phamace, LLC, a consulting firm of which Quang Pham, the Company's Chief Executive Officer, is the sole member, for an initial term of January 25, 2022 through February 28, 2022. Pursuant to the agreement, the Company shall pay the sum of $115,000 to Phamace, LLC for advisory and administrative services rendered relating to preparing the Company to launch as an operating company, which is due and payable on September 30, 2022.

On January 25, 2022, the Company issued 7,500,000 shares of common stock, pursuant to a subscription agreement, to Quang Pham, our Chief Executive Officer, of which 4,500,000 were subsequently transferred to related trusts, of which Mr. Pham's child is a beneficiary and Mr. Pham is the trustee with sole voting and disposition power with respect to the shares owned by the Trusts. Mr. Pham paid a total of $7,500 for such founders shares.

On March 1, 2022, the Company issued a convertible promissory note in the amount of $500,000 to John Murphy, a member of the Company's board of directors, which bears interest at 5% and matures on March 1, 2025. The note is convertible into shares of Common Stock at a conversion price equal to the quotient obtained by dividing (i) the entire principal amount of the note plus (if applicable) any accrued but unpaid interest under the Note by (ii) eighty percent (80%) of the price per share of the equity securities issued and sold at the initial closing of our next equity financing. See Note 6 for further discussion.

On May 17, 2022, the Company issued 450,000 shares of restricted common stock, pursuant to a restricted stock purchase agreement, to Matthew Szot its Acting Chief Financial Officer, which shares shall vest quarterly over a period of two years, subject to certain adjustments, as provided in the Restricted Stock Purchase Agreement dated May 17, 2022.

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 8. Commitments and Contingencies

Operating Lease

As of September 30, 2022, the Company has not entered into any long-term operating lease agreements.

Contingencies

In the normal course of business, the Company enters into contracts and agreements that contain a variety of representations and warranties and provide for general indemnifications. The Company's exposure under these agreements is unknown, because it involves claims that may be made against the Company in the future, but have not yet been made. The Company accrues a liability for such matters when it is probable that future expenditures will be made and such expenditures can be reasonably estimated.

Indemnification

In accordance with the Company's certificate of incorporation and bylaws, the Company indemnifies its officers and directors for certain events or occurrences, subject to certain limits, while they are serving in such capacity. There have been no claims to date, and the Company has a directors and officers liability insurance policy that may enable it to recover a portion of any amounts paid for future claims.

Note 9. Stockholders' Equity

Common Stock

Pursuant to the Certificate of Incorporation filed on January 25, 2022, the Company is authorized to issue a total of 10,000,000 shares of common stock with a par value of $0.001 per share. As of September 30, 2022, 8,126,875 shares of common stock were issued and outstanding. As of June 30, 2022, 7,950,000 shares of common stock were issued and outstanding.

Holders of common stock are entitled to one vote for each share of common stock held of record for the election of the Company's directors and all other matters requiring stockholder action. Holders of common stock will be entitled to receive such dividends, if any, as may be declared from time to time by the Company's Board in its discretion out of funds legally available therefor.

Note 10. Equity-Based Compensation

The Company adopted the Cadrenal Therapeutics, Inc. 2022 Equity Incentive Plan, or the Initial Plan, on July 11, 2022, which was later amended and restated on October 16, 2022, for purposes of clarifying the application of certain of the rules of the Initial Plan to awards approved before such amendment and restatement of the Initial Plan and to facilitate the transition to the Cadrenal Therapeutics, Inc. 2022 Successor Equity Incentive Plan (the "Successor Plan") for the issuance and approval of awards after consummation of this offering. On October 16, 2022, the Board adopted and the Company's stockholders approved the Cadrenal Therapeutics, Inc. 2022 Successor Equity Incentive Plan, or the 2022 Plan, which will be a successor to and continuation of the Initial Plan and will become effective upon the effective date of the registration statement of which this prospectus forms a part. Upon the effectiveness of the 2022 Plan, it will replace the Initial Plan, except with respect to awards outstanding under the Initial Plan, and no further awards will be available for grant under the Initial Plan.

Subject to certain adjustments, the maximum number of shares of common stock that could have been issued under the Plans in connection with awards was 2,000,000 shares, of which 810,000 remained available for issuance as September 30, 2022. The maximum number of shares of common stock that may be issued under the 2022 Plan will automatically increase on January 1 of each calendar year for a period of ten years commencing on January 1, 2023 and ending on (and including) January 1, 2032, to a number of shares of common stock equal to 20% of the total number of shares of common stock outstanding on December 31 of the preceding calendar year; provided, however

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 10. Equity-Based Compensation (cont.)

that the board of directors, or the compensation committee, may act prior to January 1 of a given calendar year to provide that the increase for such year will be a lesser number of shares of common stock. All available shares may be utilized toward the grant of any type of award under the 2022 Plan.

The Company measures its stock-based awards granted to employees, consultants and directors based on the estimated fair values of the awards and recognizes the compensation over the requisite service period. The Company uses the Black-Scholes option-pricing model to estimate the fair value of its stock option awards. Stock-based compensation is recognized using the straight-line method. As the stock compensation expense is based on awards ultimately expected to vest, it is reduced by forfeitures. The Company accounts for forfeitures as they occur.

Weighted average assumptions used in the Black-Scholes-Merton model are set forth below:

	September 30, 2022
Risk-free interest rate	2.98% – 4.10%
Dividend yield	—
Expected term (years)	5.27 – 5.81
Volatility	62.4% – 62.5%

On July 11, 2022, the Company granted stock options to purchase 650,000 shares of common stock at an exercise price of $0.64 per common share. The stock option grants were issued to directors, members of the Scientific Advisory Board, and other consultants to the Company and vest in periods ranging over one to three years.

On August 18, 2022, the Company issued stock options to purchase 300,000 shares of common stock to our Acting Chief Medical Officer. The options have an exercise price of $0.64 per common share and will vest quarterly over a period of three years, subject to certain adjustments.

On August 18, 2022, the Company granted stock options to purchase 100,000 shares of common stock at an exercise price of $0.64 per common share. The stock option grants were issued to members of the Scientific Advisory Board and vest over three years.

Activity under the Plans for the period from January 25, 2022 (inception) to September 30, 2022 is set forth below:

	Number Outstanding	Weighted-Average Exercise Price Per Share	Weighted-Average Remaining Contractual Life (Years)	Aggregate Intrinsic Value
Outstanding at January 25, 2022	—	$ —	—	$ —
Granted	1,050,000	0.64	9.82	4,578,000
Exercised	—	—	—	—
Canceled/forfeited/expired	—	—	—	—
Outstanding at September 30, 2022	1,050,000	$ 0.64	9.82	$ 4,578,000
Options vested and exercisable at September 30, 2022	33,333	$ 0.64	9.78	$ 145,333
Options vested and expected to vest as of September 30, 2022	1,050,000	$ 0.64	9.82	$ 4,578,000

The weighted average grant date fair value of options granted to date was $1.02. At September 30, 2022, the Company had $1,012,866 of unrecognized stock-based compensation expense related to stock options which will be recognized over the weighted average remaining requisite service period of 2.4 years. The Company settles employee stock option exercises with newly issued shares of common stock.

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 10. Equity-Based Compensation (cont.)

On July 11, 2022, the Company issued restricted stock of 46,875 shares of common stock to a consultant. These shares fully vested on October 1, 2022.

On August 18, 2022, the Company issued 40,000 shares of common stock to a consultant. These shares were fully vested upon the date of grant.

On August 18, 2022, the Company issued 40,000 shares of common stock to Steven Zelenkofske, a director, for recruiting services performed. These shares were fully vested upon the date of grant.

On September 16, 2022, the Company issued 50,000 shares of common stock to John Murphy, a director nominee. These shares were fully vested upon the date of grant.

Total stock compensation expense for the three months ended September 30, 2022 and the period from January 25, 2022 (inception) to September 30, 2022 was $478,684.

The allocation of stock-based compensation for the periods presented below was as follows:

	Three Months Ended September 30, 2022	January 25, 2022 (Inception) to September 30, 2022
General and administrative	$ 435,642	$ 435,642
Research and development	43,042	43,042
Total stock-based compensation	$ 478,684	$ 478,684

Note 11. Net Loss Per Share

The following table sets forth the computation of the basic and diluted net loss per common share:

	Three Months Ended September 30, 2022	January 25, 2022 (Inception) to September 30, 2022
Numerator:		
Net loss	$ (1,026,114)	$ (1,704,902)
Denominator:		
Weighted average common shares outstanding	8,037,218	7,778,778
Net loss per share, basic and diluted	$ (0.13)	$ (0.22)

Since the Company was in a loss position for the periods presented, basic net loss per share is the same as diluted net loss per share as the inclusion of all potential dilutive securities would have been anti-dilutive. For the three months ended September 30, 2022 and for the period from inception to September 30, 2022, there were no potential dilutive securities other than the convertible notes, stock options, and warrants.

Note 12. Subsequent Events

The Company has evaluated events that occurred through December 5, 2022, the date that the financial statements were issued and determined that there have been no events that have occurred that would require adjustments to our disclosures in the financial statements except for the transactions described below:

On October 3, 2022, the Company issued 137,000 shares of common stock to various consultants for services rendered. These shares were fully vested upon the date of grant. 70,000 of the shares were subsequently returned and cancelled.

CADRENAL THERAPEUTICS, INC.
Notes to Unaudited Financial Statements

Note 12. Subsequent Events (cont.)

On October 3, 2022, the Company issued stock options to purchase 50,000 shares of common stock to a director nominee. The options have an exercise price of $5.00 per common share and will vest over a three-year period, with 33.33% vesting on the first anniversary of the effective date, and thereafter 1/36 on a monthly basis.

On November 30, 2022, the Company closed a $250,000 note offering (the "November Private Placement") pursuant to which the Company sold to two accredited investors units consisting of (i) notes in the aggregate principal amount of $250,000, which bear interest at the rate of 10%, repayable at the time of the completion of this offering (the "November Notes") and (ii) warrants to purchase up to an aggregate of 250,000 shares of common stock exercisable at $1.00 per share which are automatically exercisable into shares of our common stock upon the consummation of this offering (the "November Warrants"). The principal amount of the November Notes will be repaid upon the consummation of this offering out of the proceeds of the November Warrants exercise. The issuance of the November Warrants triggered an adjustment to the conversion price of the Private Placement Notes to $1.00 per share. As a result of the adjustment of the conversion price of the Private Placement Notes, the five-year warrants issued to Boustead Securities, LLC in connection with the Private Placement, were amended to provide for the purchase of an aggregate of 11,500 shares of the Company's common stock at an exercise price equal to 60% of the initial public offering price per share.

On December 5, 2022, the Company filed an Amended and Restated Certificate of Incorporation to increase the authorized capital stock of the Company to 82,500,000 shares, consisting of 75,000,000 shares of common stock, par value $0.001 per share, and 7,500,000 shares of preferred stock, par value $0.001 per share, which was approved by the Company's Board of Directors, as well as a majority of the Company's shareholders, on December 5, 2022.

On December 5, 2022, the Company's Amended and Restated Bylaws, which were approved by the Company's Board of Directors on December 5, 2022, became effective upon the filing and effectiveness of the Company's Amended and Restated Certificate of Incorporation.

In December 2022, the Company entered into amendments to the March 2022 Note and the June 2022 Note to adjust the conversion price of the March 2022 Note and the June 2022 Note to $1.00 per common share.

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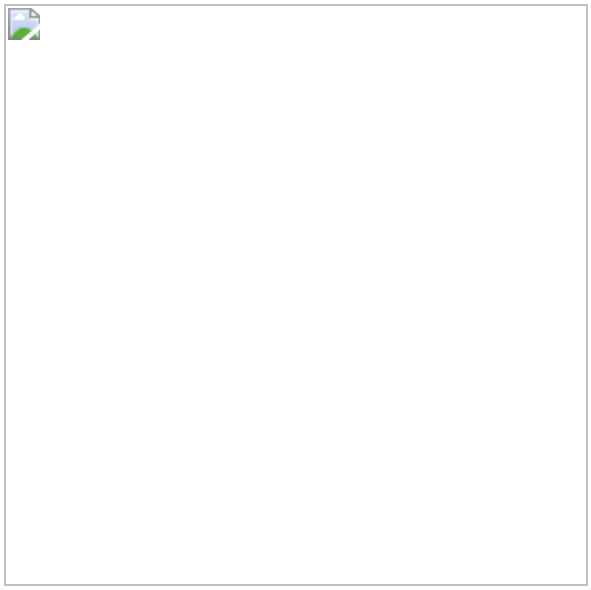

Cadrenal Therapeutics, Inc.

1,400,000 Shares

Common Stock

PROSPECTUS

Boustead Securities, LLC **Sutter Securities, Inc.**

Revere Securities LLC

January 19, 2023

Through and including February 13, 2023 (the 25th day after the date of this prospectus), all dealers effecting transactions in these securities, whether or not participating in this offering, may be required to deliver a prospectus. This is in addition to a dealer's obligation to deliver a prospectus when acting as an underwriter and with respect to an unsold allotment or subscription.